# INVARIANT RINGS OF ADJACENT-QUADRATIC TRIANGULAR DERIVATIONS

A PREPRINT

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## ABSTRACT

Let $k$ be a field of characteristic zero and $D_5$ the triangular locally nilpotent derivation $D_5(a_n) = a_{n+1}a_{n+2}$ on $A_5 = k[a_0, \dots, a_5]$. We prove

$$\boxed{A_5^{D_5} = C_5[F_{46}^{\mathrm{int}}],}$$

hence finitely generated, where $C_5 = k[p, q, J_1, J_2, F_0, H]$ is the normal hypersurface $q^4H - 11025F_0^2 - 105q^2J_2^5J_1 - 9J_2^7 = 0$. Deleting $a_0$ gives the polynomial kernel $B^{D_5} = k[p, q, J_1, J_2]$, whose localization at $pq$ is the polynomial line $K_{pq}[W]$. Global descent fails twice, each time for an explicit reason. First, the weight-24, $a_0$-degree-2 invariant $H$ lies outside $k[p, q, J_1, J_2, F_0]$, and weight 24 is the first such failure. Second, the weight-46, $a_0$-degree-4 invariant $F_{46}^{\mathrm{int}}$ lies outside $C_5$, and weight 46 is the first such failure, with equality of graded pieces through weight 45. Both failures come from algebraic residue initial forms at the primes dividing the pole-clearing denominator; this mechanism is isolated in a general multfloor valuation bound, sharp in both instances. A second slice chart from the $q$-primitive plinth element $f_{13} = pJ_2^3/35 + pq^2J_1J_2/4$ covers the locus where the first chart degenerates, and exact $q$- and $p$-saturation contract the resulting $p$-localized kernel to the closed ring. The closed ring is a codimension-two complete intersection with exact Hilbert series, torsion filtration, singular locus, and normality; the sharp $p$-adic top envelope is $\lceil 3m/2 \rceil$. The pole-cancellation mechanism is compared with the Daigle–Freudenburg example. At $N = 6$ the $B$-level ring is exact by transport while the $A_6$ ring remains open.

***Keywords*** locally nilpotent derivation · invariant ring · triangular derivation · Hilbert's fourteenth problem · formal verification

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# 1. Introduction

## 1.1 The derivation and the main theorem

Over a characteristic-zero field $k$, with $A_N = k[a_0, \dots, a_N]$ (Krull dimension $N+1$) and boundary conditions $a_{N+1} = a_{N+2} = 0$, the adjacent-quadratic derivation

$$D_N(a_n) = a_{n+1}a_{n+2}, \qquad 0 \le n \le N,$$

is locally nilpotent and triangular. Its kernel $A_N^{D_N}$ is the ring of invariants of the corresponding $\mathbb{G}_a$-action. We determine the first case whose global structure is nontrivial:

$$\boxed{A_5^{D_5} = C_5[F_{46}^{\mathrm{int}}],}$$

finitely generated (Theorem 13.5). The proof follows a single local-to-global architecture, summarized by

$$\boxed{K \;\longrightarrow\; K[F_0] \;\longrightarrow\; C_5 \;\longrightarrow\; R = C_5[F_{46}^{\mathrm{int}}] \;\longrightarrow\; A_5^{D_5},}$$

where $K$ is the $a_0$-free kernel, $F_0$ the first global pole-clearing invariant, $C_5$ the corrected model after the first obstruction, and $R$ the final model after the second. Each arrow is forced by an explicit obstruction or contraction, as described below.

## 1.2 Why $N = 5$ is the first hard case

At $N \le 4$ no hidden direction is detected: the $N = 2$ kernel is polynomial, and $N = 3, 4$ admit explicit polynomial invariants with open-set normal forms (Theorems 3.2, 3.3). At $N = 5$, deleting $a_0$ gives $B = k[a_1, \dots, a_5]$ with explicitly determined kernel $B^{D_5} = k[p, q, J_1, J_2]$ (Theorem 4.4), and localization at $pq$ yields the polynomial local form $A_{pq}^{D_5} = K_{pq}[W]$. The entire subtlety lies in restoring the divisor $pq = 0$: the localized picture does not descend naively, and each natural global model fails at a computable obstruction.

## 1.3 The two obstructions

The first restoration attempt, the five-generator algebra $K[F_0]$, fails at weight $24$: the explicit invariant $H$ has $a_0$-degree $2$ where the model allows at most $1$. The failure is a mechanism, not an accident of computation — algebraic residue cancellation at the pole primes, $r_q^2 \in L_q$ — which explains exactly why the naive saturation cannot close. Adjoining the forced generator gives the hypersurface $C_5$, and the correction is sharp: exact ranks agree with the full kernel through weight $45$. Yet $C_5$ still fails, at weight $46$ and $a_0$-degree $4$, where $F_{46}^{\mathrm{int}}$ lies outside the top ideal $J_4$. At $N = 5$, $H$ is therefore the first proved hidden generator for the $K[F_0]$-model, and $F_{46}^{\mathrm{int}}$ the second for the $C_5$-model (the $N = 6$ hidden direction $H_{20}$ is a separate $B_6$-level phenomenon).

## 1.4 Closure by two charts and saturation

With $R = C_5[F_{46}^{\mathrm{int}}]$ identified, generator discovery is over; what remains is proving that no further generator exists. A second slice chart, supplied by the $q$-primitive plinth element $f_{13}$ outside $(pq)K$, identifies the $p$-localized kernel together with the first chart; exact $q$- and $p$-saturation contract it back to $R$. No universal $a_0$-top bound is needed.

## 1.5 Structure of the closed ring

The closed ring is then determined as an exact object: a codimension-two complete intersection with explicit Hilbert series, torsion filtration, normality, and singular locus. Its top theory — Q-top-strong established, the $2m$ $p$-bound refuted, the sharp $\lceil 3m/2 \rceil$ envelope proved — separates what valuation methods can and cannot do. The residue mechanism behind the two hidden directions generalizes to the multfloor transfer theorem; the Daigle–Freudenburg comparison and the escape/climb/degeneration vocabulary interpret the proof rather than participate in it.

### 1.6 Relation to prior work, status, and the $N = 6$ frontier

The natural comparison is the Daigle–Freudenburg dimension-five example [5], whose invariant ring is not finitely generated. The present $N = 5$ derivation acts on six variables, and the comparison is one of mechanisms, not equivalence: triangularity and dimension alone do not make the two examples equivalent. We are not aware of prior work treating this specific adjacent-quadratic family in the form studied here; Fibonacci weights are used only as a descriptive grading. At $N = 6$ the $B$-level structure is exact by transport, while the $A_6$ denominator module is certified through total degree 13 (degree 14 computationally blocked, not claimed) and the $A_6$ ring itself remains open (Appendix A ff.).

**Status convention.** "Established" denotes a mathematical statement proved in the paper; "exact computation" a reproducible characteristic-zero calculation with stated range; "Lean-verified" the narrower set of claims formally checked in the stated Lean environment; "Refuted" an explicit counterexample; "Open", "Conjectural", or "Blocked" a statement not proved here. Theorems whose proof cites an exact Sage elimination/saturation log are proved modulo the stated machine certificate; the log path, timing, and Gröbner-basis size are recorded at the use site. The status matrix (Table 3) is the authoritative ledger: section prose states mathematical content directly, and explicit status language is reserved for genuine boundaries — what is proved, what is refuted, and what remains open. Exploratory computations never serve as premises for established theorems.

The paper follows the dependency of the proof rather than the chronology of experimentation: the family and its base cases; the $a_0$-free kernel and localized slice; the first obstruction and the residue mechanism it reveals; the corrected hypersurface; the top-ideal toolkit; the second obstruction; the plinth element that supplies a second chart; the two-chart closure; the exact algebra of the closed ring and its top theory; the computational trust model; then, as consequences and context rather than premises, the transfer theorem, the literature comparison, the mechanism vocabulary, the open conjectures, and the $N = 6$ frontier. Appendices collect the status ledger, the failed valuation routes, the historical saturation arguments, certificate mechanics, and the rank tables. Failed routes appear only where the failure is itself the obstruction being proved.

**Supplementary Material.** The computational verification scripts, output logs, and the Lean 4 formalization are provided as supplementary material accompanying this submission. All CAS verifications use exact rational arithmetic over $\mathbb{Q}$; all finite-field computations (prime $\ell = 1000003$) are consistency checks only and are not used as proofs. The Lean 4 formalization uses core Lean (without Mathlib); identities are discharged by native evaluation via `native_decide`, so each such theorem depends on the standard `propext`, `Classical.choice`, `Quot.sound` plus one `_native.native_decide` axiom per theorem (see the formal-verification manifest for exact `#print axioms` outputs). The supplement index maps each theorem to its certifying script, log, witness, and Lean check (`SUPPLEMENT_README.md`), with the computational environment in `ENVIRONMENT` and SHA-256 hashes of all scripts and logs in `CHECKSUMS`. The decisive closure markers reproduce with `sh scripts/reproduce_closure.sh`. Repository link: `https://github.com/AhmedMostafa16/adjquad` (convenience mirror; the submission ships the full scripts/logs/checksums, so closure does not depend on repository availability). Pinned publication commit: `e1f87c96efc8da07f23550b4b41c46844f89814b`.

## 2. The adjacent-quadratic family

Let $k$ be a field of characteristic zero, $A_N = k[a_0, \dots, a_N]$, with boundary conditions $a_{N+1} = a_{N+2} = 0$.

**Definition 2.1** (Family)**.** The adjacent-quadratic shift is the discrete map $P_{\lambda,N} \colon A_N \to A_N$ given on coordinates by

$$P_{\lambda,N}(a)_n = a_n + \lambda a_{n+1} a_{n+2}, \qquad 0 \le n \le N.$$

We identify the point map with its pullback automorphism on $A_N$ and write $P_{\lambda,N}$ (resp. $P^*_{\lambda,N}$) where the point (resp. ring) reading is meant; bare $P$ defaults to the point map. The associated triangular derivation is $D_N(a_n) = a_{n+1}a_{n+2}$, with flow $\Psi_{t,N} = \exp(tD_N)$.

**Lemma 2.2** (Triangular inverse)**.** *The adjacent-quadratic shift $P_{\lambda,N}$ is a triangular automorphism of $A_N$. Given $b = P_{\lambda,N}(a)$, the inverse is computed by descending recursion $a_N = b_N$, $a_{N-1} = b_{N-1}$, and $a_n = b_n - \lambda a_{n+1}a_{n+2}$ for $n = N - 2$ down to 0. Its Jacobian matrix is unitriangular, so $\det DP_{\lambda,N} = 1$; it fixes the top two coefficients $a_{N-1}$ and $a_N$. (It does not preserve total degree: $P^*_{\lambda,N}(a_0) = a_0 + \lambda a_1 a_2$ has total degree 2 for $\lambda \neq 0$.)*

*Remark* 2.3 (Observation: fixed-point variety, not a theorem). For $\lambda \neq 0$ (and $N \geq 2$; at $N = 1$ the map is the identity and at $\lambda = 0$ every point is fixed), the fixed-point locus of $P_{\lambda,N}$ is cut out by the products $a_1a_2, \ldots, a_{N-1}a_N$, that is $V(a_1a_2, \ldots, a_{N-1}a_N)$, which has the combinatorial shape of the Stanley–Reisner variety of a path. This is recorded as an observation guiding later sections, not as a theorem.

*Remark* 2.4 (Characteristic zero). Every result is over a field of characteristic zero, and the hypothesis is load-bearing: the invariant $F_0$ has denominators $3, 5, 35$, the Jacobian minors contribute scalars $4$ and $420 = 2^2 \cdot 3 \cdot 5 \cdot 7$, the slice theorem and factorial-closedness arguments require characteristic zero, and the multfloor proof divides by binomial coefficients. No positive-characteristic statement for $N \geq 4$ is claimed here.

**Proposition 2.5** (Discrete-versus-continuous defect)**.** *The discrete map $P_{\lambda,N}$ is the first-order coordinate update of the flow $\Psi_{t,N}$ evaluated at $t = \lambda$. For $N \geq 3$ and $\lambda \neq 0$ it is never the exponential:*

$$P_{\lambda,N} \neq \exp(\lambda D_N).$$

*(The bound is sharp: at $N = 1$, $D_1 = 0$ so both sides are the identity; at $N = 2$, $D_2^2 = 0$ on generators, so $\exp(\lambda D_2)$ and $P_{\lambda,2}$ agree as algebra automorphisms — both send each generator $a_i$ to $a_i + \lambda D_2(a_i)$, which determines each map — even though $D_2^2 \neq 0$ as an operator, e.g. $D_2^2(a_0^2) = 2(a_1a_2)^2$.) The divergence appears at order $\lambda^2$: at $N = 3$ the gap is the explicit term $(\lambda^2/2)a_2^2a_3$, and at $N = 4$ an explicit nonzero $\lambda^2$-coefficient gives a counterexample to equality, detailed in the $N \leq 4$ base section to follow.*

*Remark* 2.6 (Hadamard form). With $U$ the forward shift on coefficient sequences, the update can be written $P[f] = f + \lambda(Uf) \odot (U^2 f)$, where $\odot$ is the Hadamard (entrywise) product. The Hadamard product is classical; this rewriting is a remark only.

**Notation**

$V_w$ is the Fibonacci-weight-$w$ homogeneous component of the ambient polynomial ring, so that the invariant piece is $(A_N^{D_N})_w = \ker(D_N|_{V_w})$. The series $s_6(w)$ counts monomials of weight $w$ in the six generators (Section 6); $s_5(w)$ and $s_K(w)$ are the analogous counts in the five $K[F_0]$-generators and in $K = k[p, q, J_1, J_2]$ respectively. Pole orders along prime divisors are $e_i = \mathrm{ord}_{\pi_i}(c)$ in the saturation procedure, with $e_p = 2$ and $e_q = 4$ the $p$- and $q$-pole orders of $\rho$; $e_m$ with $m := N - 2$ is the degree recurrence of Conjecture 21.19, and $e_s$ counts directed paths in Conjecture 21.15. Denominators are $c = p^2q^4$ at $N = 5$, $c_6 = a_5^2a_6^4$ and $c' = c_6^2$ at $N = 6$.

**Notation table (single stable meaning per symbol)**

**Table 1. Notation with a single stable meaning per symbol.**

| symbol | meaning, first defined |
|---|---|
| $A_N = k[a_0, \dots, a_N]$, $a_{N+1} = a_{N+2} = 0$ | ambient ring (Definition 2.1) |
| $D_N(a_n) = a_{n+1}a_{n+2}$ | adjacent-quadratic derivation |
| $B = k[a_1, \dots, a_5]$, $x = a_3, y = a_2, z = a_1$ | five-variable base (§4) |
| $K = k[p, q, J_1, J_2] = B^{D_5}$ | base kernel (Theorem 4.4) |
| $p = a_4$, $q = a_5$ | invariant variables, Fibonacci weight 1 |
| $J_1 = 2pz - y^2$, $J_2 = 2qy - x^2$ | kernel generators, weights $6, 4$ |
| $F_0$, $c = p^2q^4$, $G_0 = F_0 - ca_0$ | fifth invariant, denominator, $B$-part (Definition 5.1) |
| $W = F_0/c = a_0 + \rho$ | invariant pole element (not a slice); $\rho = G_0/c$ |
| $s = x/(pq)$ | local slice, $D_5(s) = 1$ (Lemma 4.2) |
| $H$ ($H_B$ at $N = 6$) | hidden invariant, weight 24, $a_0$-degree 2 |
| $C_5 = k[p, q, J_1, J_2, F_0, H]$ | six-generator subalgebra (Theorem 7.1) |
| $R = C_5[F_{46}^{\mathrm{int}}]$, $E = p^2F_{46}^{\mathrm{int}}$, $r = p^2T - E$ | closed ring and torsion relation (Theorems 10.7, 14.2); $R$ never denotes $K$ |
| $S$, $\mathrm{wt}(S) = 28$ | weight-28 relation cutting out $C_5$ |
| $J_m = (p^{2m}q^{4\lceil m/2 \rceil})$ | top ideal at $a_0$-degree $m$ (Theorem N.4) |
| $F_{46}^{\mathrm{int}}, F_{46}^{\mathrm{rat}} = F_{46}^{\mathrm{int}}/26460$ | second witness, weight 46, $a_0$-degree 4; $T \mapsto F_{46}^{\mathrm{int}}$ (Theorem 10.3) |
| $G_1, \dots, G_5$, $S_B$, $H_{20}$, $F_0^{(6)}$, $D_2$ | $N = 6$ generators/relation/directions (§§12–18) |
| $v_p, v_q$ ($v_5, v_6$ at $N = 6$) | prime-divisor valuations on $\mathrm{Frac}(B)$ |
| $\deg_{\mathrm{ord}}$, wt | ordinary total degree; Fibonacci weight $(8, 5, 3, 2, 1, 1)$ on $(a_0, \dots, a_5)$, extended by $\mathrm{wt}(a_0) = 13$ at $N = 6$ (so $(13, 8, 5, 3, 2, 1, 1)$ on $(a_0, \dots, a_6)$) |
| $\deg_{a_0}$ | $a_0$-degree (exact on $B[a_0]$; tops multiply); $W$-degree coincides with it ($W = a_0 + \rho$ monic) |
| $s_6, s_5, s_K$ | generator-monomial counts of weights $(1, 1, 6, 4, 14, 24)$, five, $K$ |
| $L_w$ | denominator-clearing l.c.m. for weight-$w$ certificates |
| $m$ ($W$-/$a_0$-degree), $j$ ($W$-index) | filtration indices; $m := N - 2$ only inside Conjecture 21.19 |
| $\rho_q, \rho_p$ | per-prime resonance degrees (Definition 21.1) |

The words slice (only $s$, with $D_5(s) = 1$), pole element (only $W$, with $D_5(W) = 0$), and denominator (always with its formula: $c$, $c_6$, $c'$, $L_w$) are never interchanged.

## 3. The $N \leq 4$ base

Throughout, $k$ has characteristic zero unless a finite field is stated explicitly, and $P_{\lambda,N}$ denotes the adjacent-quadratic shift of Definition 2.1. The discrete map is the first-order coordinate update of the flow $\Psi_{t,N} = \exp(tD_N)$; for $N \geq 3$ and $\lambda \neq 0$ it is never the exponential itself (Proposition 2.5):

$$P_{\lambda,N} \neq \exp(\lambda D_N),$$

and the two diverge at order $\lambda^2$ in every continuous case below.

**Proposition 3.1** ($N = 2$ kernel)**.** *$A_2^{D_2} = k[a_1, a_2]$, a polynomial ring.*

*Proof.* At $N = 2$, $a_3 = a_4 = 0$, so $D_2$ kills $a_1$ and $a_2$ while $D_2(a_0) = a_1a_2 \neq 0$. Write $F = \sum_{k=0}^{m} C_k a_0^k$ with $C_k \in k[a_1, a_2]$; freeness gives $D_2(F) = \sum_{k \geq 1} kC_k a_1 a_2 a_0^{k-1}$, since $D_2$ vanishes on the coefficients. In the domain $A_2$, $mC_m a_1 a_2 = 0$ with $a_1a_2 \neq 0$ forces $C_m = 0$ for $m \geq 1$ (characteristic zero), so $F = C_0 \in k[a_1, a_2]$. □

**Theorem 3.2** ($N = 3$ invariants and Euler gap)**.** *For $N = 3$, with $a_4 = a_5 = 0$, the derivation acts by $D_3(a_0) = a_1a_2$, $D_3(a_1) = a_2a_3$, $D_3(a_2) = D_3(a_3) = 0$.*

1. *The rational function*
$$H_3 = a_0 - \frac{a_1^2}{2a_3}$$
*satisfies $D_3(H_3) = 0$; it is a continuous invariant of the flow $\Psi_{t,3}$ on the open set $a_3 \neq 0$. (The subscript distinguishes this $N = 3$ rational invariant from the weight-24 polynomial hidden invariant $H$ of the $N = 5$ theory.)*

2. *The discrete map fixes $a_2$ and $a_3$ and sends $b_0 = a_0 + \lambda a_1 a_2$, $b_1 = a_1 + \lambda a_2 a_3$. The rational function*
$$I_0 = a_0 - \frac{a_1^2}{2a_3} + \frac{\lambda}{2} a_1 a_2$$
*satisfies $I_0 \circ P_{\lambda,3} = I_0$ identically on $a_3 \neq 0$, and $I_0 \to H_3$ as $\lambda \to 0$.*

3. *Since $D_3^2(a_0) = a_2^2 a_3$ and $D_3^3(a_0) = 0$, the exact flow on the $a_0$-coordinate is $\Psi_{\lambda,3}(a_0) = a_0 + \lambda a_1 a_2 + \frac{\lambda^2}{2} a_2^2 a_3$, while $P_{\lambda,3}(a_0) = a_0 + \lambda a_1 a_2$. The Euler gap is exactly*
$$\Psi_{\lambda,3}(a_0) - P_{\lambda,3}(a_0) = \frac{\lambda^2}{2} a_2^2 a_3.$$

**Theorem 3.3** ($N = 4$ invariants and commutator)**.** *For $N = 4$, $D_4(a_0) = a_1 a_2$, $D_4(a_1) = a_2 a_3$, $D_4(a_2) = a_3 a_4$, $D_4(a_3) = D_4(a_4) = 0$.*

1. *The polynomials $2a_0 a_3 - a_1^2$ and $2a_1 a_4 - a_2^2$ satisfy $D_4(2a_0 a_3 - a_1^2) = D_4(2a_1 a_4 - a_2^2) = 0$; they are continuous invariants of $\Psi_{t,4}$.*

2. *On the open set $a_3 a_4 \neq 0$, with $u = a_2$ and $c = \lambda a_3 a_4 \neq 0$, the rational function*
$$I_1 = a_1 - \frac{a_2^2}{2a_4} + \frac{\lambda}{2} a_2 a_3$$
*satisfies $I_1 \circ P_{\lambda,4} = I_1$ identically. Its $\lambda \to 0$ limit is the continuous invariant $a_1 - a_2^2/(2a_4)$.*

3. *Let $\Delta_c F(u) = F(u + c) - F(u)$ and let*
$$S_1(u) = \frac{u^2 - cu}{2c}, \qquad S_2(u) = \frac{u^3}{3c} - \frac{u^2}{2} + \frac{cu}{6}, \qquad S_3(u) = \frac{u^4}{4c} - \frac{u^3}{2} + \frac{cu^2}{4},$$
*so that $\Delta_c S_k(u) = u^k$ for $k = 1, 2, 3$. Then*
$$I_0 = a_0 - \lambda I_1 S_1(a_2) - \frac{\lambda}{2a_4} S_3(a_2) + \frac{\lambda^2 a_3}{2} S_2(a_2)$$
*satisfies $I_0 \circ P_{\lambda,4} = I_0$ identically on $a_3 a_4 \neq 0$; the degenerate loci $a_3 a_4 = 0$ are excluded here.*

4. *The derivation orbit of $a_0$ is $D_4^2(a_0) = a_2^2 a_3 + a_1 a_3 a_4$, $D_4^3(a_0) = 3a_2 a_3^2 a_4$, $D_4^4(a_0) = 3a_3^3 a_4^2$, $D_4^5(a_0) = 0$, so $\exp(\lambda D_4)(a_0)$ carries nonzero $\lambda^2$-, $\lambda^3$-, and $\lambda^4$-terms while $P_{\lambda,4}(a_0) = a_0 + \lambda a_1 a_2$.*

5. *Writing $[P_{\lambda,4}, P_{\mu,4}]_0$ for the ordered point difference $(P_{\mu,4} \circ P_{\lambda,4} - P_{\lambda,4} \circ P_{\mu,4})_0$,*
$$[P_{\lambda,4}, P_{\mu,4}]_0 = \lambda\mu(\lambda - \mu) a_2 a_3^2 a_4,$$
*and components $1$ through $4$ of the two compositions agree. The reverse ordering gives the negation.*

*All identities above are verified by exact computation (see Supplementary Material).*

**Lemma 3.4** (Discrete-to-continuous bridge)**.** *Work in characteristic $\neq 2, 3$. Let $U$ be a nonempty open set on which all denominators below are nonzero, and let $I_\lambda$, rational in $(\lambda, a)$ and regular at $\lambda = 0$ on $U$ after clearing denominators, satisfy $I_\lambda \circ P_{\lambda,N} = I_\lambda$ on $U$ as a rational identity. Let $H = I_\lambda|_{\lambda=0}$ on $U$. Then $D_N(H) = 0$ on $U$.*

*Proof.* Clear denominators to a polynomial identity over $k[\lambda, a, \mathrm{denom}^{-1}]$ and differentiate formally in $\lambda$ (a $k$-linear derivation; the divisions by $2, 3$ in $S_1, S_2, S_3$ use the hypothesis), then set $\lambda = 0$. With $P_0 = \mathrm{id}$ and formal derivative $\frac{d}{d\lambda}|_0 P_{\lambda,N}(a)_n = a_{n+1} a_{n+2} = D_N(a_n)$, the formal chain rule gives $(\partial_\lambda I)_0 + dH(D_N) = (\partial_\lambda I)_0$, hence $D_N(H) = 0$. No topology or limits are used. Theorems 3.2 and 3.3 are the instances: the $N = 3$ invariant is polynomial in $\lambda$, and the $N = 4$ cleared forms $\lambda S_1 = u^2/(2a_3 a_4) - \lambda u/2$, $\lambda S_3/(2a_4) = u^4/(8a_3 a_4^2) - \lambda u^3/(4a_4) + \lambda^2 u^2 a_3/8$ and $\lambda^2 S_2$ vanishes at $\lambda = 0$ are regular at $\lambda = 0$ with limit $L = a_0 - a_1 a_2^2/(2a_3 a_4) + a_2^4/(8a_3 a_4^2)$ satisfying $D_4(L) = 0$. □

**Theorem 3.5** (Finite-field $N = 3$ cycles)**.** *Let $P_{\lambda,3}$ act over a finite field $\mathbb{F}_q$ of characteristic $p$, with $\lambda \neq 0$ and $c = \lambda a_2 a_3$. Then $a_3' = a_3$, $a_2' = a_2$, $a_1' = a_1 + c$, $a_0' = a_0 + \lambda a_1 a_2$, and:*

1. *If $c = 0$, then $a_1$ is fixed; the orbit length is $1$ if $\lambda a_1 a_2 = 0$ and exactly $p$ otherwise, since the additive group of $\mathbb{F}_q$ has exponent $p$.*
2. *If $c \neq 0$ and $p$ is odd, the orbit length is exactly $p$: over one full $a_1$-period the accumulated shift of $a_0$ is $\lambda a_2(pa_1 + c\,p(p-1)/2) = 0$.*
3. *If $c \neq 0$ and $p = 2$, the orbit length is exactly $4$: over two steps the $a_0$-shift is $\lambda^2 a_2^2 a_3 \neq 0$, and doubling it gives $0$ in characteristic $2$.*

*The exact counts for $\lambda \neq 0$, $q = p^f$, are: $q^3 + q(q-1)$ fixed points; $q(q-1)^2/p$ cycles of length $p$ from the $c = 0$ stratum; and $(q-1)^2 q^2/L$ cycles of length $L$ from the $c \neq 0$ stratum, with $L = p$ for $p$ odd and $L = 4$ for $p = 2$. The classification is proved by the algebraic argument above for every finite field. Exhaustive enumeration over the tested fields $\mathbb{F}_3$, $\mathbb{F}_5$, $\mathbb{F}_7$, $\mathbb{F}_{11}$, $\mathbb{F}_2$, and $\mathbb{F}_4$ confirms the lengths and counts on those fields; enumeration over tested fields does not imply all-$\mathbb{F}_q$ results beyond the proof.*

Below $N = 5$ nothing hides. The next section computes the $a_0$-free base on which everything turns.

## 4. The $N = 5$ five-variable kernel

This section proves the five-variable kernel, the local slice, and descent. The localized $A$-presentation $A_{pq}^{D_5} = K_{pq}[W]$ is proved in Section 5 once $F_0$ is constructed. We work over the characteristic-zero field $k$ of Definition 2.1, with $A_5 = k[a_0, \dots, a_5]$ and $D_5(a_n) = a_{n+1}a_{n+2}$. This section treats the five-variable subproblem obtained by deleting $a_0$. Put

$$x = a_3, \qquad y = a_2, \qquad z = a_1, \qquad p = a_4, \qquad q = a_5,$$
$$B = k[x, y, z, p, q] = k[a_1, \dots, a_5], \qquad K = B^{D_5},$$

with

$$D_5(x) = pq, \qquad D_5(y) = px, \qquad D_5(z) = xy, \qquad D_5(p) = D_5(q) = 0.$$

Let

$$J_1 = 2pz - y^2 = 2a_1a_4 - a_2^2, \qquad J_2 = 2qy - x^2 = 2a_2a_5 - a_3^2,$$

and write $K_0 = k[p, q, J_1, J_2]$ for the subalgebra they generate (henceforth $K = K_0$ once Theorem 4.4 identifies it with $B^{D_5}$; the symbol $R$ is reserved for the closed ring $R = C_5[F_{46}^{\mathrm{int}}]$).

**Lemma 4.1** (Kernel generators)**.** *The four polynomials $p$, $q$, $J_1$, $J_2$ lie in $K$:*

$$D_5(p) = D_5(q) = D_5(J_1) = D_5(J_2) = 0,$$

*since $D_5(J_1) = 2xy \cdot p - 2y \cdot px = 0$ and $D_5(J_2) = 2pq \cdot x - 2x \cdot pq = 0$. Hence $K_0 \subseteq K$. Moreover $p$, $q$, $J_1$, $J_2$ are algebraically independent: the $4 \times 4$ Jacobian minor in columns $(a_1, a_2, a_4, a_5)$ is $4a_4a_5 \neq 0$ over $k$ (the scalar $4$ records the standing characteristic-zero hypothesis). So $K_0$ is a polynomial ring in four variables. All identities are verified by exact computation (see Supplementary Material).*

**Lemma 4.2** (Local slice and localized kernel)**.** *On the localization $B_{pq}$ the element*

$$s = \frac{x}{pq}$$

*satisfies $D_5(s) = 1$; it is a local slice. With $Y = J_2/(2q)$ and $Z = (J_1 + Y^2)/(2p)$, one has $D_5(Y) = D_5(Z) = 0$ and the reconstruction identities*

$$x = pqs, \qquad y = Y + \tfrac{1}{2}p^2qs^2, \qquad z = Z + \tfrac{1}{2}pqYs^2 + \tfrac{1}{8}p^3q^2s^4,$$

*checked by direct expansion in exact computation (see Supplementary Material). Hence $B_{pq} = K_{0,pq}[s]$, and the standard local-slice theorem for locally nilpotent derivations gives the localized kernel*

$$B_{pq}^{D_5} = K_{0,pq} = k[p, q, J_1, J_2, (pq)^{-1}].$$

*Slice theorem (Freudenburg, Algebraic Theory of Locally Nilpotent Derivations [11], Corollary 1.26: if $D$ is locally nilpotent on a domain $A$ over a characteristic-zero field and $D(s) = 1$, then $A = A^D[s]$ with $s$ transcendental over $A^D$). Hypotheses here: $B_{pq}$ is a domain, $D_5$ extends locally nilpotently, $s = x/(pq)$ satisfies $D_5(s) = 1$ as computed above. In particular every $f \in K$ has the form $f = P/(p^a q^b)$ for some $P \in K_0$ and $a, b \geq 0$ (independent exponents: $1/p = q \cdot (pq)^{-1}$ and $1/q = p \cdot (pq)^{-1}$ in $K_{0,pq}$).*

The descent from $K_{0,pq}$ to $K_0$ is controlled by the prime divisors $(p)$ and $(q)$. Since $B$ is a polynomial ring, hence a unique factorization domain, each defines a discrete valuation, still denoted $v_p$ and $v_q$, on $\mathrm{Frac}(B)$, with valuation rings

$$V_p = B_{(p)} = \{h \in \mathrm{Frac}(B) : v_p(h) \geq 0\}, \qquad V_q = B_{(q)} = \{h \in \mathrm{Frac}(B) : v_q(h) \geq 0\},$$

the localizations of $B$ at the prime ideals $(p)$ and $(q)$.

**Lemma 4.3** (Contraction with divisibility identities)**.** *Let $\varphi\colon k[p, q, U, V] \to B$ be the substitution map $(p, q, U, V) \mapsto (p, q, J_1, J_2)$, injective by Lemma 4.1. Then*

$$\varphi^{-1}((p)B) = (p), \qquad \varphi^{-1}((q)B) = (q).$$

*Indeed, reducing modulo $p$ gives $J_1 \equiv -y^2$ and $J_2 \equiv 2qy - x^2$, with the divisibility identities*

$$(J_1 + y^2)/p = 2z \in B, \qquad (J_2 + x^2)/q = 2y \in B.$$

*The map $\alpha\colon k[q, U, V] \to B/(p) = k[x, y, z, q]$, $(q, U, V) \mapsto (q, -y^2, 2qy - x^2)$, is injective: writing a relation as $\sum_{i=0}^{n} f_i(q, U)V^i$ with $V = 2qy - x^2 = -x^2 + 2qy$, the leading term in $x$ is $(-1)^n f_n(q, -y^2)x^{2n}$; $f_n(q, -y^2) \neq 0$ if $f_n \neq 0$, since $\sum_j g_j(q)(-y^2)^j = 0$ forces each $g_j = 0$ by $y$-degree, so $f_n = 0$ by induction on $n$. Thus $P(0, q, J_1, J_2) = 0$ in $B/(p)$ implies $P(0, q, U, V) = 0$ as a polynomial, so $p \mid P$ already in $K_0$. Modulo $q$, $J_1 \equiv 2pz - y^2$ and $J_2 \equiv -x^2$, and $\beta\colon k[p, U, V] \to B/(q) = k[x, y, z, p]$, $(p, U, V) \mapsto (p, 2pz - y^2, -x^2)$, is injective: write $\sum_{j=0}^{m} g_j(p, V)U^j$ with $U = 2pz - y^2$; the leading term in $z$ is $g_m(p, -x^2)(2p)^m z^m$, and $g_m(p, -x^2) \neq 0$ if $g_m \neq 0$ by $x$-degree, so induction on $m$ gives injectivity, whence $q \mid P$ in $K_0$. The independence of the two specialization maps is independently corroborated through total degree 5 in exact computation (see Supplementary Material); the negative checks $p \nmid J_1, J_2, q$ and $q \nmid J_1, J_2, p$ in $B$ show the divisors are genuine, e.g. $J_1/p \notin B$.*

**Theorem 4.4** (Descent $B^{D_5} = K$)**.** *With $B$, $K$, $K_0$ as above,*

$$\boxed{B^{D_5} = K = k[p, q, J_1, J_2].}$$

*Explicitly, the intersection obligation holds:*

$$B \cap K_{0,pq} = K_0 \quad \textit{(hence } B \cap K_{pq} = K \textit{ once } K_0 = K\textit{)}.$$

*Proof.* $K_0 \subseteq K$ is Lemma 4.1 and $K \subseteq K_{0,pq}$ is Lemma 4.2. Let $f \in K$, so $f = P/(p^a q^b)$ with $P \in K_0$. Since $f \in B$, we have $P = p^a q^b f \in B$. If $a \geq 1$ then $P \in (p)B$, so by Lemma 4.3 $P = pP_1$ with $P_1 \in K_0$; cancelling and repeating gives $P \in (p)^a K_0$. The same argument with $q$ gives $P \in (p)^a(q)^b K_0$ (here $p, q$ are nonassociated primes in the UFD $K_0$, so $p^a \mid P$ and $q^b \mid P$ imply $p^a q^b \mid P$), whence $f \in K_0$. Thus $K \subseteq K_0$, and $B \cap K_{0,pq} = K_0$ follows by the same descent applied to any $f \in B \cap K_{0,pq}$. □

**Lemma 4.5** (Dixmier divisibility for the $a_1a_2$-image)**.** *Let $\mu \in K$ and $\nu \in B$ satisfy $a_1a_2\mu + D_5(\nu) = 0$. Then $c = p^2q^4$ divides $\mu$ in $K$.*

*Proof.* Localize at $pq$: $s = x/(pq)$ has $D_5(s) = 1$ (Lemma 4.2), so $B_{pq} = K_{0,pq}[s]$ freely and $D_5 = d/ds$. Write $\nu = \sum_{i=0}^{t} N_i s^i$ (unique, $N_i \in K_{0,pq}$) and $a_1a_2 = \sum_{k\in\{0,2,4,6\}} e_k s^k$ with the exact $s$-coefficients $e_k \in K_{0,pq}$ certified by exact computation (see Supplementary Material; $e_k = 0$ otherwise). Comparing $s$-coefficients in $a_1a_2\mu + \nu' = 0$ (characteristic zero for division by $k + 1$) gives $\nu = N_0 - \mu T$ with the fixed integrator $T = \sum_k e_k s^{k+1}/(k + 1)$. Expanding $T$ in the free $x$-basis ($x = a_3$) over $\mathrm{Frac}(k[a_1, a_2, a_4, a_5])$ (verified by the same exact computation; see Supplementary Material) gives

$$T = -\frac{x^7}{p^2q^4} + \frac{a_2x^5}{p^2q^3} - \frac{(a_1a_4^2a_5^2 + a_2^2a_4a_5^2)x^3}{p^3q^4} + \frac{a_1a_2x}{pq},$$

in particular $T$ has only odd $x$-powers with top coefficient $-1/(p^2q^4) \neq 0$. Since $P$, $Q$, $U = J_1$ are $x$-free and $V = J_2 = 2a_2q - x^2$ has $x$-degree 2, every element of $K_{0,pq}$ — hence $N_0$ and $\mu$ — has only even $x$-powers. Thus

the odd-$x$ part of $\nu$ is exactly $-\mu T$'s, and each odd-$x$ coefficient lies in $L := k[a_1, a_2, a_4, a_5]$ ($B = L[x]$ free). Write $\mu = \sum_{j=0}^{d} \mu_j V^j$ ($V$-filtration on the free ring $K$, $\mu_d \neq 0$, $\mu_j \in S := k[P, Q, U]$). The top odd-$x$ term of $\mu T$ is $x^{2d+7}$ with coefficient $(-1)^{d+1}\mu_d/(p^2q^4)$ (leading $x$-part of $V^d$ is $(-x^2)^d$; no other term reaches $2d+7$), so $\mu_d/(p^2q^4) \in L$. Applying $\pi\colon L \to k[a_2, P, Q]$, $a_1 \mapsto 0$, gives $\sum_k c_k(-a_2^2)^k/(p^2q^4) \in k[a_2, P, Q]$ for $\mu_d = \sum c_k U^k$; by $a_2$-freeness each $c_k \in (p^2q^4)$ in $k[P, Q]$, so $\mu_d \in (p^2q^4)S$. Peel: $\mu' = \mu - p^2q^4h_dV^d$ (with $\mu_d = p^2q^4h_d$) has $V$-degree $< d$ and still satisfies the hypothesis, since $a_1a_2 \cdot p^2q^4h_dV^d = -D_5(G_0h_dV^d)$ (from $a_1a_2c = -D_5(G_0)$ with $h_dV^d \in K$ invariant). Induction on $d$ gives $\mu \in (p^2q^4)K$ ($\mu = 0$ is trivial). The $N_i$-formulas are validated end-to-end on the known solution $(c, +G_0)$ in the same script. □

*Remark* 4.6 (Status boundary). The descent above is Established: its three steps (inclusion, localization, specialization descent) are proved here, with the localization step resting on the cited standard slice theorem plus the explicit reconstruction identities. Also Established here is the Dixmier divisibility (Lemma 4.5), which discharges the divisibility step of the $N = 6$ descent obligation O1 by the shift isomorphism (Lemma A.8); the $N = 6$ $B$-level ring itself is resolved by transfer (Theorem A.11), while the $a_5$-saturation inference stays open (Lemma D.2). What remains Open downstream of $K$ is the $a_0$-extension problem; the residue-transcendence saturation (statement E) is void, not open: Lemma P is REFUTED by $F_{46}$ (Remark I.6) while Lemma Q is REFUTED by $H$ (Remark J.4), voiding the joint induction. The global $N = 5$ ring conclusion — the proposed generation with its Hilbert series — was refuted by the explicit counterexample $H$ (weight 24; see 6.4); the six-generator hypersurface $C_5$ of Theorem 7.1 (part [D]; full-ring equality Refuted at weight 46, Section 10). No N=5 generation claim is made in this section itself; the full-ring equality is proved later in Theorem 13.5.

*Remark* 4.7 (Slice certificate and localized presentation — localized statement only). The slice data are independently certified by exact computation (see Supplementary Material), which reuses the exact forms of Lemma 4.1 verbatim: with $x = a_3$, $y = a_2$, $z = a_1$, $p = a_4$, $q = a_5$, $J_1 = 2pz - y^2$, $J_2 = 2qy - x^2$, the slice $s = x/(pq)$ satisfies $D_5(s) = 1$. Hence $D_5(s)$ is a nonzero element of $K$ (the constant 1 lies in $K_0 \subseteq K$), while $s$ itself is not invariant. Exact computation further checks that every denominator in the slice data is a power of $pq$ only, so the reconstruction identities of Lemma 4.2 exhibit $B_{pq} = K_{0,pq}[s]$ as a localized presentation (status Established, as proved there). With $A = k[a_0, \ldots, a_5]$, the same slice will give the localized presentation $A_{pq}^{D_5} = K_{pq}[W]$ once the invariant $F_0$ is constructed in Section 5; no global generation claim is made here. (The proposed global generation $A^{D_5} = K[F_0]$ fails by the counterexample $H$ of Remark 6.4.)

What survives when the divisor $pq = 0$ is restored is the entire subject of what follows.

## 5. The explicit $N = 5$ invariant

The localized model needs a fifth generator: the invariant $F_0$ that carries $a_0$ into the picture. This section constructs it and proves its independence from the base kernel. Whether adjoining it closes the ring is the question the next section answers.

We reuse the five-variable dictionary of the kernel section verbatim, with no redefinition:

$$x = a_3, \qquad y = a_2, \qquad z = a_1, \qquad p = a_4, \qquad q = a_5,$$
$$B = k[x, y, z, p, q] = k[a_1, \ldots, a_5], \qquad K = B^{D_5} = k[p, q, J_1, J_2],$$
$$J_1 = 2pz - y^2 = 2a_1a_4 - a_2^2, \qquad J_2 = 2qy - x^2 = 2a_2a_5 - a_3^2,$$
$$D_5(x) = pq, \qquad D_5(y) = px, \qquad D_5(z) = xy, \qquad D_5(p) = D_5(q) = 0.$$

**Definition 5.1** (The explicit invariant)**.** The degree-seven polynomial $F_0 \in A_5$ is

$$\boxed{F_0 = p^2q^4a_0 - pq^3xyz + \tfrac{1}{3}pq^2x^3z + \tfrac{1}{3}q^2x^3y^2 - \tfrac{1}{5}qx^5y + \tfrac{1}{35}x^7.}$$

In the $a$-variables,

$$F_0 = a_4^2a_5^4a_0 - a_4a_5^3a_3a_2a_1 + \tfrac{1}{3}a_4a_5^2a_3^3a_1 + \tfrac{1}{3}a_5^2a_3^3a_2^2 - \tfrac{1}{5}a_5a_3^5a_2 + \tfrac{1}{35}a_3^7.$$

The denominators 3, 5, 35 are covered by the standing characteristic-zero hypothesis. Write $F_0 = ca_0 + G_0$ with $c = p^2q^4$ and $G_0 \in B$ the remaining five terms.

**Theorem 5.2** ($D_5(F_0) = 0$ by direct computation)**.** *With $D_5(a_0) = zy = a_1a_2$,*

$$\boxed{D_5(F_0) = 0,}$$

*so $F_0 \in A_5^{D_5}$ and $k[p, q, J_1, J_2, F_0] \subseteq A_5^{D_5}$.*

**Corollary 5.3** (Localized $A$-presentation)**.** *With $c = p^2q^4$ and $W = F_0/c = a_0 + \rho$, $\rho = G_0/c \in B_{pq}$, $D_5(W) = 0$ and*

$$A_{pq}^{D_5} = K_{pq}[W],$$

*a polynomial ring in one variable over $K_{pq}$. This is a localized statement only.*

*Proof.* $D_5(c) = 0$ and $D_5(F_0) = 0$ give $D_5(W) = 0$. The slice $s = x/(pq)$ of Lemma 4.2 exhibits $A_{pq} = K_{pq}[W][s]$ with $D_5 = d/ds$ on the slice factor, so the local-slice theorem gives the displayed kernel. □

*Proof.* Since $D_5(p) = D_5(q) = 0$, computing term by term:

$$\begin{aligned}
D_5(p^2q^4a_0) &= p^2q^4zy,\\
D_5(-pq^3xyz) &= -p^2q^4yz - p^2q^3x^2z - pq^3x^2y^2,\\
D_5(\tfrac{1}{3}pq^2x^3z) &= p^2q^3x^2z + \tfrac{1}{3}pq^2x^4y,\\
D_5(\tfrac{1}{3}q^2x^3y^2) &= pq^3x^2y^2 + \tfrac{2}{3}pq^2x^4y,\\
D_5(-\tfrac{1}{5}qx^5y) &= -pq^2x^4y - \tfrac{1}{5}pqx^6,\\
D_5(\tfrac{1}{35}x^7) &= \tfrac{1}{5}pqx^6.
\end{aligned}$$

Grouping by monomial, $p^2q^4zy - p^2q^4yz = 0$, $-p^2q^3x^2z + p^2q^3x^2z = 0$, $-pq^3x^2y^2 + pq^3x^2y^2 = 0$, $(\frac{1}{3} + \frac{2}{3} - 1)pq^2x^4y = 0$, and $-\frac{1}{5}pqx^6 + \frac{1}{5}pqx^6 = 0$. Hence $D_5(F_0) = 0$. Every step is verified by exact computation (see Supplementary Material), which checks the whole-polynomial identity and each row of the table above. □

**Lemma 5.4** (Ordinary degree 7, Fibonacci weight 14)**.** *With the Fibonacci grading* $\mathrm{wt}(a_0, \ldots, a_5) = (8, 5, 3, 2, 1, 1)$,

$$\boxed{\deg_{\mathrm{ord}}(F_0) = 7, \qquad \mathrm{wt}(F_0) = 14.}$$

*Indeed each of the six monomials has ordinary degree 7, and*

| *monomial* | *ordinary degree* | *Fibonacci weight* |
|---|---|---|
| $p^2q^4a_0$ | $2+4+1=7$ | $2+4+8=14$ |
| $pq^3xyz$ | $1+3+1+1+1=7$ | $1+3+2+3+5=14$ |
| $pq^2x^3z$ | $1+2+3+1=7$ | $1+2+6+5=14$ |
| $q^2x^3y^2$ | $2+3+2=7$ | $2+6+6=14$ |
| $qx^5y$ | $1+5+1=7$ | $1+10+3=14$ |
| $x^7$ | $7$ | $14$ |

*all verified monomial by monomial in exact computation (see Supplementary Material).*

**Proposition 5.5** (Independence via the Jacobian criterion)**.** *The five invariants $p$, $q$, $J_1$, $J_2$, $F_0$ are algebraically independent over $k$.*

*Proof.* For the direction used, let $f_1, \ldots, f_5$ be the five polynomials and suppose $P(f_1, \ldots, f_5) = 0$ for nonzero $P$ of minimal total degree. Differentiating gives $\sum_i (\partial P/\partial X_i)(f)\, \partial f_i/\partial a_j = 0$ for every $j$. In characteristic zero some $\partial P/\partial X_i$ is a nonzero polynomial of strictly smaller degree, so by minimality $(\partial P/\partial X_i)(f) \neq 0$ for that $i$: the row vector $((\partial P/\partial X_i)(f))$ is nonzero and annihilates the Jacobian, whose rows are therefore dependent over $\mathrm{Frac}$ and whose maximal minors all vanish. Hence one nonvanishing maximal minor implies algebraic independence (classical Jacobian criterion; see e.g. Freudenburg [11]). In columns $(a_0, a_1, a_2, a_4, a_5)$,

$$\det \frac{\partial(p, q, J_1, J_2, F_0)}{\partial(a_0, a_1, a_2, a_4, a_5)} = 4a_4^3a_5^5 \neq 0$$

as a polynomial. The matrix is block-triangular: $p$, $q$, $J_1$, $J_2$ are free of $a_0$ while $\partial F_0/\partial a_0 = p^2q^4$, so the determinant is $p^2q^4$ times the $4 \times 4$ minor $4a_4a_5$ of the kernel generators. The scalar 4 records the standing characteristic-zero hypothesis (in particular characteristic $\neq 2$); the denominators in $F_0$ likewise require only characteristic zero. The minor is checked exactly, with generic rank 5 at a sample point, in exact computation (see Supplementary Material). □

*Remark* 5.6 (Observation: weight-shift, not a theorem). The quotient $W = F_0/(p^2q^4)$ has Fibonacci weight 8, while the polynomial $F_0$ has weight $14 = 8 + 6$, the summand 6 being the weight of the denominator $p^2q^4$. This separation between the ordinary grading and the Fibonacci grading is recorded as an observation guiding the later denominator analysis, not as a theorem; no minimality or generation consequence is drawn from it here.

*Remark* 5.7 (Status boundary). What is Established here is the invariant itself ($D_5(F_0) = 0$), its bidegree $(7, 14)$, and its algebraic independence from the kernel generators, hence the inclusion $k[p, q, J_1, J_2, F_0] \subseteq A_5^{D_5}$. The claim that these five polynomials generate the full invariant ring, with its Hilbert series, is REFUTED (counterexample $H$, weight 24; see 6.4); the candidate six-generator hypersurface $C_5$ of Theorem 7.1 (part [D] Established; full-ring equality Refuted at weight 46, part [F]). $F_0$ is minimal in the low-weight sense: $(\ker)_w = K[F_0]_w$ for $w \leq 23$ (Theorem 6.7(i)), so weight 14 is the lowest weight outside $K$: every homogeneous generating set contains an element of weight $\leq 14$ outside $K$, hence (modulo $K$) a nonzero multiple of $F_0$. Lemma P is REFUTED by $F_{46}$ (not open); Lemma Q is REFUTED by $H$, voiding the joint induction (statement E). No global generation claim is made here.

The five-generator candidate $K[F_0]$ is now fully assembled, and it is the natural global model. The next section tests it — and watches it fail at weight 24.

## 6. The first obstruction and the corrected hypersurface

The localized kernel is polynomial, but its global descent fails at the first weight where an $a_0^2$-direction can occur. The natural candidate $K[F_0]$ predicts no such direction before weight 28; the explicit invariant $H$ supplies one at weight 24. This section records that failure, explains the residue mechanism behind it, and builds the corrected six-generator algebra it forces.

We reuse the five-variable dictionary of the kernel section verbatim, with no redefinition:

$$x = a_3, \qquad y = a_2, \qquad z = a_1, \qquad p = a_4, \qquad q = a_5,$$
$$B = k[x, y, z, p, q] = k[a_1, \ldots, a_5], \qquad K = B^{D_5} = k[p, q, J_1, J_2],$$
$$J_1 = 2pz - y^2 = 2a_1a_4 - a_2^2, \qquad J_2 = 2qy - x^2 = 2a_2a_5 - a_3^2,$$
$$D_5(x) = pq, \qquad D_5(y) = px, \qquad D_5(z) = xy, \qquad D_5(p) = D_5(q) = 0.$$

We reuse the invariant of Definition 5.1 verbatim:

$$F_0 = p^2q^4a_0 - pq^3xyz + \tfrac{1}{3}pq^2x^3z + \tfrac{1}{3}q^2x^3y^2 - \tfrac{1}{5}qx^5y + \tfrac{1}{35}x^7,$$

with $c = p^2q^4$, $G_0 = F_0 - ca_0 \in B$, $W = F_0/c = a_0 + \rho$ and $\rho = G_0/c \in B_{pq}$. The localized presentation $A_{pq}^{D_5} = K_{pq}[W]$ localizes only and remains Established; the global claim $A_5^{D_5} = K[F_0]$ is REFUTED by the counterexample $H$ (Remark 6.4), not merely Conditional, and is not part of the notation.

**Corollary 6.1** (The proposed global ring (Refuted))**.** ***[REFUTED by explicit counterexample*** *$H$* ***(weight 24,*** *$a_0$****-degree 2;*** *$D_5(H) = 0$****;*** *$H \notin K[F_0]$****).]*** *The proposed equality, stated here for the record, ran as follows: under the $p$-adic and $q$-adic saturation lemmas in full (Lemmas I.3 and J.3 with their leading-coefficient forms Lemmas I.5 and J.9), together with the descent intersection $B \cap K_{pq} = K$ of Theorem 4.4,*

$$A_5^{D_5} = K[F_0] = k[p, q, J_1, J_2, F_0].$$

*It is false unconditionally: the counterexample $H$ of Remark 6.4 satisfies $D_5(H) = 0$ (hence $H \in A_5^{D_5}$) and $H \notin K[F_0]$ at Fibonacci weight 24, with no dependence on the assumptions above.*

*The refuted argument.* The inclusion $k[p, q, J_1, J_2, F_0] \subseteq A_5^{D_5}$ is Established: the kernel generators are invariant by Lemma 4.1 and $D_5(F_0) = 0$ by Theorem 5.2. The claimed reverse inclusion was the $W$-degree induction of Theorem K.5: under the assumed bounds the joint denominator estimate of Lemma K.3 ($H_j \in c^j K_{pq}$ with the full

modules $M_m = c^m K$ of Definition K.2) and the explicit $H_j \in B$ descent of Lemma K.4 place each $W$-coefficient in $B \cap c^j K_{pq}$, whence descent gives $H_m W^m = G_m F_0^m \in K[F_0]$ and subtraction lowers the $W$-degree to the base case $m = 0$ in $K$. This argument CONTAINS A GENUINE ERROR identified by the weight-24 computation of Remark 6.4: its conclusion is false, since $H \in A_5^{D_5} \setminus K[F_0]$; the descent-induction and global-intersection steps must be repaired so that the missing $a_0^2$ direction is produced, which they do not as stated; the precise missing power-contraction hypothesis is identified in Remark K.6. □

**Corollary 6.2** (The proposed five-generator polynomial ring (Refuted))**.** ***[REFUTED by $H$.]*** *The five-generator structure $k[p, q, J_1, J_2, F_0]$ is false by the counterexample $H$ of Remark 6.4 ($D_5(H) = 0$, $H \notin K[F_0]$); the full ring is not a polynomial ring — the candidate is the six-generator hypersurface $C_5$ of Theorem 7.1 (part [D]), cut out of $k[p, q, J_1, J_2, F_0, H]$ by the single weight-28 relation $S$. Under the same (now refuted) proposed generation as Corollary 6.1, the claim was that $A_5^{D_5}$ is a polynomial ring in the five $K[F_0]$-generators. It fails: $H$ lies in $A_5^{D_5}$ but not in $k[p, q, J_1, J_2, F_0]$.*

*Recorded for completeness.* The input side remains Established: by Proposition 5.5 (via the Jacobian criterion with minor $4a_4^3a_5^5 \neq 0$), the five generators $p$, $q$, $J_1$, $J_2$, $F_0$ are algebraically independent over $k$, so $k[p, q, J_1, J_2, F_0]$ is a polynomial ring in five variables. The identification of this ring with $A_5^{D_5}$ rested on the refuted generation of Corollary 6.1 and is therefore refuted as stated; no shortcut via ideal generation is used in either direction. □

**Corollary 6.3** (The proposed Hilbert series of $A_5^{D_5}$ (Refuted))**.** ***[REFUTED by explicit counterexample $H$ (weight 24, $a_0$-degree 2; $D_5(H) = 0$; $H \notin K[F_0]$).]*** *The input side is Established: with ordinary degrees*

$$\deg_{\mathrm{ord}}(p, q, J_1, J_2, F_0) = (1, 1, 2, 2, 7)$$

*(Lemma 4.1 and Lemma 5.4) and Fibonacci weights*

$$\mathrm{wt}(p, q, J_1, J_2, F_0) = (1, 1, 6, 4, 14),$$

*the five generators are algebraically independent (Proposition 5.5), so $k[p, q, J_1, J_2, F_0]$ is a polynomial ring with ordinary Hilbert series*

$$H(t) = \frac{1}{(1-t)^2(1-t^2)^2(1-t^7)} = 1 + 2t + 5t^2 + 8t^3 + 14t^4 + \cdots$$

*and Fibonacci-weighted Hilbert series*

$$H_{\mathrm{wt}}(u) = \frac{1}{(1-u)^2(1-u^4)(1-u^6)(1-u^{14})}.$$

*The proposed identification of these series with those of $A_5^{D_5}$ is false: the exact kernel computation of Remark 6.4 gives $\dim \ker(D_5|_{V_w})$ equal to the Fibonacci series through weight 23 and strictly larger from weight 24 on ($213 > 212$, then $238, 267, 296, 330, 364, 404$ against $236, 264, 292, 325, 358, 396$ over weights 25–30). Direct expansion of the rational functions matches the monomial count in $k[p, q, J_1, J_2, F_0]$ at every degree, as a consistency check on the polynomial ring $K[F_0]$ alone; that ring is genuinely smaller than the invariant ring. The exact Hilbert series of the candidate subalgebra $C_5$ is that of the six-generator hypersurface,*

$$H_{\mathrm{wt}}(u) = \frac{1 - u^{28}}{(1-u)^2(1-u^4)(1-u^6)(1-u^{14})(1-u^{24})} \qquad \textit{(Theorem 7.1).}$$

*Remark* 6.4 (Refuting counterexample $H$). The refutation is certified, not asserted. An explicit invariant $H \in A_5$ of Fibonacci weight 24 with 17 terms, content 1, and $a_0$-degree 2 satisfies $D_5(H) = 0$ and $H \notin K[F_0]$. Its Lean formalization (definition, $D_5(H) = 0$ proved by computation, uniform weight 24, and the $a_0^2$ term $11025\, a_0^2 a_4^4 a_5^4$; checked in core Lean without Mathlib by native evaluation, see the formal-verification manifest) is formally verified in Lean 4. The 17 terms are listed in full in the Supplementary Material; per-weight kernel dims are verified by exact computation (see Supplementary Material).

Non-membership $H \notin K[F_0]$ is structural: at weight 24 every monomial of $K[F_0] = k[p, q, J_1, J_2, F_0]$ has $a_0$-degree at most 1 (only $F_0$ carries $a_0$, in degree 1, and $F_0^2$ already weighs $28 > 24$), while $H$ has $a_0$-degree 2; the same exclusion holds linear-algebraically, since the 212 generator monomials of $K[F_0]_{24}$ span a 212-dimensional subspace to which $H$ is independent (rank 213, exact $\mathbb{Q}$ and $\mathbb{F}_p$). Firstness is exact: $\ker(D_5|_{V_w}) = K[F_0]_w$ for every $w \leq 23$ (Lean-verified

through 14; exact $\mathbb{Q}$-ranks 15–23 in exact computation (see Supplementary Material)), so weight 24 is the smallest weight with an invariant outside $K[F_0]$: $H$ represents the first nonzero graded obstruction to $K[F_0]$, occurring in weight 24. The formal non-membership $H \notin K[F_0]$ is proved by an inductive characterization of the smallest $k$-subalgebra of $A_5$ containing $\{p, q, J_1, J_2, F_0\}$ (i.e. $K[F_0]$), carrying by induction the safety invariant $\mathrm{wt}(m) \geq 14 \cdot \deg_{a_0}(m)$ for every term $m$, which the $a_0$-degree-2, weight-24 term $11025\, a_0^2 a_4^4 a_5^4$ of $H$ violates ($14 \cdot 2 \leq 24$ fails); hence the formal non-membership theorem, with strict-superset witness $H$ (checked in core Lean without Mathlib by native evaluation; see the formal-verification manifest).

Consequences, all unconditional (no dependence on Lemmas P and Q): $H$ is the minimal obstruction to the $K[F_0]$-saturation module and the realization of the $r_q^2$-dependence (Theorem 6.7); the corrected claim is $A_5^{D_5} \supsetneq K[F_0]$, with candidate the six-generator hypersurface $C_5$ cut out by the single weight-28 relation $S$ (Theorem 7.1, part [D]); and the failed globalization is the $q$-side transcendence inference, the $p$-side bound failing separately at weight 46.

**Proposition 6.5** (First obstruction, existence)**.** *$H \in A_5^{D_5} \setminus K[F_0]$. Indeed $D_5(H) = 0$ by direct expansion (checked in exact computation and in Lean by native evaluation), and every monomial of $K[F_0]$ of weight 24 has $a_0$-degree at most 1 (only $F_0$ carries $a_0$, and $F_0^2$ already weighs 28), while $H$ has $a_0$-degree 2 with term $11025 a_0^2 a_4^4 a_5^4$. Hence $A_5^{D_5} \supsetneq K[F_0]$. Lean formally verifies the corresponding inductively generated subalgebra exclusion predicate (`H_notin_K_F0`); the mathematical conclusion $H \notin k[p, q, J_1, J_2, F_0]$ is proved independently here.*

**Proposition 6.6** (First obstruction, minimality)**.** *$(A_5^{D_5})_w = (K[F_0])_w$ for all $w < 24$. This uses the $a_0$-degree-one closure (every invariant of $a_0$-degree $\leq 1$ lies in $K[F_0]$) together with exact rational ranks for $15 \leq w \leq 23$ and Lean-checked spanning for $w \leq 14$. In particular 24 is the first weight at which $K[F_0]$ fails.*

**Theorem 6.7** ($H$ as intrinsic obstruction)**.** *The first obstruction space $(A_5^{D_5})_{24}/(K[F_0])_{24}$ is one-dimensional (213 versus 212), and the class of $H$ is a distinguished nonzero generator of this quotient (Propositions 6.5–6.6), characterized without reference to its 17-term expression by: the minimal obstruction to the $K[F_0]$-saturation module $M_2 = c^2 K$ (its $W$-top $H_2 = 11025 p^4 q^4$ lies outside $M_2$, $v_q = 4 < 8$, while every local regularity condition holds), and the realization of the residue dependence $r_q^2 \in L_q$ as a global invariant (Section 7.1). Cohomological, conductor, or canonical-module characterizations are not claimed.*

## 7. The corrected intermediate algebra

Once $H$ is admitted, the correct intermediate algebra is no longer polynomial; it is the hypersurface obtained by adjoining the forced generator. The refuted five-generator claims above are the exact measure of what had to change.

**Theorem 7.1** (Hypersurface presentation of the six-generator subalgebra)**.** *Put $C_5 = k[p, q, J_1, J_2, F_0, H] \subseteq A_5^{D_5}$. The defining relation, invariance identities, weights, and exclusion witnesses are formally verified in core Lean (see the formal-verification manifest); the presentation ideal $\ker \pi = (S)$ is proved mathematically by the primitivity/height-one argument below. This theorem concerns the subalgebra $C_5$ only: global generation $C_5 = A_5^{D_5}$ is false (see Section 10). Exact computation (see Supplementary Material) cross-checks the bare identity $S = 0$: all six generators are invariant with weights $(1, 1, 6, 4, 14, 24)$, the bare identity $S = 0$ holds with uniform weight 28 of each summand, and $H$ lies strictly outside the five-generator ring. Since $D_5 \neq 0$ on six variables, $\mathrm{trdeg}_k\, A_5^{D_5} = 5$; with the five algebraically independent generators (Proposition 5.5), $\mathrm{trdeg}_k\, C_5 = 5$. Let $\pi\colon k[P, Q, J_1, J_2, F, H] \to C_5$ be the presentation map: $\ker \pi$ is prime ($C_5$ a domain) of height $6 - 5 = 1$. Work in the abstract polynomial ring, with $P, Q, J_1, J_2, F, H$ indeterminates and $S = Q^4 H - R_0$ where $R_0 = 11025F^2 + 105Q^2 J_2^5 J_1 + 9J_2^7$. Then $R_0|_{Q=0}^{\mathrm{abs}} = 11025F^2 + 9J_2^7 \neq 0$ as an abstract polynomial, so the prime $Q$ does not divide $R_0$ in $k[P, Q, J_1, J_2, F]$; hence $\gcd(Q^4, R_0) = 1$ and $S$, linear in $H$ with coprime coefficients, is primitive, hence irreducible, and $(S)$ is prime of height 1 (distinguish this from evaluation in the image, where $R_0|_{q=0} = 9x^{14} - 9x^{14} = 0$ is exactly the residue cancellation producing $H$). With $S = 0$ on the generators, $\ker \pi = (S)$. Thus*

$$\boxed{C_5 \;\cong\; k[P, Q, J_1, J_2, F, H] \,/\, (S),}$$

*where $H$ is the weight-24 invariant of Remark 6.4 and $S$ is the weight-28 relation*

$$\boxed{S \;=\; q^4 H \;-\; 11025\, F_0^2 \;-\; 105\, q^2 J_2^5 J_1 \;-\; 9\, J_2^7.}$$

*The six graded generators carry Fibonacci weights*

$$\mathrm{wt}(p, q, J_1, J_2, F_0, H) = (1, 1, 6, 4, 14, 24)$$

*and $S$ has weight 28. As a complete intersection, $C_5$ has the exact Fibonacci-weighted Hilbert series*

$$\boxed{H_{\mathrm{wt}}(u) \;=\; \frac{1-u^{28}}{(1-u)^2(1-u^4)(1-u^6)(1-u^{14})(1-u^{24})}.}$$

*The series is rational in closed form: the numerator records the single weight-28 relation, the denominator the six generator weights.* ***[CAS-verified [E]:]*** *anchor equality. Equivalently, with $s_6(w)$ the monomial count in the six polynomial generators, $(C_5)_w$ has dimension $s_6(w) - s_6(w-28)$ for $w \geq 28$, and $(C_5)_w = (A_5^{D_5})_w$ at anchor weights $5, 15, 23, 24, 25, 30, 35, 40$ (see Supplementary Material). The refuted five-generator series $1/((1-u)^2(1-u^4)(1-u^6)(1-u^{14}))$ is exceeded by $+1, +2, +3, +4, +5, +6, +8$ over weights 24–30 (Remark 6.4), the first excess being the weight-24 generator $H$ itself. The five generators remain algebraically independent (Proposition 5.5), so the failure is precisely the missing sixth generator modulo the single weight-28 relation, not any dependence among them.* ***[Refuted [F]:]*** *global identity. The equality $C_5 = A_5^{D_5}$ at every weight, and hence the exact Hilbert series of the full ring, is false: the weight-46 invariant $F_{46}$ of Section 10 lies outside $C_5$, and is minimal with this property (spanning holds exactly through weight 45). Conjecture 21.2 is refuted, not open.*

*Remark* 7.2 (Geometry of the hypersurface $X_5 = \mathrm{Spec}(C_5)$). $X_5 = \mathrm{Spec}(C_5)$, with $C_5$ of Theorem 7.1, part [D], is the hypersurface $V(S) \subset \mathbb{A}^6$ cut out by $S = q^4H - 11025F_0^2 - 105q^2J_2^5J_1 - 9J_2^7$. Its identification with $\mathrm{Spec}(A_5^{D_5})$ is Refuted at weight 46 (Section 10); everything below is the geometry of $C_5$ itself. $C_5$ is a domain of dimension 5 and transcendence degree 5, finitely generated by construction, with Hilbert series of part [D]. Since $S$ is linear in $H$, $\partial S/\partial H = q^4$, $\partial S/\partial F_0 = -22050F_0$, $\partial S/\partial J_1 = -105q^2J_2^5$, $\partial S/\partial J_2 = -525q^2J_2^4J_1 - 63J_2^6$, $\partial S/\partial q = 4q^3H - 210qJ_2^5J_1$, $\partial S/\partial p = 0$: $\partial_H S = 0$ gives $q = 0$, then $\partial_{F_0} S = 0$ gives $F_0 = 0$, then $\partial_{J_2} S = -J_2^4(525q^2J_1 + 63J_2^2)$ gives $J_2 = 0$; the remaining partials vanish automatically there, and $S$ vanishes on $V(q, F_0, J_2)$. Hence the singular locus is exactly the linear space $\mathrm{Sing}(X_5) = V(q, F_0, J_2) \cong \mathbb{A}^3_{p,J_1,H}$, of dimension 3 (Singular retained only as an independent cross-check; see Supplementary Material). Since $C_5 \cong k[P, Q, J_1, J_2, F, H]/(S)$, $C_5$ is a hypersurface in the regular polynomial ring $k[P, Q, J_1, J_2, F, H]$. Hence $C_5$ is Cohen–Macaulay, and with singular locus of codimension $5 - 3 = 2$ it is normal by Serre's criterion (R1+S2); see e.g. Bruns–Herzog [3]. By graded Nakayama (loc. cit.), the six generators are minimal as a homogeneous generating set: $S$ lies in $\mathfrak{m}^2$ (every term is a product of at least two positive-weight elements), and the six residue classes are linearly independent in $\mathfrak{m}/\mathfrak{m}^2$ (at weights 1, 4, 6 by independence of $p, q, J_1, J_2$; at 14 since $F_0 \notin K$; at 24 since $H \notin K[F_0]$), so $\dim_k \mathfrak{m}/\mathfrak{m}^2 = 6$. No quotient-geometric claim about $\mathrm{Spec}(A_5^{D_5})$ (orbits, stabilizers, fibers, Makar–Limanov) is made: the quotient geometry of the full ring is not established here.

*Remark* 7.3 (Status boundary). Refuted and unconditional (all by the counterexample $H$ of Remark 6.4): the ring equality (Corollary 6.1), the five-generator polynomiality (Corollary 6.2), and the five-generator Hilbert series (Corollary 6.3), together with the $q$-saturation lemma (Lemma J.3) on which they rested. The descent-induction and global-intersection steps of the saturation argument (Theorem K.5) are void, since one of their two premises is false. What is now *Established* is the *corrected* presentation: the candidate six-generator hypersurface $C_5 \cong k[P, Q, J_1, J_2, F, H]/(S)$ with its weight-28 relator and exact Hilbert series of $C_5$ (Theorem 7.1, part [D]), in addition to the input side (the kernel generators, the invariant $F_0$ with bidegree $(7, 14)$ and its independence from $K$, and the plinth containment of Lemma 11.1). The two saturation lemmas play no further role in the corrected statement: the $p$-lemma (Lemma I.3) is not contradicted by $H$ and remains a legitimate local statement, while the $q$-lemma is refuted; neither is needed to identify the ring, which the hypersurface theorem does directly. No claim for $A_5^{D_5}$ beyond the hypersurface of Theorem 7.1 is made here.

### 7.1 The Failure of Residue-Field Transcendence

The corrected global statement is not a mere repair of a failed computation; it identifies a specific obstruction to a local-to-global principle for $\mathbb{G}_a$-actions, and the counterexample $H$ is its sharpest witness. We isolate the mechanism.

*The descent that works.* On the five-variable base $B = k[a_1, \ldots, a_5]$ the slice is clean: the kernel is $K = B^{D_5} = k[p, q, J_1, J_2]$, the localized extension $A_{pq}^{D_5} = K_{pq}[W]$ is a polynomial ring in the single new coordinate $W = F_0/c$, and the descent $B \cap K_{pq} = K$ (Theorem 4.4) holds exactly. Nothing in this descent is wrong.

*The step that fails.* The $a_0$-extension is where the local-to-global passage breaks. The pole-clearing denominator $c = p^2q^4$ has pole order $e_q = 4$ at the prime $q$ (and $e_p = 2$ at $p$); clearing it to form $W = F_0/c = a_0 + \rho$ with $\rho = G_0/c \in B_{pq}$ passes through the residue field $L_q = k(p, u, x^2)$ at $q$. The residue-transcendence argument required

the initial form of $\rho$,

$$r_q = \mathrm{in}_q(q^4\rho) = \frac{x^7}{35p^2},$$

to be *transcendental* over $L_q$, so that a single initial-form relation could force every pole-clearing coefficient to be regular. That hypothesis is false. Because $x^2 \in L_q$ but $x \notin L_q$, the extension $L_q \subset k(p,u,x)$ is a single quadratic step, and $r_q$ sits in it with the explicit relation

$$r_q^{\,2} = \frac{(x^2)^7}{1225\,p^4} \in L_q,$$

(Remark J.5). The initial form is algebraic of degree two, not transcendental; the required linear independence of $1, r_q, r_q^2$ over $L_q$ fails exactly at $r_q^2$.

*How $H$ slips through the net.* The counterexample is the smallest invariant that exploits this dependence. Write $H = H_2W^2 + H_0$ with $H_2 = 11025\,p^4q^4$ and $H_0 = 105J_2^5J_1\,q^{-2} + 9J_2^7\,q^{-4}$. Every pole-clearing coefficient is regular — in particular $C_0 = H_0 + H_2\rho^2 \in B$ — yet the two $q^{-4}$ leading terms cancel *in the residue field*, not in the ring:

$$11025\,G_0^2 \equiv 9x^{14}, \qquad 9J_2^7 \equiv -9x^{14}, \qquad \text{so } 11025\,G_0^2 + 9J_2^7 \equiv 0 \pmod{q}.$$

The cancellation is the quadratic relation $r_q^2 \in L_q$ doing exactly the work transcendence was supposed to forbid. Hence $D_5(H) = 0$ and $H$ is invariant, and because its $a_0$- degree is 2 while the weight-24 slice of $K[F_0]$ has $a_0$- degree at most 1, $H \notin K[F_0]$ (Remark 6.4). The invariant $H$ satisfies every local regularity condition the descent can check and still escapes the proposed ring.

*The structural reading.* The obstruction is a local-to-global one for the $\mathbb{G}_a$-action with triangular derivation $D_5$. Its plinth contains $pq$ (Lemma 11.1; $px$ and $xy$ lie in the image but are not invariant), and the pole-clearing denominator $c = p^2q^4$ lies in the plinth ideal $(pq, px)$, so the pole divisor $\{c = 0\}$ of the pole-element coordinate $W$ sits over the plinth locus where the localized presentation $K_{pq}[W]$ degenerates ($W$ is $D_5$-invariant, not a slice; the slice is $s$). The computations are consistent with the interpretation that the hidden $q$-adic dependence accounts for the first hidden generator: the five-generator $K[F_0]$ undercounts the exact kernel dimensions by the single $a_0^2$-direction generator $H$ with its weight-28 relator $S$, and the second hidden generator $F_{46}$ accounts for the further excess from weight 46 (Section 10). The hypersurface (Theorem 7.1, part [D]) is the exact accounting of that plinth degeneration at the established level: the local slice is faithful, the residue field is exactly $k(p,u,x^2)$, and the only thing the five-generator guess missed was that the pole-clearing coordinate acquires a quadratic algebraic shadow over the $q$-residue field.

We record this as a structural observation on the $N = 5$ member (generalized as Conjecture 21.16): for a triangular derivation whose pole-clearing denominator has a residue-field initial form algebraic of degree $> 1$ over the residue field, the residue-transcendence saturation argument does not go through, and the invariant ring strictly exceeds the naive one-generator-over-$K$ extension. The realization here is the single invariant $H$ above. The mechanism is classified as escape in Section 20 (pole cancellation via the residue relation, against climb with stable tops), and its geometric shadow is the orbit collapse over $V(q)$ of Section 12.

**Theorem 7.4** (Corrected global status: four components)**.** *No theorem of this section asserts anything about the full invariant ring $A_5^{D_5}$ beyond strict containment. The global picture consists of exactly four components of distinct status:*

- *(A)* ***Proved hypersurface for*** *$C_5$.* *$C_5 = k[p,q,J_1,J_2,F_0,H] \cong k[\mathbf{x}]/(S)$ with $S$ the weight-28 relation of Theorem 7.1, part [D]. Its exact Hilbert series is $(1-t^{28})/((1-t)^2(1-t^4)(1-t^6)(1-t^{14})(1-t^{24}))$, a series of $C_5$, verified against kernel dimensions through weight 40 (see Supplementary Material). The six generators are minimal (graded Nakayama; Remark 7.2).*
- *(B)* ***Refutation of*** *$C_5 = A_5^{D_5}$.* *The weight-46 invariant $F_{46}$ satisfies $D_5(F_{46}) = 0$ and $F_{46} \notin C_5$ ($v_p = 6 < 8$, outside $J_4 = (p^8q^8)$), with $\dim\ker_{46} = 1519 > 1518$ (Theorem 10.1). Propagation: $F_{46} \cdot H \notin C_5$ and $F_{46} \cdot F_0^2 \notin C_5$ via $J_6$ ($v_p = 10 < 12$; Corollary 10.10). Torsion: $F_{46} \cdot p^2 \in C_5$ (Theorem 10.7). Spanning $C_5 = \ker$ holds exactly through weight 45 (Theorem 10.2).*
- *(C)* ***Module pattern (Refuted/Superseded).*** *The excess over $C_5$ was conjectured to be the $F_{46} \cdot K/(p^2)$ tower with three-module prediction $P(w)$ (Conjecture 10.11): prior exact-$\mathbb{Q}$ lower-bound computations match $P(w)$ below weight 84, where the fourth family refutes it (Theorem 14.5); the exact Hilbert series (Theorem 14.4) is its replacement. No second $C_5$-relation is established at any exactly-checked weight.*

*(D)* ***Tower closure Established; finite generation Established.*** *$A_5^{D_5} = C_5[F_{46}^{\mathrm{int}}]$ by Theorem 13.5 (p-local slice atlas plus p- and q-saturation); finite generation follows (hypersurface plus one generator). The four valuation routes remain obstructed at universal top-control, now bypassed rather than completed.*

The corrected algebra is in hand; the next section takes stock of what each failure revealed, and the section after assembles the top-ideal toolkit the second obstruction will need.

## 8. Six obstructions and what each reveals

Each item below has the form claim → counterexample → mechanism → status/replacement, with detailed calculations left in the sections cited.

**1. Five-generator model.** Claim: $W$-degree induction closes with $M_m = c^m K$. Counterexample: $H$ at weight 24. Mechanism: $r_q^2 \in L_q$, so pole-clearing coefficients cancel in the residue field. Replacement: the hypersurface $C_5$ (Theorem 7.1).

**2. $q$-adic saturation bound.** Claim: $v_q(H_j) \geq 4j$. Counterexample: $H$ ($v_q(H_2) = 4 < 8$, $v_q(H_0) = -4$). Mechanism: the constant coefficient cannot see past degree 2. Replacement: R1 ($\lambda_q \geq -2m$, Theorem J.7); Q-top-strong is now established (Theorem 15.1) by normal forms, so the $q$-side is closed and the surviving obstruction is $p$-side only (Rp conjectured).

**3. $p$-adic saturation bound.** Claim: $v_p(H_j) \geq 2j$. Counterexample: $F_{46}$ ($v_p(H_4) = 6 < 8$). Mechanism: $r_p$ algebraic of degree 4. Replacement: none proved; Rp ($\lambda_p \geq -4m$) is Conjecture 21.6.

**4. Unrestricted forest polynomiality.** Claim: out-degree $\leq 1$ implies polynomial kernel. Counterexample: the 4-vertex Weitzenböck path (Theorem 21.13). Mechanism: path length, not out-degree. Replacement: verified small cases only (Proposition 21.12).

**6. Principal plinth.** Claim: $\mathrm{pl}(D_5) = (pq)K$, so the resonance inventory $\{p, q\}$ is exhaustive and the tower must terminate. Counterexample: the weight-13 element $f_{13} = \frac{1}{35}pJ_2^3 + \frac{1}{4}pq^2J_1J_2 = D_5(g_{13})$ with $q \nmid f_{13}$ (Theorem 11.2). Mechanism: slice degeneration — mod $q$ the slice $s_0 = x/(pq)$ dies and $x$ becomes a new residual invariant ($\ker \delta_q = k[p, x, u] \supsetneq K/qK$, Lemma 12.4), and the class $-px^6/35$ lifts to $f_{13}$. Replacement: the plinth-module conjecture $\mathrm{pl}(D_5) \stackrel{?}{=} pqK + f_{13}K$ (Conjecture 11.5) and the two-chart slice atlas (Theorem 12.7).

**5. Degree recurrence and single-denominator module.** Claim: $e_m = e_{m-1} + e_{m-2} + 1$ with $F_0^{(6)}$ of degree 12, and a principal $A_6$-denominator module. Counterexamples: $F_0^{(6)}$ has degree 13, $H_{20}$ degree 8; the $A_6$ module is not principal. Mechanism: multiple $a_0$-directions with incompatible denominators. Replacement: minimal-direction formulation (open).

The toolkit that turns these lessons into the second exclusion — top ideals, birth windows, and the $W$-coefficient detector — is assembled in the next section.

## 9. The $a_0$-filtration and the birth window

The corrected hypersurface still fails, and the failure will again be detected in the $a_0$-filtration: by a top coefficient outside a top ideal. This section assembles that toolkit — the $a_0$-degree windows in which hidden generators can be born, and the $W$-coefficient detector that exposes them.

Status of this section: the safety invariant below is **[LEAN-VERIFIED]** (the formal weight-bound theorem in the exclusion module); corollaries are proved in the formal $a_0$-obstruction module with no CAS dependency. The $a_0$ variable weighs 8 (`weights` $= [8, 5, 3, 2, 1, 1]$).

**Theorem 9.1** ($a_0$-obstruction for $R_5 = k[p, q, J_1, J_2, F_0]$)**.** *Weights are* $(1, 1, 6, 4, 14)$ *and only* $F_0$ *carries* $a_0$ *(degree* 1*).*

*(1) Every* $R_5$*-monomial with* $a_0$*-degree* 0 *lies in* $k[p, q, J_1, J_2]$ *(generator inspection; weights in the semigroup of* $(1, 4, 6)$*). **[PROVED]***

*(2) Every* $R_5$*-monomial with* $a_0$*-degree* 1 *weighs* $\geq 14$ *(*$14 \cdot 1 \leq w$*). **[LEAN-VERIFIED]***

*(3) Every* $R_5$*-monomial with* $a_0$*-degree* $\geq 2$ *weighs* $\geq 28$ *(*$14 \cdot 2 \leq w$*). **[LEAN-VERIFIED]***

*(4) For* $w \leq 13$ *no* $R_5$*-element with* $a_0$*-degree* $\geq 1$ *exists at weight* $w$*; with Theorem L.1,* $(A_5^D)_w = k[p, q, J_1, J_2]_w$*. **[PROVED + LEAN-VERIFIED]***

*(5) At* $w = 14$ *exactly one new direction appears:* $F_0$ *(first fractional* $L_w = 105$*). **[LEAN-VERIFIED]***

*(6) For* $15 \leq w \leq 27$ *any invariant with* $a_0$*-degree* $\geq 2$ *lies outside* $R_5$*, by (3) with no computation. **[PROVED]***

**Corollary 9.2** (H-birth window $[16, 27]$)**.** *The monomial* $a_0^2$ *weighs* 16*, so* $a_0$*-degree* 2 *is monomially possible only at* $w \geq 16$ ***[PROVED]****;* $R_5$ *forbids it below* 28 *by (3), so a hidden* $a_0^2$*-generator can only be born in* $[16, 27]$*.* $H$ *(weight* 24*,* $a_0$*-degree* 2*) lies strictly inside. Weight* 15 *admits* $F_0 \cdot p$ *(*$a_0$*-degree* 1*) but no hidden direction. Computational echoes **[CAS-VERIFIED]****:* $L_w = 1$ *below* 14 *and* 105 *at* $14, 15, 24$ *(no new denominators from* $H$*);* $s_5 = s_6$ *through* 23 *with* $s_6(24) - s_5(24) = 1$ *(exact computation (see Supplementary Material)).*

**Theorem 9.3** (R3: W-coefficient detection principle)**.** *Let* $F \in K[F_0]$ *with W-expansion* $F = \sum H_j W^j$ *(*$H_j \in K_{pq}$*,* $W = F_0/c$*). Then* $H_j = h_j c^j$ *for unique* $h_j \in K$*, and* $v_q(H_j) \geq 4j$*,* $v_p(H_j) \geq 2j$*. In particular, any invariant whose W-expansion violates* $v_q(H_j) \geq 4j$ *lies outside* $K[F_0]$*.*

Status: **[PROVED]** at paper level (proof sketch below); Lean status is split (Remark 9.5).

*Remark* 9.4 (Second birth window: the $C_5$-side $a_0$-degree-4 window). The same monomial-window method that births $H$ in $[16, 27]$ births $F_{46}$. $C_5$-monomials of $a_0$-degree 4 are $F_0^4$ (weight 56), $F_0^2 H$ (weight 52), and $H^2$ (weight 48): the $C_5$ $a_0$-degree-4 window is $[48, \infty)$. The monomial $a_0^4$ weighs 32, so $a_0$-degree 4 is monomially possible from 32; $F_{46}$ (weight 46, $a_0$-degree 4) is born in the gap $[32, 47]$, strictly below the $C_5$ window — the exact analogue of $H$ below the $R_5$ window. Its top ideal $J_4 = (p^8 q^8)$ (Lemma 10.4) governs which multiples re-enter $C_5$ ($F_{46} p^2$ does; Theorem 10.7). **[PROVED from the weight table; the $F_{46} \cdot K/(p^2)$ pattern is Conjecture 10.11, refuted at weight 84 and superseded by the exact series.]**

*Proof sketch.* $W = a_0 + \rho$ is monic of $a_0$-degree 1, so W-expansions are unique by descending $a_0$-degree (top coefficient first; no slice theorem needed). Writing $F = \sum h_e F_0^e = \sum h_e c^e W^e$ and comparing coefficients gives $H_j = h_j c^j$; $v_q(h_j) \geq 0$ on $K \subset B$ yields the bounds, and the contrapositive is the detector. □

*Remark* 9.5 (Lean status and first instance). The fraction-free homogeneous corollaries are **[LEAN-VERIFIED]** in the formal $a_0$-obstruction module (the formal homogeneous corollaries). The full W-valuation form is OPEN in Lean, blocked on localized fraction-field infrastructure ($K_{pq}$, $W$, $v_p/v_q$ absent from the core-only API). First detected instance: $H$ with $(m = 2, \lambda_q = -4, \lambda_p = 0)$ **[CAS-VERIFIED]** by the exact computation (see Supplementary Material).

With windows and detector in place, the second obstruction can be stated and proved: the next section finds the invariant this machinery was built to catch.

## 10. Second hidden invariant: the six-generator ring is not the full ring

The corrected hypersurface still fails, and the failure is sharper than the first: at weight 46 and $a_0$-degree 4, the explicit invariant $F_{46}$ lies outside $C_5$, minimal with this property — spanning holds exactly through weight 45. The same witness refutes the $p$-adic bound and the global $p$-bound. What survives the wreckage is the dependency $F_{46} p^2 \in C_5$, the torsion relation on which the closure proof will turn. We work over the characteristic-zero field $k$ of Definition 2.1.

### 10.1 Exact dimensions and the refutation

**Theorem 10.1** (All-weights spanning (Refuted)). *The identity $C_5 = A_5^{D_5}$ is false. Over $\mathbb{Q}$ (exact sparse rational arithmetic, exact computation (see Supplementary Material); see Supplementary Material):*

$$\dim \ker(D_5|_{V_{46}}) = 1519 > 1518 = \dim(C_5)_{46},$$

*with $\dim V_{46} = 18597$, $\operatorname{rank}(D_5|_{V_{46}}) = 17078$, and the $1618$ weight-$46$ $C_5$-monomial images of full rank $1518 = s_6(46) - s_6(18)$. Hence the inclusion $(C_5)_{46} \hookrightarrow \ker(D_5|_{V_{46}})$ is not surjective: there exists $F \in A_5^{D_5}$ of weight $46$ with $F \notin C_5$. Conjecture 21.2 is refuted, and with it the former Conditional part [F] of Theorem 7.1 (now Refuted).*

**Theorem 10.2** (Spanning through weight 45; minimality). *For every weight $w \le 45$, $\ker(D_5|_{V_w}) = (C_5)_w$: through $w \le 14$ by the Lean-verified direct spanning (Theorem L.1), and for $15 \le w \le 45$ by exact $\mathbb{Q}$-rank comparison ($\dim\ker = \operatorname{rank}(G_w) = s_6(w) - s_6(w-28)$ with $G_w$ the $C_5$-monomial images; see Supplementary Material). Transfer of these $\mathbb{Q}$-rank equalities to arbitrary characteristic-zero $k$ is Lemma 16.1. In particular no second relation appears through weight $57$ ($\operatorname{rank}(G_w) = s_6(w) - s_6(w-28)$ everywhere checked), and weight $46$ is the smallest weight with an invariant outside $C_5$.*

### 10.2 The explicit witness

**Theorem 10.3** (Second hidden invariant $F_{46}$). *There is an explicit invariant $F_{46}^{\mathrm{int}} \in A_5$ of Fibonacci weight $46$ with $79$ terms, content $1$, and $a_0$-degree exactly $4$, satisfying $D_5(F_{46}^{\mathrm{int}}) = 0$ and $F_{46}^{\mathrm{int}} \notin C_5$. The full term list is (see Supplementary Material); the Lean formalization (definition, $D_5(F_{46}^{\mathrm{int}}) = 0$, uniform weight $46$, top term; native evaluation in core Lean without Mathlib (see the formal-verification manifest)) is formally verified in Lean 4 (see Supplementary Material). Write $F_{46}^{\mathrm{int}}$ for the cleared integral form (content $1$, top $121550625\, a_0^4 a_4^6 a_5^8$) and $F_{46}^{\mathrm{rat}} = F_{46}^{\mathrm{int}}/26460$ for the rationally normalized form (top $\frac{18375}{4}\, a_0^4 a_4^6 a_5^8$). Every exact identity in this paper ($p^2 F_{46}^{\mathrm{int}} = E$ of Theorem 10.7, $\bar{F}_{46}^{\mathrm{int}} = B\theta$ of Remark 13.9, $\ker\Phi = (S, p^2T - E)$ of Theorem 14.2 with $T \mapsto F_{46}^{\mathrm{int}}$) uses $F_{46}^{\mathrm{int}}$, while valuation, top-ideal, and dimension arguments are scale-invariant and use either form. Its $W$-expansion $F_{46}^{\mathrm{int}} = \sum_{j=0}^{4} H_j W^j$ has $H_3 = 0$ and*

| $j$ | $0$ | $1$ | $2$ | $4$ |
|---|---|---|---|---|
| $(v_q(H_j),\, v_p(H_j))$ | $(-8,-2)$ | $(-4,0)$ | $(0,2)$ | $(8,6)$ |

Consequences. *With $m = 4$: $\lambda_q(F_{46}) = \min_j(v_q(H_j) - 4j) = -8 = -2m$, so the relaxed bound R1 (Theorem J.7) holds with equality at every nonzero $W$-coefficient — $F_{46}$ is R1-sharp. But $v_p(H_4) = 6 < 8 = 2 \cdot 4$: the $p$-top bound is violated. Hence Lemma I.3 (in particular its leading-coefficient form) and Conjecture 21.5 are **[REFUTED by** $F_{46}$**]**. The strengthened $q$-top bound $v_q(C_m) \ge 4\lceil m/2 \rceil$ of Theorem N.5 below holds with equality ($8 = 8$) and stays Open.*

**Lemma 10.4** (The $C_5$ top ideal at $a_0$-degree $4$). *Let $J_4 = (p^8 q^8) \subset K$. Every $G \in C_5$ of $a_0$-degree exactly $4$ has its top $a_0$-coefficient in $J_4$.*

*Proof.* Let $G \in C_5$ have $a_0$-degree exactly $4$ and put $C_4 = [a_0^4]G \in B$, read directly from the polynomial $G$; no choice of representative modulo $S$ is involved. By definition $G$ is a $K$-linear combination of monomials $\mu F_0^e H^r$ with $\mu$ a $K$-monomial, since only $F_0$ (degree 1) and $H$ (degree 2) carry $a_0$ and $p, q, J_1, J_2 \in K$ carry none. As $B[a_0]$ is a domain over $B$ with $\operatorname{top}(F_0) = c = p^2 q^4 \neq 0$ and $\operatorname{top}(H) = h_2 = 11025 p^4 q^4 \neq 0$, $a_0$-degrees add and tops multiply: $\mu F_0^e H^r$ has $a_0$-degree exactly $e + 2r$ with top $\mu c^e h_2^r$. Hence only monomials with $e + 2r = 4$ contribute to $C_4$, and

$$C_4 = \sum_{e+2r=4} \alpha_{e,r}\, c^e h_2^r, \qquad \alpha_{e,r} \in K,$$

which lies in the ideal $(c^4, c^2 h_2, h_2^2)$. Since $11025$ is a unit, $(c^4, c^2 h_2, h_2^2) = (p^8 q^{16},\, p^8 q^{12},\, p^8 q^8) = (p^8 q^8) = J_4$, principal, with the minimum attained at $(e, r) = (0, 2)$. □

**Corollary 10.5** ($F_{46} \notin C_5$). *$F_{46}$ has $a_0$-degree exactly $4$ with top $\frac{18375}{4} p^6 q^8 \notin J_4$ ($v_p = 6 < 8$; $K$ a polynomial ring, hence a UFD, with $(p)$ prime). Any $G \in C_5$ equal to $F_{46}$ would share its $a_0$-degree and top, contradicting Lemma 10.4.*

**Proposition 10.6** (Exact graded frontier).

$$(C_5)_w = (A_5^{D_5})_w \ \ (w \le 45), \qquad \dim \ker(D_5|_{V_{46}}) = 1519 > 1518 = \dim(C_5)_{46}.$$

*In particular $46$ is the first weight with an invariant outside $C_5$.*

*Proof.* Exact rational sparse ranks give equality of graded pieces for $15 \le w \le 45$ (Appendix P; Lean-checked spanning for $w \le 14$), and at 46 the $C_5$-monomial images span a 1518-dimensional subspace of the 1519-dimensional kernel, the missing direction being $F_{46}$. □

### 10.3 The 46–48 quotient and the $p^2$-torsion

The $J_4$ lens predicts exactly which $F_{46}$-multiples can lie in $C_5$: $F_{46}\mu$ has top in $J_4$ iff $v_p(\mu) \ge 2$. This is what the exact computation finds (see Supplementary Material):

**Theorem 10.7** (The 46–48 quotient)**.** *With $F_{46} = F_{46}^{\mathrm{int}}$ throughout (so $p^2 F_{46}^{\mathrm{int}} = E$ exactly as printed), (i) At 47 (excess 2), $F_{46}p$ and $F_{46}q$ lie outside $C_5$ and span the 2-dimensional quotient $\ker_{47}/(C_5)_{47}$ over $\mathbb{Q}$. (ii) At 48 (excess 2), $F_{46}p^2 \in C_5$, explicitly (with generator exponents $(p, q, J_1, J_2, F_0, H)$)*

$$F_{46}p^2 = 40000\, q^6 J_1^7 + 193200\, q^4 J_1^6 J_2^2 + 325164\, q^2 J_1^5 J_2^4 + 148225\, J_1^4 J_2^6 + 840\, q^2 J_1^3 J_2 H + 770\, J_1^2 J_2^3 H + H^2,$$

*verified $G_{48}^T y = t$ exactly over $\mathbb{Q}$ (with $F_{46} = F_{46}^{\mathrm{int}}$ the cleared integral form, so the $H^2$-coefficient is 1; formally verified in Lean 4 by native evaluation in core Lean without Mathlib, module `AdjQuad.TorsionCheck: p2F46_eq_E46`); while $F_{46}pq$ and $F_{46}q^2$ lie outside $C_5$ and span $\ker_{48}/(C_5)_{48}$.*

**Theorem 10.8** (Exact $p$-torsion table)**.** *Write $T_p^{(r)} = (C_5 : p^r)/C_5$ for the $p$-power torsion filtration. Exact rational linear algebra gives:*

- *(i) $(C_5 : p)_w = C_5$ for $w \le 46$; $T_p^{(1)}$ is 1-dimensional at $47, 48, 49, 50$ (spanned by $pF_{46}$, $qpF_{46}$, $q^2pF_{46}$, $q^3pF_{46}$) and 2-dimensional at $51, 52$: the layer $pF_{46} \cdot (K/(p))$.*
- *(ii) $(C_5 : p^2)_w = C_5$ for $w \le 45$; $T_p^{(2)}$ is 1-dimensional at 46 (spanned by $F_{46}$), 2-dimensional at $47, 48, 49$ ($F_{46} \cdot K_1$, resp. $F_{46} \cdot K_2$ mod $p^2$), and 3-dimensional at 50: the layer $F_{46} \cdot (K/(p^2))$.*
- *(iii) $(C_5 : p^3)_w = C_5$ for $w \le 45$; at $46, 47, 48$ the excess is $1, 2, 2$ (the same $F_{46}$-classes visible at higher $r$; no new classes).*

*The counts are exact over $\mathbb{Q}$ (scripts/cas/torsion_exact.sage; log data/torsion_exact.log); the witnesses lie outside $C_5$ by the $J_4$ top argument (Lemma 10.4: $pF_{46}$-multiples with $v_p(\mu) = 0$ have top $v_p = 7 < 8$; $F_{46}$-multiples with $v_p(\mu) \le 1$ have top $v_p < 8$), and their $p$-power multiples land in $C_5$ via $p^2 F_{46}^{\mathrm{int}} \in C_5$ (Theorem 10.7). The $r = 1$ layer $pF_{46} \notin C_5$ is new: $F_{46}$ has exact $p$-torsion order 2. Above the checked range the $F_{46} \cdot K/(p^2)$ pattern was Conjecture 10.11, refuted at weight 84 (Theorem 14.5); the exact Hilbert series (Theorem 14.4) governs all weights.*

**Corollary 10.9** ($F_{46} \cdot F_0 \notin C_5$)**.** *$F_{46} \cdot F_0$ (Fibonacci weight $60 = 46 + 14$, $a_0$-degree $5 = 4 + 1$) is a $D_5$-invariant lying outside $C_5$. Indeed $D_5(F_{46} \cdot F_0) = 0$ by Leibniz, and tops multiply over the domain $B[a_0]$, so its top $a_0$-term is $c \cdot \frac{18375}{4} a_0^4 a_4^6 a_5^8$ with coefficient $\frac{18375}{4} p^8 q^{12}$; since $v_p = 8 < 10 = 2 \cdot 5$, this top lies outside $J_5 = (p^{10}q^{12})$ (Definition N.4). Any $G \in C_5$ equal to it would share its $a_0$-degree and top, contradicting the triangular-top argument of Corollary N.6.*

**Corollary 10.10** ($F_{46} \cdot H$ and $F_{46} \cdot F_0^2$ lie outside $C_5$)**.** *$F_{46} \cdot H$ (weight $70 = 46 + 24$, $a_0$-degree $6 = 4 + 2$) and $F_{46} \cdot F_0^2$ (weight $74 = 46 + 28$, $a_0$-degree $6 = 4 + 2$) are $D_5$-invariants outside $C_5$. Tops multiply over $B[a_0]$: $\mathrm{top}(F_{46} \cdot H) \propto p^{10}q^{12}$ with $v_p = 10 < 12 = 2 \cdot 6$, and $\mathrm{top}(F_{46} \cdot F_0^2) \propto p^{10}q^{16}$ with $v_p = 10 < 12$, both outside $J_6 = (p^{12}q^{12})$ (Definition N.4; exponent checks in exact top-ideal computation: $v_q = 12, 16 \ge 12$ hold, so in both cases it is the $p$-side that governs). Neither is primitive: each is an $F_{46}$-multiple, seeding the second and third $K/(p^2)$-families of the tower below.*

**Conjecture 10.11** (the $F_{46} \cdot K/(p^2)$ module pattern (superseded as stated))**. [REFUTED as an all-weights claim at $w = 84$ by the exact Hilbert series (Theorem 14.4); preserved as the first successful approximation. Valid through 83; from 84 further families (Theorem 14.5) and the exact series take over.]** Write $s_K$ for the $K$-monomial count. For $w \ge 46$ the excess $\dim \ker_w - (s_6(w) - s_6(w - 28))$ equals $\dim(K/(p^2))_{w-46} = s_K(w - 46) - s_K(w - 48)$, plus, from $w \ge 60$, the $F_{46} \cdot F_0$-family contribution $s_K(w - 60) - s_K(w - 62)$ (uniformly zero for $w < 60$), plus, from $w \ge 70$, the $F_{46} \cdot H$-family contribution $s_K(w - 70) - s_K(w - 72)$ (Corollary 10.10; zero for $w < 70$): each new direction contributes a free $K/(p^2)$-module, with $F_{46}p^2 \in C_5$ the $p^2$-torsion relation of the first family. The torsion relations of the higher families are established by transport of the first: $F_{46} \cdot F_0 \cdot p^2 = (F_{46}p^2) \cdot F_0 \in C_5$

and $F_{46} \cdot H \cdot p^2 = (F_{46}p^2) \cdot H \in C_5$. Evidence: exact over $\mathbb{Q}$ at 46–48, 54–61, 65, 69 (exact computation; see Supplementary Material), finite-field computations at 49–53 and 58–69 (exact pattern computations with sweep appendix; see Supplementary Material), and exact-$\mathbb{Q}$ lower bounds at 70–74 (no second $C_5$-relation; $F$-family independence meets prediction; exact computation (see Supplementary Material)); the fit matches prediction at every weight 46–69 checked (exact over $\mathbb{Q}$ at 46–48, 54–61, 65, 69; mod-$p$ agreement at 49–53, 62–64, 66–68), with no second $C_5$-relation at any exactly-checked weight. No excess deviation from the two-module series through 69 (exact G-rank at 58–61, 65, 69; mod-$p$ only at 62–64, 66–68): every deviation from the bare series is the $F \cdot F_0$ correction (weight-60 seed, $a_0$-degree 5, top $\frac{18375}{4}p^8q^{12}$, classified by the $J_5$ lens of Corollary 10.9; the $J_5$-membership fingerprint $\mathrm{top}(F_{46} \cdot F_0 \cdot \mu) \in J_5 \iff v_p(\mu) \geq 2$ predicts exactly the observed $K/(p^2)$ growth). The mod-$p$ sweep stopped at 69 (693s, per-weight time-box); sparse LinBox rank over $\mathbb{F}_{1000003}$ likewise exceeds the box at 70 ($n = 112241$, over 60 minutes for both $\mathbb{Q}$ and mod-$p$ full ranks). What is established at 70 by the squeeze machinery (exact computation (see Supplementary Material)) is the exact $\mathbb{Q}$ lower bound: $\mathrm{rank}(G_{70}) = 6210$ (no second $C_5$-relation) and the 45 predicted $F$-family vectors span mod $C_5$ ($\mathrm{rank} = 6255$), so $N_{\mathbb{Q}}(70) \geq 6255$; the upper bound ($N_{\mathbb{Q}}(70) \leq 6255$) is beyond current compute, leaving shortfall-type deviations ruled out and excess-type deviations (a third primitive hiding above) open. 70+ is otherwise unprobed. The three-module form predicts excesses $45, 50, 54, 58, 63$ at weights 70–74 (bare $35, 38, 40, 42, 45$ plus $F \cdot F_0$-family $9, 10, 12, 14, 15$ plus $F \cdot H$-family $1, 2, 2, 2, 3$). Falsifier: any weight with excess $\neq s_K(w-46) + s_K(w-60) + s_K(w-70) - s_K(w-48) - s_K(w-62) - s_K(w-72)$. The question of further $C_5$-relations on $F_{46}J_1$, $F_{46}J_2$ beyond the $p^2$-torsion belongs to the finer $C_5$-module decomposition, which remains open; the full presentation as a $C_5$-algebra is established by Theorem 14.2.

At this point generator discovery is over. With $R = C_5[F_{46}^{\mathrm{int}}]$ identified and the finite range exact through weight 45 with its 46-excess explained, the remaining question is no longer which generators exist but whether this algebra is already the entire kernel. Answering it requires no new invariant: it requires proving that no further generator exists. The plinth element $f_{13}$ supplies the second chart on which that proof turns.

## 11. The plinth and the second slice

The pole divisor that defeats naive descent leaves a concrete trace: the plinth ideal. Its $q$-primitive element $f_{13}$ outside $(pq)K$ will supply the second slice chart on which the closure proof turns (Section 12); the conjectures at the end of this section record what remains open about the plinth itself.

**Lemma 11.1** (Plinth support)**.** *For $D_5$ on $A = A_5$,*

$$\boxed{pq = D_5(x), \qquad px = D_5(y), \qquad xy = D_5(z).}$$

*Hence $pq$, $px$, $xy$ lie in the image $D_5(A)$. Of the three, only $pq$ is annihilated by $D_5$ (since $D_5(p) = D_5(q) = 0$); it lies in $K = k[p, q, J_1, J_2] \subseteq A^{D_5}$ by Lemma 4.1 and hence in the plinth ideal $\mathrm{pl}(D_5) = A^{D_5} \cap D_5(A)$, which is nonzero. In fact the full pole-clearing denominator lies in the plinth: $c = p^2q^4 = D_5(a_3a_4a_5^3)$, since $D_5(a_3) = a_4a_5$ while $a_4, a_5$ are killed, so $c \in A_5^{D_5} \cap D_5(A)$. By contrast $D_5(px) = p^2q \neq 0$ and $D_5(xy) = pqy + px^2 \neq 0$, so $px$ and $xy$ are in the image but not invariant (in particular neither lies in $R$). All three values and the preimage identity are machine-checked (formally verified in Lean 4 by native evaluation in core Lean without Mathlib (see the formal-verification manifest); exact computation, see Supplementary Material). The denominator $c = p^2q^4 = (pq)(pq^3)$ lies in the ideal $(pq, px) \subset B$ as an ideal membership, so the pole divisor $\{c = 0\}$ of the rational slice coordinate $W = F_0/c$ sits over the locus cut out by image elements. No generation consequence for $A_5^{D_5}$ is drawn from this containment here.*

**Theorem 11.2** (A $q$-primitive plinth element)**.** *With $J_1 = 2a_1a_4 - a_2^2$, $J_2 = 2a_2a_5 - a_3^2$, put*

$$\boxed{f_{13} = \tfrac{1}{35}pJ_2^3 + \tfrac{1}{4}pq^2J_1J_2, \qquad p = a_4,\ q = a_5.}$$

*In the $a$-variables,*

$$f_{13} = a_1a_2a_4^2a_5^3 - \tfrac{1}{2}a_1a_3^2a_4^2a_5^2 - \tfrac{19}{70}a_2^3a_4a_5^3 - \tfrac{13}{140}a_2^2a_3^2a_4a_5^2 + \tfrac{6}{35}a_2a_3^4a_4a_5 - \tfrac{1}{35}a_3^6a_4,$$

*of Fibonacci weight 13. Then:*

1. *$f_{13} \in K = k[p, q, J_1, J_2]$, so $D_5(f_{13}) = 0$.*

2. *With*

$$\boxed{g_{13} = a_0 p^2 q^3 - \tfrac{1}{6} a_1 a_3^3 pq - \tfrac{19}{70} a_2^3 a_3 q^2 + \tfrac{101}{420} a_2^2 a_3^3 q - \tfrac{1}{35} a_2 a_3^5,}$$

$$\boxed{D_5(g_{13}) = f_{13},}$$

*hence* $f_{13} \in \mathrm{pl}(D_5) = A_5^{D_5} \cap D_5(A_5)$.

3. $q \nmid f_{13}$*: setting* $q = 0$*,*

$$f_{13}|_{q=0} = \tfrac{1}{35} p J_2^3|_{q=0} = -\tfrac{1}{35} p x^6 \neq 0, \qquad x = a_3,$$

*since* $J_2|_{q=0} = -x^2$*. So* $f_{13} \notin (pq)K$*, and hence* $\mathrm{pl}(D_5) \neq (pq)K$*; i.e.* $f_{13}$ *is a* $q$*-primitive plinth element outside* $(pq)K$ *(full nonprincipality of* $\mathrm{pl}(D_5)$ *is not claimed here).*

*All three items are verified by exact computation (scripts/cas/check_plinth.py, items P3; log data/plinth.log): the* $D_5(g_{13}) - f_{13} = 0$ *identity by direct expansion, and the non-divisibility by evaluation at* $q = 0$ *(both also formally verified in Lean 4 by native evaluation in core Lean without Mathlib, module* `AdjQuad.PlinthCheck`*:* `D_F13_zero`*,* `D_G13_eq_F13`*,* `F13_res_q0`*,* `F13_qfree`*).*

*Remark* 11.3 (Why the $c$-divisibility does not extend). The Dixmier divisibility of Lemma 4.5 applies only to the specific relation $a_1 a_2 \mu + D_5(\nu) = 0$, concluding $c \mid \mu$. A plinth element $f = D_5(g)$ instead satisfies $1 \cdot f + D_5(-g) = 0$, with coefficient 1 rather than $a_1 a_2$, so the lemma does not apply. Indeed $pq = D_5(x)$ is already a counterexample to any such extrapolation, since $p^2 q^4 \nmid pq$. The element $f_{13}$ is the first plinth element outside even the weaker ideal $(pq)K$.

**Lemma 11.4** (Directness of the two plinth directions)**.** $(pq)K \cap (f_{13}) = 0$ *in* $K$*. Hence* $pq \cdot K_{w-2} + f_{13} \cdot K_{w-13} \subseteq \mathrm{pl}(D_5)_w$ *is a direct sum at every weight* $w$*.*

*Proof.* Let $f_{13} u = pqv$ with $u, v \in K$. Reduce modulo $q$: $\bar{f}_{13} \bar{u} = 0$ in $\bar{K} = K/qK$. Now $\bar{f}_{13} = p\bar{J}_2^3/35 = -px^6/35 \neq 0$, and $\bar{K} \cong k[p, 2pz - y^2, -x^2]$ is a domain by Lemma 4.3, so $\bar{u} = 0$, i.e. $q \mid u$. Write $u = qu_1$; then $f_{13} u_1 = pv$. Since $f_{13} = p \cdot e$ with $e = J_2^3/35 + q^2 J_1 J_2/4$, cancelling $p$ gives $e u_1 = qv$; reducing mod $q$ gives $\bar{e}\,\bar{u}_1 = 0$ with $\bar{e} = \bar{J}_2^3/35 \neq 0$, so $q \mid u_1$. Descent on $v_q(u)$ forces $u = 0$ ($K$ is a domain, $q$ prime). □

**Conjecture 11.5** (Plinth module: principal conjecture)**.** With $h = f_{13}/p = J_2^3/35 + q^2 J_1 J_2/4$,

$$\boxed{\mathrm{pl}(D_5) \stackrel{?}{=} pqK + f_{13}K = p(q, h)K.}$$

In words: the entire plinth is supported on $p = 0$, but inside $D(p)$ its residual structure is governed by the codimension-two ideal $(q, h)$. Evidence: exact rational linear algebra per Fibonacci weight gives $\dim(\mathrm{im}\, D_5 \cap K_w) = \dim K_{w-2} + \dim K_{w-13}$ at every $w \in [13, 20]$, with $\dim(\mathrm{im}\, D_5 \cap K_w) = \dim K_{w-2}$ for $w \leq 12$ (weights $\leq 12$ are clean; weight 13 is the first defect). With Lemma 11.4 the dimension count is exact: $\mathrm{pl}(D_5)_w = pqK_{w-2} \oplus f_{13} K_{w-13}$ for $w \leq 20$ (scripts/cas/check_plinth_module.py; log data/plinth_module.log). Falsifier: any $p$-primitive plinth class outside $(q, h)$, or any $w$ with $\dim(\mathrm{im}\, D_5 \cap K_w) > \dim K_{w-2} + \dim K_{w-13}$. Known obstruction: no inductive step extends the count past the computed range (bounded ansatz only); a proof needs a mechanism, not a longer census.

**Conjecture 11.6** ($p$-divisibility of the plinth)**.** Every $f \in \mathrm{pl}(D_5)$ satisfies $p \mid f$ in $K$. Evidence: no $p$-primitive plinth element exists through Fibonacci weight 18 (exact computation as above). Known obstruction: same bounded-range limit as Conjecture 11.5. Falsifier: an explicit $p$-primitive $f \in \mathrm{pl}(D_5)$. If true, $\mathrm{pl}(D_5) = pI$ with $I \stackrel{?}{=} (q, f_{13}/p)$ by Conjecture 11.5.

The established half of this section — the element $f_{13}$ — is what the closure proof needs; the conjectural half stays open. The next section builds the chart $f_{13}$ makes possible.

## 12. The second chart: collapse, construction, and cover

The first chart degenerates over the plinth locus, and the degeneration is itself geometric: orbits collapse over $V(q)$. This section constructs the second chart that covers the collapse locus — the chart on which the closure proof turns — and reads off what the degeneration means.

The partial quotient below is to $\mathbb{A}^5$ via the five $K[F_0]$-generators, not to $X_5 = \mathrm{Spec}(C_5)$; no claim about $\mathrm{Spec}(A_5^{D_5})$ is made here. The collapse is witnessed by explicit rational points checked by direct computation.

Let $\pi_5 \colon \mathbb{A}^6 \to \mathbb{A}^5$ be the morphism given by the five generators $(p, q, J_1, J_2, F_0)$.

**Theorem 12.1** (Exceptional locus of the five-generator quotient)**.** *Over $D(pq)$, $H$ is a regular function of $\pi_5$:*

$$H = 11025p^4q^4(F_0/c)^2 + 105J_2^5J_1/q^2 + 9J_2^7/q^4, \qquad c = p^2q^4,$$

*so $\pi_5$ loses no $H$-information off the plinth divisor $V(pq)$. Over $V(q)$, $\pi_5$ collapses distinct $\mathbb{G}_a$-orbits: the rational points*

$$a_t = (0,\ t^2/2,\ t,\ 1,\ 1,\ 0) \quad (a_0, \ldots, a_5), \qquad t = 0, 1, 2,$$

*satisfy $\pi_5(a_0) = \pi_5(a_1) = \pi_5(a_2) = (1, 0, 0, -1, 1/35)$ while $H(a_0), H(a_1), H(a_2) = 0, -315/4, -1260$ respectively. Since $H$ is $D_5$-invariant, the three points lie in three distinct orbits with one $\pi_5$-image.*

*Proof.* The $W$-expansion $H = H_2W^2 + H_0$ with $H_1 = 0$ (Remark J.4) is an identity in $K_{pq}[W]$, regular on $D(pq)$, giving (i). At $q = 0$: $J_1 = 2z - y^2 = 2(t^2/2) - t^2 = 0$, $J_2 = -x^2 = -1$, $F_0 = x^7/35 = 1/35$ (every other $F_0$-term carries $q$ or $a_0$ via $p^2q^4a_0$), with $p = 1$, $q = 0$; direct substitution of the 17-term $H$ gives $0, -315/4, -1260$ at $t = 0, 1, 2$. □

**Corollary 12.2** (Valuation failure as geometric shadow)**.** *The refutation $v_q(H_0) = -4 < 0$ of Lemma J.3 is the algebraic shadow of the collapse above: the $q^{-4}$ pole that the saturation argument forbids is exactly what allows $H$ to vary along the $J_1$-fiber $2z - y^2 = 0$ over $V(q)$ while $\pi_5$ stays fixed. The hidden generator separates orbits that the five-generator quotient identifies; hence $H \notin K[F_0]$, independently of the $a_0$-degree count.*

### Slice degeneration and the second chart

The collapse above is not an isolated accident: it is the shadow of the slice $s_0 = x/(pq)$ degenerating at $V(q)$. The plinth element $f_{13}$ of Theorem 11.2 supplies a second slice chart that covers exactly the collapsed locus.

**Definition 12.3** (Slice degeneration, residual and liftable plinth)**.** Let $D$ be triangular with local slice $s$, $D(s) = 1$, whose denominators vanish along a boundary divisor $V(\pi)$. Write $\delta_\pi$ for the residual derivation on $\bar{A} = A/\pi A$ and $\bar{K} = A^D/\pi A^D$ for the residual kernel. *Slice degeneration* is the phenomenon $\ker \delta_\pi \supsetneq \bar{K}$: the residual derivation acquires new invariant coordinates invisible generically. The *residual plinth* is $\bar{K} \cap \delta_\pi(\bar{A})$; the *liftable residual plinth* is

$$\boxed{\mathcal{L}_\pi := \mathrm{im}(\mathfrak{p}_D \to \bar{K}), \qquad \mathfrak{p}_D = A^D \cap D(A),}$$

the classes that actually lift to global plinth elements. In general $\mathcal{L}_\pi \subseteq \bar{K} \cap \delta_\pi(\bar{A})$, with equality an open question at each divisor.

**Lemma 12.4** (Residual kernel at $q$)**.** *Let $\delta_q = D_5 \bmod q$ on $\bar{B} = k[x, y, z, p]$, so $\delta_q(x) = 0$, $\delta_q(y) = px$, $\delta_q(z) = xy$. With $u = 2pz - y^2 = J_1 \bmod q$,*

$$\boxed{\ker \delta_q = k[p, x, u],} \qquad \textit{whereas} \qquad K/qK = k[p, x^2, u].$$

*Hence the boundary acquires the new invariant coordinate $x$, in a quadratic extension $k[p, x^2, u] \subset k[p, x, u]$ of degree 2.*

*Proof.* Write $\delta_q = x\delta'$ with $\delta' = p\partial_y + y\partial_z$; $\bar{B}$ is a domain and $x \neq 0$, so $\ker \delta_q = \ker \delta'$. Clearly $x, p, u \in \ker \delta'$. Conversely localize at $p$: $s = y/p$ satisfies $\delta'(s) = 1$, $\delta'$ stays locally nilpotent, and with $z = u/(2p) + ps^2/2$, $S_p = k[x, p, 1/p, u][s]$ with $s$ transcendental over $K_p = k[x, p, 1/p, u]$ (apply $\delta' = d/ds$ to any relation). So $F \in \ker \delta'$ lies in $K_p$: $F = H/p^a$ with $H \in k[x, p, u]$, $a \geq 0$ minimal. If $a \geq 1$, then $H \in pS_p$; reducing mod $p$ gives $H(x, 0, -y^2) = 0$ in $k[x, y]$, whence $H = 0$ as a polynomial by $y$-degree (the specialization $u \mapsto -y^2$ is injective), so $p \mid H$ in $k[x, p, u]$, contradicting minimality. Thus $a = 0$. The description of $K/qK$ is Lemma 4.3. □

*Remark* 12.5 (Liftable versus residual at $q$). $\bar{K} \cap \delta_q(\bar{B}) \ni -px^6/35 = \delta_q(-yx^5/35)$, and this class is liftable: it is $\bar{f}_{13}$ for the global plinth element of Theorem 11.2. So $0 \neq -px^6/35 \in \mathcal{L}_q$. No $q$-adic lifting proof of a bound $q \mid f$ can succeed past this class: the obstruction is already present in the residual problem.

**Conjecture 12.6** (Liftable residual plinth, $q$-side)**.**

$$\boxed{\mathcal{L}_q \stackrel{?}{=} (p\bar{J}_2^3)\bar{K},}$$

the scalar $1/35$ being irrelevant. Evidence: the $q$-residue ranks of the exact plinth computation equal $\dim \bar{K}_{w-13}$ at every $w \in [13, 20]$ (ranks $1, 1, 1, 1, 2, 2, 3, 3$; scripts/cas/check_plinth_module.py, log data/plinth_module.log), so every liftable class in range is an $f_{13}$-multiple. Falsifier: a liftable residual class outside $(p\bar{J}_2^3)$. Known obstruction: no lifting mechanism extends the rank match past weight 20.

**Theorem 12.7** (Second slice chart)**.** *With $f_{13}, g_{13}$ from Theorem 11.2, $s_1 = g_{13}/f_{13}$ satisfies $D_5(s_1) = 1$ on $D(f_{13})$ (by the quotient rule from $D_5(g_{13}) = f_{13}$ and $D_5(f_{13}) = 0$). The slice projection $\pi_1(b) = \sum_{i\geq 0}(-s_1)^i D_5^i(b)/i!$ gives $D_5(\pi_1(b)) = 0$ for every ambient variable; explicitly $\pi_1(b) = \mathrm{NUM}_b/f_{13}^{N_b}$ with $N_{a_3} = 1$, $N_{a_2} = 2$, $N_{a_1} = 4$, $N_{a_0} = 7$ ($\mathrm{NUM}_{a_0}$ has 456 terms), each $D_5(\mathrm{NUM}_b) = 0$ by direct expansion (scripts/cas/chart_cert.py; log data/chart_cert.log). The two slice charts are related by the exact transition*

$$\boxed{s_1 - s_0 = \frac{F_0}{qf_{13}}, \qquad s_0 = \frac{x}{pq},}$$

*verified by exact rational simplification (scripts/cas/check_plinth.py, item P3; log data/plinth.log). With $h = f_{13}/p$, the union of the two principal cylinders is $D(pq) \cup D(f_{13})$, and inside $D(p)$ the uncovered $q$-boundary is the codimension-two stratum $V(q, h)$. The cylinder/plinth-zero-locus correspondence is the framework of [23]; our contribution here is the family-specific explicit algebraic realization (charts, transition, orbit separation), not the general cylinder/plinth equality.*

**Lemma 12.8** (Two-chart intersection)**.**

$$\boxed{A_{pq} \cap A_{f_{13}} = A_p}$$

*inside* $\mathrm{Frac}(A)$. *Consequently* $(A_{pq} \cap A_{f_{13}})^{D_5} = (A^{D_5})_p$*: gluing the two charts computes exactly the $p$-localized kernel.*

*Proof.* $A$ is a unique factorization domain and $p, q$ are prime (variables). Write $f_{13} = p \cdot h$ with $h = J_2(J_2^2/35 + q^2J_1/4)$. Direct evaluation gives $q \nmid J_2, h$ (at $q = 0$: $-x^2$, $-x^6/35$) and $p \nmid J_1, J_2, h$ (at $p = 0$: $-y^2$, $2qy - x^2 \neq 0$, and a nonzero polynomial), all certified by exact computation (scripts/cas/check_plinth_module.py, item M5). Let $x \in A_{pq} \cap A_{f_{13}}$, written in lowest terms $u/v$ with $\gcd(u, v) = 1$. Then $v \mid (pq)^a$ and $v \mid f_{13}^b$ for some $a, b$, so $v \mid \gcd((pq)^a, f_{13}^b) = p^{\min(a,b)}$: any prime divisor of both is $p$ or $q$, and $q \nmid f_{13}$ excludes $q$. Hence $x \in A_p$; the reverse inclusion holds since $p \mid pq$ and $p \mid f_{13}$. For invariants, $(A_{pq} \cap A_{f_{13}})^D = (A_p)^D = (A^D)_p$ because $p$ is $D$-invariant. □

*Remark* 12.9 (Corrected proof architecture). The atlas therefore splits tower closure into two precise problems: (i) the *$D(p)$-atlas problem* $(A^{D_5})_p \stackrel{?}{=} C_5[F_{46}^{\mathrm{int}}]_p$, and (ii) the *$p$-power contraction problem* $(C_5[F_{46}^{\mathrm{int}}])_p \cap A \stackrel{?}{=} C_5[F_{46}^{\mathrm{int}}]$, whose denominators are $p$-powers only. Jointly they imply $A_5^{D_5} = C_5[F_{46}^{\mathrm{int}}]$. Note the correction: the atlas alone does not reduce the remainder to the $p$-boundary divisor, since contraction to $A$ must be checked at every height-one prime; what it does is confine all denominators to $p$-powers, so only $p$-adic divisibility remains. The element $F_{46}^{\mathrm{int}}$ already satisfies this pattern ($p^2F_{46}^{\mathrm{int}} \in C_5$, Theorem 10.7).

*Remark* 12.10 (Shape of the degeneration stratum). $h = J_2 \cdot e_2$ with $e_2 = J_2^2/35 + q^2J_1/4$ is reducible, but both components have the same radical over $q$: $J_2|_{q=0} = -x^2$ and $e_2|_{q=0} = x^4/35$, so $V(q, h)_{\mathrm{red}} = V(q, x)$, the $q = x = 0$ threefold. The degeneration stratum is set-theoretically a linear space. This is an observation guiding the gluing analysis, not a theorem about the kernel.

**Theorem 12.11** (Terminal stratum: no residual plinth on $V(q, x)$)**.** *On $V(q, x)$, $D_5$ descends to $\delta = a_1a_2\partial_{a_0}$ on $k[a_0, a_1, a_2, a_4]$, with $\ker \delta = k[a_1, a_2, a_4]$. The image of $K$ on the stratum is $k[a_4, 2a_4a_1 - a_2^2]$, and the residual plinth there is zero: no nonzero element of $k[a_4, 2a_4a_1 - a_2^2]$ is divisible by $a_1a_2$. Hence the $q$-branch of the boundary tree terminates at $f_{13}$: no further slice lifts over $V(q, x)$.*

*Proof.* $D_5(q) = 0$ and $D_5(x) = pq \in (q)$, so $(q, x)$ is $D$-stable and $D_5$ descends. On the quotient, $\delta(a_0) = a_1a_2$ while $\delta(a_1) = a_2x = 0$ and $\delta(a_2) = xa_4 = 0$. The $a_0$-filtration in the domain gives $\ker \delta = k[a_1, a_2, a_4]$ (characteristic zero, $a_1a_2 \neq 0$). The $K$-image is $p \mapsto a_4$, $q \mapsto 0$, $J_1 \mapsto 2a_4a_1 - a_2^2$, $J_2 \mapsto 0$. If $G(a_4, U) \in (a_1a_2)$ with $U = 2a_4a_1 - a_2^2$,

setting $a_1 = 0$ gives $G(a_4, -a_2^2) = 0$ as a polynomial, whence $G = 0$ by $a_2$-degree. Descent identities and bounded kernel-dimension confirmation are certified by exact computation (scripts/cas/check_plinth_module.py, item M7; log data/plinth_module.log); the vanishing itself is the argument above. □

**Theorem 12.12** (The second chart resolves the collapse)**.** *At the three collapse points* $a_t = (0, t^2/2, t, 1, 1, 0)$, $t = 0, 1, 2$*, of Theorem 12.1,*

$$f_{13}(a_t) = -1/35 \neq 0,$$

*so the* $f_{13}$*-chart covers the exact locus where the* $pq$*-chart collapses. Moreover the invariant coordinate* $\pi_1(a_0)$ *separates the three orbits:*

$$\pi_1(a_0)(a_0),\ \pi_1(a_0)(a_1),\ \pi_1(a_0)(a_2) = 0,\ -1/8,\ -2,$$

*evaluated as* $\mathrm{NUM}_{a_0}(a_t)/f_{13}(a_t)^7$ *(no simplification needed; scripts/cas/check_plinth_module.py, item M6; log data/plinth_module.log), matching the* $H$*-values* $0, -315/4, -1260$*.*

**Theorem 12.13** (Local structure of the second chart)**.** *With* $Z = \pi_1(a_0)$*,*

$$\boxed{A_{f_{13}}^{D_5} = R_{f_{13}}[Z],}$$

*since* $\pi_1(a_4) = p$*,* $\pi_1(a_5) = q$ *and* $\pi_1(a_3), \pi_1(a_2), \pi_1(a_1) \in R_{f_{13}}$ *(Theorem 12.7 and Route E1), and the* $\pi_1(a_i)$ *generate the kernel by Lemma 13.6. Moreover, with* $\rho = G_0/c \in K_{pq}[s_0]$ *the pole polynomial (*$s_0 = x/(pq)$*) and* $\tau = s_1 - s_0$*,*

$$\boxed{Z = W - \rho(-\tau),}$$

*by uniqueness of the* $s_1$*-slice decomposition in* $A_{pqf_{13}}$ *(*$D_5(\tau) = 0$*, coefficients in* $K_{pq}$*). The pole polynomial has exact* $s_0$*-degree 7 with leading coefficient* $\rho_7 = -p^5q^3/112 \neq 0$ *(exact computation: all five* $G_0$*-terms contribute* $-1/16$*,* $1/24$*,* $1/12$*,* $-1/10$*,* $1/35$*).*

*Remark* 12.14 (No translation: $Z$ has exact $W$-degree 7). Since $s_1 = \alpha W + \beta$ with $\alpha = p^2q^3/f_{13} \neq 0$ and $\beta$ of $W$-degree 0, $\tau = s_1 - s_0$ has $W$-degree exactly 1, so $\Delta := Z - W = -\rho(-\tau)$ has exact $W$-degree 7 over $\mathrm{Frac}(B)$, with leading coefficient $\rho_7\alpha^7 = (-p^5q^3/112)(p^2q^3/f_{13})^7 \neq 0$. In particular $\Delta \notin \mathrm{Frac}(K)$: the $f_{13}$-chart line is a finite cover of the $W$-line of degree at most 7, not a translate. Any program of the form $K_{pq}[W] \cap R_{f_{13}}[W + \Delta]$ with $\Delta$ in the base field is therefore void; the $W \leftrightarrow Z$ change of basis carries the degree-7 relation, and a coefficientwise intersection theorem must go through it (open).

**Theorem 12.15** (Cover polynomial and Newton polygon)**.** *With* $\alpha = p^2q^3/f_{13}$ *the transition is* $\tau = s_1 - s_0 = \alpha W$ *(i.e.* $F_0/(qf_{13})$*, Theorem 12.7), so* $Z = P(W)$ *for the explicit* $P(T) = T - \rho(-\alpha T) \in \mathrm{Frac}(K)[T]$ *of exact degree 7 (odd:* $P_{2j} = 0$ *since* $\rho$ *is odd). Up to the recorded exact forms,*

$$v_q(P_1), v_q(P_3), v_q(P_5), v_q(P_7) = 0, 8, 16, 24, \qquad P_7 = -\frac{p^{12}q^{24}}{112J_2^7e_2^7},$$

*certified in scripts/cas/cover_newton.py (log data/cover_newton.log). Hence* $[\mathrm{Frac}(K)(W) : \mathrm{Frac}(K)(Z)] = 7$*, and with* $v_q(Z) = 0$ *(certified:* $\mathrm{NUM}|_{q=0} \neq 0$*) the* $q$*-Newton polygon of* $P(T) - Z$ *is* $(0,0)$*–*$(1,0)$*–*$(7,24)$*: one* $q$*-regular sheet and six sheets with* $q$*-pole order exactly* $4 = e_q$*.*

**Theorem 12.16** (Discriminant support)**.**

$$\mathrm{Disc}_T(P(T) - Z) = u\,\frac{p^{60}q^{120}}{J_2^{42}e_2^{42}}\,B(Z), \qquad u \in \mathbb{Q}^\times,$$

*with* $B(Z)$ *irreducible of degree 6 in* $Z$ *(even), coefficients in* $k[p, q, J_1, J_2]$*, computed by exact resultant (scripts/cas/cover_disc.sage; log data/cover_disc.log). In particular every base prime divisor of the discriminant is among* $p, q, J_2, e_2$*; no unknown boundary divisor appears. The remaining ramification component* $\{B(Z) = 0\}$ *is horizontal (irreducible over* $\mathbb{Q}$*).*

**Theorem 12.17** (Divisorial Newton data)**.** *For* $P(T) - Z$ *over the Gauss extensions (*$Z$ *a unit), with* $P_{2j} = 0$*:*

| *prime* | $(j, v(P_j)),\ j = 1, 3, 5, 7$ | *hull* |
|---|---|---|
| $q$ | $(1,0), (3,8), (5,16), (7,24)$ | $(0,0)$–$(1,0)$–$(7,24)$ |
| $J_2$ | $(1,2), (3,-3), (5,-4), (7,-7)$ | $(0,0)$–$(7,-7)$ |
| $e_2$ | $(1,-1), (3,-3), (5,-5), (7,-7)$ | $(0,0)$–$(7,-7)$ |

*i.e. one $q$-regular sheet plus six sheets of $q$-pole order exactly $4 = e_q$ (using $v_q(Z) = 0$, certified), and uniform vanishing order 1 on all seven sheets at $J_2$ and at $e_2$. Certified in scripts/cas/cover_newton3.sage (log data/cover_newton3.log); the $q$-row also follows from the factored forms of Theorem 12.15.*

*Remark* 12.18 (Status boundary). Established: the collapse points, the off-plinth regularity, and the shadow reading (which re-proves $H \notin K[F_0]$ geometrically). Also Established: the residual kernel (Lemma 12.4), the $f_{13}$-chart with its transition identity and invariance certificates (Theorems 11.2 and 12.7), and the orbit separation on the collapse locus (Theorem 12.12). Open: whether the exceptional locus meets $V(p) \setminus V(q)$; separation properties of the six-generator map to $X_5$; everything about $\mathrm{Spec}(A_5^{D_5})$.

*Remark* 12.19 (The $p$-side is covered by neither chart). Both chart denominators, $pq$ and $f_{13} = p \cdot h$, vanish on $V(p)$. The slice atlas therefore says nothing about the $p$-boundary, where no extra plinth generator is known. This matches the observed asymmetry: the $q$-side produced an extra slice ($f_{13}$) plus a gluing correction ($H$), while the $p$-side produced only the global correction $F_{46}$. This is recorded as an observation, not a theorem.

**Conjecture 12.20** (Gluing program)**.** With the two charts $D(pq)$, $D(f_{13})$ and transition $F_0/(qf_{13})$ established, three mutually non-exclusive outcomes are open:

(A) $H$ is forced as the regularity correction gluing the two charts over $V(q, h)$: evidence in range is the orbit separation of Theorem 12.12; a forcing theorem is missing. Falsifier: an invariant separating the collapse orbits that is not a $K$-combination involving $H$.

(B) $F_{46}$ is forced by a second-order obstruction on the deeper overlap. Falsifier: a derivation of $F_{46}$ from the $q$-atlas alone.

(C) A third plinth direction (second $q$-degeneration or a $p$-degeneration) appears: bounded evidence against is the exact plinth computation through weight 20 (Conjecture 11.5); any weight above with $\dim(\mathrm{im}\, D_5 \cap K_w) > \dim K_{w-2} + \dim K_{w-13}$ is a falsifier.

Known obstruction for all three outcomes: no forcing or lifting mechanism is proved; the evidence is bounded-range exact computation.

## 13. The closure theorem

Status of this section: **[Closed by Theorem 13.5]**. Generator discovery is over; what remains is saturation. The two charts identify the $p$-localized kernel, and exact $q$- and $p$-saturation contract it back to the polynomial ring. The failed valuation programs that this argument bypasses are recorded in Appendix H; the proof below uses none of them.

**Question 13.1** (tower closure for $A_5^{D_5}$ (answered))**.** Is the full invariant ring generated over $C_5$ by the second witness:

$$\boxed{A_5^{D_5} = C_5[F_{46}^{\mathrm{int}}]}$$

Yes, by Theorem 13.5. Equivalently, for every Fibonacci weight $w$, $\dim_k \ker(D_5|_{V_w})$ equals the dimension of the weight-$w$ piece of $C_5[F_{46}^{\mathrm{int}}]$ modulo its relations; the three-module prediction $P(w)$ of Conjecture 10.11 was the corresponding guess, refuted at weight 84 (Theorem 14.5) and replaced by the exact series (Theorem 14.4). The proof bypasses the all-weights-upper-bound component entirely ($p$-localization plus saturation in place of bounding); the $P(w)$ pattern stands refuted and superseded by the exact series. Finite generation follows (hypersurface plus one generator).

**Route E: $p$-local slice atlas (ESTABLISHED via $q$-saturation).** The endgame architecture is

$$\boxed{\text{slice} \to \text{degree-7 cover} \to p\text{-graded saturation} \to \text{global equality},}$$

i.e. E1 (local closure) plus E2 (residual saturation) imply $A^{D_5} = R$. In full, the proof dependency chain is literally

$$\boxed{\begin{aligned}&\text{local slice (Lemma 4.2)} \to \text{local kernel } K_{pq}[W]\\ &\qquad\to \text{second chart (Theorems 11.2, 12.7, 12.13)}\\ &\qquad\to q\text{-saturation (Theorem 13.4)}\\ &\qquad\to p\text{-saturation (Theorem 13.12)}\\ &\qquad\to \text{closure } A_5^{D_5} = R,\end{aligned}}$$

with $R = C_5[F_{46}^{\mathrm{int}}]$. No later geometric theorem is used: the singular-locus analysis (Theorem 14.7), normality (Theorem 14.8), Hilbert series (Theorem 14.4), and torsion filtration (Theorem 14.6) are all strictly downstream consequences of the closed presentation, never premises of closure. Stop conditions: a new $p$-local invariant falsifies E1 (adjoin and restart E2); a new $p$-torsion class falsifies E2 (adjoin and recompute); both succeeding proved closure; non-stabilizing torsion would itself have become the main result (infinite $p$-primary tower). Lemma 12.8 gives $A_{pq} \cap A_{f_{13}} = A_p$, so gluing the two slice charts computes the $p$-localized kernel $(A^{D_5})_p$. Tower closure split into the $D(p)$-atlas problem $(A^{D_5})_p \stackrel{?}{=} C_5[F_{46}^{\mathrm{int}}]_p$ plus $p$-power contraction. Bounded consistency: every kernel element through weight 48 lies in $(C_5[F_{46}^{\mathrm{int}}])_p$ (in fact in $C_5[F_{46}^{\mathrm{int}}]$ itself); the input the proof supplied was $p$-local upper bounds past the exact rational wall, i.e. the same wall as Route B but with denominators confined to $p$-powers. A new invariant on this cover would falsify Conjecture 12.20(C) at the $p$-local level. The gluing alternatives (A)–(C) of Conjecture 12.20 discriminate $H$-forcing, $F_{46}$-forcing, and third-direction scenarios without assuming any of them.

**Route E1: local closure (ESTABLISHED via $q$-saturation).** Put $R = C_5[F_{46}^{\mathrm{int}}]$. The first chart is solved: $A_{pq}^{D_5} = K_{pq}[W] = (C_5)_{pq} = R_{pq}$, since $H = H_2W^2 + H_0 \in K_{pq}[W]$ and $F_{46}^{\mathrm{int}} = Eq^2/(pq)^2 \in (C_5)_{pq}$ from $p^2F_{46}^{\mathrm{int}} = E \in C_5$ (Theorem 10.7). The second chart is exact with one extra coordinate: with $Z = \pi_1(a_0)$, $A_{f_{13}}^{D_5} = R_{f_{13}}[Z]$ (Theorem 12.13), so

$$\boxed{(A^{D_5})_p = R_{pq} \cap R_{f_{13}}[Z]}$$

exactly, by restriction (Lemma 13.7 with $Z$ adjoined). The coordinate $Z$ satisfies $Z = W - \rho(-\tau)$ with $\rho$ of exact $s_0$-degree 7; $Z - W$ has exact $W$-degree 7, so the $f_{13}$-chart is a finite cover of the $W$-line, not a translate, and no base-field translation program can capture the intersection (Remark after Theorem 12.13). The equality $(A^{D_5})_p = R_p$ itself is proved by $q$-saturation (Theorem 13.5), bypassing the Newton/conductor divisor program, which remains as structural description. Whether $Z$ itself lies in $R_{f_{13}}$ (the $s = 0$ and $t \geq 1$ forms; $s \geq 1$ impossible since $J_2 \nmid \mathrm{NUM}$) is open and would strengthen the chart equality to $A_{f_{13}}^{D_5} = R_{f_{13}}$.

**Corollary 13.2** (Degree-7 chart intersection, via saturation)**.** *With $P(T) \in \mathrm{Frac}(K)[T]$ the explicit cover polynomial of Theorem 12.15 (so $Z = P(W)$),*

$$\boxed{K_{pq}[W] \cap R_{f_{13}}[P(W)] = R_p,}$$

*as the $R_{pq} \cap R_{f_{13}}[Z]$ case of Theorem 13.5. The Newton/conductor proof originally proposed for this equality remains open as structural refinement; the falsifiers (A) a discriminant base divisor outside $\{p, q, J_2, e_2\}$ (excluded by Theorem 12.16) and (B) conductor strictly larger than $R_p$ are now consistency checks on a proved equality.*

**Lemma 13.3** (Machine-certified $q$-elimination identity)**.** *In $\mathbb{Q}[a_0, a_1, a_2, a_3, a_4, p, Q, J_1, J_2, F_0, H, T]$ with lex order $a_0 > \cdots > a_4 > p > Q > J_1 > J_2 > F_0 > H > T$, let $J_q$ be the graph ideal $(p - a_4, Q, J_1 - (2a_1a_4 - a_2^2), J_2 + a_3^2, F_0 - x^7/35, H - \bar{H}_q, T - \bar{F}_{46})$ with $x = a_3$ and $\bar{H}_q, \bar{F}_{46}$ the mod-$q$ reductions at the cleared-integral scale. Then the elimination of $\{a_0, \ldots, a_4\}$ satisfies $J_q \cap \mathbb{Q}[p, Q, J_1, J_2, F_0, H, T] = (S, p^2T - E, Q)$ exactly, both inclusions checked. Certified by `eliminate_q.sage` (Gröbner basis size 24, 1.3 s; elimination size 3) with `EQUAL: True` in `eliminate_q.log`. This lemma asserts only that ideal equality. Its two algebraic consequences, proved mathematically in Theorem 13.4, are: (i) the residual map $R/qR \to A/qA$ is injective; (ii) hence $R \cap q^nA = q^nR$ for all $n$ by induction cancelling $q$ in the domain $A$.*

**Theorem 13.4** ($q$-saturation)**.**

$$\boxed{R/qR \hookrightarrow A/qA,} \qquad \textit{hence} \qquad \boxed{R \cap q^nA = q^nR \quad \forall n \geq 1.}$$

*Proof of Theorem 13.4 from Lemma 13.3.* For $R = C_5[F_{46}^{\mathrm{int}}]$ with $F_{46}^{\mathrm{int}}$ in the witness normalization ($p^2F_{46}^{\mathrm{int}} = E$ exactly as printed in Theorem 10.7), Lemma 13.3 gives, after the transfer of Lemma 16.1, $\tilde{G} = AS + B(p^2T - E) + CQ$ for every lift $\tilde{G}$ mapping to 0. Let $[G] \in R/qR$ map to 0, $G = \sum G_i(F_{46}^{\mathrm{int}})^i$ with $G_i \in C_5$ ($R$ generated by $F_{46}^{\mathrm{int}}$). Lifting to $\tilde{G} = \sum \tilde{G}_i T^i \in k[p, Q, J_1, J_2, F_0, H, T]$, $\tilde{G}$ maps to 0, so $\tilde{G} = AS + B(p^2T - E) + CQ$. Mapping to $R/qR$: $[S] = 0$ ($S = 0$ in $C_5$), $[p^2T - E] = 0$ (the relation $p^2F_{46}^{\mathrm{int}} = E$ in $R$), $[Q] = 0$ ($q = 0$ in the quotient). Hence $[G] = 0$: the map is injective, i.e. $R \cap qA = qR$; induction as in Route E2 (if $q^na \in R$ then $q^na \in qA$, so $q^na = qr_1$, cancel $q$) gives all $n$. The computation is small and double-checked: Gröbner basis of size 24 in 1.3s, with the ideal equality verified by inclusion in both directions in the log. Machine part: the elimination identity. Mathematical part: the lift-and-cancel argument above. Transfer to general $k$ is Lemma 16.1 (all inputs over $\mathbb{Q}$). □

**Theorem 13.5** (Closure)**.**

$$\boxed{A_5^{D_5} = C_5[F_{46}^{\mathrm{int}}],}$$

*finitely generated. In particular E1 holds:* $(A^{D_5})_p = R_p$ *with* $R = C_5[F_{46}^{\mathrm{int}}]$.

*Proof.* Let $F \in R_{pq} \cap R_{f_{13}}[Z]$. By Lemma 12.8, $F \in A_p$; write $F = U/p^m$. Since $R_{pq} = (R_p)_q$, $q^n F \in R_p$ for $n$ the $q$-adic denominator exponent; choose $N$ large enough to clear both the $p^m$-denominator of $F$ and the $p$-denominator of $q^n F \in R_p$, $G = p^N F \in A$ satisfies $q^n G \in R$, so Theorem 13.4 gives $G \in R$ and $F = G/p^N \in R_p$. Thus $R_{pq} \cap R_{f_{13}}[Z] \subseteq R_p$ (reverse always holds), i.e. $(A^{D_5})_p = R_p$ by Lemma 13.7 with the chart equalities. Then $A^{D_5} = A \cap (A^{D_5})_p = A \cap R_p = R$ by Theorem 13.12. Finite generation follows: $C_5$ is a hypersurface, plus one generator. □

**Route E1 by divisors.** An element $G \in R_{pq} \cap R_{f_{13}}[Z]$ has denominators supported only on $\{p, q, J_2, e_2\}$ (from $pq$ and $f_{13} = pJ_2e_2$). Since $p$ is inverted in $R_p$, E1 is the three local nonnegativity statements at $q$, $J_2$, $e_2$. The Newton data (Theorem 12.17) give: at $q$, one regular sheet plus six sheets of pole order 4; at $J_2$ and $e_2$, uniform vanishing order 1 on all sheets. The discriminant contributes no further vertical divisor (Theorem 12.16). Route E1-A (direct descent via these profiles) and Route E1-B (conductor from Newton plus discriminant data) are open. Note: the naive $7 \times 7$ change-of-basis matrix reducing $Z^j \bmod (P(W) - Z)$ is degenerate ($\det = 0$), since each $Z^j$ already lies in the coefficient field; the finite data are Newton plus discriminant, not that matrix.

**Lemma 13.6** (Slice generation)**.** *Let* $A_f$ *carry a slice* $s$ *with* $D(s) = 1$ *and put* $\pi(b) = \sum_j (-s)^j D^j(b)/j!$, $B = k[\pi(a_0), \ldots, \pi(a_5)]$. *Then* $A_f = B_f[s]$, *and* $A_f^D$ *is generated by the* $\pi(a_i)$.

*Proof sketch.* $a - \pi(a) = s \cdot b$ with $b$ of strictly smaller $D$-nilpotency index, so induction on the index puts every $a_i$ in $B[s]$, and with $f = \pi(f) \in B$ this gives $A_f = B_f[s]$. That the projected variables then generate the kernel is the classical Dixmier slice-generation lemma for LNDs with a slice (Freudenburg [11]). □

**Lemma 13.7** (Two-chart $p$-local identification)**.** *If* $A_{pq}^{D_5} = R_{pq}$ *and* $A_{f_{13}}^{D_5} = R_{f_{13}}[Z]$, *then* $(A^{D_5})_p = R_{pq} \cap R_{f_{13}}[Z]$ *exactly, with no further hypotheses:* $(A^{D_5})_p = (A_{pq} \cap A_{f_{13}})^{D_5}$ *since* $A_{pq} \cap A_{f_{13}} = A_p$, *and invariance can be checked in* $\mathrm{Frac}(A)$, *so restriction gives equality in both directions. The residual step* $R_{pq} \cap R_{f_{13}}[Z] = R_p$ *is* $R$*-internal (open:* $R$ *divisor theory) and is* not *assumed here. The special case* $Z \in R_{f_{13}}$ *recovers* $(A^{D_5})_p = R_{pq} \cap R_{f_{13}}$.

**Route E2: residual saturation (ESTABLISHED).** For $R \subset A$ a domain, $R \cap pA = pR$ is equivalent to injectivity of $R/pR \to A/pA$, and implies full $p$-saturation $R = A \cap R_p$: if $p^n a \in R$ then $p^n a \in pA$, so injectivity gives $p^n a = pr_1$, cancel $p$, induct on $n$. The required injection is Theorem 13.12 below, via the residual presentation Theorem 13.11. The residual kernel is now known exactly (Theorem 13.8 below), so the injection is attacked by normal forms inside $k[q, x, y, \theta]$, not by computing an unknown target. Evidence for the conclusion: $(R : p^r)_w = R_w$ for $r = 1$ ($w \leq 52$) and $r = 2$ ($w \leq 50$), exact over $\mathbb{Q}$ (scripts/cas/check_torsion_R.py; log data/torsion_R.log): adjoining $F_{46}$ killed all $C_5$-torsion in range, and no new torsion over $R$ appears.

**Theorem 13.8** (Residual kernel on $V(p)$)**.** *On* $A/pA \cong k[a_0, a_1, a_2, a_3, q]$ *the residual derivation is* $\delta_p(a_0) = a_1a_2$, $\delta_p(a_1) = a_2a_3$, *zero elsewhere, i.e.* $\delta_p = y(x\partial_z + z\partial_{a_0})$ *with* $x = a_3$, $y = a_2$, $z = a_1$. *With* $\theta = 2xa_0 - z^2$,

$$\boxed{(A/pA)^{\delta_p} = k[q, x, y, \theta].}$$

*Proof.* $A/pA$ is a domain and $y \neq 0$, so $\ker \delta_p = \ker \delta'$ for $\delta' = x\partial_z + z\partial_{a_0}$; $\theta \in \ker \delta'$. Let $F = \sum_{k=0}^{t} E_k a_0^k$ with $E_k \in k[x, y, q, z]$, $E_t \neq 0$. The top $a_0$-coefficient gives $x\,\partial E_t/\partial z = 0$, so $E_t \in k[x, y, q]$; the next gives $x\,\partial E_{t-1}/\partial z + tE_t z = 0$, whence $x \mid E_t$ ($x$ prime, characteristic zero), $E_t = xE_t'$. Then $F - (E_t'/2)\theta a_0^{t-1}$ has $a_0$-degree $< t$; induction gives $F \in k[x, y, q, \theta]$ (base $t = 0$: $x\partial F/\partial z = 0$). $\theta$ is linear in $a_0$ over the domain, so the presentation is polynomial; it is also transcendental over $k[q, x, y]$, since a relation $G(q, x, y, \theta) = 0$ gives vanishing $a_0$-coefficients inductively (leading coefficient $(2x)^d \neq 0$). Kernel dimensions match through total degree 8 in exact computation (scripts/cas/check_residual.py, item R4; log data/residual.log). □

*Remark* 13.9 (Reductions mod $p$ and the $\theta$-normal form). $\bar{q} = q$, $\bar{J}_1 = -y^2$, $\bar{J}_2 = 2qy - x^2$, $\bar{F}_0 = q^2x^3y^2/3 - qx^5y/5 + x^7/35$, $\bar{H}$ has 4 terms (all $a_0$-free), and $\bar{F}_{46}^{\mathrm{int}} \neq 0$ has 10 terms. All identities by exact computation

(scripts/cas/check_residual.py; log data/residual.log) and formally in Lean (module `AdjQuad.ResidualCheck`). Moreover

$$\boxed{\bar{F}_{46}^{\mathrm{int}} = B \cdot \theta,} \qquad \begin{aligned} B = {} & 235200\, y^9x^2q^5 + 1093680\, y^8x^4q^4 - 1029420\, y^7x^6q^3 \\ & + 291060\, y^6x^8q^2 - 26460\, y^5x^{10}q, \end{aligned}$$

by direct expansion (with $F_{46}^{\mathrm{int}}$ the cleared integral form). Hence, with Theorem 13.8, $\mathrm{in}_p(R) = k[q, -y^2, \bar{J}_2, \bar{F}_0, \bar{H}, B\theta]$: the only $\theta$-dependent generator is $\bar{F}_{46}^{\mathrm{int}}$. (Modulo $p$, $\delta_p$ coincides with $D_4|_{a_4=0}$ on $k[a_0, \ldots, a_3]$ with $a_5$ inert; the $N = 4$ problem with $a_4$ free stays open.)

**Lemma 13.10** (Machine-certified $p$-elimination identity)**.** *In $\mathbb{Q}[x, y, q, J_1, J_2, F_0, H]$ with lex order $x > y > q > J_1 > J_2 > F_0 > H$, let $J_p$ be the graph ideal $(J_1 + y^2, J_2 - (2qy - x^2), F_0 - \bar{F}_0, H - \bar{H})$ with $\bar{F}_0 = q^2x^3y^2/3 - qx^5y/5 + x^7/35$ and $\bar{H}$ the mod-$p$ reduction. Then the elimination of $\{x, y\}$ satisfies $J_p \cap \mathbb{Q}[q, J_1, J_2, F_0, H] = (S, \bar{c}_{46})$ exactly, both inclusions checked. Certified by `eliminate_p.sage` (Gröbner basis size 119, 0.2 s; 77-generator elimination basis printed in full) with `EQUAL: True` in `eliminate_p.log`. This lemma asserts only that ideal equality. Its two algebraic consequences, proved mathematically in Theorems 13.11 and 13.12, are: (i) $\ker(C_5/pC_5 \to A/pA) = (\bar{c}_{46})$, i.e. $C_5$ has no other $p$-defect; (ii) hence $R/pR \hookrightarrow A/pA$ and $R = A \cap R_p$.*

**Theorem 13.11** (Residual presentation: the $C_5$-defect is principal)**.** *Let $c_{46} = p^2F_{46}^{\mathrm{int}} \in C_5$ (Theorem 10.7) and $\bar{c}_{46}$ its class mod $p$. Then*

$$\boxed{\ker(C_5/pC_5 \to A/pA) = (\bar{c}_{46}),}$$

*by Lemma 13.10 (exact elimination; scripts and log cited there), with $S = q^4H - 11025F_0^2 - 105q^2J_2^5J_1 - 9J_2^7$ the hypersurface relation, transferred to general $k$ by Lemma 16.1. In particular $C_5$ has no other $p$-defect: Failure I does not occur.*

**Theorem 13.12** ($p$-saturation)**.**

$$\boxed{R/pR \hookrightarrow A/pA,} \qquad \textit{hence} \qquad \boxed{R = A \cap R_p.}$$

*Proof.* Let $[G] \in R/pR$ map to 0, $G = \sum_{i=0}^{m} G_i(F_{46}^{\mathrm{int}})^i$ with $G_i \in C_5$ ($R = C_5[F_{46}^{\mathrm{int}}]$ by definition). Reducing mod $p$ and using $\bar{F}_{46}^{\mathrm{int}} = B\theta$ gives $\sum_i (\bar{G}_i B^i)\theta^i = 0$ in $A/pA$, with coefficients in $k[q, x, y]$; $\theta$ transcendental (Theorem 13.8) and $B \neq 0$ force $\bar{G}_i = 0$ for all $i$. By Theorem 13.11, $G_i = c_{46}K_i + pL_i$ with $K_i, L_i \in C_5$. Hence $[G] = [\bar{c}_{46}] \sum [K_i][F_{46}^{\mathrm{int}}]^i = 0$, since $[c_{46}] = 0$ in $R/pR$ ($c_{46} = p^2F_{46}^{\mathrm{int}} \in pR$). So the residual map is injective, and the saturation criterion of Route E2 gives $R = A \cap R_p$. No presentation $R = C_5[T]/(p^2T - c_{46})$ is assumed: only generation by $F_{46}^{\mathrm{int}}$ is used. (The Gröbner sizes 119/77 quoted here are the residual-presentation certificate of Theorem 13.11; this proof itself is pure algebra given that certificate and Theorem 13.8.) □

**$p$-torsion formulation of Route E.** With $R = C_5[F_{46}^{\mathrm{int}}]$ and $M = A_5^{D_5}/C_5$, the class $[F_{46}^{\mathrm{int}}]$ is a nonzero $p$-primary torsion class killed by $p^2$ ($F_{46}^{\mathrm{int}} \notin C_5$, $p^2F_{46}^{\mathrm{int}} \in C_5$, Theorem 10.7). Adjoining $F_{46}^{\mathrm{int}}$ kills the first observed torsion layer; closure asks whether it kills all $p$-torsion. The two named targets are the *$p$-local closure theorem* $(A^{D_5})_p = R_p$ (established: Route E1, Theorem 13.5) and the *$p$-saturation theorem* $R = A \cap R_p$ (established: Theorem 13.12), i.e. $R$ is $p$-saturated in $A$ ($\{a \in A : p^n a \in R\} = R$ for all $n$), i.e. $A^{D_5}/R$ has no $p$-power torsion. Failure of either yields an interpretable counterexample: a $D(p)$-visible invariant, or an invariant with a genuine $p$-power denominator relative to $R$. The testable form is the $p$-adic filtration $T_p^{(r)} = (C_5 : p^r)/C_5$: $(C_5 : p) = C_5$ through 46 with the new $pF_{46}$-layer at 47–52; $(C_5 : p^2)/C_5 = \langle F_{46} \rangle$ at 46 with the exact $F_{46} \cdot K/(p^2)$ counts through 50; $(C_5 : p^3)/(C_5 : p^2) = 0$ through 48 (same classes, no new ones) (Theorem 10.8). Finite-field torsion sweeps give independent consistency (scripts/cas/torsion_sweep.sage); exact characteristic-zero certificates above the table range are open past the rational wall.

*Remark* 13.13 (Consolidated barrier: five routes, one missing input). All four valuation routes demanded one missing input, *universal top-control* (Q-top-strong plus a $p$-side replacement). The resonance theory explains why the naive tool fails: the constant-coefficient initial-form step proves bounds solely under transcendence, and at $N = 5$ *both* primes are algebraic, so that step is provably silent exactly where primitives live; no initial-form analysis at any order closes bounds in general (Remark D.10). But algebraic residue defeats only the naive argument: it does not imply that no replacement top-control can exist — Q-top-strong is now proved by normal forms (Theorem 15.1), and closure was proved by Route E (Theorem 13.5), bypassing the divisor program rather than completing it. The "iff" formulation (tower closes iff finitely many resonance degrees) is rejected: $c = p^2q^4$ has exactly the prime divisors $p, q$, so $\{\rho_q, \rho_p\} = \{2, 4\}$ is

complete and finite by construction, yet universal top-control was never obtained (Theorem N.2). A third primitive would either falsify necessity for Conjecture N.1 or demand an extended framework beyond primes dividing $c$. Route E does not evade this barrier: it reframes the missing input as $p$-local upper bounds plus $p$-power contraction, strictly weaker than universal top-control. Of the two halves, E2 ($p$-saturation) is now established (Theorem 13.12); E1 ($p$-local closure) is established via $q$-saturation (Theorem 13.5), bypassing the divisor program. Closure is proved.

## 14. The exact algebra of the closed ring

With the ring closed, we determine what kind of ring it is. The finite-range census recorded first is independent corroboration of the global presentation proved in Theorem 14.2, not its source; its consequences — Hilbert series, filtration, normality, singular locus — close the section.

**Theorem 14.1** (Relation ideal in range; no monic quadratic)**.** *Let $M_w$ be the 7-generator monomial map to $A_w$ and $K_w$ the subspace generated by the known relations $S$ (weight 28) and $p^2T - E$ (weight 48). Exact sparse ranks over $\mathbb{Q}$ give $\dim\ker M_w = \dim K_w$ at every even $46 \le w \le 90$ and at $w = 92$ (including the $F_{46}^2$ column): scripts/cas/relation_exact.sage (log data/relation_exact.log). In particular there is no new relation through weight 92, and $F_{46}^2 \notin C_5 + F_{46}\cdot C_5$ (single exact solve: $\operatorname{rank} M = 16458 < 16459$ augmented; scripts/cas/check_f46quadratic.sage, log data/f46quadratic.log), so no monic quadratic over $C_5$. For the record, $F_{46}^2$ has only 488 terms ($79^2$ monomials combining). The exact relation ideal is established by Theorem 14.2, so $F_{46}$ is linear over $\operatorname{Frac}(C_5)$; what remains open is the finer $C_5$-module decomposition of $R$ and its higher-layer interpretation. The finite-range calculation through weight 92 is retained as independent corroboration, not as the source of the global presentation.*

**Theorem 14.2** (Relation ideal)**.** *Let $\Phi\colon C_5[T] \to R$ be the homomorphism with $T \mapsto F_{46}^{\mathrm{int}}$ (cleared integral form, $p^2F_{46}^{\mathrm{int}} = E$). Then*

$$\boxed{\ker\Phi = (S,\ p^2T - E),}$$

*with $S$ and $E$ as above. In particular no finite-weight census is needed: the weight-92 agreement is corroboration only.*

*Proof.* Since the presentation of $C_5$ is established, it suffices to prove $\ker\psi = (p^2T - E)$ for $\psi\colon C_5[T] \to R$, $T \mapsto F_{46}^{\mathrm{int}}$, with $r := p^2T - E \in \ker\psi$. First, $(p) \subset C_5$ is prime: $C_5/pC_5 \cong k[q, J_1, J_2, F_0, H]/(S)$, and $S$ is irreducible there since it is linear in $H$: if $S = FG$ then (comparing $H$-coefficients) one factor divides $q^4$, but no nonconstant divisor of $q^4$ divides $S$ (at $q = 0$, $S = -11025F_0^2 - 9J_2^7 \neq 0$), so that factor is constant. Second, $E = p^2F_{46}^{\mathrm{int}} \notin pC_5$: else $E = pb$ gives $b = pF_{46}^{\mathrm{int}}$ in the domain $A$, contradicting $pF_{46}^{\mathrm{int}} \notin C_5$ (Theorem 10.7(i)). Hence $\bar{E} \neq 0$ in the domain $C_5/pC_5$. Now let $f(T) = \sum_{i=0}^{m} b_iT^i \in \ker\psi$, $b_m \neq 0$ (base $m = 0$: $\psi|_{C_5}$ is the inclusion, so $b_0 = 0$). From $\sum b_i(F_{46}^{\mathrm{int}})^i = 0$, multiply by $p^{2m}$ and use $p^2F_{46}^{\mathrm{int}} = E$ to get $\sum_{i=0}^{m} b_iE^ip^{2(m-i)} = 0$ in $C_5$, with no fractions. Mod $p$, all terms but $i = m$ vanish: $\bar{b}_m\bar{E}^{\,m} = 0$, so $p \mid b_m$. Write $b_m = pc_m$, divide by $p$ in the domain, reduce mod $p$ again: $\bar{c}_m\bar{E}^{\,m} = 0$, so $p \mid c_m$ and $b_m = p^2d_m$. Then $f(T) - d_mT^{m-1}r$ lies in $\ker\psi$ of strictly smaller $T$-degree. Induction gives $\ker\psi = (r)$, whence $\ker\Phi = (S, r)$. □

**Corollary 14.3** (Complete-intersection presentation)**.**

$$\boxed{R \cong C_5[T]/(p^2T - E) \cong k[p, q, J_1, J_2, F_0, H, T]/(S,\ p^2T - E),}$$

*a codimension-two complete intersection ($S$ prime; $p^2T - E \neq 0$ in the domain $C_5[T]$): a domain, as it is the known subring $R = C_5[F_{46}^{\mathrm{int}}] \subset A$. The seven displayed generators are minimal (the first six for $C_5$; $F_{46}^{\mathrm{int}} \notin C_5$). Weights $(1, 1, 6, 4, 14, 24, 46)$ with relations in degrees 28 and 48.*

**Theorem 14.4** (Exact Hilbert series)**.**

$$\boxed{H_R(t) = \frac{(1 - t^{28})(1 - t^{48})}{(1 - t)^2(1 - t^4)(1 - t^6)(1 - t^{14})(1 - t^{24})(1 - t^{46})},}$$

*by the complete-intersection presentation. Since $A^{D_5} = R$ (Theorem 13.5), $[t^w]H_R = \dim_k\ker(D_5|_{V_w})$ for every $w$. In particular the three-family $F$-module pattern (Conjecture 10.11) fails at $w = 84$: $[t^{84}]H_R = 11888 \neq 11887$; further families ($F_{46}\cdot H\cdot F_0$ from 84, $F_{46}^2$ from 92) are required, and the exact series supersedes the pattern. Coefficient agreement with all exactly-established kernel dimensions ($w \le 48$ spanning/quotients, 54–61, 65, 69) holds, certified in scripts/cas/check_hilbert.py (log data/hilbert.log).*

**Theorem 14.5** (Fourth family seed). *$F_{46}{\cdot}H{\cdot}F_0$ (398 terms, weight 84) lies outside the prior span $C_5+F_{46}K+F_{46}F_0K+F_{46}HK$ at weight 84 (single exact $\mathbb{Q}$ solve: rank 11887 versus 11888 augmented; scripts/cas/check_f46hf0.sage, log data/f46hf0.log). With $p^2(F_{46}HF_0) = EHF_0 \in C_5$, its $K/(p^2)$-multiples account for the Hilbert excess exactly through 91; $F_{46}^2$ (not in $C_5 + F_{46}C_5$) seeds the next layer at 92. Higher $T$-degree families and the full $C_5$-module structure of $R$ remain open.*

**Theorem 14.6** ($C_5$-module structure: the $T$-filtration). *With $R \cong C_5[T]/(p^2T-E)$, set $R_m := C_5+C_5T+\cdots+C_5T^m$. Then $R = \bigcup_{m\geq 0} R_m$ and, for every $m \geq 1$,*

$$\boxed{R_m/R_{m-1} \cong C_5/(p^2) \cdot [T^m]}$$

*via $b \mapsto [bT^m]$. Indeed the map is surjective by definition of $R_m$, and $bT^m \in R_{m-1}$ means $bT^m - G \in (p^2T - E)$ for some $G$ of degree $< m$; comparing the top $T^m$-coefficient forces $p^2 \mid b$ in $C_5$, while $p^2T^m = ET^{m-1} \in R_{m-1}$ gives the reverse inclusion. (Note $\mathrm{Ann}_{C_5}(T) = 0$ in $R$ itself since $R$ is a domain; the $(p^2)$-torsion lives in the successive layers, i.e. $\mathrm{Ann}_{C_5}([T^m]) = (p^2)$ in $R_m/R_{m-1}$.) Consequences:*

- *(i) No monic relation: $T^m \notin \sum_{i<m} C_5\, T^i$ for all $m \geq 1$ (a monic relation would give $[T^m] = 0$, i.e. $1 \in (p^2)$, absurd). In particular no monic quadratic; $F_{46}^2 \notin C_5 + F_{46}C_5$.*
- *(ii) Torsion ladder: $p^2T^m = ET^{m-1} \in \sum_{i<m} C_5\, T^i$ for all $m \geq 1$.*
- *(iii) Associated graded: $\mathrm{gr}_T(R) \cong C_5 \oplus \bigoplus_{m\geq 1}(C_5/(p^2))\tau^m$ as $C_5$-modules, $\tau^m$ the class of $T^m$. The open refinement is the graded decomposition of each $C_5/(p^2)$-layer: the observed 84-weight class $[HF_0T]$ and 92-weight class $[T^2]$ are the first visible pieces not generated by the $K$-substructure.*
- *(iv) The fourth family is genuinely new over $C_5$: $HF_0 \notin K + F_0K + HK + p^2C_5$ (exact $\mathbb{Q}$-solve, rank 827 vs augmented 828: scripts/cas/check_hf0.sage; log data/hf0.log), i.e. the class $[HF_0T]$ is not generated by the $K$-substructure in the $m = 1$ layer; else Theorem 14.5's seed would lie in the prior span, by substituting into $F_{46}^{\mathrm{int}}HF_0$ and using $p^2F_{46}^{\mathrm{int}} = E$.*

**Theorem 14.7** (Exact singular locus of $X = \mathrm{Spec}\, R$). ***(A) General-$k$ part.*** *Over any characteristic-zero field $k$, with $R = C_5[F_{46}^{\mathrm{int}}]$ as above: $\mathrm{Sing}\, X \cap D(p) = V(q, F_0, J_2) \cap D(p)$; the closed subset $V_{\mathrm{old}} := V(q, F_0, J_2, p^2T - E)$ is an irreducible 3-fold meeting $D(p)$ (on $V(q, F_0, J_2)$ one has $E = H^2$, so this is $\{0\} \times V(p^2T - H^2)$ in $(q, F_0, J_2) \times (p, J_1, H, T)$ coordinates; $p^2T - H^2$ is irreducible, linear in $T$ with coprime coefficients); and $\dim \mathrm{Sing}\, X = 3$.*

***(B) $\mathbb{Q}$-specific fiber decomposition.*** *Over $k = \mathbb{Q}$, $\mathrm{Sing}\, X$ has exactly six irreducible components, all of dimension 3: the $V_{\mathrm{old}}$ of (A) plus five fiber components $V(p, P_i) \times \mathbf{A}^1_T$ ($i = 0, \ldots, 4$), where $P_0, \ldots, P_4$ are the minimal primes of $(S, E, 2 \times 2$ minors$)$ in $\mathbb{Q}[q, J_1, J_2, F_0, H]$:*

$$\begin{aligned}
P_0 &= (F_0,\ 150q^2J_1^3J_2 + 331J_1^2J_2^3 + H,\ 25q^4J_1^2 + 51q^2J_1J_2^2 + 9J_2^4),\\
P_1 &= (100q^2J_1 + 21J_2^2,\ 1484J_1^2J_2^3 + 5H,\ 27J_2^7 + 7656250F_0^2),\\
P_2 &= (F_0,\ 4q^2J_1 + J_2^2,\ 276J_1^2J_2^3 + H),\\
P_3 &= (q,\ 385J_1^2J_2^3 + H,\ J_2^7 + 1225F_0^2),\\
P_4 &= (H,\ J_1,\ J_2^7 + 1225F_0^2).
\end{aligned}$$

*The $P_i$ are certified prime of dimension 2 over $\mathbb{Q}$ by exact primary decomposition; over general characteristic-zero $k$ the same closures hold wherever the $P_i$ remain prime, otherwise refined by Galois orbit (no extra embedded component appears over $\mathbb{Q}$).* Proof of (A). *The Jacobian of the complete intersection $(S, r)$, $r = p^2T - E$, has $2 \times 2$ minors $-p^2S_x$ in the $T$-column, 0 in the $p$-column (since $S_p = E_p = 0$: neither $S$ nor the seven-term $E$ involves $p$), and the ten 5-variable minors $S_xE_y - S_yE_x$ otherwise. On $D(p)$, $p$ is a unit so $\partial r/\partial T = p^2 \neq 0$; the Jacobian has rank $< 2$ iff all $S_x$ vanish, i.e. $q = F_0 = J_2 = 0$ by Remark 7.2. Hence $\mathrm{Sing}\, X \cap D(p) = V(q, F_0, J_2) \cap D(p) \subset V_{\mathrm{old}}$, all over any characteristic-zero $k$.* Proof of (B). *On $V(p)$ the fiber is supported over $V(p, E)$ (not all of $V(p)$): $T$ is free exactly over $V(p, E)$, and the fiber base decomposes as $V(P_0) \cup \cdots \cup V(P_4)$ with no further minimal primes (exact primary decomposition over $\mathbb{Q}$: scripts/cas/sing_fiber.sage; log data/sing_fiber.log; each $P_i$ certified prime of dimension 2, remaining associated primes strictly embedded). Irredundancy: $V_{\mathrm{old}}$ meets $D(p)$ while fiber components lie in $V(p)$, and a 3-fold fiber component cannot lie in the 2-fold $V_{\mathrm{old}} \cap V(p)$; distinct minimal primes give distinct components. In particular $\dim \mathrm{Sing}\, X = 3$ purely.*

**Theorem 14.8** (Normality of $R$)**.** *$R$ is normal over $k = \mathbb{Q}$ and hence over any characteristic-zero $k$. Indeed $R$ is a complete intersection, hence Cohen–Macaulay ($S_2$). By part (A) of Theorem 14.7,* $\dim \operatorname{Sing} X = 3$*, codimension 2 in the 5-fold $X$; this dimension bound uses only the general-$k$ Jacobian computation, not the $\mathbb{Q}$-specific fiber decomposition of part (B). Thus $R_1$ holds, and Serre's criterion gives normality. The complete-intersection property and the dimension bound* $\dim \operatorname{Sing} \leq 3$ *are stable under base change from $\mathbb{Q}$ to any $\mathbb{Q}$-algebra (component splitting preserves dimension), so normality transfers although the component count is stated over $\mathbb{Q}$.*

*Remark* 14.9 ($R_p$ record). $R_p = (A^{D_5})_p$ is established (Theorem 13.5), and $F_{46}^{\text{int}} = E/p^2$ exhibits $R_p = C_{5,p}[E/p^2]$ explicitly. No conductor conjecture is made: the conductor/exceptional ideal of the modification $X \to X_5$ awaits the exact presentation analysis above.

*Remark* 14.10 (Affine modification picture). With $X_5 = \operatorname{Spec} C_5$ and $X = \operatorname{Spec} R$, $X|_{D(p)} \cong X_5|_{D(p)}$ since $T = E/p^2$ there, and the exceptional locus lies over $V_{X_5}(p, E)$. Over $p = E = 0$ the fiber is $\mathbf{A}^1_T \times V(p, E)$, since $p^2T - E \in (p, E)$ and $T$ is otherwise free. No blow-up identification is claimed.

The presentation also fixes the top theory of the closed ring: the next section separates what valuation methods can prove from what the hidden directions refute.

## 15. The two-slope top theory

The closed ring $R = A_5^{D_5} = C_5[F_{46}^{\text{int}}]$ has presentation $R \cong C_5[T]/(p^2T - E)$. Over $K = k[p, q, J_1, J_2]$ the ring is generated by $F_0, H, F_{46}^{\text{rat}}$ of $a_0$-degrees $(1, 2, 4)$ with tops $(t_0, t_H, t_{46}) = (p^2q^4,\ 11025p^4q^4,\ \frac{18375}{4}p^6q^8)$ (the rationally normalized top; the cleared-integral file scalar does not affect degrees or valuations) of $(v_p, v_q)$-bidegrees $(2, 4)$, $(4, 4)$, $(6, 8)$ (exact anchor: scripts/cas/check_tops.py; log data/tops.log). Since $A = B[a_0]$ is a domain, $a_0$-tops multiply, extending the $F_0, H$ case of Section 10 to $F_{46}$ via the identical argument ($t_{46} \neq 0$).

### 15.1 The q-envelope

**Theorem 15.1** (Q-top-strong)**.** *Let $F \in A_5^{D_5}$ have $a_0$-degree $m$ with top $C_m \in K$. Then $v_q(C_m) \geq 4\lceil m/2 \rceil$.*

*Proof.* Write $F = P(F_0, H, F_{46})$ and let $\tilde{P}$ be the top-$a_0$-homogeneous part ($\deg_{a_0} = m$); lower parts do not affect the top. Tops multiply, so $C_m = \sum_{a+2b+4r=m} c_\alpha t_0^a t_H^b t_{46}^r$ with $c_\alpha \in K$. Each monomial has $v_q = 4a + 4b + 8r$, and $4a + 4b + 8r - 4\lceil (a + 2b + 4r)/2 \rceil = 4\lfloor a/2 \rfloor \geq 0$ ($a = 2j$ gives $4j$; $a = 2j + 1$ gives $4j$). Since $c_\alpha$ is polynomial, $v_q(c_\alpha) \geq 0$, and $v_q$ of a sum is at least the minimum. Hence $v_q(C_m) \geq 4\lceil m/2 \rceil$. □

*Remark* 15.2. The three generators meet the bound with equality exactly when the gap $4\lfloor a/2 \rfloor$ vanishes: $F_0$ ($m = 1$), $H$ ($m = 2$), $F_{46}$ ($m = 4$). The resonance that killed the superseded $4j$ $W$-coset bound (Lemma J.3) is compliant here: $H$ attains the envelope rather than violating it.

### 15.2 The p-side: impossibility and sharp envelope

**Proposition 15.3** (No bounded-defect P-top correction)**.** *There is no constant $C$ with $v_p(C_m) \geq 2m - C$ for all homogeneous invariants. Indeed $F_{46}^r$ has $m = 4r$, $C_{4r} = (p^6q^8)^r$ (tops multiply), so $v_p = 6r$ and the defect $8r - 6r = 2r \to \infty$.*

**Theorem 15.4** (Sharp p-envelope)**.** *$v_p(C_m) \geq \lceil 3m/2 \rceil$ for every $a_0$-degree-$m$ invariant top, and the constant $3/2$ is best possible: $v_p = 6r = 3m/2$ on $F_{46}^r$ ($m = 4r$) for all $r \geq 1$.*

*Proof.* Same normal form as Theorem 15.1: monomial tops have $v_p = 2a + 4b + 6r \geq 3m/2$ since $2a + 4b + 6r - 3(a + 2b + 4r)/2 = (a + 2b)/2 \geq 0$; integrality gives $\lceil 3m/2 \rceil$. Sums preserve via $v_p \geq \min$. Sharpness is the $F_{46}^r$ computation above. □

### 15.3 p-excess directions

Write $\varepsilon_p(F) = v_p(C_m) - \lceil 3m/2 \rceil \geq 0$ for the p-excess (nonnegative by Theorem 15.4). Monomial excesses are closed-form, $\varepsilon_p = 2a + 4b + 6r - \lceil 3(a + 2b + 4r)/2 \rceil$:

| | $m$ | $v_p$ | $\varepsilon_p$ |
|---|---|---|---|
| $F_0$ | 1 | 2 | 0 |
| $H$ | 2 | 4 | 1 |
| $F_{46}$ | 4 | 6 | 0 |
| $F_{46}^2$ | 8 | 12 | 0 |
| $F_{46}HF_0$ | 7 | 12 | 1 |
| $F_0^2$ | 2 | 4 | 1 |

**Proposition 15.5** (Zero-excess control)**.** *A monomial $F_0^a H^b F_{46}^r$ has $\varepsilon_p = 0$ iff $a+2b \leq 1$ (since $v_p - 3m/2 = (a+2b)/2$, so equality with the ceiling holds iff $(a + 2b)/2 < 1$). Consequently, if $\varepsilon_p(F) = 0$ then the top $C_m$ contains a nonzero term $c\, t_0^a t_H^b t_{46}^r$ with $a + 2b \leq 1$: indeed each normal-form term is itself a homogeneous invariant, so every term has $v_p \geq \lceil 3m/2 \rceil$, and $v_p(C_m) = \lceil 3m/2 \rceil$ forces some term to attain it. In particular the zero-excess directions are spanned by the $F_{46}$-tower and its $F_0$-translate; no bounded exact solve is needed, and none is claimed, beyond this monomial-plus-minimum argument. The full low-excess ($\varepsilon_p = 1$) census, where the observed $F_{46}HF_0$ seed sits, remains an open investigative direction.*

**Proposition 15.6** (Excess-one control)**.** *A monomial $F_0^a H^b F_{46}^r$ of $a_0$-degree $m$ has $\varepsilon_p = 1$ iff ($m$ even and $a + 2b = 2$) or ($m$ odd and $a + 2b = 3$): with $g = (a + 2b)/2 = v_p - 3m/2$ and $f = \lceil 3m/2 \rceil - 3m/2 \in \{0, \frac{1}{2}\}$, $\varepsilon_p = g - f = 1$ forces exactly these cases. Consequently, if $\varepsilon_p(F) = 1$ then the top $C_m$ contains a nonzero term with $a + 2b \leq 3$. The observed seed $F_{46}HF_0$ ($m = 7$, $v_p = 12$, $\varepsilon_p = 1$) sits in the odd $a + 2b = 3$ family alongside $F_0^3 F_{46}$; $F_0^2$ sits in the even $a + 2b = 2$ family (exact top checks: scripts/cas/check_excess1.py; log data/excess1.log). This closes the $\varepsilon_p = 1$ census at monomial-plus-minimum level; finer direction classification within the $a + 2b \in \{2, 3\}$ strata remains open.*

### 15.4 Program: resonance defect filtrations (conjectural)

No result is claimed here. For general $N$, the pattern established for the closed $N = 5$ ring is: a strong valuation bound either holds outright from the exact presentation (as Q-top-strong did at $N = 5$) or fails with defect concentrated along identified resonant/tower directions ($H$ against the superseded $4j$ bound; $F_{46}^r$ against $2m$). The question whether $\mathcal{E}(m) = \{\text{tops}\}/J_m$ is generated by residual resonance classes is therefore asked, not answered, beyond $N = 5$.

## 16. Computational trust model

Machine-checked claims in this paper rest on three separated tiers that are never mixed. The manifest records exact theorem names and axiom outputs; certificate mechanics are documented alongside it.

### 16.1 Trust model

Three tiers, never mixed. *Mathematical proof* is an argument appearing in the paper (e.g. the $J_m$ formula, the hypersurface primitivity argument, the $F_{46}$ top valuation). *Exact computation* is a fresh characteristic-zero calculation with reproducible dimensions and ranks (e.g. the 15–46 rank table); finite-field runs are consistency checks only. *Formal verification* is a Lean check in core Lean without Mathlib by native evaluation, recorded with exact theorem names and `#print axioms` outputs in the appendix manifest (Section M); it covers identities, weights, tops, the $H$-exclusion predicate, and per-weight spanning through 14. In particular, $F_{46} \notin C_5$ is a paper-level valuation argument, not a Lean theorem, and no upper-bound computation past the checked range is claimed. The decisive closure markers (both eliminations, second chart, plinth element and transition, chart-intersection certs, F46 gate) reproduce with one command, `sh scripts/reproduce_closure.sh` (SageMath + SymPy; 6 steps, each checking its pass marker; see the script header and SUPPLEMENT_README for markers and runtimes). Throughout, a claim is either **proved in the manuscript** (no machine premise), **proved using an explicit exact computational certificate** (machine identity stated with ring, generators, order, and log), or **conjectural/open/unverified**.

**Lemma 16.1** (Transfer of $\mathbb{Q}$-certificates to arbitrary characteristic-zero $k$)**.** *All exact linear-algebra and Gröbner-elimination certificates in this paper use input data with coefficients in $\mathbb{Q}$ (indeed in $\mathbb{Z}[1/420]$). Let $k$ be any characteristic-zero field, so $k$ is a field extension of $\mathbb{Q}$, hence a faithfully flat $\mathbb{Q}$-module. Then: (i)* ranks transfer: *a matrix over $\mathbb{Q}$ and its image over $k$ have the same rank, since a nonzero rational minor stays nonzero in $k$; exact $\mathbb{Q}$-rank equalities (spanning, quotient dimensions, torsion counts) therefore hold over $k$. (ii)* elimination ideals commute with this base change: *for $I \subset \mathbb{Q}[X]$ and $J = I \cdot k[X]$, contraction along a coordinate projection satisfies $J \cap k[Y] = (I \cap \mathbb{Q}[Y]) \cdot k[Y]$, because $- \otimes_{\mathbb{Q}} k$ is exact; an elimination identity certified over $\mathbb{Q}$ thus yields the same identity over $k$. What does* not *transfer is primality and primary-decomposition component counts: $D \otimes_{\mathbb{Q}} k$ for a $\mathbb{Q}$-domain $D$ need not be a domain, and minimal primes can split. Hence component counts (notably the six singular components) are stated over $k = \mathbb{Q}$, with an explicit geometric-prime note wherever a general-$k$ reading would require one.*

With the tiers separated, the residue mechanism behind the hidden directions can be isolated as a general theorem: the next section extracts it from the $N = 5$ setting.

## 17. The multfloor transfer theorem

The theorem below is a general statement about triangular derivations with a pole element and finite residue data; its proof is the multiplicity-transfer argument of Theorem J.7, isolated from the $N = 5$ setting. The $q$-side instantiation recovers Theorem J.7; the $p$-side instantiation is conditional (inputs unproved); the Daigle–Freudenburg instantiation is computed here and unconditional (no priority claim beyond this paper). We work in characteristic zero throughout.

### 17.1 Setup and hypotheses

Let $D$ be triangular on a domain $A = B[t]$ with $t$ transcendental over $B$ and $D(t) \in B$. Let $K = B^D$ and assume:

- (H1) $K$ is factorially closed in $B$ (standard for LND kernels).
- (H2) There is a pole element $u = t + \rho$ with $0 \neq \rho \in B_f$ for some $0 \neq f \in K$, $D(u) = 0$, and $u$ transcendental over $\mathrm{Frac}(B)$.
- (H3) Every invariant $F \in A^D$ expands uniquely $F = \sum_{j=0}^{m} H_j u^j$ with $H_j \in K_f$ ($m$ the $u$-degree). This holds wherever the localized slice presentation $A_f^D = K_f[u]$ is established; it is verified per instance below.
- (H4) For each prime $\pi_i \mid (f)$ in $K$ (height-one, with $\pi_i \in K$), the valuation $v_i$ extends to $\mathrm{Frac}(B)$ with $v_i|_B \geq 0$; the pole orders $e_i = -v_i(\rho) > 0$ are exact; residues of $K_f$-elements land in $L_i = \mathrm{Frac}(K/\pi_i)$. This landing is an independent hypothesis: it does not follow from factorial closure (H1) alone.
- (H5) Each initial form $r_i = \mathrm{in}_i(\pi_i^{e_i}\rho)$ has separable minimal polynomial $m_i$ of exact degree $d_i < \infty$ over $L_i$.

**Theorem 17.1** (Multfloor transfer)**.** *Under* (H1)–(H5)*, every $D$-invariant $F$ of $u$-degree $m$ satisfies, at each $i$,*

$$\boxed{\lambda_i(F) := \min_j (v_i(H_j) - e_i j) \geq -(e_i/d_i)m,}$$

*with deficit $+\infty$ for $H_j = 0$.*

*Proof.* The minimum is attained ($H_m \neq 0$); if $\lambda := \lambda_i(F) \geq 0$ there is nothing to prove. Assume $\lambda < 0$ and put $S = \{j : v_i(H_j) - e_i j = \lambda\} \neq \emptyset$, $P(T) = \sum_{j \in S} h_j T^j$ with $0 \neq h_j = \mathrm{in}_i(H_j) \in L_i$ (nonzero by definition of the initial form); $P \neq 0$ has $\deg P \leq m$. The $t$-coefficients $C_k = \sum_{j \geq k} \binom{j}{k} H_j \rho^{j-k}$ lie in $B$ (pure algebra: $u$ is monic of $t$-degree 1), so $v_i(C_k) \geq 0$ for all $k$. Since $v_i(\rho) = -e_i$ exactly, each $(j,k)$-summand has $v_i \geq \lambda + e_i k$, with equality iff $j \in S$; its initial form is then $\binom{j}{k} h_j r_i^{j-k}$. Hence for every $k$ with $\lambda + e_i k < 0$ the leading coefficient of $C_k$ vanishes: $P^{(k)}(r_i) = 0$, using $P^{(k)}(T)/k! = \sum_{j \geq k} \binom{j}{k} h_j T^{j-k}$ in characteristic zero. The qualifying $k \in [0, m]$ form an initial segment $\{0, \ldots, c-1\}$ with $c = \#\{0 \leq k \leq m : \lambda + e_i k < 0\}$; thus $r_i$ is a root of $P$ of multiplicity at least $c$, so $m_i^c \mid P$ in $L_i[T]$ (separable minimal polynomial), giving $d_i c \leq \deg P \leq m$. If $\lambda < -(e_i/d_i)m$ then $-\lambda/e_i > m/d_i$, so every $0 \leq k \leq m/d_i$ qualifies and $c > m/d_i$, contradicting $d_i c \leq m$. Hence $\lambda \geq -(e_i/d_i)m$. □

### 17.2 Instantiations

$N = 5$, *q-side (recovers R1).* With $e_q = 4$, $d_q = 2$ the theorem gives $\lambda_q \geq -2m$: this is Theorem J.7, now seen as the $q$-instantiation of the transfer. All inputs are established (Lemma J.6, descent, contraction). For (H4): $K = k[p, q, J_1, J_2]$ is polynomial with $(p), (q)$ non-associated primes by Lemma 4.3, so $y = A/p^a \in K_{pq}$ has $v_q(y) = v_q(A)$ with residue $(A \bmod q)/(p^a \bmod q) \in L_q = \mathrm{Frac}(K/qK)$; the exact degree $d_q = 2$ is Lemma J.6.

$N = 5$, *p-side (conditional).* With $e_p = 2$, $d_p = 4$ the theorem would give $\lambda_p \geq -m/2$. The inputs are unproved: exact $\rho_p = 4$ (computational residue evidence; see Supplementary Material) and $K_{pq}$-residue landing. Hence $-m/2$ is a conditional floor, not an established bound; it does not replace the conjectured weak floor $\lambda_p \geq -4m$ of Conjecture 21.6, which both witnesses clear loosely.

*Daigle–Freudenburg $x$-side (computed here, unconditional).* For the derivation of Section 19 below, (H1)–(H5) hold with a single prime, $e_x = 1$, $d_x = 1$: the $H_j$-descent into $(S^\Delta)_x$ is proved there, $t = -s \in L_x$ has minpoly $T - t$. The theorem gives $\lambda_x \geq -m$ for every $u$-degree-$m$ invariant. Verified tight on the literature degree-1 tower: $\lambda(\beta_1) = 0$, $\lambda(\gamma_1) = \lambda(\delta_1) = -1$ (exact SymPy $u$-expansions with $u = v - s/x$). The bound coexists with infinite generation: valuation floors do not control finite generation.

*Remark* 17.2 (When (H4) holds). (H4) holds whenever the residue field is accessible by specialization: it is verified above for the $N = 5$ $q$-side ($K$ polynomial, $(p), (q)$ non-associated primes) and for the Daigle–Freudenburg $x$-side (descent into $(S^\Delta)_x$ proved in Section 19). It fails to be verifiable on the $N = 5$ $p$-side, where both the exact residue degree and the landing are unproved — hence the conditional status there. In general, (H4) is the precise point at which a computational residue description must be promoted to a theorem before the transfer bound can be invoked.

*Remark* 17.3 (Status boundary). Established: the transfer theorem and its $q$-side and DF instantiations. Conditional: the $p$-side $-m/2$ floor (inputs unproved). Untouched: the weak $-4m$ conjecture, the refuted R2, and tower closure. No generation consequence is drawn from floors alone.

## 18. Comparison with neighbouring theories

The purpose is to place the adjacent-quadratic shift inside its classical superclass, to name the closest prior art, and to fence off theories that are context only.

### 18.1 Jonquières automorphisms and locally nilpotent derivations: the superclass

The maps $P_{\lambda,N}$ are triangular automorphisms with explicit descending-recursion inverses and unit Jacobian determinant (Lemma 2.2), and each $D_N$ is a triangular monomial locally nilpotent derivation. The ambient theories — the de Jonquières group of triangular automorphisms and the algebraic theory of locally nilpotent derivations (Freudenburg's monograph [11]; van den Essen [26, 27]; Dufresne–Kohls [8, 9]) — are therefore the classical superclass of everything in this paper, not a competitor to it. Our claim sits strictly inside that framework: a new structured family with an exact $N = 5$ case and a testbed for saturation algorithms. No automorphism-class novelty is claimed, and the Fibonacci-weighted grading is used only as a descriptive grading, never as a new derivation concept. This distinguishes it from the linear 'Fibonacci derivations' of Bedratyuk [1, 2].

### 18.2 Non-goals and superficial similarities

The coordinate update is superficially related to several triangular nonlinear maps in discrete mathematics and cryptography, but those objects are not used in the present analysis. The Keccak $\chi$ nonlinearity (Schoone–Daemen [22, 21]; Kriepke [18]) is cyclic over $\mathbb{F}_2$ with a complement term, while the shift is acyclic with $a_{N+1} = a_{N+2} = 0$ over characteristic zero and no constant term. Volterra-type lattices are infinite translation-invariant integrable systems; the shift is a finite nilpotent cascade whose boundary breaks translation invariance, so no lattice-integrability claim is made. Triangular unit-Jacobian maps formally resemble normalizing-flow coupling layers; no application is developed here.

### 18.3 Daigle–Freudenburg methods: method context, not evidence

The slice, plinth-ideal, and saturation techniques used in Sections 4–6 belong to the lineage of Daigle–Freudenburg dimension-five invariant theory [5, 12] and the triangular kernel algorithms of Maubach [20, 19] and van den Essen [26]. Within that lineage, van den Essen's algorithm computes invariants by Gröbner elimination after localizing at a slice, while Maubach's variant [19] works degree by degree with linear algebra over the $D$-grading, avoiding Gröbner bases; both terminate only given finite generation. The finite-generation ideal perspective of Hart [16, 17] (finite-generation ideal of the Daigle–Freudenburg five-variable kernel as the radical of three infinite families, which together with one further invariant form a SAGBI basis) is the closest structural cousin: it measures the locus on which finite generation holds rather than assuming it. The present family is distinct: its $a_0$-free base kernel is polynomial, and the subtlety lies entirely in the $a_0$-extension with two finite residue degrees $\{2, 4\}$ and a proved hypersurface piece $C_5$, versus the Daigle–Freudenburg infinite climb with stable tops and unbounded degree. That work concerns the Daigle–Freudenburg example and its finite-generation ideal/SAGBI structure; ours gives the exact invariant-ring structure (presentation and closure) for the adjacent-quadratic $N = 5$ case, a different derivation. No slice, Jacobian, UFD-descent, or SAGBI novelty is claimed; novelty is claimed only for this family, the explicit $H$ and $F_{46}$ obstructions, the $C_5$ hypersurface, the residue-algebraicity mechanism, and the multfloor bound. These are cited strictly as method context: they situate the residue-transcendence argument of Lemmas I.3 and J.3, and they are never evidence for the refuted five-generator claim of Corollary 6.1. Saturation-as-conceived cannot complete: its joint induction is void (both premises refuted); method novelty, if any, lies in the replacement tower methods (Theorem J.7, Section 13), not in part [E] (anchor equality).

The lineage runs both ways. The resonance computation of Section 19 gives $\rho(\mathrm{DF1999}) = \{1\}$: the DF derivation is infinitely generated despite minimal resonance. Resonance degree alone does not control finite generation; the escape/climb distinction of Section 20 identifies the structural difference (DF climbs with stable tops and unbounded degree; $N = 5$ escapes via residue relations enabling pole cancellation). The transfer theorem (Section 17) bounds both towers' valuation floors without implying anything about their finite generation.

The bounded ambient-derivation search (Makar–Limanov value open) is recorded in Appendix Q: no ML value is computed or claimed.

### 18.4 Geometry of the hypersurface quotient

The singular locus of $X_5$ is computed in Remark 7.2 (Section 6); what remains here are the open geometric questions.

*Remark* 18.1 (Open geometric questions). (i) $\mathrm{ML}(X_5)$ is open (Section Q, bounded triangular search only). (ii) Whether $X_5 \times \mathbb{A}^1 \cong \mathbb{A}^6$ (Zariski cancellation) is open; $X_5$ is recorded here as a test case, with no position taken. (iii) The $\mathbb{G}_a$-quotient map $\mathbb{A}^6 \to X_5$: the slice $s$ with $D(s) = 1$ exists only after localizing at $pq$, so principality of the bundle is claimed at most over the distinguished open $D(pq)$; the fibers over $V(pq)$ and over $\mathrm{Sing}(X_5)$ are open. No global bundle claim is made.

### 18.5 Certificate pattern and generation conjecture in the literature

The van den Essen kernel algorithm terminates *given* finite generation and states no degree bound; the refuted all-weights spanning claim would have implied termination for $D_5$ (implication only, never an equivalence to a bound) — that successor, the $F_{46} \cdot K/(p^2)$ module pattern (Conjecture 10.11), is itself refuted at weight 84; termination now follows from the proved closure $A_5^{D_5} = C_5[F_{46}^{\mathrm{int}}]$ (Theorem 13.5). Daigle–Freudenburg is negative in 5 [5] (no universal bound can exist — the blocking theorem of Remark L.2); the six-generator subalgebra $C_5$ is finitely generated by construction (it is presented), and the full kernel is now closed by Theorem 13.5, and that contrast is the point. Dufresne–Kohls [8, 9] separate finite separating sets from finite generating sets; the six are established as generators of $C_5$ (separating status untested here), and full-ring generation by the six is Refuted, not conjectured. On the formal side the GroebnerTactic [24, 14] pipeline shares the untrusted-producer plus proved-checker plus machine-checked pattern, with the honest delta that theirs are Mathlib-dependent while ours is core-only (different trust base: core axioms plus kernel-checked native evaluation). A full Gröbner discharge of the superseded $F_{46} \cdot K/(p^2)$ module pattern is no longer a target; the exact Hilbert series and presentation above are its replacement.

## 19. Resonance of the Daigle–Freudenburg derivation

The resonance computation ($\rho = 1$) is proved below with SymPy-verified identities. The infinite generation of the DF kernel is the theorem of Daigle–Freudenburg [5], cited not proved. The comparison with $N = 5$ is a proved contrast of mechanisms, not a general theorem. We work in characteristic zero.

Let $R_{\mathrm{DF}} = k[x, s, t, u, v]$ with the triangular derivation

$$D = x^3\partial_s + s\partial_t + t\partial_u + x^2\partial_v, \qquad S = k[x, s, t, u], \quad \Delta = D|_S.$$

(Within this section $R_{\mathrm{DF}}$ denotes the DF ambient ring, distinct from the $N = 5$ closed ring $R = C_5[F_{46}^{\mathrm{int}}]$.) The base kernel contains $x$ and (verified by exact expansion)

$$\gamma_0 = 2x^3t - s^2, \qquad \delta_0 = 3x^6u - 3x^3st + s^3,$$

with plinth preimages $\Delta(s) = x^3$, $\Delta(3x^3u - st) = \gamma_0$, $\Delta(3x^3su - 4x^3t^2 + s^2t) = \delta_0$ (verified by the same exact computation; see Supplementary Material). Put $\beta_1 = xv - s$; $D(\beta_1) = x \cdot x^2 - x^3 = 0$.

**Lemma 19.1** (No monic pole element)**.** *$x^2 \notin \Delta(S)$: there is no $\rho \in S$ with $\Delta(\rho) = -x^2$, hence no pole element $u = v + \rho$ with $\rho \in S$. The minimal $v$-coefficient is $x$ (witness $\beta_1$).*

*Proof.* Grade $K[s, t, u]$ by $w(s, t, u) = (1, 2, 3)$ ($x$ weight 0). Each term of $\Delta$ lowers $w$-degree by exactly 1 ($x^3\partial_s$: $-1$; $s\partial_t$: $+1-2 = -1$; $t\partial_u$: $+2-3 = -1$). If $\Delta(F) = x^2$ (w-degree 0), only the w-degree-1 part of $F$ can contribute; it equals $s \cdot p(x)$, and $\Delta(s \cdot p(x)) = x^3p(x) = x^2$ forces $x \cdot p(x) = 1$, impossible in $k[x]$. □

The canonical pole element therefore lives over $S_x$: $u = v - s/x$ with $D(u) = x^2 - x^3/x = 0$, pole order $e_x = 1$ ($v_x(s) = 0$), and initial form $t = \mathrm{in}_x(xu) = -s$.

**Theorem 19.2** (Resonance of DF1999)**.** *$\rho(\mathrm{DF1999}) = \{1\}$: with $\mathfrak{p} = (x) \cap S^\Delta$ (prime, by contraction) and $L_x = \mathrm{Frac}(S^\Delta/\mathfrak{p})$, the initial form $t = -s$ lies in $L_x$, with separable minpoly $T - t$ of exact degree 1.*

*Proof.* Modulo $x$: $\bar{\gamma}_0 = -s^2$, $\bar{\delta}_0 = s^3$ (direct substitution). Both lie in the image of $S^\Delta/\mathfrak{p}$ since $\gamma_0, \delta_0 \in S^\Delta$ (verified above), and $-\bar{\gamma}_0 = s^2 \neq 0$, so $s = [\delta_0]/[-\gamma_0] \in L_x$. Thus $t = -s \in L_x$ with minpoly $T - t$ (derivative $1 \neq 0$). No finite-generation knowledge of $S^\Delta$ is used: only $\gamma_0, \delta_0 \in S^\Delta$ enter. □

*Remark* 19.3 (Status boundary and mechanism contrast). Established here: $\rho = 1$ and the $H_j$-descent $H_j \in (S^\Delta)_x$ used in Section 17 (via factorial-closedness of LND kernels: with $H_j = P/x^k$, $P \in S \cap \mathrm{Frac}(S^\Delta) = S^\Delta$). Cited: infinite generation (Daigle–Freudenburg). Consequence: resonance degree does not order infinitude ($\rho = 1$ with infinite generation here; $\rho \in \{2, 4\}$ with unknown status at $N = 5$). The $N = 5$ obstruction is residue relations enabling pole cancellation (escape); the DF obstruction is stable tops with unbounded $v$-degree (climb, Roberts pattern). Resonance detects residue-field collapse ($\rho \geq 1$), not finite generation; see Section 20.

## 20. Two mechanisms: escape and climb

Status of this section: the definitions are new; the $N = 5$ classification (escape) and the DF classification (climb) are established on the evidence cited; the general classification is an **[Open conjecture]** with explicit falsifier. Tower closure itself is proved (Theorem 13.5 in §13); what stays open here is only the general escape/climb classification, not finite generation of $A_5^{D_5}$.

**Definition 20.1** (Escape mechanism (definition introduced here))**.** An LND $D$ with pole element $u$ and residue inventory $\{\rho_i\}$ exhibits *escape* if some $\rho_i > 1$ and the corresponding residue-field algebraic relation enables pole cancellation: a $W$-coefficient combination regular in $B$ whose leading pole terms cancel in the residue field, producing an invariant outside the naive slice extension (a hidden generator).

**Definition 20.2** (Climb mechanism (definition introduced here))**.** An LND $D$ exhibits *climb* if the top coefficients of its invariants stabilize (no pole cancellation occurs) while their degree grows without bound, producing an infinite sequence of generators.

**Definition 20.3** (Degeneration mechanism (definition introduced here))**.** An LND $D$ with local slice $s$ exhibits *degeneration* at a boundary divisor $V(\pi)$ if $\ker \delta_\pi \supsetneq A^D/\pi A^D$ for the residual derivation $\delta_\pi$: the slice dies at $\pi$ and the boundary acquires new invariant coordinates. A class in the residual plinth $\bar{K} \cap \delta_\pi(\bar{A})$ that lifts to a global plinth element (an element of the liftable residual plinth $\mathcal{L}_\pi$ of Definition 12.3) supplies a new slice chart covering the degeneration locus.

$N = 5$ *exhibits escape (Established).* The quadratic relation $r_q^2 \in L_q$ enables the $q^{-4}$ pole cancellation in $H$ ($11025\,G_0^2 + 9J_2^7 \equiv 0 \pmod q$, Section 7.1); the quartic $p$-relation enables the $p^{-2}$ cancellation in $F_{46}$ (Theorem N.2). Both produce hidden generators outside the previously generated subalgebra. Whether the tower of hidden generators terminates is Question 13.1, answered affirmatively by Theorem 13.5.

*DF1999 exhibits climb (Established as computation + cited theorem).* $\rho = 1$ (Theorem 19.2): the initial form is already rational, so no nontrivial residue relation exists to enable cancellation. The $v$-degree towers $\beta_n, \gamma_n, \delta_n$ have stable leading coefficients $(\beta_0, \gamma_0, \delta_0)$ with unbounded $v$-degree, and the kernel is infinitely generated [5]. The implication "$\rho = 1 \Rightarrow$ infinite" is not proved here; it is the conjecture below.

### 20.1 Transcendence holds: the $N = 3$ classification

At $N = 3$, $D_3(a_0) = a_1a_2$, $D_3(a_1) = a_2a_3$, $D_3(a_2) = D_3(a_3) = 0$. Put $B = k[a_1, a_2, a_3]$, $K = B^{D_3} = k[a_2, a_3]$ (from $\partial F/\partial a_1 \cdot a_2a_3 = 0$), $H' = 2a_0a_3 - a_1^2$ ($D_3(H') = 0$), $c = 2a_3$, $W = H'/c = a_0 + \rho$ with $\rho = -a_1^2/(2a_3) \in B_{a_3}$. The pole prime is $(a_3)$ with $e = 1$; $r = \mathrm{in}_{a_3}(a_3\rho) = -a_1^2/2$ is transcendental over $L = k(a_2)$ ($a_1$ is transcendental over $k(a_2)$), so $\rho = 0$. The prime $(a_2)$ is regular ($v_{a_2}(\rho) = 0$).

**Theorem 20.4** ($N = 3$ kernel is polynomial)**.** $A_3^{D_3} = k[a_2, a_3, 2a_0a_3 - a_1^2]$. *In particular transcendence ($\rho = 0$) implies the one-generator extension here.*

*Proof.* Transfer at $(a_3)$ with $d = \infty$: if $\lambda = \min_j(v_{a_3}(H_j) - j) < 0$, then $k = 0$ qualifies and $P(r) = 0$ for $P \neq 0$ over $L$, impossible since $r$ is transcendental; so $v_{a_3}(H_j) \geq j$ (residues land in $k(a_2)$ as $a_2 \bmod a_3 \neq 0$). Transfer at $(a_2)$ with $e = 0$: $\lambda_2 = \min_j v_{a_2}(H_j) \geq 0$, else $P^{(k)}(r_2) = 0$ for all $0 \leq k \leq m$ forces $P = 0$ ($r_2 = -a_1^2/(2a_3)$). With $H_j \in K_{a_2a_3}$ (polynomial $W$-expansion plus $\mathrm{Frac}(K) \cap B_{a_2a_3} = K_{a_2a_3}$, clearing $a_2, a_3 \in K$), write $H_j = A/(a_2^k a_3^l)$, $A \in K$: $v_{a_2}(A) \geq k$, $v_{a_3}(A) \geq j+l$, so $A \in (a_2^k) \cap (a_3^{j+l}) = (a_2^k a_3^{j+l})$ (distinct primes in the UFD $K$) and $H_j = c^j h_j$ with $h_j \in K$. Top $H_m = C_m \in K$ (freeness plus $D(C_m) = 0$), so $F - h_m H'^m$ drops $W$-degree; base $m = 0$ is $B^{D_3} = K$. Hence $F \in K[H']$. Kernel dimensions match $k[a_2, a_3, H']$ through total degree 8 (exact-$\mathbb{Q}$ rank check), confirming no missing generator in range. □

### 20.2 Mixed inventory: the $N = 4$ base kernel

At $N = 4$, $B = k[a_1, \dots, a_4]$, $D_4(a_1) = a_2a_3$, $D_4(a_2) = a_3a_4$. Put $J = 2a_1a_4 - a_2^2$ ($D_4(J) = 0$).

**Proposition 20.5** ($N = 4$ base kernel is polynomial)**.** $B^{D_4} = k[a_3, a_4, J]$. *Consequently with* $L = a_0 - a_1a_2^2/(2a_3a_4) + a_2^4/(8a_3a_4^2)$ *(Theorem 3.3) the* $a_0$*-pole data is:* $r_3 = -a_1a_2^2/(2a_4) + a_2^4/(8a_4^2) = -a_1^2/2 + \bar{J}^2/(8a_4^2)$ *with* $\bar{J} = 2a_1a_4 - a_2^2$*, transcendental over* $L_3 = \mathrm{Frac}(K_4/a_3K_4)$ *(*$a_1$ *is transcendental over* $L_3$ *by transcendence degree), so* $\rho = 0$ *at* $a_3$*; and* $r_4 = a_2^4/(8a_3) \in L_4 = \mathrm{Frac}(K_4/a_4K_4)$*, so* $\rho = 1$ *at* $a_4$*. The full ring* $A_4^{D_4}$ *remains open.*

*Proof sketch.* $a_1$-filtration: $F = \sum C_k a_1^k$, $C_k \in k[a_2, a_3, a_4]$, $D(C_k) + (k+1)C_{k+1}a_2a_3 = 0$. Top $C_m \in k[a_3, a_4]$ ($\ker D|_{k[a_2,a_3,a_4]} = k[a_3, a_4]$, $N = 2$ pattern). Descending $a_2$-coefficient comparison forces $C_m \in (a_4^m)$: at level $m-k$ the $a_2^k$-coefficient equals $(-1)^k\binom{m}{k}C_m/a_4^k \in k[a_3, a_4]$, so $a_4^k \mid C_m$ ($a_4$ prime, characteristic zero); after $m$ levels $C_m \in (a_4^m)$. Then $G - (C_m/(2a_4)^m)J^m$ drops $a_1$-degree ($J^m$ has top $a_1$-coefficient $(2a_4)^m$), and induction gives $K_4 = k[a_3, a_4, J]$ (base $m = 0$ is $k[a_3, a_4]$). Kernel dimensions match through total degree 8 (exact $\mathbb{Q}$ ranks). Residues: $K_4 \cong k[A_3, A_4, J]$ is polynomial ($J$ independent via minor $2a_4 \neq 0$), so $(a_3), (a_4)$ stay prime and $L_3 = k(a_4, \bar{J})$, $L_4 = k(a_3, \overline{a_2^2})$; $r_3 + a_1^2/2 \in L_3$ with $a_1 \notin^{\mathrm{alg}} L_3$ gives transcendence, and $r_4 = (\overline{a_2^2})^2/(8a_3) \in L_4$ gives minpoly $T - r_4$. □

### 20.3 Exact $N = 6$ resonance values

The $a_6$-side setup of Section 15 ($L_6 = k(z_1, z_3, z_4)$, $r_6 = a_4^7/(35a_5^2)$, $C^{D_6}$ established) gives $\rho_6 = 2$ exactly: $r_6^2 = -z_4^7/(1225z_1^4) \in L_6$, while $r_6 \notin L_6$ since every element of $k[z_1, z_3, z_4]$ has even $a_4$-degree ($z_4 = -a_4^2$ is the only $a_4$-carrier; sums preserve even maxima) whereas $r_6$ has $a_4$-degree 7 (odd), and the same parity persists in the fraction field (even minus even). Minpoly $T^2 + z_4^7/(1225z_1^4)$, separable. On the $a_5$-side, $L_5 = k(y_2, y_3, y_4)$ and $r_5 = (35a_3^2a_4^3a_6^2 - 21a_3a_4^5a_6 + 3a_4^7)/(105a_6^4)$: $r_5 \notin L_5$ by the same $a_4$-parity ($a_4$-degree 7 vs even), and exact elimination over $\mathbb{Q}(y_2, y_3, y_4)$ yields the irreducible quartic annihilator $121550625y_2^{16}T^4 + (\cdots)T^2 + (\cdots)$ (Sage Gröbner + resultant tower, factored over the rational function field), so the minpoly has exact degree 4: $\rho_5 = 4$. Thus the $N = 6$ $B$-level inventory is $\{4, 2\}$, mirroring $N = 5$, with the hypersurface finite and $H_B$ the hidden generator: a second escape instance.

### 20.4 Falsification: Freudenburg-2000 has $\rho = 2$ and climbs

Let $D = x^3\partial_s + y^3s\partial_t + y^3t\partial_u + x^2y^2\partial_v$ on $k[x, y, s, t, u, v]$, triangular, with infinitely generated kernel [10] (see also the surveys citing Theorem 3 there; leading tower $f_n = xv^n +$ lower has stable top $x$ and unbounded $v$-degree: climb-shaped). Put $S = k[x, y, s, t, u]$, $\Delta = D|_S$, $\gamma = 2x^3t - y^3s^2 \in S^\Delta$ (verified $\Delta(\gamma) = 0$). The pole element is $u = v - y^2s/x$ from the exact invariant $u_F = xv - y^2s$ ($D(u_F) = x^3y^2 - y^2x^3 = 0$, verified); $x^2y^2 \notin \Delta(S)$ (weighted-degree obstruction: with $w(x, y, s, t, u) = (0, 0, 1, 2, 3)$, $\Delta$ lowers $w$-degree by 1, so $\Delta(F) = x^2y^2$ would need $F^{(1)} = s \cdot p(x, y)$ with $x^3p = x^2y^2$, impossible), so $x$ is the minimal $v$-coefficient and $e_x = 1$ is exact. With $\mathfrak{p} = (x) \cap S^\Delta$ and $t = \mathrm{in}_x(x\rho) = -y^2s$: $t^2 = -y\bar{\gamma} \in L_x$ (verified $t^2 + y\bar{\gamma} = 0$), while $t \notin L_x$ (if $-y^2s = A/B$ with $A, B \in S^\Delta$, $B \notin \mathfrak{p}$, then $x^3B + s\Delta(B) = 0$ forces $s \mid B$, and infinite $s$-descent ($(k+1)x^3C_k + s\Delta(C_k) = 0$ in characteristic zero) forces $B = 0$, contradiction). Hence minpoly $T^2 + y\bar{\gamma}$ has exact degree 2: $\rho_x = 2$. The denominator is $x$ alone, so the inventory is $\{2\}$.

*Consequence.* An infinitely generated triangular monomial LND with $\max \rho_i = 2 > 1$ and climb-shaped generation exists. The strong form "$\rho > 1 \Rightarrow$ finite (escape)" is **[REFUTED by Freudenburg-2000]**; "climb $\Rightarrow \rho = 1$" fails likewise. Resonance degree does not order finite generation.

### 20.5 Degeneration: the $N = 5$ $q$-boundary (Established instance)

At $N = 5$ the slice $s_0 = x/(pq)$ degenerates at $V(q)$: $\ker \delta_q = k[p, x, u] \supsetneq K/qK = k[p, x^2, u]$ (Lemma 12.4). The residual class $-px^6/35$ lifts to the global plinth element $f_{13}$ (Theorem 11.2), whose slice $s_1 = g_{13}/f_{13}$ covers the collapse locus of Theorem 12.1 and separates the collapsed orbits (Theorem 12.12). So $N = 5$ exhibits degeneration *in addition to* escape: the $q$-boundary carries both a pole cancellation ($H$) and a slice degeneration ($f_{13}$), with $H$ the candidate gluing correction (Conjecture 12.20). Whether degeneration occurs at any other boundary divisor studied here ($p$-side at $N = 5$, either side at $N = 6$, DF1999, Freudenburg-2000) is open; checking the residual kernel against

the residual image at each boundary divisor is now part of the resonance inventory. The $q$-branch continues one step deeper: over $V(q,x)$ the residual plinth is zero (Theorem 12.11), so the branch terminates at $f_{13}$.

**Definition 20.6** (Boundary tree (definition introduced here))**.** The *boundary tree* $\mathcal{T}(D)$ of a triangular LND records, per boundary prime divisor: the residual derivation, its kernel versus the residual image of $A^D$, the residual plinth with its liftable part, and the next (deeper) stratum. Instance at $N=5$, $q$-branch: $(p,q) \to q \to (k[p,x,u] \supset k[p,x^2,u]) \to f_{13} \to V(q,x) \to$ terminal (Theorem 12.11); the $p$-branch is open. Generality of the tree beyond $N=5$ is open.

### 20.6 Evidence table and revised conjecture

**Table 2. Resonance inventory vs generation across six LNDs.**

| LND | $\rho$ | generation | mechanism |
|---|---|---|---|
| $N=3$ adj-quad | $\{0\}$ | polynomial (Thm 20.4) | transcendence $\Rightarrow$ extension |
| $N=4$ adj-quad | $\{0,1\}$ | base polynomial (Prop 20.5); full open | mixed |
| $N=5$ adj-quad | $\{2,4\}$ | hypersurface + $F_{46}$; full open | escape (production verified) |
| $N=6$ $B$-level | $\{4,2\}$ | hypersurface (finite) | escape |
| DF1999 | $\{1\}$ | infinite [5] | climb |
| Freudenburg-2000 | $\{2\}$ | infinite [10] | climb-shaped; refutes degree-prediction |

**Conjecture 20.7** (Resonance-transcendence extension (Revised open conjecture))**.** Grounded in Lemmas 21.24–21.25 (transcendence closes the constant-coefficient step; algebraicity of degree $d$ breaks it at $W$-degrees $\geq d$): if $\rho = 0$ at every pole prime (all initial forms transcendental over their residue fields), then the slice extension is polynomial (one-generator-over-$K$). Instance: $N=3$ (Theorem 20.4). Falsifier: $\rho = 0$ with hidden generators. No prediction is made for $\rho \geq 1$ anywhere: escape ($N=5$, $N=6$), climb (DF1999), and climb-shaped infinitude (Freudenburg-2000, $\rho = 2$) all occur. Whether "all $\rho_i = 1 \Rightarrow$ climb" holds is an open question (single instance DF).

*Remark* 20.8 (Status boundary). New: the definitions, the $N=3$ classification, the $N=4$ base classification with mixed inventory, the exact $N=6$ values, and the Freudenburg-2000 computation. Established: the mechanism instances on stated evidence ($N=5$ escape production; DF and Freudenburg-2000 climb-shape with cited infinitude). Refuted: the strong degree prediction (Freudenburg-2000: $\rho = 2$ with infinitude). Open: the revised transcendence-extension conjecture, the all-$\rho$-1 question, tower closure, and the $N=4$ full ring. The multfloor transfer theorem (Theorem 17.1) constrains floors in all mechanisms and implies neither finiteness nor infinitude.

## 21. Generalizations and conjectures

Status of this section: Open. Every statement here is a conjecture or a prediction to verify, not a result, except for the established infrastructure (Theorems 21.10 and 21.17). Computation through $N=6$ is evidence, not proof, and nothing here upgrades the Conditional statements of Sections K and 6.

### 21.1 Central conjectures: spanning and the relaxed bounds

The conjectures below are the continuity of the $N=5$ story: each carries explicit CAS evidence and a falsifier. They are ordered by dependency: spanning is the goal, R1 and R2 the relaxed bounds, and the $H \cdot K$ structure the evidence for R1.

**Residue-transcendence saturation procedure.** Status: Open method description, not a result. With triangular LND $D$ on $A = k[a_0, \ldots, a_N]$, local slice $s$ with $D(s) = 1$ over $K_f$ ($K$ the base kernel), and localized presentation $A_f^D = K_f[W]$, $W = F/c$:

1. Factor $c$ in $K$ (prime divisors $\pi_1, \ldots, \pi_r$; record pole orders $e_i = \mathrm{ord}_{\pi_i}(c)$).
2. For each $\pi_i$: residue field $L_i = \mathrm{Frac}(K/\pi_i K)$ via elimination; residue field $\kappa_i$ of $B$ at $\pi_i$; embed $L_i \hookrightarrow \kappa_i$ (certify injectivity by an explicit Jacobian minor; standing characteristic zero).

3. Initial form $r_i = \mathrm{in}_{\pi_i}(\pi_i^{e_i}\rho) \in \kappa_i$, $\rho = W - a_0$ the pole part.
4. Bounded transcendence test at degree bound $d$ (default $d = 4$, from the $N$-tower): nullspace over $L_i$ of $\{1, r_i, \ldots, r_i^d\}$ in $\kappa_i$ (clear denominators; linear algebra over $\mathrm{Frac}(K/\pi_i)$). Relation found: emit *Hidden Generators Warning* with the minimal polynomial $m_i(T)$ and the predicted hidden weight (first $w$ where the free count falls below $\ker\dim$). None through $d$: PASS at bound $d$ only.

Output PASS on all $i$ supports (but does not prove) $A^D = K[cW]$-type generation; any Warning stops generation claims and starts the hidden-invariant hunt. Dry run on $N = 5$: the $q$-side finds $m(T) = T^2 - (x^2)^7/(1225p^4)$ at $d = 2$ (Warning; $H$ found at weight 24, Remark 6.4); the $p$-side passes (the lemma then standing; since refuted by $F_{46}$). The test is sound for Warning and bounded for PASS: by Remark D.10 no initial-form analysis at any order can close the bounds in general, so PASS is never a completeness theorem.

**Definition 21.1** (Per-prime resonance (Open definition))**.** Let $D$ be triangular with $\mathfrak{p} = A^D \cap D(A)$ the plinth ideal, $s$ a local slice with $0 \neq D(s) = f \in \mathfrak{p}$, and $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$ the height-one primes of $A^D$ dividing $(f)$. At $\mathfrak{p}_i$ use the $\mathfrak{p}_i$-adic valuation $v_i$, filtration $F_n = \{g : v_i(g) \geq n\}$, associated graded gr, and initial forms $\mathrm{in}_i$; put $L_i = \mathrm{Frac}(A^D/\mathfrak{p}_i)$. For a stated pole element $u \in B_f$ with pole order $e_i = -v_i(u) > 0$ and initial form $t_i = \mathrm{in}_i(u)$, define $\rho_i(D, u) = 0$ if $t_i$ is transcendental over $L_i$, else the minimal degree of an algebraic relation of $t_i$ over $L_i$; record $\{\rho_i\}$ and aggregate $\rho(D, u) = \max_i \rho_i(D, u)$ (the worst prime governs denominator clearing). Canonical pole element: $u = W = F_0/c$ with $e_p = 2$, $e_q = 4$. $W$ is $D$-invariant ($D(F_0) = D(c) = 0$), not a Kuroda degree-1 slice — the $N = 5$ instantiation applies the per-prime initial-form machinery to the saturation pole element $W$, not to a classical slice. Slice/pole-element-independence of $\rho$ is open. $N = 5$ instances: $\rho_q = 2$ exactly ($r_q^2 = (x^2)^7/(1225p^4) \in L_q$, $r_q \notin L_q$, Remark J.5, Lemma J.6); $\rho_p = 4$ exactly ($r_p$ primitive for the full tower (exact rational computation); Remark I.4). Hence $\rho(D, W) = \max(4, 2) = 4$, exact. The p-side resonance is dominant: $H$ respects the p-bound, and the p-bound is Refuted by $F_{46}$ (Conjecture 21.5). Resonance degree is necessary but not sufficient for hidden-generator production.

**Theorem 21.2** (All-weights spanning (Refuted))**.** ***[REFUTED at weight*** 46 ***by the explicit invariant*** $F_{46}$ ***of Section 10 (Theorem 10.1).]*** *The refuted statement was: for every weight $w$, the weight-$w$ graded piece $(A_5^{D_5})_w = \ker(D_5|_{V_w})$ is spanned, over the ground field, by the weight-$w$ monomials in the six generators $p, q, J_1, J_2, F_0, H$ — equivalently, the natural map $k[p, q, J_1, J_2, F_0, H] \to A_5^{D_5}$ is surjective, in which case its kernel is the principal ideal $(S)$ of the weight-28 relation by Theorem 7.1. It holds exactly through weight 45 (*Lean-verified directly for every weight $w \leq 14$ *(Theorem L.1); exact $\mathbb{Q}$-ranks for 15–45, Theorem 10.2) and fails first at 46 (*$\dim\ker = 1519 > 1518$*). Its replacement was the $F_{46} \cdot K/(p^2)$ module pattern (Conjecture 10.11), itself refuted at weight 84; the live successor is the exact Hilbert series (Theorem 14.4) and $T$-filtration (Theorem 14.6).*

*Remark* 21.3 (R1 evidence and sharpness). Theorem R3 bounds every $K[F_0]$-element by $v_q(H_j) \geq 4j$, but $H$ violates it at $j = 0$ and $j = 2$ (parts C–D of exact computation (see Supplementary Material)). The strict bounds therefore cannot govern the full invariant ring. R1 is the relaxed envelope they sit inside: every invariant $F$ of W-degree $m$ satisfies $\lambda_q(F) \geq -2m$, where $\lambda_q = \min_j(v_q(H_j) - 4j)$. **[Proved as Theorem J.7]** by multiplicity transfer from all of $\{C_k\}$ (exact residue degree 2, Lemma J.6); the CAS evidence below is corroboration and sharpness, not the proof. The bound is tight at $H$ ($\lambda_q = -4 = -2 \cdot 2$); the slope 2 matches the residue algebraicity degree of $r_q$ over $L_q$, whose minimal polynomial has degree 2 (whether slope always equals residue degree is part of the conjecture, attested at $H$ and $F_{46}$).

Evidence **[CAS-VERIFIED]**: kernel dimensions equal $s_6(w) - s_6(w-28)$ at anchor weights $5, 15, 23, 24, 25, 30, 35, 40$ (computed over $\mathbb{F}_{1000003}$; see Supplementary Material); the values $11, 18, 27, 55, 75, 213$ at $w = 6, 8, 10, 14, 16, 24$ and $238, 267, 296, 330, 364, 404, 444, 490, 536, 588, 640, 700, 760, 828, 896, 972$ at 25–40 are the formula values (exact computation (see Supplementary Material)). Exact-$\mathbb{Q}$ deficit profiles are logged at $w =$ 29–35 and show minimum deficit $-4$ (no deficit below through the logged range, consistent with the birth window of Corollary 9.2). The floor is attained exactly on $H$-multiples: at each logged weight the $m = 2$ deficit-row count is $s_K(w-24)$ $(8, 11, 14, 18, 22, 27,$ 32 at 29–35; $1, 2, 3, 4, 6$ at 24–28 are formula values), with profiles $(\lambda_q, \lambda_p) = (v_q(\mu) - 4, v_p(\mu))$ over $K$-monomials $\mu$ exactly through $w = 31$ and mixed from $w = 32$ with counts and floors intact (Remark 21.4; exact kernel data and checker computations are provided in the Supplementary Material). No $m = 3$ row appears at logged weights 29–35, consistent with the second family $H \cdot F_0 \cdot (K$-monomials) entering at $m = 3$ from $w = 38$ (unprobed). No rows outside the $H \cdot K$ count appear at 29–35; weights 36–40 are unprobed. Counts and dimensions are basis-independent; profiles are basis-dependent observations.

Advance prediction: the $H \cdot K$ count structure holds beyond 40. This is CAS evidence, not a proof; the product-reduction program targets the structural form. The former falsifier (any invariant with $\lambda_q < -2m$) is now excluded by Theorem J.7. *Update:* the weight-46 invariant $F_{46}$ sits exactly on the R1 floor at every nonzero $W$-coefficient ($\lambda_q = -8 = -2 \cdot 4$; Theorem 10.3), the strongest confirmation of R1 to date — and it is R1-tightness, not R1-violation, that accompanies the $p$-side collapse.

*Remark* 21.4 (Why the multiset mixes at 32). $S$-multiples enter the kernel picture from $w = 28$ and rotate the arbitrary nullspace bases returned by exact rational elimination, so per-row profiles are basis-dependent while counts, minima, and W-degrees are not. The first visible rotation is at $w = 32$ (extra $(-4, 0)$ for predicted $(-2, 0)$), a profile still realized inside the $H \cdot K$-module as $H \cdot (q^2 + c)$. Structural arguments must therefore use $H \cdot \mu$ with general $\mu \in K$ past 31; monomial $\mu$ suffices through 31.

**Theorem 21.5** (Global p-bound (R2) (Refuted))**.** ***[REFUTED by the weight-46 invariant*** $F_{46}$ ***of Section 10: its*** $W$***-expansion has*** $v_p(H_4) = 6 < 8 = 2 \cdot 4$ ***(and*** $v_p(H_j) < 2j$ ***at every nonzero*** $j$***; Remark I.6).]*** *The refuted statement was: every invariant $F$ with W-expansion $\sum H_j W^j$ satisfies $v_p(H_j) \geq 2j$.*

**Conjecture 21.6** (relaxed p-side bound (Rp))**.** Every invariant $F$ of W-degree $m$ satisfies $\lambda_p(F) := \min_j(v_p(H_j) - 2j) \geq -4m$. Rationale: this is a deliberately weak floor, not the slope the mechanism implies. The multiplicity transfer of Theorem J.7 with pole order $e_p = 2$ would suggest the far tighter $\lambda_p \geq -e_p m/d_r = -m/2$ ($r_p$ primitive for the full tower (exact rational computation); Remark I.4), *provided* $r_p$ has exact degree 4 over $L_p$ and $K_{pq}$-residues land in $L_p$ across its degree-$\leq 4$ tower — both unproved (no parity argument is available for the trinomial initial form), and with no $\lambda_p$-coverage beyond $H$ and $F_{46}$ in the computed tower data. We therefore conjecture only the weak $-4m$ form, which both witnesses clear loosely: $\lambda_p(F_{46}) = -2$ against the floor $-16$ (Theorem 10.3), and $\lambda_p(H) = 0$ against $-8$ ($v_p(H_2) = 4$, $v_p(H_0) = 0$ since $v_p(J_1) = v_p(J_2) = 0$); sharpness is unknown and the true floor may be much higher. Falsifier: any invariant with $\lambda_p < -4m$. This concerns W-degree minima $\lambda_p$, distinct from the $a_0$-top $p$-envelope $v_p(C_m) \geq \lceil 3m/2 \rceil$ (Theorem 15.4); neither implies the other. The transfer theorem (Theorem 17.1) would give the far tighter conditional floor $\lambda_p \geq -m/2$ ($e_p = 2$, $d_p = 4$), pending the two unproved inputs above; it does not replace this conjecture, and the original $v_p \geq 2j$ bound stays refuted.

*Remark* 21.7 (What the refuted spanning claim leaves behind). The conjecture keeps its name and label (21.2); it is **[REFUTED]** at weight 46 (Theorem 10.1), **[LEAN-VERIFIED]** through 14, **[EXACT]** through 45. The $F_{46} \cdot K/(p^2)$ module pattern (Conjecture 10.11) was its replacement: the maximal weight of a minimal $C_5$-generator remains 24 (attained by $H$), and 46 is the birth weight of the first generator outside $C_5$, attained by $F_{46}$ — exact through the verified range. That pattern itself fails at weight 84 (fourth family, Theorem 14.5); completeness is now settled by the exact Hilbert series (Theorem 14.4), not by any module-pattern conjecture. No dimension argument can bypass computing the kernel dimensions: the series *is* the dimensions.

**Conjecture 21.8** (H·K structure of the deficit (24–28))**.** For $24 \leq w \leq 27$,

$$\ker(D|_{V_w}) = \operatorname{span}_k(\text{five-generator monomials})_w \ \oplus \ \operatorname{span}_k\{H \cdot \mu : \mu \text{ a } K\text{-monomial of weight } w - 24\},$$

and the second summand attains $(\lambda_q, \lambda_p) = (v_q(\mu) - 4, v_p(\mu))$ on each $H \cdot \mu$; at $w = 28$ the same holds modulo the single relation $S$. In particular every deficit row is an $H$-multiple on 24–28, $\max m = 2$ there (the second $H$-family $H \cdot F_0 \cdot K$ enters at $m = 3$ from $w = 38$ on), consistent with Theorem J.7, with equality exactly on this summand. Evidence **[CAS-VERIFIED]**: dimensions $213, 238, 267, 296$ at 24–27 ($330 = 331 - 1$ at 28) are the $s_6 - s_6(-28)$ formula values, anchor-checked at $24, 25$ (see Supplementary Material); exact-$\mathbb{Q}$ deficit profiles are logged at 29–35. Deficit-row counts $1, 2, 3, 4$ (6 at 28, matching $s_K(0)$–$s_K(4)$) are formula values; per-monomial profiles are observed in the logged 29–35 bases (basis-dependent observation for the profiles; dimensions basis-independent). Falsifier: any deficit row not matching some $H \cdot \mu$ profile, or any dimension mismatch.

### 21.2 Structural observations

The statements below guide future work at general $N$: established infrastructure, structural conjectures with named instances, and computational observations. None is claimed as a result.

**Graph derivations.**

**Definition 21.9** (Truncated graph derivation)**.** Let $\Gamma$ be a finite directed acyclic graph on vertices $\{0, \ldots, N+2\}$ with out-neighbours $N^+_\Gamma(i)$, and put $A_N = k[a_0, \ldots, a_N]$ with boundary $a_{N+1} = a_{N+2} = 0$. Write $\bar{a}_j = a_j$ for $j \leq N$

and $\bar{a}_j = 0$ for $j > N$. The truncated graph-derivation is

$$D_{\Gamma,N}(a_i) = \begin{cases} 0 & N^+_\Gamma(i) = \emptyset \text{ or } N^+_\Gamma(i) \ni j > N, \\ \displaystyle\prod_{j \in N^+_\Gamma(i)} \bar{a}_j & \text{otherwise,} \end{cases}$$

extended by the Leibniz rule. Vertices $N+1, N+2$ are *ghost sinks*: they exist in $\Gamma$ but evaluate to 0 in $A_N$, so any $i$ with a ghost successor is ghost-killed ($D_{\Gamma,N}(a_i) = 0$, a derived sink); a vertex with no out-neighbours is a true sink and also gives 0 by the first case (empty product is *not* used). This is the quotient $k[a_0, \ldots, a_{N+2}]/(a_{N+1}, a_{N+2})$ restricted to the first $N+1$ variables. Each $D_{\Gamma,N}$ is locally nilpotent: along every monomial step the topological index strictly decreases, and sinks kill. This restricts the dictionary to triangular monomials with no constant term (Maubach/Karaś class). The adjacent-quadratic family is the skip-1 case $N^+(i) = \{i+1, i+2\}$: at $N = 3$ this gives $D_3 = (a_1a_2, a_2a_3, 0, 0)$ (Theorem 3.2), since $N^+(2) = \{3, 4\}$ contains a ghost and gives 0; at $N = 2$, $N^+(1) = \{2, 3\}$ is ghost-killed and $N^+(2) = \{3, 4\}$ is ghost-killed. Tests: isolated vertex gives 0; sink gives 0; single edge $a_0 \mapsto a_1$ gives $D(a_0) = a_1$, $D(a_1) = 0$; path length 2, 3 give the Weitzenböck derivations with $D = 0$ at the sink.

**Theorem 21.10** (Homogeneous gradings always exist)**.** *$D_{H,N}$ is homogeneous of degree 0 for a weight vector $w$ with all $w_i > 0$ iff $w_i = \sum_{j \in N^+_H(i),\, j \le N} w_j$ for every non-ghost-killed $i$ (ghost successors contribute 0 to the sum, consistent with $\bar{a}_j = 0$). Every DAG admits a positive integer solution: set $w = 1$ on sinks and induct backwards along any reverse topological order. The Fibonacci grading is the special case of this construction. Write this degree-0 grading $w^{\mathrm{inv}}$: $D_{H,N} \colon V_k \to V_k$ preserves each graded piece, acting as a finite-dimensional linear map per piece.*

**Theorem 21.11** (Derivation grading)**.** *Let $A_N$ be the truncated adjacency matrix (ghost sinks evaluated to 0, not removed: rows of ghost-killed vertices are zero). Then $D_{H,N}$ is homogeneous of degree $-1$ for the positive integral weight $w^\partial = (I - A_N)^{-1}\mathbf{1} = (I + A_N + A_N^2 + \cdots)\mathbf{1}$ (the Neumann series terminates since $A_N$ is nilpotent; $w^\partial$ is the unique such grading once ghost-killed/sink vertices are normalized to $w_i = 1$, the minimal positive choice, since the grading condition is vacuous there). Consequence: $D_{H,N} \colon V_k \to V_{k-1}$ is the standard kernel filtration by top-degree terms. At $N = 5$, $w^\partial = (15, 9, 5, 3, 1, 1)$ versus $w^{\mathrm{inv}} = (8, 5, 3, 2, 1, 1)$. For sinks the grading condition is vacuous ($D_{H,N}(a_i) = 0$); the value $w_i = 1$ is the minimal positive integer consistent with the path-count formula.*

*Remark* (Open: grading control of nilpotency and poles). Whether $w^\partial$ controls the nilpotency index and Dixmier pole orders beyond inducing the top-degree kernel filtration is open.

**Proposition 21.12** (Verified small forest kernels)**.** *The following graph-derivation kernels are polynomial: the single edge ($D(a_0) = a_1$; kernel $k[a_1]$, by $a_0$-degree); the V-tree ($D(a_0) = a_1a_2$; kernel $k[a_1, a_2]$, same argument); and the 3-vertex path ($D(a_0) = a_1$, $D(a_1) = a_2$; kernel $k[a_2, 2a_0a_2 - a_1^2]$, classical Weitzenböck/Jordan-block theorem, with $D(2a_0a_2 - a_1^2) = 0$ checked in exact computation (see Supplementary Material)).*

**Theorem 21.13** (Unrestricted polynomial forests (Refuted))**.** *The conjecture that every rooted forest (out-degree $\le 1$) has polynomial invariant ring is false. The 4-vertex path $a_0 \to a_1 \to a_2 \to a_3$ ($D(a_i) = a_{i+1}$) has kernel with 4 minimal homogeneous generators in degrees $1, 2, 3, 4$ against transcendence degree 3, and the 5-vertex path has 5 against transcendence degree 4 (exact rational indecomposable counts, exact computation (see Supplementary Material)); neither kernel is polynomial. The killer already occurs at 4 vertices, strictly inside the out-degree-$\le$-1 class.*

*Remark* 21.14 (WARNING: no $\mathrm{codim} = b_1$ bound for general triangular derivations). The bound above is monomial-only and does not extend to general triangular derivations. Daigle–Freudenburg [6, 7] construct, for every $n \ge 3$, a triangular derivation of $k[W, X, Y, Z]$ with $n \le \#(\ker \Delta) \le n + 1$ (e.g. parameters $p = 2, q = 5, h = 4$ give $6 \le \#(\ker \Delta) \le 7$); dimension-four triangular kernels are always finitely generated in characteristic zero [7], while their minimal generator number is unbounded. Every such graph is a DAG ($b_1 = 0$), so no bound of the form $\mathrm{codim}(A^D) = b_1(G)$ can hold in general; Maubach's $\le 4$ bound is monomial-triangular in dimension 4 (covering e.g. our $N = 3$ member on $k[a_0, \ldots, a_3]$), while the unbounded family is general triangular. In short: monomial triangular in dimension 4 needs at most 4 generators; arbitrary triangular in dimension 4 is finitely generated with unbounded generator number; in dimension 5 finite generation itself can fail. Squarefree monomiality alone does not imply any such bound above dimension 4. A second, distinct killer lives inside the monomial world itself: the Weitzenböck path refutation of Theorem 21.13 shows out-degree $\le 1$ does not imply polynomial either — the obstructions are the path length (chains of 4+ vertices), not the out-degree.

**Conjecture 21.15** (denominators count paths)**.** Let $c_G$ be the minimal monomial clearing the slice denominators of $D_G$. Then $c_G = \prod_{\text{sinks } s} a_s^{e_s}$ with $e_s$ the number of directed $a_0 \to s$ paths counted with ladder multiplicity. Instances:

$N = 5$ gives $p^2q^4$ and $N = 6$ gives $a_5^2a_6^4$ (Proposition C.1); each new $N$ is a direct-expansion-style verification, not a proof.

**Conjecture 21.16** (saturation-obstruction predictor)**.** Algebraic residue initial forms obstruct the naive local-to-global saturation mechanism. Precisely, at a prime divisor $\pi \mid c_G$: a transcendent residue initial form implies no pole cancellation by the constant coefficient, while an algebraic residue initial form means the constant-coefficient test can fail, necessarily at $W$-degrees at least the algebraic degree (below that degree the step may still function), with $\deg(r_\pi)$ bounded by the tower of quadratic steps resolving the non-residue coordinates. The converse—that algebraicity necessarily produces a new invariant—is not claimed. (Scope: this governs the failure mode of the constant-coefficient residue argument, not the underlying saturation statement, which an algebraic initial form might still admit via higher-order lifting.) Instances: $N = 5$ gives $r_q^2 \in L_q$ (Remark J.5); $N = 6$ gives degree $\leq 4$ at $a_5$ and $\leq 2$ at $a_6$ (Remarks D.3–D.6). The procedure described in §21.1 is the executable form; its PASS verdicts are bounded-ansatz only, never completeness theorems (Remark D.10).

**Recursion across $N$.**

**Theorem 21.17** (Recursive shift isomorphism)**.** *For every $N \geq 6$, with $B_N = k[a_1, \ldots, a_N]$ and $D_N(a_n) = a_{n+1}a_{n+2}$ ($a_{N+1} = a_{N+2} = 0$), the shift $\varphi_N \colon A_{N-1} \to B_N$, $a_i \mapsto a_{i+1}$, satisfies $D_N(\varphi_N(a_i)) = \varphi_N(D_{N-1}(a_i))$ for all $i$: for $i \leq N - 3$ both sides equal $a_{i+2}a_{i+3}$, and at the boundary both sides vanish ($D_{N-1}$ kills $a_{N-2}, a_{N-1}$ since $a_N = 0$ in $A_{N-1}$; $D_N$ kills $a_{N-1}, a_N$ since $a_{N+1} = 0$). Fibonacci weights transfer verbatim, so $\varphi_N$ restricts to a graded isomorphism $A_{N-1}^{D_{N-1}} \xrightarrow{\sim} B_N^{D_N}$. The $N = 6$ section proves its own $B_5 \to C_6$ instance separately ((1); domain $k[a_1, \ldots, a_5]$ with $D_5$).*

**Corollary 21.18** (B-level transport; A-level hypersurface open)**.** *By Theorem 21.17, wherever $A_{N-1}^D$ is a hypersurface, $B_N^{D_N}$ is the transported hypersurface with the transported relation. Instances: $B_6$ (Theorem A.11). Open: whether $A_N^D$ is a hypersurface — $A_6^D$ is not known to be one; its $a_0$-denominator module is not principal (Lemma G.1, Lemma F.4) — so nothing is predicted at A-level for $N \geq 6$ until the $A_6$ ring is determined. Falsifier: if at some $N$ the $B_N$ kernel dimensions deviate from the transported complete-intersection series, the transport itself fails at that $N$.*

**Theorem 21.19** (Degree recurrence and predicted values (Refuted as stated))**.** ***[REFUTED as stated by the $N = 6$ data below; repaired form Open.]*** *Let $e_m$, with $m := N - 2$, be the ordinary degree of the general-$N$ analogue of the invariant $F_0$. The predicted recurrence was*

$$\boxed{e_m = e_{m-1} + e_{m-2} + 1, \qquad e_m = F_{m+3} - 1,}$$

*with $F_j$ the Fibonacci numbers, so that the predicted ordinary degree at general $N$ is $F_{N+1} - 1 = \tau_N - 1$, where $\tau_N$ is the nilpotency index of $D_N$. The established instance is $N = 5$: ordinary degree 7 and Fibonacci weight 14 (Lemma 5.4), with $F_6 - 1 = 7$. The refutation: at $N = 6$ ($m = 4$) the formula predicts $e_4 = F_7 - 1 = 12$, but the denominator-cleared $F_0^{(6)}$ has ordinary degree 13 (Lemma F.1), and the hidden generator $H_{20}$ has ordinary degree 8 (Lemma F.4) — so 12 matches no $a_0$-direction at $N = 6$. The recurrence is valid at $N = 5$ but fails for the denominator-cleared $F_0^{(6)}$ at $N = 6$; the "general-$N$ analogue of $F_0$" is not canonical once multiple $a_0$-directions exist. Repaired conjecture (Open): the recurrence governs a canonically minimal $a_0$-generator, to be defined; the predicted ordinary degrees*

| $N$ | 7 | 8 | 9 | 10 |
|---|---|---|---|---|
| $\deg F_0$ | 20 | 33 | 54 | 88 |

*are retained as predictions for that minimal generator, to verify, not results. No Fibonacci-weight predictions are made here; "weight" always means the Fibonacci weight of Lemma 5.4 as opposed to ordinary degree, and the weight-shift separation noted there remains an observation, not a theorem.*

*Remark* 21.20 (Open: denominator bounds). Denominator bounds for the general-$N$ saturation are unknown. Any sufficient characteristic bound of the form "characteristic $> N$" is retracted as unproved; the bound, if one exists, is part of the conjecture, not a lemma.

*Remark* 21.21 (Recon: $N = 7$ $B$-level and the recursion). The shift $(A_6, D_6) \to (B_7, D_7)$ intertwines on all generators (checked as in exact computation (see Supplementary Material)), so $B_7^{D_7} \cong A_6^{D_6}$ is exactly as open as the $A_6$ ring. The weighted $B_7$-probe (exact computation (see Supplementary Material)) through weight 22 fits the free count on $(1, 1, 4, 6, 14)$ through 19 with excess $+1, +2, +4$ at $20, 21, 22$ — consistent with the shifted $H_{20}$-family (shifted $H_{20}$ at 20; times $\{a_6, a_7\}$ at 21; three products plus the shifted $D_2$-direction at 22, all exact invariants by intertwining).

Completeness of this accounting (no further hidden directions through 22) is unproved. The $N = 7$ $B$-generators (ladder hunt) and everything downstream stay Open.

*Remark* 21.22 (Resolved: $F_0$-degree at $N = 6$). The $a_0$-top generator $F_0^{(6)}$ has ordinary degree 13 (Lemma F.1), one above the $F_{N+1} - 1 = 12$ prediction of Conjecture 21.19 at $N = 6$ — and 13 is best possible among sink-monomial denominators, but not minimal as an invariant: the hidden generator $H_{20}$ (Lemma F.4) has ordinary degree 8 with the smaller denominator $a_5^4 a_6^3$ — indeed $F_0^{(6)} = a_6^5 H_{20} + K_0$ is the degree-8 generator cleared through the degree-12 denominator — with $K_0 \in K_6$ (verified in exact computation (see Supplementary Material)), so $F_0^{(6)}$ adds no new $a_0$-direction beyond $H_{20}$. A sink-monomial sweep (42 candidates $a_5^a a_6^b$, $a + b \leq 11$) finds solvable denominators exactly for $a \geq 4$ and $b \geq 3$, i.e. exactly the $H_{20}$-multiples; degrees 8–11 contribute no family outside $H_{20}$. Whether a degree-12 $a_0$-top invariant exists outside the $H_{20}$-family (non-monomial denominator of total degree 11) remains open, as does the $N \geq 7$ analogue of the prediction.

### 21.3 Where the phenomena begin and break

**Theorem 21.23** (First proved occurrences of the phenomena)**.** *(i) At $N \leq 4$ the established theory gives the polynomial kernel $k[a_1, a_2]$ at $N = 2$ (Proposition 3.1) and explicit polynomial invariants with open-set normal forms at $N = 3, 4$ (Theorems 3.2, 3.3); no hidden direction is detected, and full invariant-ring presentations at $N = 3, 4$ are not attempted here (open). (ii) $N = 5$ is the first member with a proved hidden generator: $H$ at weight 24 is the first invariant outside $K[F_0]$, exactly ($\ker_w = K[F_0]_w$ for $w \leq 23$), and $F_{46}$ at weight 46 is the first invariant outside $C_5$, exactly (spanning through 45); the residue mechanism $r_q^2 \in L_q$ is identified, (P-top) is refuted, (Q-top-strong) is established (Theorem 15.1). (iii) At $N = 6$ the principal-denominator pattern already fails: the $A_6$ denominator module is not principal (Lemma F.4, Lemma G.1), while the $B$-level transports the $N = 5$ tower exactly, including its refutation by transported $F_{46}$ (Theorem A.11). The $A_6$ ring is open. What this does not say: it is not a classification (the $N \leq 4$ rings are not closed here, and nothing is claimed at $N \geq 7$); it locates the first proved occurrence of each phenomenon.*

**Resonance lemmas: what the constant-coefficient step proves.** Let $D$ be triangular locally nilpotent on $A = B[a]$ over a characteristic-zero field, with $a$ transcendental over the domain $B$; let $K_f[W]$, $W = a + \rho$ monic of $a$-degree 1, be a localized invariant presentation; let $v = \mathrm{ord}_\pi$ along a prime divisor $\pi$ with pole order $e = -v(\rho) > 0$, residue fields $L \hookrightarrow \kappa$, and initial form $r = \mathrm{in}_\pi(\pi^e \rho) \in \kappa$. Write a localized invariant $F = \sum_{j=0}^m H_j W^j$ and put $\lambda = \min_j(v(H_j) - ej)$.

**Lemma 21.24** (Transcendence closes the step)**.** *If $r$ is transcendental over $L$, then $\lambda \geq 0$, i.e. $v(H_j) \geq ej$ for every $j$.*

*Proof.* Otherwise $\lambda < 0$: dividing the constant $a$-coefficient $C_0 = \sum_j H_j \rho^j \in B$ by $\pi^\lambda$ and passing to initial forms gives $\sum_{j \in S} h_j r^j = 0$ with $0 \neq h_j \in L$ on the nonempty set $S = \{j : v(H_j) - ej = \lambda\}$, contradicting transcendence. □

**Lemma 21.25** (Algebraicity breaks the step at high degree)**.** *If $r$ is algebraic over $L$ of degree $d$, the constant $a$-coefficient alone cannot rule out $\lambda < 0$ at $W$-degrees $m \geq d$: the minimal polynomial of $r$ provides a consistent nontrivial residue pattern, so the inference stops. Below degree $d$ the step may still function.*

*Proof.* With $\deg m_r = d \leq m$, the coefficients of $m_r$ give $0 \neq h_j \in L$ with $\sum_{j=0}^d h_j r^j = 0$, a consistent negative-$\lambda$ initial-form equation the constant coefficient cannot exclude. □

What is not proved here: algebraicity does not by itself produce a hidden generator (no claim is made that it must); the residue degree does not in general equal a denominator-growth slope (the slope-2 versus degree-2 match at $r_q$ is conjectural pattern, attested at $H$ and $F_{46}$); prime interactions are open. The $N = 5$ instances are $r_q$ (degree 2, step fails, $H$ exists) and $r_p$ (degree 4, step fails, (P-top) false via $F_{46}$).

### 21.4 Speculative program

The statements below need substantial new machinery and are recorded as predictions only, with the instances that motivate them.

**Conjecture 21.26** (general-$N$ normal form)**.** Over a characteristic-zero field, on the open set $\lambda a_{N-1}a_N \neq 0$, the adjacent-quadratic shift admits a normal form obtained via $\Delta_c$-surjectivity, extending the $N = 4$ construction with $u = a_{N-2}$ and $c = \lambda a_{N-1}a_N$ of Theorem 3.3. Surjectivity of the relevant difference operators has been verified computationally through $N = 6$; that verification is evidence, not proof, of the general statement.

*Remark* 21.27 (Open: positive characteristic and singular strata). All positive-characteristic statements for $N \geq 4$ are open, as is saturation at $N \geq 6$. The singular strata $a_{N-1}a_N = 0$, where the normal-form coordinate $c = \lambda a_{N-1}a_N$ vanishes and the $\Delta_c$-construction of Theorem 3.3 breaks down, are excluded from Conjecture 21.26 and remain fully open.

*Remark* 21.28 (Future axes). Two directions are left as future work, with no results claimed: the $N \to \infty$ limit (the infinite-series inverse is restricted throughout to polynomials and truncations; the infinite case is open), and locally nilpotent derivations attached to general dependency graphs, of which the adjacent-quadratic derivations form the path-graph case.

*Remark* 21.29 (No applied claims). No machine-learning, cryptographic, or physical applications are developed or claimed here; any such direction is speculative.

### 21.5 From mechanism to conjectured principle

The $N = 5$ analysis isolates a mechanism with a conjectured wider scope. Principle (conjectured): for a triangular LND with pole-clearing denominator $c$, the local-to-global obstruction is controlled by the algebraic dependence of the initial forms $r_\pi$ in the residue fields $L_\pi$; each independent relation produces hidden generators, and transcendence at every $\pi \mid c$ is the exact condition for one-generator-over-$K$ extension. The $N = 5$ member is the first in the family in which the obstruction is established (no hidden generator is established for $N \leq 4$, whose full rings are unclassified): transcendence fails at $q$ with $r_q^2 \in L_q$, and the hypersurface structure is the exact accounting of the resulting extra generator and relation. The executable form is Conjecture 21.16 with the procedure described in §21.1. Whether the same analysis governs the Nagata derivation or the Daigle–Freudenburg examples is open: applying the procedure there is proposed future work, and no claim about those derivations is made here.

## 22. Conclusion and outlook

What has been proved is a complete finite-range picture with an exact boundary. The $N = 5$ base kernel, localization, and Dixmier divisibility are proved (Theorem 4.4, Theorem 5.2, Lemma 4.5); the six-generator subalgebra is a proved hypersurface with exact series, Cohen–Macaulay, normal, and minimal generators (Theorem 7.1, Remark 7.2); two hidden generators are proved first outside their windows ($H$ at 24 outside $K[F_0]$, $F_{46}$ at 46 outside $C_5$) with exact spanning below each; the $a_0$-top ideals $J_m$ and the bounds-generation criterion are proved (Theorem N.4, Theorem N.5); and the first proved occurrences of the phenomena are located (Theorem 21.23). The unifying principle is filtration-resonance analysis: localize with the explicit slice, read pole orders and residue initial forms, convert their algebraicity into top-ideal obstructions $J_m$, and detect hidden directions by valuation fingerprints (Lemmas 21.24–21.25). Residue algebraicity defeats naive saturation at computable degrees ($r_q$ degree 2, $r_p$ degree 4); the $q$-adic and then $p$-adic programs fail at exactly the predicted places, and the failure mechanism is proved, not merely observed (Section 8). The four-gate filtration checker certifies weight, $a_0$-degree, top, and $J_m$-verdicts with opposite-polarity controls, and the per-weight witness pipeline certifies graded kernel spanning through 14 in Lean with a truncation-rejecting decode invariant (Section M.2). The plinth ideal is not principal: the explicit $q$-primitive element $f_{13}$ supplies a second slice chart with transition $F_0/(qf_{13})$, covering the collapse locus and separating the collapsed orbits (Theorems 11.2–12.12); the residual kernel $\ker \delta_q = k[p, x, u]$ is proved, and the plinth module and gluing program are conjectured with exact bounded evidence (Conjectures 11.5–12.20); the $p$-side is closed at the residual level ($\ker \delta_p$ computed, $R/pR \hookrightarrow A/pA$ proved, hence $R = A \cap R_p$: Theorems 13.8–13.12).

What remains open is sharply delimited. The presentation of $R = C_5[F_{46}^{\mathrm{int}}]$, Q-top-strong, and the sharp $p$-side envelope are established (Sections 10–15); the $F_{46} \cdot K/(p^2)$ module pattern of Conjecture 10.11 is refuted at weight 84. What stays open: the conjectured $p$-side bound Rp (Conjecture 21.6); the $A_6$ ring; and the Lean `a0top_filter_combine` step. The tower-closure barrier and how it was bypassed are the dedicated Section 13.

At $N = 6$ the $B$-level ring is exact by transport (Theorems A.12–A.13), while the $A_6$ denominator module is certified through total degree 13 (Theorems G.6–G.7) with degree 14 computationally blocked, not claimed, and the $A_6$ ring itself remains open.

The tower-closure question is therefore answered:

$$\boxed{A_5^{D_5} = C_5[F_{46}^{\mathrm{int}}], \text{ finitely generated,}}$$

by Theorem 13.5 (slice atlas plus $p$- and $q$-saturation), bypassing the universal top-control that the valuation routes could never supply.

The argument has two halves of different character, and the paper is organized around their junction. The first half discovers: the local model fails twice, and each failure is explained by a precise residue mechanism before the next algebra is built. The second half closes: with the candidate identified, no further discovery is attempted, and saturation does the remaining work. The lesson suggested by the $N = 5$ case — stated here as an observation, not a general theorem — is the separation itself: a local slice can be exactly right about the localized algebra while missing the global ring, and in this case the missing directions were announced by algebraicity in the residue fields (whether this holds generally is Conjecture 21.16).

The program ends where the proof ends: with an exact $N = 5$ invariant ring, an exact $N = 6$ $B$-level structure, and a rigorously certified $A_6$ frontier whose remaining questions are stated for the next theorem rather than presumed to have been solved.

## A. The $N = 6$ base kernel

We work over the characteristic-zero field $k$ with $A_6 = k[a_0, \dots, a_6]$ and $D_6(a_n) = a_{n+1}a_{n+2}$ of the adjacent-quadratic shift family. This section treats the six-variable subproblem obtained by deleting $a_0$. Put

$$B_6 = k[a_1, \dots, a_6], \qquad K_6 = B_6^{D_6},$$

where, throughout this appendix, $B_6$ denotes the six-variable ring $k[a_1, \dots, a_6]$, not the five-variable $B = k[a_1, \dots, a_5]$ of the $N = 5$ sections. Likewise, in this section only, $R$ denotes the $N = 6$ candidate $R_6 = k[G_1, \dots, G_5]$ defined below, distinct from the $N = 5$ closed ring $R = C_5[F_{46}^{\mathrm{int}}]$. with

$$D_6(a_1) = a_2a_3, \quad D_6(a_2) = a_3a_4, \quad D_6(a_3) = a_4a_5, \quad D_6(a_4) = a_5a_6, \quad D_6(a_5) = D_6(a_6) = 0.$$

The Hilbert-series probes that suggested the generator degrees below are evidence, not proof; every exact identity in this section is proved by the hand computation shown plus exact computation (see Supplementary Material).

Write $C = k[a_2, \dots, a_6]$, which is $D_6$-stable. The shift isomorphism $\varphi\colon (k[a_1, \dots, a_5], D_5) \to (C, D_6)$, $a_i \mapsto a_{i+1}$, intertwines the two derivations including the boundary rule ($D_5(a_4) = 0$ since $a_6 = 0$ in $A_5$, matching $D_6(a_5) = 0$). Hence the $N = 5$ descent (Theorem 4.4) transfers verbatim:

$$C^{D_6} = k[a_5, a_6, G_3, G_4], \qquad G_3 = 2a_2a_5 - a_3^2, \quad G_4 = 2a_3a_6 - a_4^2, \tag{1}$$

a polynomial ring in four variables. This identification is Established.

*Remark* A.1 (Pattern breakdown). The $N = 5$ generator $J_1 = 2a_1a_4 - a_2^2$ does not survive: since $D_6(a_4) = a_5a_6 \neq 0$,

$$D_6(2a_1a_4 - a_2^2) = 2a_1a_5a_6 \neq 0,$$

as certified in exact computation (see Supplementary Material). The replacement involving $a_1$ is not quadratic — it has degree 7.

**Lemma A.2** (The five $N = 6$ generators)**.** *With $G_1 = a_5$, $G_2 = a_6$ and $G_3, G_4$ as in* (1)*, let*

$$G_5 = 105a_1a_5^2a_6^4 - 105a_2a_3a_4a_5a_6^3 + 35a_2a_4^3a_5a_6^2 + 35a_3^2a_4^3a_6^2 - 21a_3a_4^5a_6 + 3a_4^7. \tag{2}$$

*Then $D_6(G_i) = 0$ for $i = 1, \dots, 5$. Indeed, writing $G_5 = T_0 + \cdots + T_5$ for the six displayed terms in order, each term kills the defect left by the previous one:*

$$\begin{aligned}
D_6(T_0) &= 105a_2a_3a_5^2a_6^4,\\
D_6(T_1) &= -105a_2a_3a_5^2a_6^4 - 105a_3^2a_4^2a_5a_6^3 - 105a_2a_4^2a_5^2a_6^3,\\
D_6(T_2) &= 35a_3a_4^4a_5a_6^2 + 105a_2a_4^2a_5^2a_6^3,\\
D_6(T_3) &= 70a_3a_4^4a_5a_6^2 + 105a_3^2a_4^2a_5a_6^3,\\
D_6(T_4) &= -21a_4^6a_5a_6 - 105a_3a_4^4a_5a_6^2,\\
D_6(T_5) &= 21a_4^6a_5a_6,
\end{aligned}$$

*whose sum is 0 (defect coefficients $105 - 105$, $-105 + 105$, $-105 + 105$, $35 + 70 - 105$, $-21 + 21$). All six rungs are verified by expansion in exact computation (see Supplementary Material). Hence, with $R = k[G_1, \dots, G_5]$, $R \subseteq K_6$.*

**Lemma A.3** (Independence)**.** *$G_1, \dots, G_5$ are algebraically independent: the $5 \times 5$ Jacobian minor in columns $(a_1, a_2, a_3, a_5, a_6)$ equals*

$$420\, a_5^3a_6^5 \neq 0$$

*over $k$ (the scalar $420 = 2^2 \cdot 3 \cdot 5 \cdot 7$ records the standing characteristic-zero hypothesis), and the full $5 \times 6$ Jacobian has rank 5 at $(a_1, \dots, a_6) = (1, \dots, 6)$, both checked in exact computation (see Supplementary Material). So $R$ is a polynomial ring in five variables.*

**Lemma A.4** (Minimality of the degree-7 generator)**.** *Every element of $K_6$ of total degree at most 6 lies in $k[a_5, a_6, G_3, G_4]$, and $G_5$ is the unique new generator in degree 7 up to scale and addition of products of the other four. Indeed, writing a putative kernel element as a polynomial in $a_1$ over $C$, $G = \sum_{i=0}^{t} a_1^i N_i$ with $N_t \neq 0$, the top $a_1$-coefficients satisfy $D_6(N_t) = 0$ and $t\, a_2a_3N_t + D_6(N_{t-1}) = 0$; thus $N_t \in C^{D_6}$ and $(tN_t, N_{t-1})$ solve the $a_1$-linear system $a_2a_3M + D_6(N) = 0$ with $M \in k[a_5, a_6, G_3, G_4]$. The exact computation certifies that for $M$ of degree $m = 1, \dots, 5$ this system has solution dimensions $5, 8, 14, 20, 30$, all with $M = 0$ (hence no $a_1$-involving*

*kernel element of degree at most* 6*), while for* $m = 6$ *the dimension is* 41 *with exactly one new direction, realized by the* $a_1$*-coefficient* $105a_5^2a_6^4$ *of the pasted* $G_5$*. Consequently the generator degrees are* $1, 1, 2, 2, 7$ *— completed to* $1, 1, 2, 2, 7, 24$ *by the transported hidden generator* $H_B$ *(Remark A.15; total degree* 10*, so the classification through total degree* 7 *above is unaffected).*

*Remark* A.5 (Fibonacci-weighted dimensions through weight 32). Exact computation (see Supplementary Material) gives $\dim\ker(D_6|_{V_w})$ over $\mathbb{F}_{1000003}$ for $0 \le w \le 32$ with the correct $V_w \to V_w$ map ($D_6$ preserves Fibonacci weight): gap $+0$ against the free five-generator count through $w = 23$, then exactly the hypersurface excess $+1, +2, +3, +4, +5, +6, +8, +10, +12$ over $w = 24$–$32$ (213 at $w = 24$, 404 at $w = 30$), matching the transported Hilbert series of Theorem A.11 entry for entry (and the exact $\mathbb{Q}$ $N = 5$ kernel table, by the shift isomorphism). Exact-$\mathbb{Q}$ confirmation through weight 30 (exact computation (see Supplementary Material)): intertwining on all six generators, derivation-matrix coincidence with $A_5$, $H_B \in \ker D_6$, and $\dim\ker(D_6|_{V_w}) = s_6(w) - s_6(w-28)$ at every $0 \le w \le 30$ (47 PASS, zero failures). No weight shift is involved: $\varphi$ preserves the weights $(8, 5, 3, 2, 1, 1)$ verbatim, so the $B_6$ dimensions equal the $A_5$ ones entry for entry.

**Lemma A.6** (Local slice and localized kernel)**.** *On the localization* $B_{a_5a_6}$ *the element*

$$s = \frac{a_4}{a_5a_6}$$

*satisfies* $D_6(s) = 1$*; it is a local slice. Moreover*

$$\begin{aligned} a_4 &= a_5a_6s, \\ a_3 &= \frac{G_4}{2a_6} + \frac{a_5^2a_6s^2}{2}, \\ a_2 &= \frac{G_3}{2a_5} + \frac{a_3^2}{2a_5}, \\ a_1 &= \frac{G_5 - N_5}{105a_5^2a_6^4}, \end{aligned}$$

*where* $N_5 = G_5 - 105a_1a_5^2a_6^4$ *is the* $a_1$*-free part of* (2)*, all checked by direct expansion in exact computation (see Supplementary Material); every denominator is a power of* $a_5a_6$ *(the scalar* 105 *is a unit in characteristic zero). Hence* $B_{a_5a_6} = R_{a_5a_6}[s]$*, and the local-slice theorem (Freudenburg [11], Corollary 1.26; see also van den Essen–van Rossum [27], Proposition 2.4: a slice* $s$ *with* $D(s) = 1$ *gives* $A = A^D[s]$*) gives*

$$B_{a_5a_6}^{D_6} = R_{a_5a_6}.$$

*In particular every* $f \in K_6$ *has the form* $f = P/(a_5^a a_6^b)$ *for some* $P \in R$ *and* $a, b \ge 0$*.*

**Lemma A.7** (Certified descent obstruction)**.** *With* $Z = G_5^2 + 9G_4^7$ *one has* $Z|_{a_6=0} = 0$ *in* $B_6$*, so* $Z = a_6S$ *for an explicit polynomial* $S \in B_6$ *of degree* 13 *with* $D_6(S) = 0$*, both verified in exact computation (see Supplementary Material). Hence the mod-*$a_6$ *specialization* $(Y_1, Y_3, Y_4, Y_5) \mapsto (a_5, G_3|_{a_6=0}, G_4|_{a_6=0}, G_5|_{a_6=0})$ *has the nonzero relation* $Y_5^2 + 9Y_4^7$ *in its kernel, and contraction at the divisor* $(a_6)$ *does not follow by the one-step leading-term route used at* $N = 5$*.*

**Lemma A.8** ($N = 6$ Dixmier divisibility)**.** *Every* $M \in C^{D_6}$ *with* $a_2a_3M \in D_6(C)$ *is divisible by* $c_6 = a_5^2a_6^4$ *in* $C^{D_6}$*.*

*Proof.* Transport Lemma 4.5 along the shift isomorphism $\varphi\colon (k[a_1, \dots, a_5], D_5) \to (C, D_6)$, $a_i \mapsto a_{i+1}$, of (1) (characteristic zero throughout, as in the lemma): $\varphi$ restricts to an isomorphism $K = k[p, q, J_1, J_2] \to C^{D_6}$ sending $a_1a_2 \mapsto a_2a_3$ and $c = p^2q^4 \mapsto c_6$, and intertwines the derivations, so $a_2a_3M \in D_6(C)$ pulls back to $a_1a_2\mu \in D_5(B_5)$ with $M = \varphi(\mu)$; $c \mid \mu$ pushes forward to $c_6 \mid M$. The bounded check exact computation (see Supplementary Material) agrees through total degree 12. □

**Corollary A.9** ($a_1$-degree-one fragment)**.** *Every* $F \in K_6$ *of* $a_1$*-degree at most* 1 *lies in* $R$*.*

*Proof.* Write $F = N_1a_1 + N_0$ with $N_i \in C$: $D_6(F) = 0$ gives $D_6(N_1) = 0$ and $a_2a_3N_1 + D_6(N_0) = 0$, so $N_1 = c_6G'$ by Lemma A.8. Then $E = G'G_5/105 \in R$ has the same $a_1$-top as $F$, and $F - E$ is invariant of $a_1$-degree 0, hence in $C^{D_6} \subseteq R$. So $F \in R$. □

**Corollary A.10** (Five-generation at $N = 6$ (Refuted)). ***[REFUTED by the transported weight-24 invariant*** $H_B$ ***($D_6(H_B) = 0$, $a_1$-degree 2; Remark A.15).]*** *The proposed equality, stated here for the record, ran $B_6^{D_6} = K_6 = k[G_1, \dots, G_5]$. It is false unconditionally: $H_B \in B_6^{D_6}$ while $H_B \notin R$, since $R_{24}$ has $a_1$-degree at most 1 (only $G_5$ carries $a_1$, and $G_5^2$ already weighs 28). The $W^\sharp$-degree induction skeleton of Remark D.8 is void as a route to this conclusion.*

**Theorem A.11** (Transported hypersurface presentation). *Full-ring equality $C_{B,6} = B_6^{D_6}$ is false by the transported $F_{46}$ below; the presentation itself is proved as at $N = 5$. From Theorem 7.1 (same CAS standard): the shift $\varphi$ below is exact algebra, and every transported identity is re-verified in exact computation (see Supplementary Material). The transported identities ($H_B$ invariant of weight 24 and $a_1$-degree 2, $S_B = 0$ with uniform weight 28, and $H_B$ strictly outside $k[G_1, \dots, G_5]$) are formally verified in core Lean (see the formal-verification manifest); the transported presentation $k[G_1, \dots, G_5, H_B]/(S_B)$ follows by the same primitivity argument as at $N = 5$ ($S_B$ linear in $H_B$ with $G_2 \nmid G_5^2 + 9G_4^7$, so $S_B$ is primitive irreducible in the abstract presentation ring). Anchor equality rests on the transferred CAS certificate. As for Theorem 7.1, equality at every weight is Refuted by transport, i.e. with $R_6 = k[G_1, \dots, G_5]$, $C_{B,6} = k[G_1, \dots, G_5, H_B]$ and $K_6 = B_6^{D_6}$, $R_6 \subsetneq C_{B,6} \subsetneq K_6$ (first strictness by $H_B$, second by $\varphi(F_{46})$; hence $K_6 \cong A_5^{D_5}$ is equally unresolved): the graded shift isomorphism $\varphi\colon (k[a_0, \dots, a_5], D_5) \to (B_6, D_6)$, $a_i \mapsto a_{i+1}$ (weights $(8, 5, 3, 2, 1, 1)$ on both sides), restricts to a graded isomorphism $A_5^{D_5} \to B_6^{D_6}$, and $\varphi(F_{46})$ is a $D_6$-invariant of weight 46 and $a_1$-degree 4 with top $\frac{18375}{4} a_1^4 a_5^6 a_6^8$ outside the transported top ideal $(a_5^8 a_6^8)$, by the transported Lemma 10.4. Hence the transported candidate is the six-generator hypersurface*

$$\boxed{C_{B,6} \;=\; k[G_1, \dots, G_5, H_B] \,/\, (S_B),}$$

*with full-ring equality $C_{B,6} = B_6^{D_6}$ Refuted by the transported $F_{46}$ above, where $H_B = \varphi(H)$ is the weight-24 invariant of Remark A.15, $\varphi(F_0) = G_5/105$ (so no seventh generator arises), and $S_B$ is the transported weight-28 relation*

$$\boxed{S_B \;=\; a_6^4 H_B \;-\; G_5^2 \;-\; 105\, a_6^2 G_4^5 G_3 \;-\; 9\, G_4^7.}$$

*The six graded generators carry Fibonacci weights $(1, 1, 6, 4, 14, 24)$, and with $s_6$ the monomial count therein, $\dim_k(C_{B,6})_w = s_6(w) - s_6(w - 28)$ (exact through weight 32 in exact computation (see Supplementary Material) for the full ring; deviation from the five-generator count starts at $w = 24$).*

### A.1 The exact $B_6$-level ring by transport

**Theorem A.12** (Exact $B_6$ ring). *The shift $\varphi : a_i \mapsto a_{i+1}$ restricts to a graded isomorphism $A_5^{D_5} \to B_6^{D_6}$ (intertwining checked at the boundary: $D_5(a_4) = D_5(a_5) = 0$ match $D_6(a_5) = D_6(a_6) = 0$; the preimage of an invariant is invariant by injectivity). Hence with $C_{B,6}$ of Theorem A.11, $\boxed{K_6 = B_6^{D_6} \cong C_{B,6}[T_B]/((a_5)^2 T_B - E_B)}$, $T_B \leftrightarrow \varphi(F_{46})$, $E_B = \varphi(E)$ the transported seven-term weight-48 element (scripts/cas/check_n6_transport.py; log data/n6_transport.log). The transported invariance and top data are re-verified independently (scripts/cas/check_N6_transport_F.py; log data/n6/transport_F.log).*

**Theorem A.13** (Transported $B_6$ structure). *$H_{K_6}(t) = (1 - t^{28})(1 - t^{48})/((1 - t)^2(1 - t^4)(1 - t^6)(1 - t^{14})(1 - t^{24})(1 - t^{46}))$ in Fibonacci weights $(8, 5, 3, 2, 1, 1)$ on $(a_1, \dots, a_6)$; $K_6$ is a codim-2 complete intersection, normal, with $R_m^{(6)}/R_{m-1}^{(6)} \cong C_{B,6}/((a_5)^2)$ layers and exactly six transported singular 3-fold components. With $m$ the $a_1$-degree: $v_6(C_m) \geq 4\lceil m/2 \rceil$ and $v_5(C_m) \geq \lceil 3m/2 \rceil$, the latter sharp on $\varphi(F_{46})^r$. All by transport of the $N = 5$ presentation and two-slope bounds along $\varphi$: the generator table ($\varphi(F_0) = G_5/105$ of $a_1$-degree 1, $H_B$ of $a_1$-degree 2 with top $11025 a_5^4 a_6^4$, $\varphi(F_{46})$ of $a_1$-degree 4 with top $\frac{18375}{4} a_1^4 a_5^6 a_6^8$) carries $(v_5, v_6)$-bidegrees $(2, 4)$, $(4, 4)$, $(6, 8)$, identical to the $N = 5$ table, so the gap identities and sharpness transfer verbatim.*

**Conjecture A.14** (birth spectrum). **[Open.]** Each primitive $N$-level generator of $a_0$-degree $m$ and Fibonacci weight $w$ transports along the shift isomorphism $\varphi_N$ of Theorem 21.17 to a primitive $(N + 1)$-level $B_6$-generator of $a_1$-degree $m$ and weight $w$. Instances: $H \mapsto H_B$ (weight 24, Remark A.15) and $F_{46} \mapsto \varphi(F_{46})$ (weight 46, $a_1$-degree 4, top $\frac{18375}{4} a_1^4 a_5^6 a_6^8$ outside the transported top ideal $(a_5^8 a_6^8)$ by the transported Lemma 10.4, certified $D_6(\varphi(F_{46})) = 0$ in exact computation (see Supplementary Material)). What transport gives is $D_6$-invariance plus the top pattern; what stays open is primitivity at $N + 1$ (no earlier $B_6$-level generator explains the transported invariant). Falsifier: an $N$ where the transported top falls back inside the transported $J$-ideal.

*Remark* A.15 ($H_B$: the transported hidden generator). $H_B = \varphi(H)$ is the 17-term weight-24 invariant with $a_1$-degree 2 (leading term $11025\, a_1^2 a_5^4 a_6^4$), $D_6(H_B) = 0$ by intertwining, re-verified by expansion in exact computation (see Supplementary Material). Its $W^\sharp$-top coefficient $H_2 = 11025\, a_5^4 a_6^4$ has $v_6(H_2) = 4 < 8$, refuting Lemma D.5 with the same numbers as $H$ refutes Lemma J.3; $v_5(H_2) = 4 = 2 \cdot 2$ leaves Lemma D.2 uncontradicted. $H_B$ is a second exact boundary realization of Remark D.10 ($a_6$-side, $m = 2$, $\lambda = -4$). The invariance, weight, $a_1$-degree bound, relation, and strictness are Lean-checked in the formal transport module in core Lean without Mathlib by native evaluation (see the formal-verification manifest).

*Remark* A.16 (Why single divisibility does not close the induction). The $a_1$-top equation gives $N_t = c_6 G'$, and $E = G' G_5 a_1^{t-1}/105$ cancels the $a_1$-top of $F$ — but for $t \geq 2$, $D_6(E) = G' G_5 (t-1) a_1^{t-2} a_2 a_3/105 \neq 0$ in general, so $F - E$ exits $K_6$ and no induction on $K_6$ results; invariant subtractors with the same top would need the full power divisibility $c_6^t \mid N_t$. The $t = 1$ case is exactly Corollary A.9. The inference that single divisibility yields $K_6 \subseteq R$ by $a_1$-induction is withdrawn (it would need $c_6^t \mid N_t$), and the bounded machine check is retained as corroboration only. The generation question is settled instead by transfer (Theorem A.11).

*Remark* A.17 (Status boundary). Established in this base-kernel section: the generator invariance (Lemma A.2), the independence (Lemma A.3), the degree minimality (Lemma A.4), the localized kernel $B^{D_6}_{a_5 a_6} = R_{a_5 a_6}$ (Lemma A.6), the obstruction identity (Lemma A.7), the Dixmier divisibility (Lemma A.8) with the $a_1$-degree-one fragment (Corollary A.9), and — by transfer — the refutation of five-generation (Corollary A.10) with the true hypersurface ring (Theorem A.11). Open: the $a_5$-saturation inference (Lemma D.2, now motivationless for $B_6$-generation) and the full $A_6$ ring with its Hilbert series (Section F). No generation claim for $A_6^{D_6}$ is made here.

## B. The $N = 6$ slice and localized presentation

This slice section uses the generators $G_1, \ldots, G_5$ of Lemma A.2 verbatim (no redefinition) and isolates the localization mechanism over $B_{a_5 a_6}$. Every identity below is proved by the expansion shown plus exact computation (see Supplementary Material).

**Proposition B.1** (Slice certificate and denominator structure)**.** *With $B_6$, $R$, $D_6$ as above, the element*

$$s = \frac{a_4}{a_5 a_6}$$

*satisfies $D_6(s) = 1$ by the quotient rule from $D_6(a_4) = a_5 a_6$ and $D_6(a_5) = D_6(a_6) = 0$. Hence $D_6(s) = 1$ is a nonzero element of $K_6$ (constants are invariant), while $s$ itself is not invariant. Writing $s$ for a transcendental indeterminate over $R_{a_5 a_6}$, the right-hand sides*

$$\begin{aligned} A_4 &= G_1 G_2 s, \\ A_3 &= \frac{G_4 + G_1^2 G_2^2 s^2}{2 G_2}, \\ A_2 &= \frac{G_3 + A_3^2}{2 G_1}, \\ A_1 &= \frac{G_5 - N_5(A_2, A_3, A_4, G_1, G_2)}{105\, G_1^2 G_2^4}, \end{aligned}$$

*where $N_5 = G_5 - 105 a_1 a_5^2 a_6^4$ is the $a_1$-free part of (2), all lie in $R_{a_5 a_6}[s]$: with $s$ transcendental each denominator is $c\, a_5^u a_6^v$ with $c \in k^\times$ (the scalars 2 and 105 are units under the standing characteristic-zero hypothesis), and the substituted numerator $N_5(A_2, A_3, A_4, G_1, G_2)$ is a polynomial (constant denominator), all checked in exact computation (see Supplementary Material). Substituting $s = a_4/(a_5 a_6)$ recovers each variable, $A_i|_{s = a_4/(a_5 a_6)} = a_i$ for $i = 1, \ldots, 4$ with the index convention $A_4 \leftrightarrow a_4$, by direct expansion. The denominator exponents $(u, v)$ in $(a_5^u a_6^v)$ are $(0, 0)$ for $a_4$, $(0, 1)$ for $a_3$, $(1, 2)$ for $a_2$, and $(2, 4)$ for $a_1$. Consequently $B_{a_5 a_6} = R_{a_5 a_6}[s]$ and, by the standard local-slice theorem, $B^{D_6}_{a_5 a_6} = R_{a_5 a_6}$, localizing Lemma A.6 with explicit denominators.*

*Remark* B.2 (Slice status boundary). Established in this section: the slice certificate $D_6(s) = 1$ with $D_6(s) \in K_6$ nonzero, and the localized presentation $B_{a_5 a_6} = R_{a_5 a_6}[s]$ with the denominator exponents above (Proposition B.1). This discharges the slice item of the base-kernel status list above. The $B_6$-level candidate is the hypersurface $C_{B,6}$ of Theorem A.11 (five-generation refuted, Corollary A.10). Still Open: the $a_5$-saturation inference, and hence the global $N = 6$ ring with its Hilbert series. No generation claim for $A_6^{D_6}$ is made here.

## C. The $N = 6$ denominator and weight-shift

This denominator section uses the generators $G_1, \dots, G_5$ of Lemma A.2 verbatim (no redefinition) and the slice $s = a_4/(a_5a_6)$ with $D_6(s) = 1$ of Proposition B.1 (no redefinition), within the adjacent-quadratic shift family. Every identity below is proved by the expansion shown plus exact computation (see Supplementary Material).

**Proposition C.1** (Explicit denominator $c_6$)**.** *Put*

$$c_6 = a_5^2 a_6^4 = G_1^2 G_2^4.$$

*Then $D_6(c_6) = 0$, so $c_6 \in K_6$, and the exact $a_1$-coefficient of $G_5$ in (2) is $105\,c_6$ (the scalar 105 is a unit under the standing characteristic-zero hypothesis); write $G_5 = 105\,c_6 a_1 + N_5$ with $N_5 \in B_6$ free of $a_1$. With $A_4, A_3, A_2, A_1$ as in Proposition B.1, each product $c_6 A_i$ lies in $R[s]$: in lowest terms each has constant denominator (the residual scalars 2, 8, 105 are units), and $105\,c_6 A_1 = G_5 - N_5(A_2, A_3, A_4, G_1, G_2)$ is a polynomial exactly, all checked in exact computation (see Supplementary Material). Thus $c_6$ is the minimal monomial clearing the slice denominators $(u, v) = (0, 0), (0, 1), (1, 2), (2, 4)$: $c_6A_4, c_6A_3, c_6A_2, c_6A_1 \in R[s]$.*

**Theorem C.2** (Weight-shift holds)**.** *With the Fibonacci grading $\mathrm{wt}(a_1, \dots, a_6) = (8, 5, 3, 2, 1, 1)$, compatible with $D_6$ since $\mathrm{wt}(a_n) = \mathrm{wt}(a_{n+1}) + \mathrm{wt}(a_{n+2})$,*

$$\mathrm{wt}(G_1, G_2, G_3, G_4, G_5) = (1, 1, 6, 4, 14),$$

*each verified monomial by monomial in exact computation (see Supplementary Material) ($G_5$ has six monomials, all of weight 14; $N_5$ has five, all of weight 14). Moreover $\mathrm{wt}(c_6) = 2 + 4 = 6$, and for*

$$W = \frac{G_5}{105\,c_6} = a_1 + \frac{N_5}{105\,c_6} \in B_{a_5a_6}$$

*each term has weight 8 (the summand $a_1$ has weight 8, and each $N_5$-monomial of weight 14 drops by 6 upon division by $c_6$). Hence, with $F_0$ denoting the $a_1$-top generator $G_5$,*

$$\boxed{\mathrm{wt}(F_0) = \mathrm{wt}(W) + \mathrm{wt}(c_6), \qquad 14 = 8 + 6,}$$

*so the weight-shift verdict is HOLDS as a theorem (the $N = 5$ observation becomes a proved identity at $N = 6$ for the $B_6$-level top generator). The scalar 105 carries weight 0 and does not affect the grading.*

*Remark* C.3 (Denominator status boundary). Established in this section: the explicit denominator $c_6 = a_5^2 a_6^4$ clearing every slice denominator (Proposition C.1), and the weight-shift identity $\mathrm{wt}(F_0) = \mathrm{wt}(W) + \mathrm{wt}(c_6)$ as HOLDS (Theorem C.2). This discharges the denominator item of the base-kernel status list above. The $B_6$-level ring is the hypersurface of Theorem A.11. Still Open: the $a_5$-saturation inference, and hence the global $N = 6$ ring with its Hilbert series. No generation claim for $A_6^{D_6}$ is made here.

## D. The $N = 6$ saturation: per-divisor residue analysis

Status of this section: Central theorem under verification. Apart from the multiplicity-floor Remark D.10 (Established), no statement in this section carries status Established. We work over the characteristic-zero field $k$ with $B_6 = k[a_1, \dots, a_6]$, $C = k[a_2, \dots, a_6]$, and $D_6$ of the adjacent-quadratic shift family, reusing the generators $G_1, \dots, G_5$ of Lemma A.2 verbatim (no redefinition), the slice $s = a_4/(a_5a_6)$ with $D_6(s) = 1$ of Proposition B.1 (no redefinition), and the denominator $c_6 = a_5^2 a_6^4$ with $G_5 = 105\,c_6 a_1 + N_5$ of Proposition C.1 (no redefinition), where $N_5 \in C$ is free of $a_1$. This section treats the two prime divisors of $c_6$ separately: the $a_5$-adic half below and the $a_6$-adic half in the companion lemma. The two residue fields, pole orders, and initial forms are computed independently on each side, and no symmetry argument is used in either direction: the exponents (2 versus 4), the initial forms (trinomial versus monomial), and the algebraicity degrees (tower of degree at most 4 versus one step of degree at most 2) all differ.

Put $C^{D_6} = k[a_5, a_6, G_3, G_4]$ as in (1) and

$$W^\sharp = \frac{G_5}{105\,c_6} = a_1 + \rho^\sharp, \qquad \rho^\sharp = \frac{N_5}{105\,c_6} \in C_{a_5a_6}.$$

A general element over the localization is written

$$F = \sum_{j=0}^{m} H_j (W^\sharp)^j, \qquad H_j \in (C^{D_6})_{a_5a_6},$$

with $a_1$-coefficients

$$C_k = \sum_{j=k}^{m} \binom{j}{k} H_j (\rho^\sharp)^{j-k} \in C \qquad (0 \le k \le m)$$

whenever $F \in B_6$. For each $m \ge 0$ the $a_5$-part of the $m$-th denominator module is $M_m^5 = a_5^{2m} C^{D_6}$ and the $a_6$-part is $M_m^6 = a_6^{4m} C^{D_6}$; the full module $M_m = c_6^m C^{D_6}$ requires both bounds together. The $W^\sharp$-degree induction skeleton of Remark D.8 is Conditional on both lemmas below. Every specialization identity, exact pole order, Jacobian minor, and algebraicity witness in this section is certified by exact computation (see Supplementary Material); the verification checks setup data only and never the saturation inference itself.

*Remark* D.1 (Valuation setup on the $a_5$-side). Write $C = k[a_2, a_3, a_4, a_6][a_5]$ and let $v_5 = \mathrm{ord}_{a_5}$ on $\mathrm{Frac}(C)$ (and on $\mathrm{Frac}(B_6)$), with valuation ring $V_5 = C_{(a_5)}$ and residue field $\kappa_5 = k(a_2, a_3, a_4, a_6)$. The height-one prime $(a_5) \subset C^{D_6}$ has residue

$$C^{D_6}/a_5 C^{D_6} \cong k[a_6, -a_3^2, 2a_3a_6 - a_4^2],$$

a polynomial ring: the three residue classes are algebraically independent (Jacobian minor $4a_3a_4 \neq 0$ in variables $(a_6, a_3, a_4)$), so with $y_2 = a_6$, $y_3 = -a_3^2$, $y_4 = 2a_3a_6 - a_4^2$,

$$L_5 = \mathrm{Frac}(C^{D_6}/a_5 C^{D_6}) = k(y_2, y_3, y_4) \hookrightarrow \kappa_5.$$

Since $N_5$ has $a_5$-adic order 0 (its $a_5$-free part is the nonzero trinomial $35a_3^2a_4^3a_6^2 - 21a_3a_4^5a_6 + 3a_4^7$) while $c_6 = a_5^2a_6^4$ has order 2, $\rho^\sharp$ has exact pole order $e_5 = 2$ at $a_5$, with trinomial initial form

$$r_5 = \mathrm{in}_{a_5}(a_5^2\rho^\sharp) = \frac{35a_3^2a_4^3a_6^2 - 21a_3a_4^5a_6 + 3a_4^7}{105\, a_6^4}.$$

**Lemma D.2** (Central theorem under verification: $a_5$-adic denominator bound)**.** ***[Central theorem under verification.]*** *Let $F = \sum_{j=0}^{m} H_j (W^\sharp)^j \in B_6$ with $H_j \in (C^{D_6})_{a_5a_6}$. Then*

$$\boxed{v_5(H_j) \ge 2j \quad \textit{for every } j,}$$

*i.e. $H_j \in a_5^{2j}(C^{D_6})_{a_5a_6}$ for every $j$.*

*Reduction and explicit gap.* The constant $a_1$-coefficient is $C_0 = \sum_j H_j(\rho^\sharp)^j \in C$, so $v_5(C_0) \ge 0$. Put $\lambda_5 = \min_j\{v_5(H_j) - 2j\}$ and $S = \{j : v_5(H_j) - 2j = \lambda_5\}$. If $\lambda_5 < 0$, dividing by $a_5^{\lambda_5}$ and passing to initial forms gives the relation

$$\sum_{j \in S} h_j r_5^j = 0$$

with $0 \neq h_j \in L_5$ the residues of the normalized coefficients. The reduction to this initial-form equation is valid. What is not verified is the final inference: concluding $h_j = 0$ for all $j \in S$ from this single relation would require linear independence of arbitrarily high powers of $r_5$ over $L_5$, which is not proved; Remark D.3 records why the direct constant-coefficient route is unavailable. No parity shortcut is used as proof, no slice-power computation is used as proof, no coordinate change is used as proof, and deriving generation from a single divisor is not used as proof. Until the inference is verified, for instance via a higher-order ($\pi$-adic lifting) refinement (initial-form analysis of the full triangular system $\{C_k\}$ alone cannot suffice, Remark D.10), the lemma stays under verification. □

*Remark* D.3 (Observation: why the constant-coefficient route is unavailable on the $a_5$-side). The triple $\{y_2, y_3, y_4\}$ is algebraically independent (Jacobian minor $4a_3a_4 \neq 0$), so $L_5 = k(y_2, y_3, y_4)$ is purely transcendental of degree 3. But with $t = a_3/a_6$ one has $t^2 = -y_3/y_2^2 \in L_5$, and with $u = a_4/a_6$ one has $u^2 = 2t - y_4/y_2^2 \in L_5(t)$, so

$$L_5 \subseteq L_5(t) \subseteq L_5(t)(u) = k(a_6, a_3, a_4)$$

is a tower of degree at most 2 at each step; hence the initial form $r_5 \in k(a_3, a_4, a_6)$ is algebraic over $L_5$ (of degree at most 4), not transcendental. Consequently no argument needing $1, r_5, r_5^2, \ldots$ to be linearly independent over the full range of $W^\sharp$-degrees can succeed from the constant coefficient alone. The witnesses $t^2 + y_3/y_2^2 = 0$ and $u^2 - 2t + y_4/y_2^2 = 0$ are certified in exact computation (see Supplementary Material).

*Remark* D.4 (Valuation setup on the $a_6$-side). Write $C = k[a_2, a_3, a_4, a_5][a_6]$ and let $v_6 = \mathrm{ord}_{a_6}$ on $\mathrm{Frac}(C)$ (and on $\mathrm{Frac}(B_6)$), with valuation ring $V_6 = C_{(a_6)}$ and residue field $\kappa_6 = k(a_2, a_3, a_4, a_5)$. The height-one prime $(a_6) \subset C^{D_6}$ has residue

$$C^{D_6}/a_6C^{D_6} \cong k[a_5, 2a_2a_5 - a_3^2, -a_4^2],$$

a polynomial ring: the three residue classes are algebraically independent (Jacobian minor $-4a_4a_5 \neq 0$ in variables $(a_5, a_2, a_4)$), so with $z_1 = a_5$, $z_3 = 2a_2a_5 - a_3^2$, $z_4 = -a_4^2$,

$$L_6 = \mathrm{Frac}(C^{D_6}/a_6C^{D_6}) = k(z_1, z_3, z_4) \hookrightarrow \kappa_6.$$

Note the asymmetry with the $a_5$-side: here the surviving specialization $G_3 \bmod a_6 = 2a_2a_5 - a_3^2$ retains the variable $a_2$, while $G_4 \bmod a_6 = -a_4^2$ is a pure square; the roles of the two kernel generators are not interchanged by any symmetry. Since $N_5$ has $a_6$-adic order 0 (its $a_6$-free part is the nonzero monomial $3a_4^7$) while $c_6$ has order 4, $\rho^\sharp$ has exact pole order $e_6 = 4$ at $a_6$, with monomial initial form

$$r_6 = \mathrm{in}_{a_6}(a_6^4\rho^\sharp) = \frac{a_4^7}{35\,a_5^2}.$$

**Lemma D.5** ($a_6$-adic denominator bound (Refuted))**.** ***[REFUTED by the transported invariant $H_B$ (Remark A.15).]*** *The refuted statement is: for $F = \sum_{j=0}^{m} H_j(W^\sharp)^j \in B_6$ with $H_j \in (C^{D_6})_{a_5a_6}$,*

$$\boxed{v_6(H_j) \geq 4j \quad \textit{for every } j,}$$

*i.e. $H_j \in a_6^{4j}(C^{D_6})_{a_5a_6}$ for every $j$. It is false: the $W^\sharp$-top coefficient of $H_B$ is $H_2 = 11025\,a_5^4a_6^4$ with $v_6(H_2) = 4 < 8 = 4 \cdot 2$.*

*Reduction valid; inference refuted by the transported counterexample.* The constant $a_1$-coefficient is $C_0 = \sum_j H_j(\rho^\sharp)^j \in C$, so $v_6(C_0) \geq 0$. Put $\lambda_6 = \min_j\{v_6(H_j) - 4j\}$ and $S = \{j : v_6(H_j) - 4j = \lambda_6\}$. If $\lambda_6 < 0$, dividing by $a_6^{\lambda_6}$ and passing to initial forms gives the relation

$$\sum_{j \in S} h_j r_6^j = 0$$

with $0 \neq h_j \in L_6$ the residues of the normalized coefficients. The reduction to this initial-form equation is valid. The final inference — concluding $h_j = 0$ for all $j \in S$ — would require the powers $1, r_6, r_6^2, \ldots$ to be linearly independent over $L_6$. That independence is false: $r_6$ is algebraic of degree at most 2 over $L_6$ (Remark D.6), and the transported invariant $H_B$ provides the concrete non-trivial relation (Remark A.15). The $a_5$-side argument is not copied by symmetry: the residue field, the pole order, and the initial form above are computed independently, and no symmetry with Lemma D.2 is used in either direction. □

*Remark* D.6 (Observation: why the constant-coefficient route is unavailable on the $a_6$-side). The triple $\{z_1, z_3, z_4\}$ is algebraically independent (Jacobian minor $-4a_4a_5 \neq 0$), so $L_6 = k(z_1, z_3, z_4)$ is purely transcendental of degree 3. But with $\bar{s}_6 = a_4/a_5$ one has $\bar{s}_6^2 = -z_4/z_1^2 \in L_6$, so

$$L_6 \subseteq L_6(\bar{s}_6) = k(a_5, z_3, a_4)$$

is a single quadratic step; hence $[k(a_5, z_3, a_4) : L_6] \leq 2$. Since $r_6 = a_4^7/(35a_5^2) \in L_6(\bar{s}_6)$, the monomial initial form $r_6$ is algebraic over $L_6$ (of degree at most 2), not transcendental. Consequently no argument needing $1, r_6, r_6^2, \ldots$ to be linearly independent over the full range of $W^\sharp$-degrees can succeed from the constant coefficient alone. This $a_6$-side degree bound is independent of the $a_5$-side tower (there a two-step extension of degree at most 4 governs a trinomial initial form; here a one-step extension of degree at most 2 governs a monomial initial form), and the difference in pole orders ($e_6 = 4$ versus $e_5 = 2$) is part of why the two lemmas cannot share a proof. The witness $\bar{s}_6^2 + z_4/z_1^2 = 0$ is certified in exact computation (see Supplementary Material).

*Remark* D.7 (Observation: $R$-level specialization is not injective on either side). The two lemmas above work at $C$-level, where both specialization maps are injective. At $R$-level the one-step contraction route is blocked on both sides already at the specialization step. On the $a_6$-side this is Lemma A.7 ($G_5^2 + 9G_4^7 = a_6S$). On the $a_5$-side, with $Y_2 = a_6$,

$Y_3 = -a_3^2$, $Y_4 = 2a_3a_6 - a_4^2$ and $Y_5 = 35a_3^2a_4^3a_6^2 - 21a_3a_4^5a_6 + 3a_4^7$ the $R$-residues mod $a_5$, the nonzero polynomial

$$\begin{aligned} REL_5 &= Y_5^4 + 18Y_4^7Y_5^2 + 81Y_4^{14} + Y_2^2Y_3\,Q(Y_2, Y_3, Y_4, Y_5), \\ Q &= 40000Y_2^{12}Y_3^6 + 193200Y_2^{10}Y_3^5Y_4^2 + 325164Y_2^8Y_3^4Y_4^4 + 236425Y_2^6Y_3^3Y_4^6 \\ &\quad + 88410Y_2^4Y_3^2Y_4^8 + 840Y_2^4Y_3^2Y_4Y_5^2 + 17955Y_2^2Y_3Y_4^{10} + 770Y_2^2Y_3Y_4^3Y_5^2 \\ &\quad + 1890Y_4^{12} + 210Y_4^5Y_5^2, \end{aligned}$$

i.e. $(Y_5^2 + 9Y_4^7)^2$ plus terms vanishing to second order in $(Y_2, Y_3)$, vanishes on the residues ($REL_5 = 0$ in $B_6/(a_5)$, checked by expansion in exact computation (see Supplementary Material)). Hence neither $R$-level specialization map is injective. These kernel relations block that route only; the five-generation conclusion they guarded is refuted independently (Corollary A.10).

*Remark* D.8 (Void $W^\sharp$-degree induction skeleton). The induction below assumed the $a_5$-adic bound of Lemma D.2 together with the companion $a_6$-adic bound of Lemma D.5 in full, whence each $H_j$ lies in $C$ (top coefficient $H_m = [a_1^m]F$ in $C$ directly, descending induction via $C_{m-1} = H_{m-1} + mH_m\rho^\sharp$), the transferred $C$-level contraction gives $H_m = c_6^m G_m$, and subtracting $G_m G_5^m/105^m \in R$ lowers the $W^\sharp$-degree. It is void: Lemma D.5 is refuted by $H_B$, so one of its two premises is false, and its conclusion ($B_6^{D_6} = R$) is false (Corollary A.10). The true $B_6$-level candidate structure is $C_{B,6}$ of Theorem A.11.

*Remark* D.9 (Saturation status boundary). What this section fixes is the valuation setup on each side: the rings $V_5, V_6$, the residue embeddings $L_5 \hookrightarrow \kappa_5$ and $L_6 \hookrightarrow \kappa_6$, the exact pole orders $e_5 = 2$, $e_6 = 4$, the initial forms $r_5$ (trinomial) and $r_6$ (monomial), the denominator modules $M_m^5$, $M_m^6$, and the precise gaps, plus the $R$-level kernel witness $REL_5$ alongside Lemma A.7. The $a_6$-side inference is refuted by $H_B$ (Lemma D.5); what stays Open is the $a_5$-side inference (Lemma D.2), now motivationless for $B_6$-generation. The $B_6$-level candidate is the hypersurface $C_{B,6}$ of Theorem A.11; the $W^\sharp$-degree induction is void as a route to five-generation. No generation claim for $A_6^{D_6}$ is made here.

*Remark* D.10 (Multiplicity floor for initial-form methods). Pure algebra; explicit minpoly-power solutions verified in exact computation (see Supplementary Material). For fixed $W^\sharp$-degree $m$, the joint initial-form system from *all* of $\{C_k\}$ is exactly $P(r) = P'(r) = \cdots = P^{(K)}(r) = 0$ with $P(T) = \sum_{j\in S} h_j T^j$ (the $C_k$ equation is the $k$-th formal derivative of the $C_0$ equation), i.e. $r$ is a root of multiplicity at least $K+1$, where $K+1 = \#\{k \geq 0 : \lambda + ek < 0\}$ for deficit $\lambda < 0$ and pole order $e$. Since $\mathrm{minpoly}(r)$ is separable (characteristic zero), $\mathrm{minpoly}(r)^{K+1} \mid P$, so $(K+1)\cdot d_r \leq m$ with $d_r$ the algebraicity degree. On the $a_6$-side ($e_6 = 4$, $d_r = 2$) this gives the hard floor $\lambda \geq -2m$: no initial-form analysis, at any order of the triangular system, can establish $\lambda \geq 0$. The floor is sharp: $P_s = \mathrm{minpoly}(r_6)^s$ has exact multiplicity $s$, realizing every deficit $-4s$ at $W^\sharp$-degree $2s$ (verified for $s \leq 6$; see Supplementary Material). The $N = 5$ counterexample $H$ sits exactly on the corresponding floor ($e_q = 4$, $d_r = 2$, $m = 2$, $\lambda = -4$). Consequently the "full triangular system $\{C_k\}$" refinement route named in Lemmas D.2–D.5 cannot close the saturation bounds; only genuinely higher-order ($\pi$-adic lifting) or global input can. The $a_5$-side follows the same pattern over its degree-$\leq 4$ tower.

## E. The $N = 6$ global corollaries

Status of this section: global statements below are Conditional or Refuted; none is Established. We work over the characteristic-zero field $k$ with $B_6 = k[a_1, \ldots, a_6]$, $K_6 = B_6^{D_6}$, and $R = k[G_1, \ldots, G_5]$, reusing the generators $G_1, \ldots, G_5$ of Lemma A.2 verbatim (no redefinition), the slice $s = a_4/(a_5a_6)$ with $D_6(s) = 1$ of Proposition B.1 (no redefinition), and the denominator $c_6 = a_5^2a_6^4$ with $G_5 = 105\,c_6a_1 + N_5$ of Proposition C.1 (no redefinition) within the adjacent-quadratic shift family. No new valuation is proved here; this section records the global $B_6$-level consequences, now resolved by transfer. No generation claim for $A_6^{D_6}$ is made here: the $a_0$-level generator $F_0^{(6)}$ is constructed in Section F, and the full $N = 6$ ring stays Open.

**Corollary E.1** (Global $B_6$-level five-generation (Refuted)). ***[REFUTED by $H_B$ (Corollary A.10).]*** *The proposed equality* $\boxed{B_6^{D_6} = K_6 = k[G_1, \ldots, G_5]}$ *is false; the candidate is the transported hypersurface of Theorem A.11 (full-ring equality likewise Refuted by transported $F_{46}$, as at $N = 5$).*

*Proof.* The inclusion $R \subseteq K_6$ is the Established invariance of Lemma A.2. The claimed reverse inclusion rested on the $W^\sharp$-degree induction skeleton of Remark D.8, now void (Corollary A.10). □

**Corollary E.2** (Polynomiality (Refuted)). ***[REFUTED:** $B_6^{D_6}$ **is a polynomial ring; the candidate finite model is the six-generator hypersurface** $C_{B,6}$ **of Theorem A.11, with full-ring equality** $C_{B,6} = B_6^{D_6}$ **refuted by transport, not the full ring.]** The five generators remain algebraically independent (Lemma A.3); the failure is precisely the missing sixth generator $H_B$ modulo the single weight-28 relation $S_B$.*

**Corollary E.3** (Five-generator Hilbert series (Refuted)). ***[REFUTED by** $H_B$**; the exact series is that of Theorem A.11.]** The input side is Established: with ordinary degrees*

$$\deg_{\mathrm{ord}}(G_1, \ldots, G_5) = (1, 1, 2, 2, 7)$$

*(Lemma A.4) and Fibonacci weights*

$$\mathrm{wt}(G_1, \ldots, G_5) = (1, 1, 6, 4, 14)$$

*(Theorem C.2), $k[G_1, \ldots, G_5]$ is a polynomial ring with the stated rational functions as* its *Hilbert series. Their identification with those of $B_6^{D_6}$ is false from weight 24 ($213 > 212$; exact computation (see Supplementary Material)). The exact Fibonacci-weighted Hilbert series of the candidate transported hypersurface is*

$$\boxed{H_{\mathrm{wt}}(u) \;=\; \frac{1 - u^{28}}{(1-u)^2(1-u^4)(1-u^6)(1-u^{14})(1-u^{24})}} \qquad \textit{(Theorem A.11).}$$

*Remark* E.4 (Global status boundary). Refuted and unconditional (all by the transported counterexample $H_B$): five-generation (Corollary E.1), polynomiality (Corollary E.2), and the five-generator Hilbert series (Corollary E.3), together with the $a_6$-saturation lemma on which they rested (Lemma D.5). What is now *Established* is the *corrected* $B_6$-level candidate presentation: the six-generator hypersurface $C_{B,6} = k[G_1, \ldots, G_5, H_B]/(S_B)$ with its weight-28 relator and exact Hilbert series of $C_{B,6}$ (Theorem A.11); full-ring equality $C_{B,6} = B_6^{D_6}$ is Refuted by the transported $F_{46}$ (same theorem). What stays Open is the $a_5$-saturation inference (Lemma D.2, now motivationless for $B_6$-generation) and, with it, every claim for $A_6^{D_6}$; in particular the full $N = 6$ ring with its Hilbert series remains Open.

## F. The $N = 6$ $a_0$-top generator

Status of this section: the top generator is Established; everything downstream (the $a_0$-saturation, hence the full ring $A_6^{D_6}$) is Open. We work over the characteristic-zero field $k$ with $A_6 = k[a_0, \ldots, a_6]$ and $D_6$ of the adjacent-quadratic shift family, reusing $B_6$, $R$, $c_6$ verbatim. The Fibonacci grading extends by $\mathrm{wt}(a_0) = \mathrm{wt}(a_1) + \mathrm{wt}(a_2) = 13$, compatible with $D_6(a_0) = a_1 a_2$. Every identity below is proved by the expansion shown plus exact computation (see Supplementary Material).

**Lemma F.1** (The $a_0$-top generator $F_0^{(6)}$). *With $c_6 = a_5^2 a_6^4$, put $c' = c_6^2 = a_5^4 a_6^8$ and*

$$\begin{aligned} F_0^{(6)} = {} & c' a_0 - a_1 a_2 a_4 a_5^3 a_6^7 + \tfrac{1}{3} a_1 a_3 a_4^3 a_5^2 a_6^6 - \tfrac{1}{15} a_1 a_4^5 a_5^2 a_6^5 \\ & + \tfrac{1}{2} a_2^2 a_3 a_4^2 a_5^2 a_6^6 - \tfrac{1}{8} a_2^2 a_4^4 a_5^2 a_6^5 - \tfrac{1}{3} a_2 a_3^2 a_4^4 a_5 a_6^5 + \tfrac{59}{360} a_2 a_3 a_4^6 a_5 a_6^4 \\ & - \tfrac{59}{2880} a_2 a_4^8 a_5 a_6^3 + \tfrac{1}{18} a_3^3 a_4^6 a_6^4 - \tfrac{119}{2880} a_3^2 a_4^8 a_6^3 \\ & + \tfrac{33}{3200} a_3 a_4^{10} a_6^2 - \tfrac{11}{12800} a_4^{12} a_6, \end{aligned}$$

*so $F_0^{(6)} = c' a_0 + G'$ with $G' \in B_6$ $a_0$-free of 12 terms. Then $D_6(F_0^{(6)}) = 0$ by direct expansion; $F_0^{(6)}$ has ordinary degree 13, uniform Fibonacci weight $25 = 12 + 13$, and $a_0$-coefficient exactly $c'$. The $a_0$-free part $G'$ was found by elimination ($c_6$, $c_6 a_5$, $c_6 a_6$ admit no $G'$ with $D_6(G') = -c' a_1 a_2$; $c_6^2$ is the first denominator that does, with a unique direction) and the identity $D_6(G') + c' a_1 a_2 = 0$ is verified exactly over $\mathbb{Q}$ in exact computation (see Supplementary Material).*

*Remark* F.2 (Denominator-squared pattern). The $a_0$-denominator is the square of the $B_6$-level slice denominator: $c' = c_6^2$. The same squaring holds at $N = 5$, where the $N = 5$ slice denominator $a_4 a_5^2$ squares to $c = p^2 q^4$. This refines the path-count guess of Section 21: the $a_0$-denominator exponents are twice the $B_6$-slice-denominator exponents, not raw path counts. The squaring is not minimal, however: $c_6^2$ is itself an $H_{20}$-multiple ($c_6^2 = a_5^4 a_6^3 \cdot a_6^5$), and indeed $F_0^{(6)} = a_6^5 H_{20} + K_0$ with $K_0 \in K_6$ ($a_0$-free, $D_6$-invariant, verified in exact computation (see Supplementary Material)) — so $F_0^{(6)}$ adds no new $a_0$-direction beyond the minimal denominator $a_5^4 a_6^3$. A 42-candidate sink-monomial sweep finds solvable denominators exactly for $a \geq 4$, $b \geq 3$, i.e. exactly the $H_{20}$-multiples. The $a_0$-level Dixmier game (Section G) starts from $H_{20}$, not from $F_0^{(6)}$.

*Remark* F.3 (Top-generator status boundary). Established in this section: the $a_0$-top generator $F_0^{(6)}$ with bidegree $(13, 25)$ and denominator $c_6^2$ (Lemma F.1). Open: the $a_0$-saturation (denominator bounds for the $a_0$-expansion coefficients), and hence the full ring $A_6^{D_6}$ with its Hilbert series. No generation claim for $A_6^{D_6}$ is made here.

**Lemma F.4** (Hidden $A_6$-generator $H_{20}$)**.** *Exact computation (see Supplementary Material); Lean-proven invariance, weight, split, and strictness in the formal $H_{20}$ module (formally verified certificate, native evaluation in core Lean without Mathlib (see the formal-verification manifest)). There is an explicit invariant $H_{20} \in A_6$ of Fibonacci weight 20 with 12 terms and $a_0$-degree 1 satisfying $D_6(H_{20}) = 0$ with $a_0$-coefficient $a_5^4 a_6^3$; the full record is provided in the Supplementary Material (terms not pasted here). At weight 20 the free span of $k[G_1, \ldots, G_5, F_0^{(6)}]$ is $a_0$-free ($F_0^{(6)}$ weighs $25 > 20$; $G_1, \ldots, G_5$ are $a_0$-free), while $H_{20}$ has nonzero $a_0$-part — hence $H_{20} \notin k[G_1, \ldots, G_5, F_0^{(6)}]$ unconditionally (no $B_6$-generation needed), and $A_6^{D_6} \supsetneq k[G, F_0^{(6)}]$. The first excess over the free count is exactly at weight 20 (gap $+1$; exact-$\mathbb{Q}$ $A_6$ probe through weight 20).*

*Remark* F.5 (Disentangling the $A_6$ gaps). The $A_6$ weighted probe (exact computation (see Supplementary Material)) conflates two excesses: the $B_6$-level hypersurface excess (starting at weight 24 with $+1, +2, \ldots$, Theorem A.11) and genuine $a_0$-content. At weight 20 the $B_6$-level matches the free count exactly (exact $\mathbb{Q}$), so the $+1$ there is purely the $a_0$-direction $H_{20}$ above. At weights 21–23 the probe is clean (gap $+0$ mod-prime); from 24 on, the two excesses mix, and the $a_0$-level analysis (Section G) must subtract the known $B_6$-hypersurface contribution first. The $a_0$-coefficient $a_5^4 a_6^3$ of $H_{20}$ is divisible by neither $c_6$ nor $c_6^2$: no $c_6^2$-type $a_0$-denominator bound holds in general.

## G. The $N = 6$ $a_0$-denominator module (partial results)

Status of this section: two denominator directions Established; the full module presentation is Open. The $a_0^1$-system is $a_1 a_2 M + D_6(N) = 0$ with $M \in K_6$ of total degree $m$ and $N \in B_6$ of total degree $m + 1$ ($D_6$ is total-degree $+1$ homogeneous). Through total degree 12 the spanning set of $R$-monomials plus $H_B$-times-$R$-monomials is free (the relation $S_B$ has total degree 28), so the computation below is exact. Exact computation (see Supplementary Material); nullities mod-1000003 cross-checked at a second prime, witnesses exact over $\mathbb{Q}$.

**Lemma G.1** (Second $a_0$-direction)**.** *Exact computation (see Supplementary Material). With $M = a_5^4 a_6 G_4$ (total degree 7) there is an explicit 10-term $N$ with $a_1 a_2 M + D_6(N) = 0$ (full record in the Supplementary Material), i.e. $F = a_0 M + N$ satisfies $D_6(F) = 0$ at Fibonacci weight 22. $M$ is not a multiple of $a_5^4 a_6^3$ ($a_6$-exponents $1, 2 < 3$), so this direction is not in the $H_{20}$-family. That $F$ lies outside $k[G, F_0^{(6)}, H_{20}]$ is Proposition G.2 below.*

**Proposition G.2** (D2-novelty)**.** *The weight-22 invariant $F_{22} = a_0 M + N$ of Lemma G.1 lies outside $k[G_1, \ldots, G_5, H_{20}]$: the weight-22 slice of the subalgebra ($F_0^{(6)}$ weighs 25 and $H_B$ weighs 24, both excluded by weight) has exact rank 171 vs augmented 172 (scripts/cas/check_n6_D2novel.sage; log data/n6_D2novel.log). Hence the $D_2$-direction is a new generator direction over $k[G, H_{20}]$, and the $A_6$ obstruction is not inherited transport baggage.*

*Remark* G.3 (The denominator module is not principal). The solution projection of the $a_0^1$-system strictly exceeds the $a_5^4 a_6^3$-family from total degree 7: at $m = 7$ three directions ($a_5^4 a_6^3$ exact via $H_{20}$; $a_5^4 a_6 G_4$ exact via Lemma G.1; one mixed direction mod-prime only, exact lift open), growing to $+35$ excess dimensions at $m = 12$ (two primes agree). The unbounded module presentation (all generators, all relations among denominator directions) is Open; it is the input the $a_0$-Dixmier game over $R[H_B]/(S_B)$ needs. In particular the $a_0$-saturation cannot take the form of a single-monomial denominator bound.

**Theorem G.4** (No single-monomial $a_0$-denominator bound)**.** *No monomial $d \in K_6$ bounds all $a_0^1$-denominator directions: the $H_{20}$-direction $a_5^4 a_6^3$ (Lemma F.4) and the $D_2$-direction $a_5^4 a_6 G_4$ (Lemma G.1) are divisibility-incomparable in $K_6$, so no single $d$ is divided by, hence bounds, both. Indeed divisibility in the subring $K_6$ implies divisibility in $B_6$, and in $B_6$: $a_6^2 \nmid G_4$ ($G_4$ has the $a_6$-free term $-a_4^2$), and $G_4 \nmid a_6^2$ (exact-division remainder $a_6^2 \neq 0$; $G_4$ is irreducible; scripts/cas/check_n6_G4.py; log data/n6_G4.log). In particular $a_0$-saturation at $N = 6$ cannot take the $c_6$-power form of any one-monomial conjecture.*

*Remark* G.5 (Denominator-module status boundary). Established in this section: three exact $m = 7$ denominator directions ($a_5^4 a_6^3$ via $H_{20}$, $a_5^4 a_6 G_4$ via Lemma G.1, $M_3$ via Theorem G.9), one new exact direction at $m = 8$, and the exact denominator module with closure through total degree 13 (Theorems G.6–G.7). Open: the full (unbounded) denominator-module presentation, and with it the $A_6$ ring (relations, series) and its saturation; degree 14 is computationally blocked, not claimed.

**Theorem G.6** (Bounded denominator module)**.** *Let $\mathcal{M}_m$ be the $M$-projection of $\{(M,N) : a_1a_2M + D_6(N) = 0\}$ at total degree $m$ over $\mathbb{Q}$, and $\mathcal{G}_m$ the cumulative $K_6$-span of exact directions below $m$. Exact QQ nullspaces give, for $7 \le m \le 12$:*

| $m$ | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|
| $\dim \mathcal{M}_m$ | 3 | 7 | 14 | 23 | 35 | 50 |
| $\dim \mathcal{G}_m$ | 0 | 6 | 14 | 23 | 35 | 50 |
| *new* | 3 | 1 | 0 | 0 | 0 | 0 |

*The three $m = 7$ directions have Fibonacci weights $7, 9, 11$; the single new $m = 8$ direction is $M_8 = a_4^4a_5^4 - 4a_3a_4^2a_5^4a_6 + \frac{17}{3}a_3^2a_5^4a_6^2 - \frac{10}{3}a_2a_5^5a_6^2$ of Fibonacci weight 12, with exact 13-term $N_8$ and $a_1a_2M_8 + D_6(N_8) = 0$ by direct expansion (scripts/cas/check_n6_m8wit.sage; log data/n6_m8wit.log). Every direction through 12 is $K_6$-generated by these four; the four generator $M$-vectors are jointly independent, so there are no relations among them within the truncation (scripts/cas/check_n6_denmod.py; log data/n6_denmod.log). No claim is made beyond total degree 12.*

**Theorem G.7** (Closure through $m = 13$)**.** $\mathcal{Q}_{13} = 0$*: at total degree 13 the exact solution space has dimension 68, all $K_6$-generated by the four exact directions below it (quotient dimension 0 over $\mathbb{Q}$; scripts/cas/check_n6_m13.py; log data/n6_m13.log). The exact closure window extends through 13 with no new generator. No theorem-grade conclusion is drawn beyond total degree 13; degree 14 is diagnostic-only (remark below).*

*Remark* G.8 (Degree 14: computationally blocked, not claimed). The exact denominator-module computation is certified through total degree 13: the solution module is generated in the certified window by four independent characteristic-zero directions of degrees $7, 7, 7, 8$, with no relations in that range. An exploratory computation at degree 14 produced a stable kernel-count signal across independent checks, but the corresponding projection-dimension extraction was not reconciled with the exact nullity calculation. We therefore make no degree-14 structural claim and do not use this computation in any theorem. The diagnostic logs are retained as unresolved material only.

**Theorem G.9** (Third $m = 7$ denominator direction)**.** *The mod-prime third $m = 7$ direction lifts: with $M_3 = a_4^4a_5^2a_6 - 4a_3a_4^2a_5^2a_6^2 - \frac{16}{21}a_3^2a_5^2a_6^3 + \frac{200}{21}a_2a_5^3a_6^3$ and $N_3$ the explicit 14-term Fibonacci-24 degree-8 polynomial of the log, $a_1a_2M_3 + D_6(N_3) = 0$ over $\mathbb{Q}$ by direct expansion, and $M_3$ lies outside $\mathrm{span}\{a_5^4a_6^3, a_5^4a_6G_4\}$ (exact rank test) (scripts/cas/check_n6_m7lift.sage; log data/n6_m7lift.log; mod-prime baseline $60/57/3$ in data/n6/A0sat.log reproduced exactly over $\mathbb{Q}$). Hence $F_3 = a_0M_3 + N_3 \in A_6^{D_6}$ of Fibonacci weight 24 and $a_0$-degree 1 is a third exact $a_0$-direction. The degree-12 module program now starts from three exact $m = 7$ generators.*

### G.1 $A_6$ attack scaffolding (investigation, no claims)

No result is claimed here. With $K_6$ exact (Theorem A.12), $A_6 = B_6[a_0]$, $D_6(a_0) = a_1a_2$: $a_0$-filtration tops land in the known ring $K_6$, and the denominator module needs $\ge 2$ generators (Theorem G.4). Exit gates, in order: (i) top-ideal computation in $K_6$ (bounded, exact); (ii) resonance hunt (second hidden direction or proof of none to stated bounds); (iii) either obstruction theorem #2 or a documented census with data tables. Finite generation of $A_6^{D_6}$ is not assumed at any gate. Dimension six plus triangular does not determine the outcome either way: the Freudenburg dimension-six derivation has non-finitely-generated kernel [10], so the $A_6$ ring stays open on its own evidence. Precommitted non-goals: reviving $W^\sharp$-induction, single-monomial saturation, any use of D.2 as premise.

## Summary of proof and computational status

Every main claim, where it is proved, what computational or formal check covers it, and what it depends on. Evidence uses four rigid categories. **Mathematical proof** means the argument appears in the paper. **Formal verification** means checked by Lean in core Lean without Mathlib by native evaluation, with the exact theorem name and `#print axioms` output recorded (standard foundation plus theorem-specific native-evaluation axioms). **Exact computation** means a fresh exact characteristic-zero computation with reproducible dimensions/ranks. **Conjectural/open** means not proved. In the table, "Established" = mathematical proof (possibly plus exact computation or formal verification as stated); "Refuted" = explicit counterexample; "Open" = conjectural. Finite-field counts are consistency checks only.

Table 3: Main claims with proof location and supporting computation.

| claim | status | proof location | computation / depends on |
|---|---|---|---|
| $B^{D_5} = K$ | Established | 4.4 §3 | exact computation (see Supplementary Material); slice Cor 1.26 |
| $D_5(F_0) = 0$, bidegree, independence | Established | §4 | exact computation (see Supplementary Material); Jacobian minor |
| $D_5(H) = 0$, weight 24, $a_0^2$ top | Established | 6.4 §6 | formal verification (`H_D_zero` etc.) + exact computation |
| $H \notin K[F_0]$ | Established | Prop. 6.5, `H_notin_K_F0` | formal verification of the inductive predicate (API bridge: paper-level) |
| first at 24 | Established | Prop. 6.6 | exact computation ($< 24$ coverage; App. P) |
| $H$ intrinsic | Established | 6.7 | above + $J$-ideal + $r_q^2$ mechanism |
| $N$-boundary (first occurrences) | Established | 21.23 | 3.1; resonance/N6 items as cited |
| $C_5 \cong k[\mathbf{x}]/(S)$, series of $C_5$ | Established | Thm 7.1[D] | the formal hypersurface module, exact computation (see Supplementary Material) ($S = 0$); primitivity |
| five-generator ring and series | Refuted | 6.1, 6.3 | witness $H$ |
| $q$-lemma $v_q \geq 4j$ | Refuted | J.3 | $H$ $W$-profile |
| $p$-lemma $v_p \geq 2j$ | Refuted | I.3 | $F_{46}$ top $v_p = 6 < 8$ |
| R2 global $p$-bound | Refuted | 21.5 | $F_{46}$ $\lambda_p = -2$ |
| Rp relaxed $p$-bound | Open (conjectured) | 21.6 | margins loose on $H, F_{46}$; bounded-defect $2m$ refuted (15.3), sharp envelope $3/2$ (15.4) |
| No bounded-defect P-top | Established (negative) | 15.3 | $F_{46}^r$ tower, pure |
| Sharp p-envelope $3/2$ | Established | 15.4 | pure proof; sharpness exact |
| Zero-excess control ($a + 2b \leq 1$) | Established | 15.5 | pure (monomial + minimum); eps-1 census open |
| Excess-one control (parity families) | Established | 15.6 | pure; exact top checks (excess1 log) |
| R1 relaxed $q$-bound | Established | J.7 | mathematical proof (multiplicity transfer); sharp at $H, F_{46}$ |
| Q-top-strong | Established | 15.1 | pure proof from presentation; generator data (tops log) |
| all-weights spanning (11.2) | Refuted | 10.1 | $N_{\mathbb{Q}}(46) = 1519 > 1518$ exact |
| spanning through 45 | Established | 10.2 | formal verification $\leq 14$ + fresh exact-$\mathbb{Q}$ 15–45 (App. P) |
| $F_{46} \notin C_5$ | Established | Cor + 10.4 | mathematical proof (top valuation $6 < 8$); Lean covers $D = 0$/weight/top only |

(continued)

| claim | status | proof location | computation / depends on |
|---|---|---|---|
| 46–48 quotients, $F_{46}p^2 \in C_5$ | Established | 10.7 | exact Sage computation (see Supplementary Material); Lean (TorsionCheck) |
| $F_{46}\cdot K/(p^2)$ tower (3 families) | Refuted/Superseded | 10.11 | weight-84 counterexample (f46hf0 log); replaced by exact series + filtration |
| relation ideal $= (S, p^2T - E)$ through 92; no monic quadratic | Established in range | 14.1 | exact $\mathbb{Q}$ (relation logs); $F_{46}^2$ 488 terms |
| fourth family $F_{46}HF_0$ seed; Hilbert series exact | Established | 14.5, 14.4 | exact solve + integer cert (hilbert log) |
| $C_5$-module structure ($T$-filtration, $C_5/(p^2)$ layers, gr) | Established | 14.6 | pure proof from relation ideal; HF0 solve (hf0 log) |
| $\operatorname{Sing} X$ exact (6 components, all dim 3, over $\mathbb{Q}$) | Established (Q; geom. prime note for general $k$) | 14.7 | exact primdec (sing_fiber log); primes certified over $\mathbb{Q}$ |
| $R$ normal (CI + $\dim \operatorname{Sing} = 3$ pure) | Established | 14.8 | Theorem 14.7; Serre |
| $a_0$-degree-1 closure; reduction | Established | N.3, N.5 | pure (Dixmier, descent, UFD) |
| $X_5$ CM/HS/minimal gens | Established | 7.2 | Singular cross-check; [3] |
| $X_5$ sing-dim 3; normality | Established | 7.2 | mathematical proof (Jacobian + Serre); Singular cross-check |
| $N = 6$ $B$-level presentation / equality | Est. / Refuted | A.11 | the formal transport module; transported $F_{46}$ |
| $K_6$ exact presentation (transport) | Established | A.12 | shift iso + torsion re-verify (n6_transport log) |
| $K_6$ structure (HS/filtration/Sing/two-slope) | Established | A.13 | pure transport; generator table hand-checked |
| No monomial a0-bound (N=6) | Established | G.4 | incomparability (n6_G4 log) |
| D2 new direction over k[G,H20] | Established | G.2 | exact weight-22 solve (n6_D2novel log) |
| Third m7 direction (lift) | Established | G.9 | exact QQ witness (n6_m7lift log) |
| Denominator module through m=12 | Established (bounded) | G.6 | exact QQ quotients + witnesses (logs) |
| n6-denmod13-closure | Established | G.7 | Q13 = 0 exact (n6_m13 log); window through 13 |
| n6-denmod14 | Blocked (inconclusive computation) | — | kernel counts stable, projection unreconciled; no theorem use |
| $A_6$ denominator module (unbounded) | Open | — | bounded window only; full presentation open |
| $N = 6$ $A$-level objects / ring | Est. objects / Open | F.1, F.4, G.1 | exact computation (see Supplementary Material) |
| forest/degrees/general-$N$ | Refuted as stated | §8 | exact computation (see Supplementary Material) |
| tower closure $= C_5[F_{46}^{\mathrm{int}}]$; finite gen. | Established | 13.5 | E1 (charts + $q$-sat) + E2 ($p$-sat); machine certs as cited |

(continued)

| claim | status | proof location | computation / depends on |
|---|---|---|---|
| multfloor transfer (general); DF instance | Established | 17.1 | proof (R1 argument isolated); SymPy DF checks |
| DF resonance $\rho = 1$; $x^2 \notin \Delta(S)$ | Established | 19.2, 19.1 | $w$-grading proof; SymPy |
| escape/climb defs + classifications; general | New/Open conj. | 20.1, 20.2, 20.7 | $N = 5$ + DF instances; falsifier stated |
| orbit collapse over $V(q)$; shadow | Established | 12.1, 12.2 | explicit points $a_t$ |
| $q$-primitive plinth $f_{13} = D_5(g_{13})$, $q \nmid f_{13}$ | Established | 11.2 | formal verification (`D_F13_zero` etc.) + exact computation |
| plinth module $= pqK + f_{13}K$; $p$-divisibility | Open (conjectured) | 11.5, 11.6 | exact dim match $w \leq 20$; no $p$-primitive $w \leq 18$ |
| second chart, transition, orbit separation | Established | 12.7, 12.12 | exact expansion ($\mathrm{NUM}_{a_0}$ 456 terms); values $0, -1/8, -2$ |
| degeneration def + $N = 5$ instance; gluing | New/Open conj. | 20.3, 12.20 | 12.4; falsifiers stated |
| $A_{pq} \cap A_{f13} = A_p$ (atlas = $p$-local kernel) | Established | 12.8 | exact non-divisibility certs (M5) |
| liftable residual plinth $= (p\bar{J}_2^3)$ | Open (conjectured) | 12.6 | residue ranks $= \dim \bar{K}_{w-13}$, $13 \leq w \leq 20$ |
| terminal stratum $V(q, x)$: residual plinth 0 | Established | 12.11 | proof ($a_0$-filtration + evaluation); certs (M7) |
| $p$-torsion table $T_p^{(r)}$ exact through 52 | Established | 10.8 | exact $\mathbb{Q}$ (torsion_exact, torsion_qq logs) |
| $(R : p^r)_w = R_w$ ($r = 1$, $w \leq 52$; $r = 2$, $w \leq 50$) | Established | §13 (Route E2) | exact $\mathbb{Q}$ (torsion_R log) |
| $f_{13}$-chart coordinates in $R_{f_{13}}$ | Est. $a_1, a_2, a_3$; Open $a_0$ | 12.7, Route E1 | exact sparse certificates (chart_R log) |
| $A^D_{f_{13}} = R_{f_{13}}[Z]$; $Z = W - \rho(-\tau)$, $\deg_W 7$ | Established | 12.13 | generation lemma; $\rho_7 = -p^5q^3/112$ exact |
| cover $P(T)$, Newton slope 4, discriminant support | Established | 12.15, 12.16 | exact (cover logs); no new divisor |
| $J_2/e_2$ Newton: uniform vanishing order 1 | Established | 12.17 | exact (cover_newton3 log) |
| cover intersection $= R_p$ | Established (via $q$-saturation) | 13.5 | Newton/conductor program bypassed; E1 by $q$-sat |
| $q$-elimination $J_q \cap \mathbb{Q}[\text{kept}] = (S, p^2T - E, Q)$ | Proved using explicit exact certificate | 13.3 | exact elimination (eliminate_q log, GB 24, EQUAL) |
| $q$-saturation $R/qR \hookrightarrow A/qA$ | Proved using explicit exact certificate | 13.4 | cert lemma + lift-and-cancel; transfer lem |
| $p$-elimination $J_p \cap \mathbb{Q}[\text{kept}] = (S, \bar{c}_{46})$ | Proved using explicit exact certificate | 13.10 | exact elimination (eliminate_p log, GB 119, EQUAL) |
| residual kernel $= k[q, x, y, \theta]$; reductions; $\bar{F}_{46} = B\theta$ | Established | 13.8, 13.9 | proof + exact computation + Lean (ResidualCheck) |

(continued)

| claim | status | proof location | computation / depends on |
|---|---|---|---|
| $\ker(C_5/pC_5 \to A/pA) = (\bar{c}_{46})$ | Proved using explicit exact certificate | 13.11 | cert lemma (eliminate_p log) |
| $p$-saturation $R = A \cap R_p$ | Proved using explicit exact certificate | 13.12 | residual presentation + criterion |
| $\mathrm{in}_p(R)$ SAGBI normal forms | Open (optional) | §13 | presentation now available |
| $p$-local closure + saturation | Proved using explicit exact certificates | 13.5 | charts + $q$-sat + $p$-sat; repro: `reproduce_closure.sh` |

## H. Failed valuation routes and superseded approaches (historical record)

The valuation programs below fail for the precise algebraic reasons stated; they are retained because each failure explains a constraint the successful Route E proof had to satisfy. Nothing here is a premise of Theorem 13.5.

**Route A: resonance necessity (OBSTRUCTED).** Necessity for Conjecture N.1 (no primitive at W-degree $m \notin \{2, 4\}$) cannot be proved by valuation: with $\rho_q = 2$, $\rho_p = 4$ exact (Definition 21.1), the constant-coefficient step is silent at every $m \geq 2$ on the $q$-side and every $m \geq 4$ on the $p$-side (Lemmas 21.24–21.25; below degree $d$ the step only "may" function, which is unproved). At $m = 2, 3$ the possibly-functioning $p$-side supplies at most half of $J_m$-membership ($v_p$ without $v_q$ does not force a top into $J_m$). Birth windows exclude nothing: $\min C_5(m) = 14, 24, 38, 48, 62, 72, 86, 96$ for $m = 1, \ldots, 8$ (exhaustive check in exact top-ideal computation), so every gap $[8m, \min C_5(m) - 1]$ is non-empty; only $m = 1$ is closed, by the independent Theorem N.3. The $J$-ideal form of the exclusion needs both top bounds, of which the $p$-side global form is Refuted and the $q$-side strong form is Established (Theorem 15.1).

**Route B: associated-graded induction (OBSTRUCTED at the step).** $\mathrm{gr}_{a_0}(C_5) = \bigoplus J_m t^m$ is Established (Theorem N.7); its analogue for $A_5^{D_5}$ is open from $m = 4$. The graded pieces are known for $m \leq 1$ (descent; Theorem N.3), evidenced through weight 45 at $m = 2$ and through 85 at $m = 4$, and open above. The induction step at arbitrary $M$ is Theorem N.5(i) generalized to $C_5[F_{46}^{\mathrm{int}}]$, whose hypotheses (Q-top-strong plus the conjectured $p$-side Rp, both unproved) are open. Since $C_5$-generation is false, at least one hypothesis of the ungeneralized form is false. The induction premise is the conclusion.

**Route C: structural degree bound (OBSTRUCTED).** No imported bound applies: Maubach's $\leq 4$ is dimension-4 monomial-triangular; $N = 5$ is dimension 6; Daigle–Freudenburg kills universal bounds for general triangular derivations (Remark 21.14). Growth-rate comparison is circular: proving the tower Hilbert series equals the kernel series *is* component (1). The three-module excesses $45, 50, 54, 58, 63$ at weights 70–74 follow from $s_K$-arithmetic (internal consistency of the conjecture, not evidence for it).

**Route D: W-expansion product-reduction (OBSTRUCTED at top reduction).** W-expansions are unique ($W$ monic of $a_0$-degree 1), but top-coefficient reduction is unproved at every $m \geq 2$ as a general statement: W-degree-2 tops generated by $H$ (above weight 45), W-degree-4 tops generated by $F_{46}$ modulo $J_4$ (past 85), and $H_m$ forced into $C_5$ for $m \notin \{2, 4\}$ (needs both bounds) are all open. The descending induction inherits the gap.

**Superseded approaches (historical record).** The following inputs would have overcome the barrier; closure has since been proved by other means (Theorem 13.5), and they remain as proof infrastructure and independent open problems. (i) A $p$-side replacement bound: R1 ($\lambda_q \geq -2m$) is now Established (Theorem J.7); the relaxed $p$-side form $\lambda_p \geq -4m$ is now conjectured as a deliberately weak floor (Conjecture 21.6; the multiplicity mechanism would suggest $-m/2$, margins loose on $H$ and $F_{46}$) but unproved — its proof needs exact degree 4 of $r_p$ over $L_p$ plus residue landing, neither

available. Without it the $J_m$-membership requirement ($v_p \geq 2m$ AND $v_q \geq 4\lceil m/2 \rceil$) cannot be met. (ii) A finiteness or degree-bound theorem for squarefree-monomial triangular LNDs in dimension $\geq 5$. (iii) A compute breakthrough pushing full-kernel upper bounds past weight 70 (infeasible over $\mathbb{Q}$ and LinBox at $n = 112241$); this extends the verified range only. A non-resonance source (a primitive outside the $c$-prime framework) is not covered by any route: possibility A (W-degree outside $\{2, 4\}$) is a concrete falsifier to hunt in the birth windows of Theorem N.8; possibility B (extended framework) has no candidate mechanism. Falsifiers: any $w$ with excess $\neq P(w) - (s_6(w) - s_6(w - 28))$; any $w$ with $N_J(w) < N_{\leq 3}(w)$ at $a_0$-degree 3 (Remark N.9).

**Degree-$m$ census (evidence, not proof).** For each $a_0$-degree $m$ with $J_m = (p^{2m}q^{4\lceil m/2 \rceil})$ principal, the exact-$\mathbb{Q}$ nullity on $V_w^{a_0 \leq m}$ is compared against the $J_m$-constrained chain nullity (top $C_m$ parametrized in $J_m$); a gap counts $J_m$-violating degree-$m$ tops, compared against the $F$-module bracket prediction. Completed prior computations (supplement logs; outside the fresh 15–46 rerun): exact-$\mathbb{Q}$ for $m = 4$ over 46–75 dense (30 weights, all match), $m = 5$ over 60–72 (13 weights incl. the $F_{46} \cdot F_0$ seed fingerprint), $m = 6$ over 70–74 ($F_{46} \cdot H$ seed at 70), $m = 3$ clean (0 violations) through the mid-70s, and $m = 7$ at 84 with exactly one violator (the $F_{46} \cdot F_0 \cdot H$ seed); finite-field consistency checks cover the remaining stated ranges. No excess (no new primitive) and no deficit (no extra torsion) anywhere checked. Remaining ranges ($m = 3$ tail, $m = 6$ upper, $m = 7$ upper) are being computed; any excess escalates to a new primitive.

*Remark* H.1 (Sharp reformulation via gap-born detection). A primitive $a_0$-degree-$m$ direction born below the $C_5$ window ($w < \min C_5(m)$) has top of weight $w - 8m < \min C_5(m) - 8m = 2m + 4\lceil m/2 \rceil = \mathrm{wt}(j_m)$ (direct check on $e + 2r = m$; minimum at maximal $r$), hence top outside $J_m$: every gap-born primitive is $J$-violating and census-visible. So closure $\iff$ no $J$-violating new tops at any $(m, w)$ beyond checked ranges, plus no above-window $J$-clean primitives (the latter confined above $\min C_5(m)$ per degree).

## I. Residue-field saturation: the $p$-adic lemma

Status of this section: the $p$-adic denominator bound (Lemma I.3) is **[REFUTED]** by the explicit weight-46 invariant $F_{46}$ of Section 10: its top $W$-coefficient $H_4 = \frac{18375}{4}p^6q^8$ has $v_p = 6 < 8 = 2 \cdot 4$. The valuation setup (residue field, pole order, initial form) stands; the inference does not. No statement in this section carries status Established. We work over the characteristic-zero field $k$ of Definition 2.1, with $A_5 = k[a_0, \ldots, a_5]$ and $D_5(a_n) = a_{n+1}a_{n+2}$. This section treats the $p$-adic half of the residue-transcendence saturation for the adjacent-quadratic shift at $N = 5$. Here the residue initial form has degree 4 over its residue field, and $F_{46}$ gives the first observed failure of the corresponding naive bound. The $q$-adic half is stated separately in the companion $q$-lemma section: the two residue fields are not symmetric, and no symmetry argument is used here.

We reuse the five-variable dictionary of the kernel section verbatim, with no redefinition:

$$x = a_3, \qquad y = a_2, \qquad z = a_1, \qquad p = a_4, \qquad q = a_5,$$
$$B = k[x, y, z, p, q] = k[a_1, \ldots, a_5], \qquad K = B^{D_5} = k[p, q, J_1, J_2],$$
$$J_1 = 2pz - y^2 = 2a_1a_4 - a_2^2, \qquad J_2 = 2qy - x^2 = 2a_2a_5 - a_3^2,$$
$$D_5(x) = pq, \qquad D_5(y) = px, \qquad D_5(z) = xy, \qquad D_5(p) = D_5(q) = 0.$$

We reuse the invariant of Definition 5.1 verbatim:

$$F_0 = p^2q^4a_0 - pq^3xyz + \tfrac{1}{3}pq^2x^3z + \tfrac{1}{3}q^2x^3y^2 - \tfrac{1}{5}qx^5y + \tfrac{1}{35}x^7,$$

and write $c = p^2q^4$, $G_0 = F_0 - ca_0 \in B$,

$$G_0 = -pq^3xyz + \tfrac{1}{3}pq^2x^3z + \tfrac{1}{3}q^2x^3y^2 - \tfrac{1}{5}qx^5y + \tfrac{1}{35}x^7,$$

$W = F_0/c = a_0 + \rho$ with $\rho = G_0/c \in B_{pq}$. A general localized invariant is written

$$F = \sum_{j=0}^{m} H_j W^j, \qquad H_j \in K_{pq},$$

with $a_0$-coefficients

$$C_k = \sum_{j=k}^{m} \binom{j}{k} H_j \rho^{j-k} \in B \qquad (0 \leq k \leq m).$$

*Remark* I.1 (Valuation setup and the correct $p$-adic residue field). Write $B = k[x, y, z, q][p]$ and let $v_p = \mathrm{ord}_p$ on $\mathrm{Frac}(B)$, with valuation ring $V_p = B_{(p)}$ and residue field $\kappa_p = k(x, y, z, q)$. The height-one prime $(p) \subset K$ has residue

$$K/pK \cong k[q, -y^2, 2qy - x^2],$$

by the injectivity of the specialization map of Lemma 4.3, so with $w = 2qy - x^2$,

$$L_p = \mathrm{Frac}(K/pK) = k(q, y^2, w) \hookrightarrow \kappa_p.$$

In the $a$-variables this is $k(a_5, -a_2^2, 2a_2a_5 - a_3^2)$. All residue computations in this section use the dictionary above; The residue computation under a transposed variable naming is retracted; only the corrected computation below is used. Since $G_0$ has $p$-adic order 0 while $c = p^2q^4$ has order 2, $\rho$ has exact pole order $e_p = 2$ at $p$, with initial form

$$r_p = \mathrm{in}_p(p^2\rho) = \frac{1}{q^4}\left(\tfrac{1}{3}q^2x^3y^2 - \tfrac{1}{5}qx^5y + \tfrac{1}{35}x^7\right).$$

The specialization identities, the exact pole order, and the Jacobian independence of $\{q, y^2, w\}$ (minor $4xy \neq 0$ in columns $(x, y, q)$) are certified by exact computation (see Supplementary Material).

**Definition I.2** (Denominator modules, $p$-part)**.** For each $m \geq 0$ the $p$-part of the $m$-th denominator module is

$$M_m^p = p^{2m}K.$$

The full module $M_m = p^{2m}q^{4m}K = c^mK$ additionally requires the companion $q$-adic bound, stated separately.

**Lemma I.3** ($p$-adic denominator bound (Refuted))**.** ***[REFUTED by the explicit weight-46 invariant $F_{46}$ of Section 10 (Remark I.6 below).]*** *The refuted statement is: Let $F = \sum_{j=0}^{m} H_jW^j \in A = k[a_0, \ldots, a_5]$ with $H_j \in K_{pq}$. Then*

$$\boxed{v_p(H_j) \geq 2j \quad \text{for every } j,}$$

*i.e. $H_j \in p^{2j}K_{pq}$ for every $j$.*

*Reduction and explicit gap.* The constant $a_0$-coefficient is $C_0 = \sum_j H_j\rho^j \in B$, so $v_p(C_0) \geq 0$. Put $\lambda_p = \min_j\{v_p(H_j) - 2j\}$ and $S = \{j : v_p(H_j) - 2j = \lambda_p\}$. If $\lambda_p < 0$, dividing by $p^{\lambda_p}$ and passing to initial forms gives the relation

$$\sum_{j \in S} h_j r_p^j = 0$$

with $0 \neq h_j \in L_p$ the residues of the normalized coefficients. The reduction to this initial-form equation is valid. The final inference — concluding $h_j = 0$ for all $j \in S$ — would require linear independence of arbitrarily high powers of $r_p$ over $L_p$, which is false in the needed range: $r_p$ is algebraic of degree 4 (Remark I.4), and the explicit invariant $F_{46}$ realizes $\lambda_p = -2 < 0$ (Remark I.6). Earlier heuristic routes — a parity shortcut, a slice-power computation, and the transposed-naming residue computation — are retracted as proofs and retained as heuristics only. □

*Remark* I.4 (Observation: why the constant-coefficient route is unavailable). The triple $\{q, y^2, w\}$ is algebraically independent (Jacobian minor $4xy \neq 0$), so $L_p = k(q, y^2, w)$ is purely transcendental of degree 3. But $y^2 \in L_p$ and $x^2 = 2qy - w \in L_p(y)$, so

$$L_p \subseteq L_p(y) \subseteq L_p(y)(x) = k(q, x, y)$$

is a tower of degree at most 2 at each step; hence $[k(q, x, y) : L_p] \leq 4$. Since $r_p \in k(q, x, y)$, the initial form $r_p$ is algebraic over $L_p$ (of degree exactly 4: a primitive element of the full tower, by exact-$\mathbb{Q}$ computation (see Supplementary Material), not transcendental. The same witness check on the $q$-side gives $x^2 \in L_q$, so $r_q = x^7/(35p^2)$ is algebraic over $L_q$ of degree at most 2. Consequently no argument needing $1, r_p, r_p^2, \ldots$ (respectively $1, r_q, r_q^2, \ldots$) to be linearly independent over the full range of $W$-degrees can succeed from the constant coefficient alone. This precise obstacle is part of the verification task; the algebraicity witnesses $y^2 = -(J_1 \bmod p)$, $x^2 = 2qy - (J_2 \bmod p)$, $x^2 = -(J_2 \bmod q)$ are certified in exact computation (see Supplementary Material).

**Lemma I.5** ($p$-adic leading-coefficient form (Refuted))**.** ***[REFUTED by the same invariant $F_{46}$: its top $W$-coefficient $H_4 = \frac{18375}{4}p^6q^8$ has $v_p = 6 < 8 = 2 \cdot 4$, so $H_4 \notin p^8K_{pq}$ (Remark I.6).]*** *With $F$, $m$, $H_j$ as in Lemma I.3, the top coefficient $H_m = [a_0^m]F$ was claimed to lie in $B$ and satisfy $H_m \in p^{2m}K_{pq}$. It fails by the same invariant.*

*Remark* I.6 (Refutation by the weight-46 invariant $F_{46}$). The invariant $F_{46}$ of Theorem 10.3 ($a_0$-degree 4, $W$-expansion $F_{46} = \sum_{j=0}^{4} H_j W^j$ with $H_3 = 0$) has $(v_q(H_j), v_p(H_j)) = (-8,-2), (-4,0), (0,2), (8,6)$ at $j = 0, 1, 2, 4$. The bound $v_p(H_j) \geq 2j$ fails at every nonzero $j$: $0 < 2$, $2 < 4$, $6 < 8$. The $p$-adic story is over: where $H$ spared the $p$-side ($v_p(H_2) = 4 = 2 \cdot 2$), $F_{46}$ breaks it at the top coefficient itself.

*Remark* I.7 (Conditional $W$-degree induction skeleton). Assume the $p$-adic bound of Lemma I.3 together with the companion $q$-adic bound $v_q(H_j) \geq 4j$ in full. Then each $H_j$ lies in $B$: the top coefficient $H_m = [a_0^m]F$ is in $B$ directly, and descending induction via $C_{m-1} = H_{m-1} + mH_m\rho$ with $H_m\rho \in B$ places every $H_j$ in $B$. The intersection $B \cap K_{pq} = K$ of Theorem 4.4 then gives $H_m = c^m G_m$ with $G_m \in K$, whence $H_m W^m = G_m F_0^m \in K[F_0]$; subtracting lowers the $W$-degree. This induction skeleton is Conditional (on E), i.e. conditional on both saturation lemmas, and void since both are refuted. The global generation $A^{D_5} = K[F_0]$ and its Hilbert series are REFUTED (counterexample $H$, weight 24; see 6.4); the all-weights six-generator identity is likewise REFUTED (counterexample $F_{46}$, weight 46; see Section 10). Lemma P is REFUTED by $F_{46}$ and Lemma Q is REFUTED by $H$, voiding the joint induction.

*Remark* I.8 (Status boundary). What this section fixes is the valuation setup: the ring $V_p$, the residue embedding $L_p \hookrightarrow \kappa_p$, the exact pole order $e_p = 2$, the initial form $r_p$, the denominator modules $M_m^p$, and the precise mechanism of failure. What is now *refuted* is the saturation inference itself (Lemma I.3) and its leading-coefficient form (Lemma I.5), both by the invariant $F_{46}$ of Remark I.6: the inference needs $r_p$ to be transcendental over $L_p$, but it is algebraic of degree 4 (Remark I.4), and $F_{46}$ realizes the resulting pole. No N=5 generation claim is made in this section itself; the full-ring equality is proved later in Theorem 13.5.

## J. Residue-field saturation: the $q$-adic lemma

Status of this section: the $q$-adic denominator bound (Lemma J.3) is **[REFUTED]** by the explicit weight-24 invariant $H$ of Remark 6.4 (Section 6): $H$ is $D_5$-invariant with $W$-cosets $H_2 = 11025\, p^4 q^4$ ($v_q = 4 < 8 = 4 \cdot 2$) and $H_0 \in q^{-4} K_{pq}$ ($v_q = -4 < 0$), directly violating $v_q(H_j) \geq 4j$. The reduction is valid; the residue-field inference is where it fails. Two statements are **[Established]** here: the exact residue degree (Lemma J.6) and the relaxed $q$-bound R1 (Theorem J.7), proved by multiplicity transfer from all of $\{C_k\}$; the original denominator bound stays Refuted. We work over the characteristic-zero field $k$ of Definition 2.1, with $A_5 = k[a_0, \ldots, a_5]$ and $D_5(a_n) = a_{n+1}a_{n+2}$. This section treats the $q$-adic half of the residue-transcendence saturation for the adjacent-quadratic shift at $N = 5$. Here the residue initial form has degree 2 over its residue field, and this is exactly the first degree at which the naive saturation argument can fail; $H$ realizes that failure. The asymmetry between the two prime divisors requires separate treatment: the $q$-adic residue field, pole order, and initial form below are computed independently on the $q$-side, and no symmetry argument with the companion $p$-adic section is used in either direction.

We reuse the five-variable dictionary of the kernel section verbatim, with no redefinition:

$$x = a_3, \qquad y = a_2, \qquad z = a_1, \qquad p = a_4, \qquad q = a_5,$$
$$B = k[x, y, z, p, q] = k[a_1, \ldots, a_5], \qquad K = B^{D_5} = k[p, q, J_1, J_2],$$
$$J_1 = 2pz - y^2 = 2a_1a_4 - a_2^2, \qquad J_2 = 2qy - x^2 = 2a_2a_5 - a_3^2,$$
$$D_5(x) = pq, \qquad D_5(y) = px, \qquad D_5(z) = xy, \qquad D_5(p) = D_5(q) = 0.$$

We reuse the invariant of Definition 5.1 verbatim:

$$F_0 = p^2q^4a_0 - pq^3xyz + \tfrac{1}{3}pq^2x^3z + \tfrac{1}{3}q^2x^3y^2 - \tfrac{1}{5}qx^5y + \tfrac{1}{35}x^7,$$

and write $c = p^2q^4$, $G_0 = F_0 - ca_0 \in B$,

$$G_0 = -pq^3xyz + \tfrac{1}{3}pq^2x^3z + \tfrac{1}{3}q^2x^3y^2 - \tfrac{1}{5}qx^5y + \tfrac{1}{35}x^7,$$

$W = F_0/c = a_0 + \rho$ with $\rho = G_0/c \in B_{pq}$. A general localized invariant is written

$$F = \sum_{j=0}^{m} H_j W^j, \qquad H_j \in K_{pq},$$

with $a_0$-coefficients

$$C_k = \sum_{j=k}^{m} \binom{j}{k} H_j \rho^{j-k} \in B \qquad (0 \leq k \leq m).$$

*Remark* J.1 (Valuation setup and the correct $q$-adic residue field). Write $B = k[x, y, z, p][q]$ and let $v_q = \mathrm{ord}_q$ on $\mathrm{Frac}(B)$, with valuation ring $V_q = B_{(q)}$ and residue field $\kappa_q = k(x, y, z, p)$. The height-one prime $(q) \subset K$ has residue

$$K/qK \cong k[p, 2pz - y^2, -x^2],$$

by the injectivity of the specialization map of Lemma 4.3, so with $u = 2pz - y^2$,

$$L_q = \mathrm{Frac}(K/qK) = k(p, u, x^2) \hookrightarrow \kappa_q.$$

In the $a$-variables this is $k(a_4, 2a_1a_4 - a_2^2, -a_3^2)$. Note the asymmetry with the $p$-side: here the surviving specialization $J_1 \bmod q = 2pz - y^2$ retains the variable $z$, while $J_2 \bmod q = -x^2$ is a pure square; the roles of the two kernel generators are not interchanged by any symmetry. All residue computations in this section use the dictionary above; The symmetrized copy of the $p$-side argument is retracted; the $q$-side computation below is independent. Since $G_0$ has $q$-adic order 0 while $c = p^2q^4$ has order 4, $\rho$ has exact pole order $e_q = 4$ at $q$, with monomial initial form

$$r_q = \mathrm{in}_q(q^4\rho) = \frac{x^7}{35p^2}.$$

The specialization identities, the exact pole order, and the Jacobian independence of $\{p, u, x^2\}$ (minor $4xy \neq 0$ in variables $(p, x, y)$) are certified by exact computation (see Supplementary Material), which checks the $q$-side rows independently of the $p$-side rows.

**Definition J.2** (Denominator modules, $q$-part)**.** For each $m \geq 0$ the $q$-part of the $m$-th denominator module is

$$M_m^q = q^{4m}K.$$

The full module $M_m = p^{2m}q^{4m}K = c^mK$ additionally requires the companion $p$-adic bound, stated separately.

**Lemma J.3** ($q$-adic denominator bound (Refuted))**.** ***[REFUTED by the explicit weight-24 invariant $H$ of Remark 6.4 (Remark J.4 below).]*** *The refuted statement is: for $F = \sum_{j=0}^{m} H_j W^j \in A = k[a_0, \ldots, a_5]$ with $H_j \in K_{pq}$,*

$$\boxed{v_q(H_j) \geq 4j \quad \textit{for every } j,}$$

*i.e. $H_j \in q^{4j}K_{pq}$ for every $j$.*

*Reduction valid; inference refuted by the counterexample.* The constant $a_0$-coefficient is $C_0 = \sum_j H_j\rho^j \in B$, so $v_q(C_0) \geq 0$. Put $\lambda_q = \min_j\{v_q(H_j) - 4j\}$ and $S = \{j : v_q(H_j) - 4j = \lambda_q\}$. If $\lambda_q < 0$, dividing by $q^{\lambda_q}$ and passing to initial forms gives the single relation in the residue field $L_q$

$$\sum_{j \in S} h_j r_q^j = 0, \qquad 0 \neq h_j \in L_q,$$

with $r_q$ the initial form of Remark J.5. The reduction to this one equation is valid. The final inference — concluding $h_j = 0$ for all $j \in S$ — would require the powers $1, r_q, r_q^2, \ldots$ to be linearly independent over $L_q$. That independence is false: $r_q$ is algebraic of degree at most 2 over $L_q$ (Remark J.5), and the explicit invariant $H$ provides the concrete non-trivial relation (Lemma J.4). Earlier heuristic routes — a symmetrized copy of the $p$-side argument, a slice-power computation, and the transposed-naming residue computation — are retracted as proofs and retained as heuristics only. □

*Remark* J.4 (Refutation by the weight-24 invariant $H$). The counterexample is the explicit $D_5$-invariant $H$ of Remark 6.4 (weight 24, $a_0$-degree 2, $11025\, a_0^2a_4^4a_5^4$ as leading $a_0^2$-term), certified in exact computation (see Supplementary Material). Its $W$-basis expansion (obtained from the weight-28 syzygy of Theorem 7.1) is

$$H = \underbrace{11025\, p^4q^4}_{H_2}\, W^2 + \underbrace{\left(\tfrac{105\, J_2^5 J_1}{q^2} + \tfrac{9\, J_2^7}{q^4}\right)}_{H_0}, \qquad H_1 = 0,$$

so the $W$-degree is 2 with $H_2 = 11025\, p^4q^4$ and $H_0 = 105\, J_2^5 J_1\, q^{-2} + 9\, J_2^7\, q^{-4}$ (a genuine element of $K_{pq}$). The bound $v_q(H_j) \geq 4j$ is violated at both non-zero $W$-degrees:

$$v_q(H_2) = \mathrm{ord}_q(11025\, p^4q^4) = 4 < 8 = 4 \cdot 2, \qquad v_q(H_0) = -4 < 0 = 4 \cdot 0$$

(the two summands of $H_0$ have $q$-orders $-2$ and $-4$, so no cancellation occurs at the leading $q^{-4}$ pole and the order of the sum is $\min(-2,-4)=-4$). Thus $v_q(H_2)<8$ and $v_q(H_0)<0$; the lemma fails unconditionally. The constant $a_0$-coefficient $C_0=H_0+H_2\rho^2$ is nevertheless a genuine polynomial ($v_q(C_0)\geq 0$), because its $q^{-4}$ pole cancels: the residue-field relation forced by $C_0\in B$ is exactly the quadratic one on $r_q$ (cf. Remark J.5, and the cancellation certificate exact computation (see Supplementary Material)). The $a_0$-degree-2 term $11025\,a_0^2a_4^4a_5^4$ is what keeps $H\notin K[F_0]$ (Remark 6.4) while the $W$-cosets above break the $q$-bound — the two failures are the same invariant.

*Remark* J.5 (Observation: why the constant-coefficient route is unavailable on the $q$-side). The triple $\{p,u,x^2\}$ is algebraically independent (Jacobian minor $4xy\neq 0$ in variables $(p,x,y)$), so $L_q=k(p,u,x^2)$ is purely transcendental of degree 3. But $x^2\in L_q$, so

$$L_q\subseteq L_q(x)=k(p,u,x)$$

is a single quadratic step; hence $[k(p,u,x):L_q]\leq 2$. Since $r_q=x^7/(35p^2)\in k(p,x)\subseteq L_q(x)$, the initial form $r_q$ is algebraic over $L_q$ of degree at most 2, not transcendental — indeed it satisfies the explicit quadratic relation

$$r_q^2=\Big(\frac{x^7}{35\,p^2}\Big)^2=\frac{(x^2)^7}{1225\,p^4}\in k(p,u,x^2)=L_q.$$

This is the residue-field shadow of the weight-28 syzygy (Theorem 7.1): the $q^{-4}$ pole of the constant term $C_0=H_0+H_2\rho^2$ of the counterexample (Remark J.4) cancels in $L_q$ *precisely because* $r_q^2$ already lies in $L_q$, so the two $q^{-4}$ leading terms $11025\,G_0^2\equiv 9x^{14}$ and $9J_2^7\equiv -9x^{14}$ sum to zero modulo $q$. Consequently no argument needing $1,r_q,r_q^2,\dots$ to be linearly independent over the full range of $W$-degrees can succeed from the constant coefficient alone; the dependence at $r_q^2$ is the exact obstruction. This $q$-side degree bound is independent of the $p$-side tower (there a two-step extension of degree at most 4 governs a trinomial initial form; here a one-step extension of degree at most 2 governs a monomial initial form), and the difference in pole orders ($e_q=4$ versus $e_p=2$) is part of why the two lemmas cannot share a proof. The algebraicity witness $x^2=-(J_2 \bmod q)$ is certified in exact computation (see Supplementary Material).

**Lemma J.6** ($r_q$ has exact degree 2 over $L_q$)**.** *$r_q=x^7/(35p^2)$ satisfies $r_q^2=(x^2)^7/(1225p^4)\in L_q$ (exact computation; see Supplementary Material) and $r_q\notin L_q$. Hence its minimal polynomial over $L_q$ is $m_q(T)=T^2-(x^2)^7/(1225p^4)$, of exact degree 2 with simple root $r_q$ (separable; characteristic zero).*

*Proof.* If $r_q\in L_q=k(p,u,x^2)$, write $r_q=A/B$ with $A,B\in k[p,u,x^2]$, $B\neq 0$. Then $x^7B=35p^2A$ in $k[p,y,z,x]$ (via $u=2pz-y^2$). As polynomials in $x$, $A$ and $B$ have even degree ($x$ occurs only through $x^2$), so the left side has odd $x$-degree $7+\deg_x B$ while the right side has even $x$-degree $\deg_x A$ — impossible since $x^7B\neq 0$. Thus $r_q\notin L_q$; with $r_q^2\in L_q$ the minimal polynomial is the stated quadratic, separable with $m_q'(r_q)=2r_q\neq 0$. □

**Theorem J.7** (Relaxed $q$-bound (R1))**.** *Let $F\in A_5^{D_5}$ have $W$-degree $m$, with unique $W$-expansion $F=\sum_{j=0}^m H_jW^j$, $H_j\in K_{pq}$ (Lemma 4.2). Then*

$$\boxed{\lambda_q(F):=\min_j(v_q(H_j)-4j)\geq -2m,}$$

*with deficit $+\infty$ for $H_j=0$. The bound is sharp: $H$ attains $\lambda_q=-4$ at $m=2$ (Remark J.4) and $F_{46}$ attains $\lambda_q=-8$ at $m=4$ (Theorem 10.3).*

*Proof.* The minimum is attained ($H_m\neq 0$); if $\lambda:=\lambda_q(F)\geq 0$ there is nothing to prove. Assume $\lambda<0$ and put $S=\{j: v_q(H_j)-4j=\lambda\}\neq\emptyset$. The $a_0$-coefficients $C_k=\sum_{j\geq k}\binom{j}{k}H_j\rho^{j-k}$ lie in $B$, so $v_q(C_k)\geq 0$ for all $k$. Since $v_q(\rho)=-4$ exactly with initial form $r_q$, each $(j,k)$-summand has $v_q\geq\lambda+4j-4(j-k)=\lambda+4k$, with equality iff $j\in S$; its initial form is then $\binom{j}{k}h_jr_q^{j-k}$ with $0\neq h_j=\mathrm{in}_q(H_j)\in L_q$. Residues of $K_{pq}$-elements land in $L_q=\mathrm{Frac}(K/qK)$: $q$-adic values on $K$ agree in $B$ (if $q\mid A$ in $B$ with $A\in K$, descent puts the quotient in $K$ by Theorem 4.4, since $\mathrm{Frac}(K)\subset K_{pq}$), and denominators are $p$-powers with $p \bmod q\neq 0$ since $(p),(q)$ are non-associated primes of $K$ (Lemma 4.3). Hence for every $k$ with $\lambda+4k<0$ the leading $q^{\lambda+4k}$-coefficient of $C_k$ must vanish: with $P(T)=\sum_{j\in S}h_jT^j$ (nonzero, $\deg P\leq m$, over $L_q$), $P^{(k)}(r_q)=0$, using $P^{(k)}(T)/k!=\sum_{j\geq k}\binom{j}{k}h_jT^{j-k}$ in characteristic zero. The qualifying $k\in[0,m]$ form an initial segment $\{0,\dots,c-1\}$ with $c=\#\{0\leq k\leq m:\lambda+4k<0\}$; thus $r_q$ is a root of $P$ of multiplicity at least $c$, so $m_q^c\mid P$ in $L_q[T]$ (separable minimal polynomial, Lemma J.6), giving $2c\leq\deg P\leq m$. If $\lambda<-2m$ then $-\lambda/4>m/2$, so every $0\leq k\leq m/2$ qualifies and $c>m/2$, contradicting $2c\leq m$. Hence $\lambda\geq -2m$. The combinatorial implication is checked exhaustively for $m\leq 12$ in exact computation (see Supplementary Material). Lean status: paper-level proof; the full $W$-valuation form stays open in Lean on the localized fraction-field infrastructure (Remark 9.5). □

*Remark* J.8 (R1 as transfer instance). The proof above is the $q$-instantiation ($e_q = 4$, $d_q = 2$) of the multfloor transfer theorem (Theorem 17.1), which isolates the argument for general triangular LNDs with a pole element and finite residue data. Status unchanged (Established); the transfer only recontextualizes it.

**Lemma J.9** ($q$-adic leading-coefficient form (Refuted))**.** ***[REFUTED by the same counterexample*** $H$***: its top*** $W$***-coefficient*** $H_2 = 11025\,p^4q^4$ ***has*** $v_q = 4 < 8 = 4 \cdot 2$***, so*** $H_2 \notin q^8 K_{pq}$ ***(Remark J.4).]*** *With $F$, $m$, $H_j$ as in Lemma J.3, the top coefficient $H_m = [a_0^m]F$ was claimed to lie in $B$ and to satisfy $H_m \in q^{4m}K_{pq}$. It fails by the same counterexample.*

*Remark* J.10 (Status boundary). What this section fixes is the valuation setup: the ring $V_q$, the residue embedding $L_q \hookrightarrow \kappa_q$, the exact pole order $e_q = 4$, the monomial initial form $r_q$, the denominator modules $M_m^q$, and the reduction to the residue-field initial-form equation, which remains valid. What is now *refuted* is the $q$-adic denominator bound (Lemma J.3) and its leading-coefficient form (Lemma J.9), both by the counterexample $H$ of Remark J.4: the inference needs $r_q$ to be transcendental over $L_q$, but it is algebraic (Remark J.5). The companion $p$-adic bound (Lemma I.3) is *not* contradicted by $H$ ($v_p(H_2) = 4 = 2 \cdot 2$), and the joint $W$-degree induction (Theorem K.5) is void because one of its two premises (Lemma J.3) is false; the global invariant ring, now identified as the six-generator hypersurface of Section 6, no longer depends on either saturation lemma. No N=5 generation claim is made in this section itself; the full-ring equality is proved later in Theorem 13.5.

## K. Residue-field saturation: denominator modules and $W$-degree induction

Status of this section: *moot*. Its two premises are Lemma P (the $p$-side, Section I) and Lemma Q (the $q$-side, Section J); both are now refuted — $Q$ by the counterexample $H$ (Section J), $P$ by $F_{46}$ (Section 10) — so the joint $W$-degree induction of Theorem K.5 is void. The wrapper is retained only as a record of the attempted proof; no statement in this section carries status Established. The relaxed $q$-bound that survives is proved separately (Theorem J.7).

*Remark* K.1 (Status aliases). Throughout, $P$ denotes Lemma I.3 (the $p$-adic denominator bound, **[REFUTED by** $F_{46}$**]**, Section 10), $Q$ denotes Lemma J.3 (the $q$-adic bound, REFUTED by $H$), and $E$ denotes their joint $W$-degree induction (Theorem K.5), now void. The surviving relaxed $q$-bound R1 is Theorem J.7, not a conditional statement of this section.

We work over the characteristic-zero field $k$ of Definition 2.1, with $A_5 = k[a_0, \dots, a_5]$ and $D_5(a_n) = a_{n+1}a_{n+2}$. This section combines the two halves of the residue-transcendence saturation for the adjacent-quadratic shift at $N = 5$: the $p$-adic bound of Lemma I.3 in Section I and the $q$-adic bound of Lemma J.3 in Section J. No new valuation is proved here; the wrapper records the joint denominator modules, the explicit $H_j \in B$ requirement, and the $W$-degree induction that closes global generation conditionally.

We reuse the five-variable dictionary of the kernel section verbatim, with no redefinition:

$$x = a_3, \qquad y = a_2, \qquad z = a_1, \qquad p = a_4, \qquad q = a_5,$$
$$B = k[x, y, z, p, q] = k[a_1, \dots, a_5], \qquad K = B^{D_5} = k[p, q, J_1, J_2],$$
$$J_1 = 2pz - y^2 = 2a_1a_4 - a_2^2, \qquad J_2 = 2qy - x^2 = 2a_2a_5 - a_3^2,$$
$$D_5(x) = pq, \qquad D_5(y) = px, \qquad D_5(z) = xy, \qquad D_5(p) = D_5(q) = 0.$$

We reuse the invariant of Definition 5.1 verbatim:

$$F_0 = p^2q^4a_0 - pq^3xyz + \tfrac{1}{3}pq^2x^3z + \tfrac{1}{3}q^2x^3y^2 - \tfrac{1}{5}qx^5y + \tfrac{1}{35}x^7,$$

and write $c = p^2q^4$, $G_0 = F_0 - ca_0 \in B$, $W = F_0/c = a_0 + \rho$ with $\rho = G_0/c \in B_{pq}$. A general localized invariant is written

$$F = \sum_{j=0}^{m} H_j W^j, \qquad H_j \in K_{pq},$$

with $a_0$-coefficients

$$C_k = \sum_{j=k}^{m} \binom{j}{k} H_j \rho^{j-k} \in B \qquad (0 \le k \le m).$$

The localized presentation $A_{pq}^{D_5} = K_{pq}[W]$ used here is the Conditional localized statement recorded with the slice certificate; it localizes only, and no global generation claim is built into the notation.

**Definition K.2** (The full denominator modules (Conditional))**.** For each $m \geq 0$ the $m$-th (full) denominator module is

$$\boxed{M_m = p^{2m} q^{4m} K = c^m K.}$$

It is the intersection of the two one-sided modules from the companion sections: $M_m^p = p^{2m} K$ of the $p$-adic section and $M_m^q = q^{4m} K$ of the $q$-adic section. The exponents record the exact pole orders $e_p = 2$ and $e_q = 4$ of $\rho$ at the two prime divisors; the asymmetry $2 \neq 4$ is why the two bounds cannot share a proof.

**Lemma K.3** (Joint denominator bound (Conditional on Lemmas P and Q))**.** *Assume the $p$-adic bound of Lemma I.3, $v_p(H_j) \geq 2j$ for every $j$, together with the companion $q$-adic bound of Lemma J.3, $v_q(H_j) \geq 4j$ for every $j$. Then for every $j$,*

$$\boxed{H_j \in p^{2j} q^{4j} K_{pq} = c^j K_{pq}.}$$

*Proof.* The two assumed bounds place $H_j$ in $p^{2j} K_{pq}$ and in $q^{4j} K_{pq}$ respectively. Since $p$ and $q$ are non-associated primes of the polynomial ring $B$ (hence of $K$ by the contraction identities of Lemma 4.3), the two divisibility conditions combine to $H_j / c^j \in K_{pq}$. □

**Lemma K.4** (The $H_j \in B$ requirement (Conditional on Lemmas P and Q))**.** *Assume the joint bound of Lemma K.3, so each $H_j = c^j \tilde{G}_j$ with $\tilde{G}_j \in K_{pq}$. Then every coefficient lies in $B$:*

$$\boxed{H_j \in B \quad \text{for every } j.}$$

*Explicitly, the top coefficient $H_m = [a_0^m]F$ lies in $B$ directly, and descending induction via the triangular system $\{C_k\}$ places each $H_j$ in $B$.*

*Proof.* $F \in A = k[a_0, \ldots, a_5]$ is a polynomial, so its $a_0$-coefficient $H_m = [a_0^m]F$ lies in $B$ directly; this is the leading-coefficient content of Lemmas I.5 and J.9 in joint form. For the descent, fix $k < m$ and assume $H_m, \ldots, H_{k+1}$ already known to be of the form $c^j G_j$ with $G_j \in K \subset B$. From $C_k = H_k + \sum_{j=k+1}^{m} \binom{j}{k} H_j \rho^{j-k} \in B$, each summand satisfies

$$H_j \rho^{j-k} = c^j G_j (G_0/c)^{j-k} = c^k G_j G_0^{j-k} \in B,$$

since $c$, $G_j$, and $G_0$ all lie in $B$ and $k \geq 0$. Hence $H_k = C_k - \sum_{j>k} \binom{j}{k} H_j \rho^{j-k} \in B$. Descending from $k = m-1$ to $k = 0$ gives $H_j \in B$ for all $j$. □

**Theorem K.5** ($W$-degree induction (Conditional on Lemmas P and Q))**.** *Assume Lemmas P and Q in full (Lemmas I.3 and J.3 with their leading-coefficient forms Lemmas I.5 and J.9), together with the descent intersection $B \cap K_{pq} = K$ of Theorem 4.4. Let $F \in A^{D_5}$ have $W$-degree $m$. Then $F \in K[F_0]$, by induction on $m$ following the skeleton of Remark I.7.*

*Proof.* By Lemma K.4 each $H_j$ lies in $B \cap c^j K_{pq}$. The descent intersection $B \cap K_{pq} = K$ of Theorem 4.4 gives $H_m = c^m G_m$ with $G_m \in K$, whence

$$H_m W^m = G_m (cW)^m = G_m F_0^m \in K[F_0].$$

The difference $F - G_m F_0^m$ is $D_5$-invariant of $W$-degree at most $m-1$. Subtracting lowers the $W$-degree; the base case $m = 0$ is $F = H_0 \in B \cap K_{pq} = K$. Induction on $m$ closes global generation conditionally: every invariant is a polynomial in $F_0$ over $K$. □

*Remark* K.6 (Status boundary). What this section attempted to prove, conditionally on Lemmas P and Q, is the combination step: the full modules $M_m = c^m K$, the explicit $H_j \in B$ descent through the triangular system $\{C_k\}$, and the $W$-degree subtraction closing $A^{D_5} = K[F_0]$. This attempt is now *retired, not repaired*: the $q$-premise (Lemma J.3) is refuted by the counterexample $H$ (Section J), so the induction (Theorem K.5) and the generation claim it was meant to close are void. The $p$-premise (Lemma I.3) is not contradicted by $H$ but is no longer needed, since the candidate hypersurface is identified directly (Theorem 7.1, part [D]). No N=5 generation claim is made in this section itself; the full-ring equality is proved later in Theorem 13.5. For the record, the gap is structural, not merely the falsity of the premises: even the conditional proof above is invalid as written, since $H_m \in B \cap c^m K_{pq}$ together with the descent intersection $B \cap K_{pq} = K$ does not yield $H_m = c^m G_m$ — that step needs the strictly stronger power-contraction $B \cap c^m K_{pq} = c^m K$ for every $m$, which is not among the hypotheses.

## L. Generation certificates

This section records what the machine establishes, in two layers. *Layer 1 (CAS):* the identity $S = 0$ and $D_5(H) = 0$, weight-28 uniformity (irreducibility of $S$ as the full relation ideal is now proved in Theorem 7.1, part [D], via primitivity of the $H$-linear $S$; the CAS log is retained as an independent cross-check), the $W$-expansion $H = H_2W^2 + H_1W + H_0$ with exact $p/q$-valuations, the residue cancellation $11025\,G_0^2 + 9J_2^7 \equiv 0 \pmod q$, and the kernel-dimension match $\dim\ker(D_5|_{V_w}) = s_6(w) - s_6(w-28)$ over $\mathbb{F}_{1000003}$ at anchor weights $5, 15, 23, 24, 25, 30, 35, 40$, with the first excess $+1$ at $w = 24$, rest on exact computation with rerunnable verification scripts (see Supplementary Material). The finite-field prime 1000003 anchors are consistency checks only and prove no characteristic-zero rank; the theorem-grade graded comparison is the exact-$\mathbf{Q}$ computation of Appendix P; the identity parts A–D are over $\mathbb{Q}$ exactly. *Layer 2 (Lean, new):* for every weight $w \le 14$, per-weight generation is now *directly verified* in Lean core (no Mathlib) against the formal $D_5$, by machine-checked integer witness certificates (Section L.1) — replacing the dimension argument in this regime — including a soundness fix for a latent CAS truncation bug (Section L.2).

### L.1 Direct per-weight generation through $W_{\max} = 14$ (Lean-verified)

**Theorem L.1** (Direct per-weight generation, $w \le 14$)**.** *Let $0 \le w \le 14$. Every invariant $F \in A_5^{D_5}$ of exact Fibonacci weight $w$ lies in the $k$-span of the weight-$w$ monomials in the five weight-$\le 14$ generators $p, q, J_1, J_2, F_0$ (equivalently, in the weight-$w$ monomials in the six generators, since $H$ has weight 24 and therefore does not occur in any weight-$w$ monomial with $w \le 14$). **[Lean-verified, native evaluation in core Lean without Mathlib (see the formal-verification manifest).]** For each weight $w \in \{0, 1, 4, 6\}$ (Band 1) and $w \in \{2, 3, 5, 7, 8, 9, 10, 11, 12, 13, 14\}$ (Band 2), the emitted integer witness bundle is re-verified entry by entry by machine-checked integer arithmetic against the matrix of $D_5|_{V_w}$ computed in Lean from the formal definition of $D_5$ (the formal weight-data module); the kernel of $D_5|_{V_w}$ is then* proved *to be the span of the selected generator-monomial columns by a residual argument using the verified left-inverse of $Q = M[S] \amalg K_g$ (the formal kernel-spanning module), and the coordinate vector is lifted back to an exact polynomial identity (the formal generation-spanning and generation-verification modules). The generic machinery depends only on the standard propositional-extensionality and choice foundation; the per-weight facts are exact machine-checked computations on the emitted data.*

**How the verification works.** For each weight $w$ exact rational computation (see Supplementary Material) emits an integer witness bundle: $M$, the integer matrix of $D_5|_{V_w}$ in the weight-$w$ monomial basis; $S$, the pivot-row index set (row-echelon form) with right inverse $T$ and row-reconstruction $C$ pinning the row space; $N$, a kernel basis with left inverse $K$ pinning the kernel dimension from both sides ($MN = 0$, $KN = d_3 I$); and $G_{\text{sel}}$, the columns of the selected weight-$w$ generator monomials, with left inverse $K_g$ pinning them as an independent set spanning the kernel ($K_g G_{\text{sel}} = d_g I$). *Every entry of the bundle is re-verified in Lean by machine-checked integer arithmetic against the matrix computed from the formal $D_5$* — the CAS output is untrusted input; trust sits in the checker. The kernel of $D_5|_{V_w}$ is then proved to be the span of the columns of $G_{\text{sel}}$ by the residual argument of the formal kernel-spanning module; the columns of $G_{\text{sel}}$ are, by the decode invariant of Section L.2, exactly the (denominator- cleared) rational coefficient vectors of the weight-$w$ generator monomials; and the lift from the coordinate vector back to a polynomial is an exact equality of normalized polynomials. Hence every weight-$w$ invariant is a scaled linear combination of weight-$w$ monomials, i.e. lies in their $k$-span. The verification is carried over $\mathbb{Q}$; all certificate data and the derivation have rational coefficients, and the weight-$w$ kernel over any characteristic-zero field $k$ is the $k$-extension of the one over $\mathbb{Q}$, so the statement holds over any such $k$.

### L.2 Soundness via denominator clearing

The certificates are integral, but the *true* coefficient vectors of the weight-$w$ generator monomials are rational and in general non-integral — e.g. a coefficient of $F_0$ at weight 14 is $\frac{1}{3}$. A naive `int()` truncation would *silently break the spanning claim* (turning $\frac{1}{3} \mapsto 0$ and dropping the whole $F_0$ column) and emit a certificate that *falsely* proves the weight-14 piece is spanned without $F_0$ — a vacuous, unsound "proof" at the most critical weight. All entries would still be integral and all matrix products would still hold; the certificate would merely be wrong.

The fix is exact denominator clearing. The producer computes the denominator-clearing factor

$$L_w \;=\; \operatorname{lcm}\big\{\text{denominators of all coefficients of the weight-}w\text{ generator monomials}\big\}$$

(called `D0` in the scripts and Lean modules) and emits $G_{\mathrm{sel}} = L_w \cdot$ (true rational vectors), integral by construction, folding $L_w$ into the scaling of $K_g$ so that $K_g G_{\mathrm{sel}} = d_g I$ stays integral. Lean then checks the *cross-multiplied exact* equality

$$G_{\mathrm{sel}}[i][j] \cdot c_{ij}^{den} \;=\; L_w \cdot c_{ij}^{num},$$

where $c_{ij}$ is the true rational coefficient of the $j$-th selected generator monomial in the $i$-th basis monomial ($c = c^{num}/c^{den}$, $c^{den} > 0$), together with $L_w \neq 0$, the row count, and index nodup/in-range (the formal decode invariant). Any truncated entry violates this exact equality, so the soundness fix is *enforced by the checker, not by convention*: the decode invariant asks for exact rational equality, not mere integrality. The same contract is what would keep the weight-24 $H$ columns sound if the pipeline is ever extended past $W_{\max} = 14$ (Section L.3). The general integer-certificate form of this contract (faithfulness versus spanning) is recorded in the graph-derivation dictionary (see Supplementary Material).

**Methodological point.** This is a concrete instance of the general principle that a formal checker, with the trust boundary drawn at *exact* integer equality rather than at integrality, converts a silent computational soundness bug into a hard compile-time error. The truncation bug is real: it is present in the obvious first draft of the producer and produces a certificate that looks valid (integral entries, all products holding) but is mathematically wrong. The only reason it is caught is that the decode invariant asks for exact rational equality. This is the methodological headline of the formalization: the latent bug is not that the CAS produced a wrong answer, but that an unchecked integrality test would have *certified* it.

### L.3 Frontier: what remains, and what is impossible

*Remark* L.2 (Frontier: six-generator spanning refuted at 46). Theorem L.1 establishes generation *directly* through $W_{\max} = 14$; exact $\mathbb{Q}$-ranks extend $C_5$-spanning exactly through weight 45 (Theorem 10.2). At weight 46 the six-generator claim fails: $\dim \ker = 1519 > 1518$ with explicit witness $F_{46}$ (Section 10). Two negative observations delimit the frontier precisely. *(i) No universal bound.* No explicit generator-degree bound $B$ exists for kernels of triangular locally nilpotent derivations in $\geq 5$ variables: the dimension-5 Daigle–Freudenburg kernel is not finitely generated [5], so finite generation — a fortiori any degree bound — fails in general. The proved closure $A_5^{D_5} = C_5[F_{46}^{\mathrm{int}}]$ (Theorem 13.5) is therefore specific to this derivation, not an instance of a general theorem; likewise the refuted $F_{46} \cdot K/(p^2)$ pattern (Conjecture 10.11) failed as a general tower law at weight 84. *(ii) The reduction premise cannot be linear.* A natural way to close generation by induction on weight would be the premise "every invariant of weight $> B$ is a scaled linear combination of strictly lower-weight invariants." This premise is *unrealizable for any finite* $B$: $p^{B+1}$ is an invariant of weight $B+1$ that is a monomial in the weight-1 generator $p$, hence cannot be a nonzero linear combination of invariants of weight $< B+1$ (their coefficient at weight $B+1$ is zero). A valid inductive reduction must allow *products* of lower-weight invariants — a genuinely different, and harder, statement than the linear spanning that the witness pipeline verifies. The formal closure assembly records exactly this: the closure theorem is a conditional implication whose displayed linear-reduction premise is unsatisfiable, and it is *not* presented as an unconditional generation theorem.

*Remark* L.3 (Why no direct generation proof is attempted). There is no direct generation proof to give: the weight-46 refutation (Section 10) settles the six-generator claim, and generation itself is closed by Theorem 13.5 ($A_5^{D_5} = C_5[F_{46}^{\mathrm{int}}]$ via slice atlas plus saturations): (i) uniqueness of $S$ as the full relation ideal holds at every exactly-checked weight (prior computations found no second relation through weight 57; the fresh exact-$\mathbb{Q}$ rerun reported here covers weights 15–46), now superseded by the all-weights relation ideal (Theorem 14.2); (ii) no all-weights six-generator argument exists, since the claim is false. The Hilbert series of $A_5^{D_5}$ is the kernel dimensions themselves, so no dimension argument can bypass computing them; the exact content is the $F_{46}$-descent and the exact presentation and Hilbert series (Theorems 14.2–14.4), with the $a_0$-degree-1 closure (Theorem N.3) and the bounds-generation characterization (Theorem N.5) as further proved structural theorems. *Update: the $w \leq 14$ part is a direct Lean-verified spanning (Theorem L.1), and 15–45 an exact $\mathbb{Q}$-rank spanning (Theorem 10.2).* What was "the remainder" is now a refutation with an exact successor structure, not a gap in the certificate.

*Remark* L.4 (Formalized generation outline). The Lean core now covers, in addition to the relation, the six weights, and strictness (the formal hypersurface, exclusion, and counterexample modules with native evaluation in core Lean without Mathlib (see the formal-verification manifest)), the full per-weight witness pipeline through $W_{\max} = 14$: the formal witness module (integer matrix API, corrected Steinitz exchange), the formal weight-data module (formal weight-$w$

matrix of $D_5$), the formal witness-data modules (emitted integer certificates, machine-checked), and the assembly modules (kernel span, generation span, closure, bundle, and verification), and the second witness (formally verified $F_{46}$ invariant of weight 46). What remains for a full formalization is the closure/saturation machinery (slice charts, $p$-/$q$-saturation, relation ideal), now proved at paper level: the superseded successor module pattern (Conjecture 10.11, refuted at weight 84) is no longer a formalization target; the open Lean remainder is case-by-case extension of the per-weight certificates past 14 (the $w = 15$ pilot is emitted and its explicit target is Lean-checked; weights $15 \dots 25$ remain open in Lean, though exact $\mathbb{Q}$-ranks cover 15–45 at paper level) and the `a0top_filter_combine` step.

*Remark* L.5 (Status at $w = 24$). The kernel dimension at 24 is 213, CAS-verified; Lean verification reaches only weight 14. The full $w = 24$ bundle ($n = 1461$, $k = 213$, roughly $150\times$ the $w = 14$ cost) is beyond the current certificate range. Recorded substitutes: $L_w(24) = 105$ stability, $s_6 - s_5 = 1$ at 24, and the $a_0$-window $[16, 27]$ (Corollary 9.2).

## M. A certification principle for graded kernel spanning

Status of this section: reusable method. The per-weight certificate used for Theorem L.1 is stated here independently of $N = 5$. Hypotheses: triangular derivation (finite-dimensional weight spaces), positive grading, $L_w \neq 0$, and the integer left-inverse identities below (over $\mathbb{Z}$, not merely $\mathbb{Q}$).

**Theorem M.1** (Q-left-inverse kernel spanning)**.** *Assume $M$ is $n \times n$; $G_{\mathrm{sel}}$ is $n \times k$ with $k > 0$ and $MG_{\mathrm{sel}} = 0$; $K_g$ is $k \times n$ with $K_g G_{\mathrm{sel}} = d_g I_k$ for $d_g \neq 0$; and there are $Y$ and $d_Y \neq 0$ with $YQ = d_Y I_n$, where $Q$ is the vertical stacking of the pivot rows $M[S]$ on top of $K_g$ (row-block concatenation, never matrix addition). Then every integer $v$ with $Mv = 0$ satisfies $d_g v = G_{\mathrm{sel}}(K_g v)$.*

Status: checked in core Lean without Mathlib by native evaluation (see the formal-verification manifest below); the residual $z = d_g v - G_{\mathrm{sel}}(K_g v)$ satisfies $Mz = 0$ and $K_g z = 0$, whence $d_Y z = 0$ and $z = 0$ over the characteristic-zero field. This follows the certificate-checking pattern (external producer plus proved checker [15, 24, 14]): Gröbner/elimination facts used by the CAS side are classical [4, 13, 25]. Unlike [14], which formalizes general Gröbner-basis theory in Lean with Mathlib, our formal layer is a small theorem-specific core-Lean certification of individual identities, not a general formalization.

### M.1 Formal-verification manifest

Lean 4.33.1, Lake 5.0.0, core Lean without Mathlib, base commit `7da6b63`. Every proposition below is discharged by `native_decide`; kernel reduction by `decide` was attempted for the flagship $F_0$ identity and fails on the custom normalization/rational API, so the native-evaluation trust model is retained. `#print axioms` reports the standard foundation plus theorem-specific native-evaluation axioms in each case:

| theorem | module | source file | build | axioms beyond `propext`, `Classical.choice`, `Quot.sound` |
|---|---|---|---|---|
| `D_F0_zero` | `CompLemmas` | `AdjQuad/CompLemmas.lean` | `lake build` | one native axiom |
| `H_D_zero` | `Counterexample` | `AdjQuad/Counterexample.lean` | `lake build` | one native axiom |
| `weightedDegree_H`, `H_has_a0_sq` | `Counterexample` | `AdjQuad/Counterexample.lean` | `lake build` | one native axiom each |
| `H_notin_K_F0` | `Exclusion` | `AdjQuad/Exclusion.lean` | `lake build` | native axioms of the listed base facts + two native axioms (§G.1 records the full output) |
| `S_rel` | `Hypersurface` | `AdjQuad/Hypersurface.lean` | `lake build` | one native axiom |
| `F46_D_zero`, `weightedDegree_F46`, `F46_has_a0_fourth` | `SecondWitness` | `AdjQuad/SecondWitness.lean` | `lake build` | one native axiom each |
| per-weight spanning $w \leq 14$ | `GenerationVerified` et al. | `AdjQuad/Generation*.lean` | `lake build` | native witness-check axioms per weight |

`H_notin_K_F0` proves nonmembership in the inductively generated predicate `inK_F0` (the smallest subalgebra: 0, 1, the five generators, addition, multiplication) via the safety invariant $14\deg_{a_0} \le \mathrm{wt}$ carried by structural induction; its mathematical reading as $H \notin k[p,q,J_1,J_2,F_0]$ additionally uses the paper-level API bridge (every mathematical subalgebra expression is an inductive construction; normalization faithfulness is proved by the `normalize_mem_exp` family). By contrast $F_{46} \notin C_5$ is not a Lean claim: Lean covers definition, $D_5(F_{46}) = 0$, weight, and top, while nonmembership is the paper-level $J_4$ valuation. `AdjQuad.Counterexample` and `AdjQuad.SecondWitness` were rebuilt fresh (`lake build`, success); remaining modules are reported from present `.olean` artifacts with unmodified sources.

*Remark* M.2 (Denominator soundness rejects truncation). Let $E$ be the true rational coefficient matrix of the selected generator monomials, $L_w$ the l.c.m. of all its denominators ($L_w \neq 0$), and $G_{\mathrm{sel}} = L_w E$ integral. The checker enforces entrywise $G_{\mathrm{sel}}[i][j] \cdot c^{den} = L_w \cdot c^{num}$ by machine-checked integer arithmetic (decode-entry and cross-multiplication checks). The naive truncation $1/3 \mapsto 0$ in $F_0$ at weight 14 fails this check ($0 \cdot 3 \neq 1 \cdot 1$); the scaled entry 35 passes ($35 \cdot 3 = 105 \cdot 1$). The interface is pinned by the formal certification-principle module (with three type-checked interface lines; an upstream rename breaks the build loudly). General form: faithfulness (cross-multiplied representation) versus spanning (left-inverse identities) are separated as Theorems F and S of the graph-derivation dictionary (see Supplementary Material); this remark is the faithfulness half.

### M.2 Filtration certificates

Status of this subsection: reusable method (Theorem IV of the filtration tower). It fixes a producer/checker split for every future filtration claim of the form "$F$ is an invariant of weight $w$ and $a_0$-degree $m$ with top outside $J_m$" (Definition N.4, Corollary N.6). The split is the same philosophy as the decode invariant of Section L.2: the producer is untrusted and only the checker's exact integer/rational equalities carry weight.

**Trust boundary.** The producer — exact Sage solves and SymPy $W$-expansion and witness-emission computation — suggests a candidate $F$. Nothing it outputs is trusted: supports may be truncated, denominators dropped, tops misread. The checker (exact SymPy polynomial arithmetic over $\mathbb{Q}$, stdlib only; see Supplementary Material) re-verifies four gates from the polynomial itself:

(i) $D_5(F) = 0$ *exactly*: the triangular image is accumulated monomial by monomial ($D_5(a_0) = a_1a_2$, $D_5(a_1) = a_2a_3$, $D_5(a_2) = a_3p$, $D_5(a_3) = pq$, $D_5(p) = D_5(q) = 0$) and every coefficient of the image must be identically zero over $\mathbb{Q}$.

(ii) Uniform Fibonacci weight: every monomial of $F$ weighs $w$ under $(8,5,3,2,1,1)$.

(iii) $a_0$-degree plus top: $\mathtt{Poly}(F, a_0)$ has degree exactly $m$ and its top coefficient $C_m = [a_0^m]F$ is reported verbatim.

(iv) $J_m$ (non-)membership by exact divisibility: with $J_m = (p^{2m}q^{4\lceil m/2\rceil})$ principal (Definition N.4; exponent-vector check of exact top-ideal computation), a top in $k[p,q]$ lies in $J_m$ iff every monomial's $(p,q)$-exponents dominate the target, i.e. $v_p \ge 2m$ and $v_q \ge 4\lceil m/2\rceil$. Coefficients are ignored (units in characteristic zero); only exponent dominance counts. The zero top is vacuously in every $J_m$.

A certificate is the quadruple (D-zero proof, weight uniformity, degree + top, divisibility verdict); rerunnable with self-test (see Supplementary Material; the final verdict line reads `filtration-cert OK`).

**First certified instance: $F_{46}$.** The witness (see Supplementary Material) (79 terms, integral cleared form) passes all four gates at $(w,m) = (46,4)$: $D_5(F_{46}) = 0$ exactly, uniform weight 46, $a_0$-degree exactly 4 with top $121550625\,p^6q^8$ (normalized $\frac{18375}{4}p^6q^8$, Theorem 10.3), and $v_p = 6 < 8$ so the top lies outside $J_4 = (p^8q^8)$ (Lemma 10.4). The Lean formalization covers definition, $D_5(F_{46}) = 0$, uniform weight, and top, checked in core Lean without Mathlib by native evaluation (see the formal-verification manifest); non-membership is the paper-level $J_4$ divisibility step recorded here, machine-reproducible through the checker.

**Controls (polarity).** The self-test ships two doctored-input controls with fixed expected polarity, so a vacuous checker cannot pass. (a) The top altered to $p^8q^8$ *must be accepted* by `in_Jm` at $m = 4$ (it is the $J_4$ generator itself), at both the top-expression level and the poly level with the top term swapped. (b) $F_{46}$ plus a weight-homogeneous $a_0^5p^3q^3$ term

($5 \cdot 8 + 3 + 3 = 46$, so the weight gate still passes) *must be rejected* by the degree gate expecting $m = 4$ (observed degree 5). Both controls pass with the specified polarity; inverting either turns the final verdict to `filtration-cert FAILED`.

**Reuse contract.** Future tower claims (degree-3 census extensions, $F_{46} \cdot F_0$-family members at $m = 5$ with $J_5 = (p^{10}q^{12})$, any new primitive direction) certify by instantiating the same four gates at their $(w, m)$ with the same script: only the expected weight, degree, and $J_m$ target change. The $J_m$ formula itself stays pinned by exact top-ideal computation.

### M.3 Proof-strategy classification

**Table 4. Proof strategies: what each proves here, what it cannot prove, and its failure mode.**

| method | hypotheses | proves here | cannot prove | failure mode |
|---|---|---|---|---|
| slice/localization | LND + slice element | $B_{pq} = K_{0,pq}[s]$, localized kernels | global generation | pole divisor $\{c = 0\}$ |
| divisor contraction | UFD + injective specialization | descent $B \cap K_{0,pq} = K_0$ | missing directions | non-injective specialization |
| residue-field analysis | valuation + initial form | transcendence⇒bound; setup data | bounds past algebraicity degree | $r_q^2$, deg-4 $r_p$ minpoly dependence |
| $a_0$-filtration induction | tops in $J_m$ | generation $\iff$ bounds | the bounds themselves | $F_{46}$ top outside $J_4$ |
| denominator modules | explicit $c$, $W$ | $C_5$-side $J_m$ exactly | full $\mathcal{M}_m$ | $H$, $F_{46}$ denominators |
| Gröbner elimination | finite ideal data | $S = 0$, exact per-weight dims | all-weights statements | $n = 112241$ wall at 70 |
| SAGBI | monomial order + finite basis | not attempted here | — | unknown termination |
| representation theory | reductive group action | not applicable (unipotent triangular) | — | — |
| graded certificates | per-weight matrices | exact spanning/dims per weight | all-weights | exponential growth (wall at 69) |
| formal proof (Lean core) | list-matrix API | identities, spanning $\leq 14$ | fraction fields, `a0top` combine | infrastructure gaps |

The $N = 5$ $q$-adic failure is the canonical example of a proof-method failure: a valid reduction (constant coefficient in $B$) followed by an inference (transcendence) whose hypothesis is false ($r_q$ algebraic), producing a bound refuted by $H$ at the first possible degree. The $p$-adic program fails the same way at degree 4 (witness $F_{46}$).

## N. Top-ideal machinery and birth-window census (supporting record)

This appendix collects the $J_m$ filtration theory and birth-window census supporting the main text. Conjectural items are marked as such; nothing here is a premise of the closure theorem.

**Conjecture N.1** (resonance degree equals birth degree)**.** Let $\rho_q = 2$ and $\rho_p = 4$ be the exact per-prime resonance degrees (Definition 21.1). The first $W$-degree carrying an invariant outside the previously generated subalgebra equals a resonance degree: $m = 2$ on the $q$-side (witness $H$, whose $W$-expansion realizes the $r_q^2$-dependence) and $m = 4$ on the $p$-side (witness $F_{46}$, with $v_p = 6 < 8$ at $m = 4$). Consistent instances: $m = 1$ lies below both resonance degrees and is closed (Theorem N.3); $m = 3$ matches no resonance degree and shows no $J_3$-violating top through weight 60 (Remark N.9). No production mechanism is claimed: resonance does not by itself produce the generator. Falsifier: a primitive at $W$-degree $m$ with $m$ equal to no exact resonance degree, or an exact resonance degree $d$ with the $W$-degree-$d$ slice fully verified and no new direction.

**Theorem N.2** ($F_{46}$ realizes the $\rho_p$-cancellation)**.** *Let $F_{46} = \sum_{j=0}^{4} H_j W^j$ be the $W$-expansion ($H_3 = 0$). Every nonzero $W$-coefficient has $p$-pole order exactly $-2$ in the constant term: $v_p(H_j) - 2j = -2$ for $j = 4, 2, 1, 0$ (exact: $6-8$, $2-4$, $0-2$, $-2$), and the four per-term $p$-residues $E_j = [p^2 H_j \rho^j]_{p=0}$ are well defined with $E_4 \neq 0$ and $\sum_j E_j = 0$ exactly (exact computation (see Supplementary Material)). Since $\rho_p = 4$ exactly (, exact rational computation), $1, r_p, r_p^2, r_p^3$ are independent over $L_p$, so this quartic relation is an $L_p$-unit multiple of the minimal polynomial $m_p$: the $p$-side failure has the same shape as the $q$-side, with parallel data*

| | $m$ | $e$ | *top* | $C_0$*-order* |
|---|---|---|---|---|
| $H/q$ | 2 | 4 | $v_q = 4$ | $-4$ |
| $F_{46}/p$ | 4 | 2 | $v_p = 6$ | $-2$ |

*(constant-term pole cancellation governed by the minimal polynomial: quadratic at $q$, quartic at $p$). What is not shown: that resonance produces the generator (the existence direction remains open; $F_{46}$ itself is the existence witness, not its consequence). In particular, tower closure is independent of resonance finiteness: $c = p^2q^4$ has exactly the prime divisors $p$ and $q$, so the resonance inventory $\{\rho_q, \rho_p\}$ is already complete and finite, yet the tower question stays open. A third primitive, if any, would either falsify the necessity direction of Conjecture N.1 or demand an extended resonance framework beyond primes dividing $c$.*

### N.1 Strongest correct structural theorems

The refutation leaves two unconditional structural theorems. The first closes $a_0$-degree 1 with no saturation input.

**Theorem N.3** ($a_0$-degree-1 closure)**.** *Every $D_5$-invariant of $a_0$-degree at most 1 lies in $K[F_0] \subseteq C_5$.*

*Proof.* Let $F = C_1 a_0 + C_0$ with $C_i \in B$. From $D_5(F) = 0$, freeness of $B[a_0]$ gives $D_5(C_1) = 0$ (so $C_1 \in K$ by Theorem 4.4) and $D_5(C_0) + C_1 a_1 a_2 = 0$. The Established Dixmier divisibility (Lemma 4.5) gives $c \mid C_1$ in $K$, $C_1 = ce_1$. Then $F - e_1 F_0 = C_0 - e_1 G_0 \in B$ is invariant, hence in $K$. So $F \in K[F_0]$. □

The second turns the failed induction into an exact characterization. It is stated inside the $W$-expansion setup only: with $c = p^2q^4$, $W = F_0/c = a_0 + \rho$, every localized invariant writes uniquely $F = \sum_{j=0}^{m} H_j W^j$ with $H_j \in K_{pq}$ ($W$ monic of $a_0$-degree 1, so $W$-degree equals $a_0$-degree $m$ and the top $W$-coefficient equals the top $a_0$-coefficient $C_m = [a_0^m]F$, which lies in $K$ unconditionally by Theorem 4.4). Valuations $v_p, v_q$ are the UFD valuations along $(p), (q)$, restricted to $K$ via Lemma 4.3 ($K$ a polynomial ring, $(p), (q)$ non-associated primes). Put $h_2 = 11025p^4q^4 = \mathrm{top}(H)$ and $c = \mathrm{top}(F_0)$.

**Theorem N.4** (Top ideals of $C_5$)**.** *Let $c = \mathrm{top}(F_0) = p^2q^4$ and $h_2 = \mathrm{top}(H) = 11025p^4q^4$, with $\deg_{a_0} F_0 = 1$ and $\deg_{a_0} H = 2$. For $m \geq 0$ let $J_m \subset K$ be the $K$-span of the products $c^e h_2^r$ with $e + 2r = m$. Then*

$$J_m = (p^{2m} q^{4\lceil m/2 \rceil}),$$

*principal for every $m$. In particular $C \in J_m$ iff $v_p(C) \geq 2m$ and $v_q(C) \geq 4\lceil m/2 \rceil$.*

*Proof.* $B[a_0]$ with $B = k[a_1, \ldots, a_5]$ is a domain, and $c, h_2 \neq 0$. Hence for $F_0^e H^r$ the $a_0$-degree adds and tops multiply:

$$\mathrm{top}(F_0^e H^r) = c^e h_2^r = 11025^r p^{2e+4r} q^{4e+4r}.$$

If $e + 2r = m$ then $2e + 4r = 2(e + 2r) = 2m$ and $4e + 4r = 4m - 4r$, i.e.

$$\mathrm{top}(F_0^e H^r) = 11025^r p^{2m} q^{4m-4r}.$$

Since $0 \leq r \leq \lfloor m/2 \rfloor$, $v_p = 2m$ for every generator. For $v_q = 4m - 4r$, the minimum over admissible $r$ occurs at maximal $r = \lfloor m/2 \rfloor$, giving $4m - 4\lfloor m/2 \rfloor = 4\lceil m/2 \rceil$. The monomial $F_0^{m \bmod 2} H^{\lfloor m/2 \rfloor}$ satisfies $e + 2r = m$ and attains $(2m, 4\lceil m/2 \rceil)$: if $m = 2k$, $(e, r) = (0, k)$ gives $p^{4k}q^{4k}$; if $m = 2k + 1$, $(e, r) = (1, k)$ gives $p^{4k+2}q^{4k+4}$. Every other generator differs by $(0, 4(r^\star - r))$ in $(p, q)$-exponents with $r^\star = \lfloor m/2 \rfloor$, hence is a $K$-multiple of the extremal one (11025 is a unit in characteristic zero). Thus the $K$-span is the principal ideal displayed. No division by $q$ and no use of the hypersurface relation is made. □

**Theorem N.5** (Bounds-generation characterization)**.** *Over the Established base (descent, Dixmier, $D_5(F_0) = D_5(H) = 0$, presentation; characteristic zero for Dixmier), with (P-top) $v_p(C_m) \geq 2m$ and (Q-top-strong) $v_q(C_m) \geq 4\lceil m/2 \rceil$ for the top $a_0$-coefficient $C_m \in K$ of every invariant of $a_0$-degree $m$: (i) if (P-top) and (Q-top-strong) hold for every invariant, then $A_5^{D_5} = C_5$; (ii) if $A_5^{D_5} = C_5$, then (P-top) and (Q-top-strong) hold for every invariant.*

*Proof.* (i) Induction on $m$. Let $F$ have $a_0$-degree $m$ with top $C_m \in K$. The bounds give $C_m \in J_m$, so $C_m = \sum_{e+2r=m} \alpha_{e,r} c^e h_2^r$ with $\alpha_{e,r} \in K$. Then $G = \sum \alpha_{e,r} F_0^e H^r \in C_5$ has the same top $a_0^m$-term (tops multiply over the domain $B[a_0]$), so $F - G$ is invariant of $a_0$-degree $< m$; by induction $F - G \in C_5$, hence $F \in C_5$. Base $m = 0$ is Theorem 4.4. (ii) Let $F \in A_5^{D_5} = C_5$ have $a_0$-degree $m$. Write $F$ as a $K$-combination of $C_5$-monomials; those of $a_0$-degree $m$ contribute tops in $J_m$ and lower-degree monomials do not affect the top, so $C_m \in J_m$, i.e. both bounds hold by Definition N.4. □

**Corollary N.6** (Obstruction criterion)**.** *If some invariant has top $a_0$-coefficient outside $J_m$ for some $m$, then $A_5^{D_5} \neq C_5$. $F_{46}$ realizes this at $m = 4$: $\frac{18375}{4} p^6 q^8 \notin J_4 = (p^8 q^8)$. Since generation is false (Theorem 10.1), at least one bound is false: (P-top) is, witnessed by $F_{46}$ ($v_p = 6 < 8$ at $m = 4$). (Q-top-strong) holds (Theorem 15.1), so the obstruction loads entirely onto the $p$-side.*

Lean covers invariance + fingerprint; non-membership is paper-level + Sage-exact; the blocking formal step is filter-through-normalize commutation for the $a_0$-top slice (`a0top_filter_combine`).

**Theorem N.7** (Associated graded of $C_5$)**.** *Filter $C_5$ by $a_0$-degree, $F_{\leq m} = \{G \in C_5 : \deg_{a_0} G \leq m\}$. Its associated graded $K$-algebra is*

$$\mathrm{gr}_{a_0}(C_5) \;=\; \bigoplus_{m \geq 0} J_m \, t^m \quad \subseteq \quad K[t],$$

*with the principal ideals $J_m = (p^{2m} q^{4\lceil m/2 \rceil})$ of Definition N.4 and multiplication induced by tops: the degree-$m$ piece is exactly the $K$-span of the tops $c^e h_2^r$, $e + 2r = m$, and tops multiply over the domain $B[a_0]$, so $J_m \cdot J_n \subseteq J_{m+n}$. A finite presentation of this graded algebra is not established here. The analogue for $A_5^{D_5}$ is open: its graded pieces strictly contain the $J_m$ from $m = 4$ (witness $F_{46}$).*

**Theorem N.8** (Primitive birth windows)**.** *A primitive $a_0$-degree-$m$ direction outside $C_5$, if born at weight $w$, satisfies $8m \leq w \leq \min C_5(m) - 1$, where $\min C_5(m) = \min_{e+2r=m}(14e + 24r)$ is the lightest $C_5$-monomial weight at that $a_0$-degree. The values (exhaustive integer check in exact top-ideal computation):*

| $m$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| $\min C_5(m)$ | 14 | 24 | 38 | 48 | 62 | 72 | 86 | 96 |
| *gap* | $[8, 13]$ | $[16, 23]$ | $[24, 37]$ | $[32, 47]$ | $[40, 61]$ | $[48, 71]$ | $[56, 85]$ | $[64, 95]$ |

*Known window examples (mostly non-primitive): $m = 1$ is closed by Theorem N.3 (no primitive survives the gap); $m = 2$ holds $H$ at 24 ($H \in C_5$ defines the minimum); $m = 3$ holds no $J_3$-violating top through 60 (Remark N.9); $m = 4$ holds $F_{46}$ at 46; $m = 5$ holds $F_{46} \cdot F_0$ at 60 (non-primitive); $m = 6$ holds $F_{46} \cdot H$ at 70 (non-primitive). A third primitive direction of degree 5 would be born in $[40, 61]$, of degree 6 in $[48, 71]$, of degree 7 in $[56, 85]$; no such weights are excluded. This replaces weight-difference numerology: the windows come from monomial arithmetic, not from extrapolating $10, 22, \ldots$.*

*Remark* N.9 (Degree-3 census (EXACT-$\mathbb{Q}$ through 38–60)). For each weight $38 \leq w \leq 60$, let $N_{\leq 3}(w)$ be the exact $\mathbb{Q}$-nullity of $D_5$ on $V_w^{a_0 \leq 3}$ (blocked $B_w, B_{w-8}, B_{w-16}, B_{w-24}$ construction) and $N_J(w)$ the nullity of the $J_3$-constrained $a_0$-chain (top $C_3$ parametrized as $p^6 q^8 u$); a $J_3$-violating $a_0$-degree-3 top exists iff $N_J(w) < N_{\leq 3}(w)$ (exact computation (see Supplementary Material); divisibility by the monomial $p^6 q^8$ is exact remainder-zero division). Outcome: $N_J(w) = N_{\leq 3}(w)$ at every $w$ (no violation). With $\dim T(w) = N_{\leq 3}(w) - N_{\leq 2}(w)$ counting the degree-3 tops present:

**Table 5. Degree-3 census counts and $J_3$-violation verdicts, weights 38–60.**

| $w$ | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\dim T$ | 1 | 2 | 3 | 4 | 6 | 8 | 11 | 14 | 18 | 22 | 27 | 32 |
| viol. | no | no | no | no | no | no | no | no | no | no | no | no |

| $w$ | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $\dim T$ | 39 | 46 | 54 | 62 | 72 | 82 | 94 | 106 | 120 | 134 | 150 |
| viol. | no | no | no | no | no | no | no | no | no | no | no |

i.e. every $a_0$-degree-3 invariant top through weight 60 lies in $J_3 = (p^6q^8)$ (at $w = 46$, $N_{\le 3} = 1518$ reproduces $\dim(C_5)_{46}$, since every weight-46 $C_5$-member has $a_0$-degree $\le 3$). This is evidence, not proof: no new primitive direction of $a_0$-degree 3 occurs through 60; falsifier: any $w$ with $N_J(w) < N_{\le 3}(w)$.

## O. Explicit witnesses $H$ and $F_{46}$

This appendix records the two hidden invariants in auditable form. The main text proves what they imply; here the reader can check what they are. Coefficient vectors are in $(a_0, \dots, a_5)$-exponents with Fibonacci weights $(8, 5, 3, 2, 1, 1)$.

### O.1 The weight-24 invariant $H$

$H$ has 17 terms, content 1, uniform Fibonacci weight 24, ordinary degree 10, and $a_0$-degree 2:

| coefficient | $(a_0, a_1, a_2, a_3, a_4, a_5)$-exponents |
|---|---|
| 11025 | $(2, 0, 0, 0, 4, 4)$ |
| $-22050$ | $(1, 1, 1, 1, 3, 3)$ |
| 7350 | $(1, 1, 0, 3, 3, 2)$ |
| 7350 | $(1, 0, 2, 3, 2, 2)$ |
| $-4410$ | $(1, 0, 1, 5, 2, 1)$ |
| 630 | $(1, 0, 0, 7, 2, 0)$ |
| 11025 | $(0, 2, 2, 2, 2, 2)$ |
| $-7350$ | $(0, 2, 1, 4, 2, 1)$ |
| 1225 | $(0, 2, 0, 6, 2, 0)$ |
| 6720 | $(0, 1, 5, 0, 1, 3)$ |
| $-16800$ | $(0, 1, 4, 2, 1, 2)$ |
| 9450 | $(0, 1, 3, 4, 1, 1)$ |
| $-1540$ | $(0, 1, 2, 6, 1, 0)$ |
| $-2208$ | $(0, 0, 7, 0, 0, 3)$ |
| 4368 | $(0, 0, 6, 2, 0, 2)$ |
| $-2352$ | $(0, 0, 5, 4, 0, 1)$ |
| 385 | $(0, 0, 4, 6, 0, 0)$ |

Direct expansion gives $D_5(H) = 0$; every monomial weighs 24; the unique $a_0^2$-term is $11025a_0^2a_4^4a_5^4$. With $c = p^2q^4$, $W = F_0/c$, $H_2 = 11025p^4q^4$ and $H_0 = 105J_2^5J_1/q^2 + 9J_2^7/q^4$, $H = H_2W^2 + H_0$ with $H_1 = 0$. The Lean source is `def H` in `formalize/AdjQuad/Counterexample.lean`.

### O.2 The weight-46 invariant $F_{46}$

$F_{46}$ has 79 terms, content 1, uniform Fibonacci weight 46, ordinary degree 18, and $a_0$-degree exactly 4. Direct expansion gives $D_5(F_{46}) = 0$. Its unique $a_0^4$-term is

$$121550625\, a_0^4a_4^6a_5^8, \qquad \text{i.e.} \quad [a_0^4]F_{46} = \frac{18375}{4}p^6q^8 \text{ normalized},$$

with $v_p = 6 < 8$, hence outside $J_4 = (p^8q^8)$. The full 79-term support is machine-readable as `def F46` in `formalize/AdjQuad/SecondWitness.lean`; the four-gate filtration certificate (derivation image, uniform weight, $a_0$-degree plus top, $J_4$-divisibility with opposite-polarity controls) reproduces the verdict `filtration-cert OK` from the polynomial itself. No compact closed form is known; the file plus the checksum recorded in the reproducibility supplement is the canonical reference.

## P. Exact graded comparison, weights 15–46

For each Fibonacci weight $w$ let $V_w$ be the weight-$w$ monomial space, $M_w$ the matrix of $D_5|_{V_w}$ over $\mathbf{Q}$, $N(w) = \dim\ker M_w$, and $G(w)$ the rank of the $C_5$-monomial images in $V_w$. The $C_5$-formula is $s_6(w) - s_6(w-28)$ with $s_6$ the monomial count in generator weights $(1,1,6,4,14,24)$. Ranks are exact rational sparse ranks (Sage, LinBox). Finite-field runs at $\ell = 1000003$ are independent consistency checks only and prove nothing.

| $w$ | $\dim V_w$ | $\operatorname{rank} M_w$ | $N(w)$ | $\operatorname{rank} G(w)$ | excess |
|---|---|---|---|---|---|
| 15 | 293 | 229 | 64 | 64 | 0 |
| 16 | 361 | 286 | 75 | 75 | 0 |
| 17 | 440 | 354 | 86 | 86 | 0 |
| 18 | 533 | 433 | 100 | 100 | 0 |
| 19 | 640 | 526 | 114 | 114 | 0 |
| 20 | 764 | 633 | 131 | 131 | 0 |
| 21 | 906 | 758 | 148 | 148 | 0 |
| 22 | 1068 | 900 | 168 | 168 | 0 |
| 23 | 1252 | 1064 | 188 | 188 | 0 |
| 24 | 1461 | 1248 | 213 | 213 | 0 |
| 25 | 1696 | 1458 | 238 | 238 | 0 |
| 26 | 1961 | 1694 | 267 | 267 | 0 |
| 27 | 2257 | 1961 | 296 | 296 | 0 |
| 28 | 2588 | 2258 | 330 | 330 | 0 |
| 29 | 2956 | 2592 | 364 | 364 | 0 |
| 30 | 3365 | 2961 | 404 | 404 | 0 |
| 31 | 3817 | 3373 | 444 | 444 | 0 |
| 32 | 4317 | 3827 | 490 | 490 | 0 |
| 33 | 4867 | 4331 | 536 | 536 | 0 |
| 34 | 5472 | 4884 | 588 | 588 | 0 |
| 35 | 6135 | 5495 | 640 | 640 | 0 |
| 36 | 6861 | 6161 | 700 | 700 | 0 |
| 37 | 7653 | 6893 | 760 | 760 | 0 |
| 38 | 8517 | 7689 | 828 | 828 | 0 |
| 39 | 9456 | 8560 | 896 | 896 | 0 |
| 40 | 10477 | 9505 | 972 | 972 | 0 |
| 41 | 11583 | 10535 | 1048 | 1048 | 0 |
| 42 | 12781 | 11647 | 1134 | 1134 | 0 |
| 43 | 14075 | 12855 | 1220 | 1220 | 0 |
| 44 | 15472 | 14156 | 1316 | 1316 | 0 |
| 45 | 16977 | 15565 | 1412 | 1412 | 0 |
| 46 | 18597 | 17078 | 1519 | 1518 | 1 |

At $w = 24$ the five-generator count is $s_5(24) = 212 < 213 = N(24)$, the $H$-excess; at $w = 46$ the six-generator formula gives $1518 < 1519$. Hence 24 is the first five-generator failure and 46 the first six-generator failure. The consequence used in the main text is the boundary pair (45 equality, 46 excess one); the interior rows are retained here so the firstness claim is auditable at every weight.

## Q. Bounded ambient-derivation search

$\mathrm{ML}(A_5^{D_5})$ is Open. Every triangular derivation is locally nilpotent; among triangular derivations of $A_5$ of total degree at most 2 with $E(a_5) = 0$ (55 parameters, as $D_5$ itself kills $a_5$), the subspace preserving $K = k[p, q, J_1, J_2]$ has

dimension 32 (exact $\mathbb{Q}$ nullspace, exact computation (see Supplementary Material)), and none of its directions kills $F_0$ or $H$. Hence no triangular degree-$\leq 2$ locally nilpotent derivation moves the hypersurface generators: a weak bounded negative. Whether any preserving direction sends $F_0$ or $H$ to a nonzero element of the hypersurface ring (which would shrink the invariant), and everything about non-triangular or higher-degree derivations, is untested. Restriction of the tested derivations to $A_5^{D_5}$ (or $C_5$) is not established, so this is a bounded search among ambient derivations, not a computation of $\mathrm{ML}$. This is a bounded negative result: no $\mathrm{ML}$ value is computed or claimed.

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